YASKAWA

LA500

AC Micro Drives for Industrial Applications

Technical Manual

Type **CIPR-LA50Axxxxxxxx**

Models **400 V Class, Three-Phase Input: 3.0 to 22 kW**

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Table of Contents

i. Preface and General Precautions 11
i.1 Receiving 12
Glossary 12
About Registered Trademarks 12
i.2 Using the Product Safely 13
Explanation of Signal Words 13
Section Safety 13
Warning Label Content and Location 15
i.3 Warranty Information 17
Exclusion of Liability 17
1. Receiving 19
1.1 Model Number and Nameplate Check 20
Nameplate 20
How to Read the Model Number 20
1.2 Features and Advantages of Control Methods 22
2. Mechanical Installation 23
2.1 Section Safety 24
2.2 Installation Environment 25
2.3 Installation Position and Clearances 26
Install Single Drive 26
Install the Drive Horizontally 26
Install Drives Side-by-Side 28
2.4 Moving the Drive 29
2.5 Drive Models and Drive Watt Loss 30
Drive Watt Loss (without Built-in EMC Filter) 30
Drive Watt Loss (with Built-in EMC Filter) 30
2.6 Removing/Reattaching Covers 31
Remove the Front Cover 31
Reattach the Front Cover 31
2.7 Remove and Reattach the Keypad 33
Remove the Keypad 33
Reattach the Keypad 33
2.8 Install the Keypad in a Control Panel or Another Device 34
Operate the Keypad from a Remote Location 34
3. Electrical Installation 35
3.1 Section Safety 36
3.2 Electrical Installation 38
Standard Connection Diagram 38

3.3 Main Circuit Wiring . . . 41
Motor and Main Circuit Connections . . . 41
Configuration of Main Circuit Terminal Block . . . 42
Main Circuit Terminal Functions . . . 45
Wire Selection . . . 45
Main Circuit Terminal and Motor Wiring . . . 45
3.4 Main Circuit Terminal Block Wiring Procedure . . . 48
Wire to the Main Circuit Terminal Block . . . 48
3.5 Control Circuit Wiring . . . 52
Control Circuit Connection Diagram . . . 52
Control Circuit Terminal Block Functions . . . 53
Control Circuit Terminal Configuration . . . 55
Wiring the Control Circuit Terminal . . . 57
Switches and Jumpers on the Terminal Board . . . 58
3.6 Control I/O Connections . . . 60
Set Sinking Mode/Sourcing Mode . . . 60
Set the Input Signal for the MFAI Terminal A2 . . . 60
Set the Output Signal for the MFAO Terminal AM . . . 61
Switch ON Termination Resistor for MEMOBUS/Modbus Communications . . . 61
3.7 Connect the Drive to a PC . . . 63
3.8 Braking Resistor Installation . . . 64
Install a Braking Resistor: ERF-Type . . . 64
Install a Braking Resistor Unit: LKEB-Type . . . 64
Dynamic Braking Option Overload Protection . . . 65
3.9 Drive Wiring Protection . . . 66
Installing a Residual Current Monitoring/Detection (RCM/RCD) . . . 66
Installing a Molded-Case Circuit Breaker (MCCB) or Residual Current Monitor/Device (RCM/RCD) . . . 66
3.10 Dynamic Braking Option, Motor Protection . . . 67
Install an Electromagnetic Contactor (MC) at the Input Side of the Drive . . . 67
Installing a Thermal Overload Relay on the Drive Output . . . 67
3.11 Improve the Power Factor . . . 69
Connect an AC Reactor or a DC Reactor . . . 69
3.12 Prevent Switching Surge . . . 70
3.13 Decrease Noise . . . 71
Connect a Noise Filter to the Input Side (Primary Side) . . . 71
Connect a Noise Filter to the Output Side (Secondary Side) . . . 71
3.14 Protect the Drive during Failures . . . 73
Factory-Recommended Branch Circuit Protection for UL Listing . . . 73
3.15 Wiring Checklist . . . 74
3.16 Motor Application Precautions . . . 75
Precautions for Existing Standard Motors . . . 75
Precautions for Specialized Motors . . . 76
Notes on the Power Transmission Mechanism . . . 76
4. Startup Procedure and Test Run . . . 77
4.1 Section Safety . . . 78
4.2 Overview of Keypad Components and Functions . . . 79
Indicator flashing statuses . . . 80
Keypad Mode and Menu Displays . . . 81
4.3 Set up the Drive with User-Parameter Menu . . . 82
4.4 Drive Mode and Programming Mode . . . 84
Drive Mode . . . 84
Programming Mode . . . 85

Change Parameter Settings . . . 86
Verify and Set the Changed Parameters (Verify Menu) . . . 86
How to Switch between LOCAL and REMOTE . . . 87
4.5 Start-up Procedures . . . 88
Flowchart A: Connect and Run the Motor with Minimal Setting Changes . . . 88
Power On . . . 88
Control Method Selection . . . 89
Motor Rotation Direction Setup . . . 89
Flowchart B: Auto-Tuning for Induction Motors . . . 89
4.6 Auto-Tuning . . . 91
Types of Auto-Tuning . . . 91
Before Auto-Tuning the Drive . . . 91
Auto-Tuning Interruption and Fault Codes . . . 92
Auto-Tuning Operation Example . . . 92
Input Data for Motor Auto-Tuning: T1 . . . 93
4.7 Setup Procedure for Elevator Applications . . . 94
Up and Down Commands And Speed Reference Selection . . . 94
Speed Selection Using Digital Inputs . . . 94
Multi-Function Terminal Setup . . . 96
Accel/Decel Ramp and Jerk Settings . . . 96
Inspection Operation . . . 97
Brake Sequence . . . 97
Rescue Operation . . . 98
4.8 Items to Check before Starting Up the Drive . . . 102
Check before You Energize the Drive . . . 102
Check after You Energize the Drive . . . 102
4.9 Keypad Operation . . . 103
Show the Monitor . . . 103
Check Modified Parameters . . . 103
Set and View Quick Setup Parameters . . . 103
Change Parameter Settings . . . 104
Save a Backup of Parameters . . . 104
Write Backed-up Parameters to the Drive . . . 105
Verify Keypad Parameters and Drive Parameters . . . 105
Delete Parameters Backed Up to the Keypad . . . 106
4.10 Auto-Tuning . . . 107
Precautions before Auto-Tuning . . . 107
4.11 Fine Tuning during Test Runs (Adjust the Control Function) . . . 109
V/f Control . . . 109
Open Loop Vector Control Method . . . 109
5. Standards Compliance . . . 111
5.1 Section Safety . . . 112
5.2 European Standards . . . 114
EU Declaration of Conformity . . . 115
CE Low Voltage Directive Compliance . . . 120
EMC Directive . . . 126
5.3 UL Standards . . . 132
Area of Use . . . 132
Wire the Main Circuit Terminal Block . . . 132
Low Voltage Wiring for Control Circuit Terminals . . . 136
Drive Motor Overload and Overheat Protection . . . 137
6. Troubleshooting . . . 139
6.1 Section Safety . . . 140
6.2 Types of Faults, Minor Faults, Alarms, and Errors . . . 142

6.3 List of Fault, Minor Fault, Alarm, and Error Codes . . . 143
6.4 Fault . . . 146
6.5 Minor Faults/Alarms . . . 158
6.6 Parameter Setting Errors . . . 166
6.7 Auto-Tuning Errors . . . 168
6.8 Backup Function Operating Mode Display and Errors . . . 171
6.9 Diagnosing and Resetting Faults . . . 172
Fault and Power Loss Occur at the Same Time . . . 172
Fault Occurs Without Power Loss . . . 172
Fault Reset Procedure . . . 172
6.10 Troubleshooting Without Fault Display . . . 173
The Parameter Settings Will Not Change . . . 173
The Motor Does Not Rotate after You Enter a Run Command . . . 173
The Motor Rotates in the Opposite Direction from the Up/Down Command . . . 174
The Motor Rotates in Only One Direction . . . 174
The Motor Is Too Hot . . . 174
The Correct Auto-Tuning Mode Is Not Available . . . 175
The Motor Stalls during Acceleration or Accel/Decel Time Is Too Long . . . 175
There Is Too Much Motor Oscillation and the Rotation Is Irregular . . . 176
Deceleration Takes Longer Than Expected When Dynamic Braking Is Enabled . . . 176
There Is Audible Noise from the Drive or Motor Cables When the Drive Is Energized . . . 176
Residual Current Monitoring/Detection (RCM/RCD) Trips During Run . . . 176
Motor Rotation Causes Unexpected Audible Noise from Connected Machinery . . . 177
Motor Rotation Causes Oscillation or Hunting . . . 177
The Starting Torque Is Not Sufficient . . . 177
The Output Frequency Is Lower Than the Frequency Reference . . . 177
The Motor Will Not Restart after a Loss of Power . . . 177

7. Periodic Inspection and Maintenance . . . 179
7.1 Section Safety . . . 180
7.2 Inspection . . . 182
Recommended Daily Inspection . . . 182
Recommended Periodic Inspection . . . 182
7.3 Maintenance . . . 184
7.4 Replace Cooling Fans . . . 186
Replace the Cooling Fan (Procedure A) . . . 186
Replace the Cooling Fan (Procedure B) . . . 188
7.5 Replace the Drive . . . 191
About the Control Circuit Board . . . 191
Replace the Drive . . . 191
7.6 Storage Guidelines . . . 193

8. Disposal . . . 195
8.1 Section Safety . . . 196
8.2 Disposal Instructions . . . 197
8.3 WEEE Directive . . . 198

9. Specifications . . . 199
9.1 Model Specifications (Three-Phase 400 V Class) . . . 200
9.2 Drive Specifications . . . 201
9.3 Drive Derating . . . 203
Carrier Frequency Settings and Rated Current Values . . . 203
Derating Depending on Ambient Temperature . . . 203

Altitude Derating . . . 203
9.4 Drive Exterior and Mounting Dimensions . . . 204
IP20/UL Open Type . . . 204
9.5 Peripheral Devices and Options . . . 206
10. Parameter List . . . 207
10.1 How to Read the Parameter List . . . 208
Icons and Terms that Identify Parameters and Control Modes . . . 208
10.2 Parameter Groups . . . 209
10.3 A: Initialization Parameters . . . 210
A1: Initialization . . . 210
A2: User Parameters . . . 210
10.4 b: Application . . . 213
b1: Operation Mode Selection . . . 213
b4: Timer Function . . . 213
b6: Dwell Function . . . 213
10.5 C: Tuning . . . 214
C1: Accel & Decel Time . . . 214
C2: Jerk Characteristics . . . 214
C3: Slip Compensation . . . 215
C4: Torque Compensation . . . 215
C6: Duty & Carrier Frequency . . . 215
10.6 d: Reference Settings . . . 216
d1: Speed Reference . . . 216
10.7 E: Motor Parameters . . . 218
E1: V/f Pattern for Motor 1 . . . 218
E2: Motor Parameters . . . 218
10.8 H: Terminal Functions . . . 220
H1: Digital Inputs . . . 220
H2: Digital Outputs . . . 222
H4: Analog Outputs . . . 225
10.9 L: Protection Functions . . . 226
L1: Motor Protection . . . 226
L2: Power Loss Ride Through . . . 226
L3: Stall Prevention . . . 226
L4: Speed Detection . . . 227
L5: Fault Restart . . . 227
L6: Torque Detection . . . 227
L7: Torque Limit . . . 228
L8: Drive Protection . . . 228
10.10 n: Special Adjustment . . . 231
n2: Speed Feedback Detection Control (AFR) Tuning . . . 231
n6: Online Tuning . . . 231
10.11 o: Keypad-Related Settings . . . 232
o1: Keypad Display . . . 232
o2: Keypad Operation . . . 232
o3: Copy Keypad Function . . . 233
o4: Maintenance Monitor Settings . . . 233
10.12 S: Elevator Parameters . . . 235
S1: Brake Sequence . . . 235
S2: Slip Compensation for Elevators . . . 235
S4: Rescue Operation . . . 236
S6: Error Detection . . . 236
10.13 T: Motor Tuning . . . 237

T1: Induction Motor Auto-Tuning . . . 237
10.14 U: Monitors . . . 238
U1: Operation Status Monitors . . . 238
U2: Fault Trace . . . 239
U3: Fault History . . . 241
U4: Maintenance Monitors . . . 243
U6: Operation Status Monitors . . . 245
10.15 Parameters that Change from the Default Settings with A1-02 [Control Method Selection] . . . 247
10.16 Defaults by Drive Model . . . 248
Three-Phase 400 V Class . . . 248
11. Parameter Details . . . 249
11.1 A: Initialization Parameters . . . 250
A1: Initialization . . . 250
A2: User Parameters . . . 253
11.2 b: Application . . . 254
b1: Operation Mode Selection . . . 254
b4: Timer Function . . . 255
b6: Dwell Function . . . 256
11.3 C: Tuning . . . 258
C1: Accel & Decel Speed . . . 258
C2: Jerk Characteristics . . . 261
C3: Slip Compensation . . . 262
C4: Torque Compensation . . . 263
C6: Carrier Frequency . . . 265
11.4 d: References . . . 266
d1: Speed Reference . . . 266
11.5 E: Motor Parameters . . . 269
E1: V/f Pattern for Motor 1 . . . 269
E2: Motor Parameters . . . 270
11.6 H: Terminal Function Selection . . . 274
H1: Digital Inputs . . . 274
MFDI Setting Values . . . 275
H2: Digital Outputs . . . 280
MFDO Setting Values . . . 281
H4: Analog Outputs . . . 290
11.7 L: Protection Functions . . . 293
L1: Motor Protection . . . 293
L2: Undervoltage Detection . . . 295
L3: Stall Prevention . . . 296
L4: Speed Detection . . . 298
L5: Fault Restart . . . 299
L6: Torque Detection . . . 300
L7: Torque Limit . . . 304
L8: Drive Protection . . . 306
11.8 n: Special Adjustment . . . 312
n2: Speed Feedback Detection Control (AFR) Tuning . . . 312
n6: Online Tuning . . . 313
11.9 o: Keypad-Related Settings . . . 314
o1: Keypad Display Selection . . . 314
o2: Keypad Operation . . . 316
o3: Copy Function . . . 319
o4: Maintenance Mon Settings . . . 319
11.10 S: Elevator Parameters . . . 323

S1: Brake Sequence 323
S2: Slip Compensation for Elevators 324
S4: Rescue Operation 325
S6: Faults for Elevator Applications 325
11.11 T: Auto-Tuning 327
T1: Induction Motor Auto-Tuning 327
Index 329
Revision History 334

i

Preface and General Precautions

This chapter gives information about important safety precautions for the use of this product. Failure to obey these precautions can cause serious injury or death, or damage to the product or related devices and systems. Yaskawa must not be held responsible for any injury or equipment damage as a result of the failure to observe these precautions and instructions.

i.1 Receiving 12
i.2 Using the Product Safely 13
i.3 Warranty Information 17

i.1 Receiving

These instructions contain the information necessary to use the product correctly. Read and understand the safety information and precautions before you start to use the product.

◆ Glossary

Phrase	Definition
Drive	YASKAWA AC Drive LA500
EDM	External Device Monitor
HD	Heavy Duty
MFAI	Multi-Function Analog Input
MFAO	Multi-Function Analog Output
MFDI	Multi-Function Digital Input
MFDO	Multi-Function Digital Output
OLV	Open Loop Vector Control
SIL	Safety Integrity Level
V/f	V/f Control

◆ About Registered Trademarks

- All company names and product names in this document are trademarks or registered trademarks of the respective companies.

i.2 Using the Product Safely

◆ Explanation of Signal Words

Read and understand this manual before you install, operate, or do maintenance on the drive. Install the drive as specified by this manual and local codes.

The symbols in this section identify safety messages in this manual. If you do not obey these safety messages, the hazards can cause serious injury, death, or damage to the products and related equipment and systems. These identifier words categorize and emphasize important safety precautions in these instructions.

⚠DANGER

This signal word identifies a hazard that will cause serious injury or death if you do not prevent it.

⚠WARNING

This signal word identifies a hazard that can cause death or serious injuries if you do not prevent it.

⚠CAUTION

Identifies a hazardous situation, which, if not avoided, can cause minor or moderate injury.

NOTICE

This signal word identifies a property damage message that is not related to personal injury.

◆ Section Safety

- Some figures in the instructions include options and drives without covers or safety shields to more clearly show the inside of the drive. Replace covers and shields before operation. Use options and drives only as specified by the instructions.
- The figures in this manual are examples only. All figures do not apply to all products included in this manual.
- The manufacturer can change the products, specifications, and content of the instructions without notice to make the product and/or the instructions better.
- If you damage or lose these instructions, contact a representative or the nearest sales office of the manufacturer. You can find the contact information on the rear cover of the manual, and tell them the document number to order new copies. You can also download the manual from the manufacturer's website.

⚠DANGER

Do not ignore the safety messages in this manual.

If you ignore the safety messages in this manual, it will cause serious injury or death. The manufacturer is not responsible for injuries or damage to equipment.

Electrical Shock Hazard

Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe.

If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.

⚠WARNING

Crush Hazard

Test the system to make sure that the drive operates safely after you wire the drive and set parameters.

If you do not test the system, it can cause damage to equipment or serious injury or death.

Sudden Movement Hazard

Before you do a test run, make sure that the setting values for virtual input and output function parameters are correct. Virtual input and output functions can have different default settings and operation than wired input and output functions.

Incorrect function settings can cause serious injury or death.

Remove all personnel and objects from the area around the drive, motor, and machine and attach covers, couplings, shaft keys, and machine loads before you energize the drive.

If personnel are too close or if there are missing parts, it can cause serious injury or death.

Examine the I/O signals and internal sequence with the engineer who made the DriveWorksEZ program before you operate the drive.

If you do not know how the drive will operate, it can cause serious injury or death. When you use DriveWorksEZ to make custom programming, the drive I/O terminal functions change from factory settings and the drive will not operate as written in this manual.

Electrical Shock Hazard

Do not modify the drive body or drive circuitry.

Modifications to drive body and circuitry can cause serious injury or death, will cause damage to the drive, and will void the warranty. Yaskawa is not responsible for modifications of the product made by the user.

Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive.

If personnel are not approved, it can cause serious injury or death.

Do not remove covers or touch circuit boards while the drive is energized.

If you touch the internal components of an energized drive, it can cause serious injury or death.

After the drive blows a fuse or trips an RCM/RCD, do not immediately energize the drive or operate peripheral devices. Wait for the time specified on the warning label at a minimum and make sure that all indicators are OFF. Then check the wiring and peripheral device ratings to find the cause of the problem. If you do not know the cause of the problem, contact Yaskawa before you energize the drive or peripheral devices.

If you do not fix the problem before you operate the drive or peripheral devices, it can cause serious injury or death.

Damage to Equipment

Do not apply incorrect voltage to the main circuit of the drive. Operate the drive in the specified range of the input voltage on the drive nameplate.

Voltages that are higher than the permitted nameplate tolerance can cause damage to the drive.

Fire Hazard

Install sufficient branch circuit short circuit protection as specified by applicable codes and this manual. The drive is suited for circuits that supply not more than 31,000 RMS symmetrical amperes, 240 Vac maximum (200 V Class), 480 Vac maximum (400 V Class).

Incorrect branch circuit short circuit protection can cause serious injury or death.

⚠CAUTION

Crush Hazard

Tighten terminal cover screws and hold the case safely when you move the drive.

If the drive or covers fall, it can cause moderate injury.

NOTICE

Use an inverter-duty motor or vector-duty motor with reinforced insulation and windings applicable for use with an AC drive.

If the motor does not have the correct insulation, it can cause a short circuit or ground fault from insulation deterioration.

Damage to Equipment

When you touch the drive and circuit boards, make sure that you observe correct electrostatic discharge (ESD) procedures.

If you do not follow procedures, it can cause ESD damage to the drive circuitry.

Do not do a withstand voltage test or use a megohmmeter or megger insulation tester on the drive.

These tests can cause damage to the drive.

Do not operate a drive or connected equipment that has damaged or missing parts.

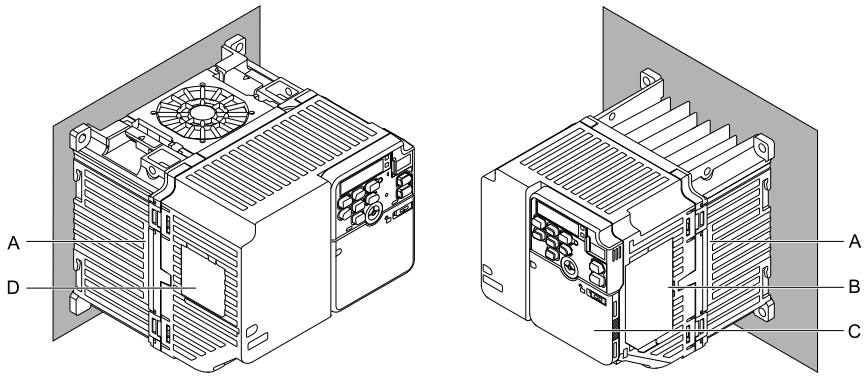
You can cause damage to the drive and connected equipment.

Do not use steam or other disinfectants to fumigate wood for packaging the drive. Use alternative methods, for example heat treatment, before you package the components.

Gas from wood packaging fumigated with halogen disinfectants, for example fluorine, chlorine, bromine, iodine or DOP gas (phthalic acid ester), can cause damage to the drive.

◆ Warning Label Content and Location

The drive warning label is in the location shown in Figure i.1. Use the drive as specified by this information. Replace unreadable or missing labels.



A - Warning label “Risk of electric shock”
B - Nameplate
C - Warning label “Hot surface”
D - Warning label “Application”

Figure i.1 Warning Label Content and Location

The labels show the following content:

Figure i.2 Warning Label “Risk of electric shock”

Figure i.3 Warning Label “Hot surface”

Figure i.4 Warning Label “Application”

i.3 Warranty Information

◆ Exclusion of Liability

- This product is not designed and manufactured for use in life-support machines or systems.
- Contact a Yaskawa representative or your Yaskawa sales representative if you are considering the application of this product for special purposes, such as machines or systems used for passenger cars, medicine, airplanes and aerospace, nuclear power, electric power, or undersea relaying.

⚠WARNING

Injury to Personnel

When you use this product in applications where its failure could cause the loss of human life, a serious accident, or physical injury, you must install applicable safety devices.

If you do not correctly install safety devices, it can cause serious injury or death.

1

Receiving

This chapter gives information about the different drive models and features, and how to examine the drive when you receive it.

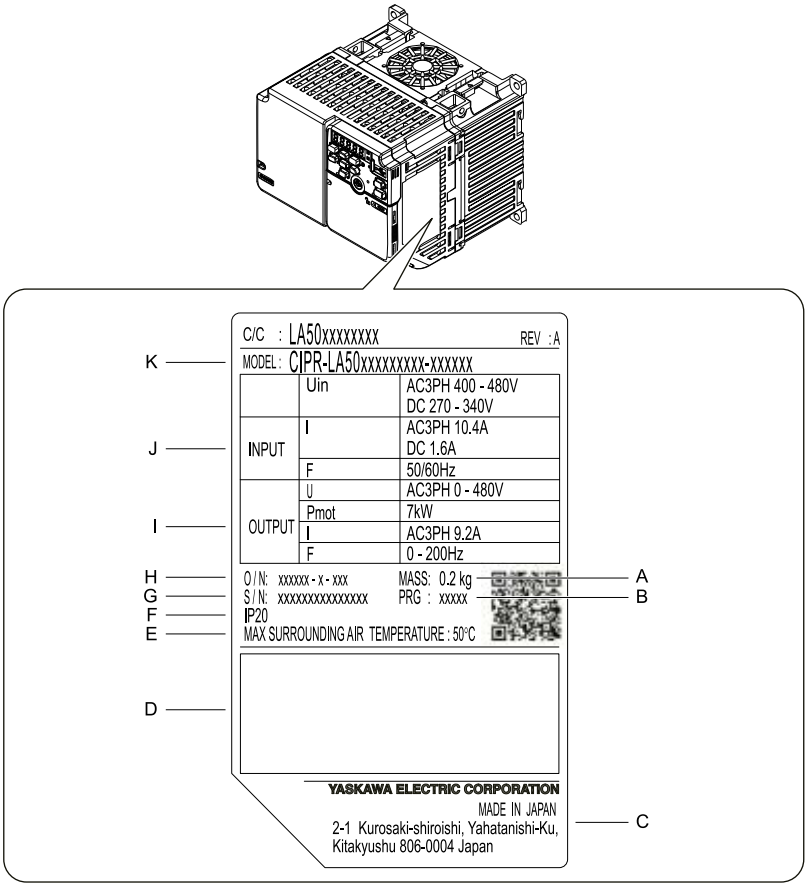
1.1 Model Number and Nameplate Check 20
1.2 Features and Advantages of Control Methods 22

1.1 Model Number and Nameplate Check

Please examine these items after you receive the drive:

- Examine the drive for damage. Immediately contact the shipping company if the drive is damaged. The Yaskawa warranty does not cover damage from shipping.
- Examine the drive model number to make sure that you received the correct model. Examine the model number in the “MODEL” section of the drive nameplate to make sure that you received the correct model.
- If you received the incorrect product or a product with a defect, contact Yaskawa or your nearest sales representative.

◆ Nameplate



A - Weight
B - Drive software version
C - The address of the head office of Yaskawa Electric Corporation
D - Accreditation standards
E - Ambient temperature setting
F - Enclosure protection design
G - Product number
H - Serial number
I - Output specifications
J - Input specifications
K - Drive model

Figure 1.1 Nameplate Information Example

◆ How to Read the Model Number

Use the information in Figure 1.2 and Table 1.1 to read the drive model numbers.

CIPR	-	LA50	A	4	009	A	B	A	A	-	C	A	A	A	S	A
1		2	3	4	5	6	7	8	9		10	11	12	13	14	15

Figure 1.2 Drive model

Table 1.1 Model Number Details

No.	Description
1	Drive
2	Product series
3	Region code • C: Europe
4	Input power supply voltage • 4: Three-Phase AC 400 V Class
5	Rated Output Current **Note:** Refer to page 21 for more information.
6	EMC noise filter • A: No internal EMC filter • E: Built-in EMC Filter
7	Enclosure protection design • B: IP20/UL Open Type • F: IP20/UL Type 1
8	Environmental specification
9	Design revision order
10	Control circuit board
11	Option
12	A: Standard
13	A: Standard
14	Keypad
15	Special applications

■ Rated Output Current

The following tables give the rated output current values.

Note:
- Rated output current values are applicable for drives that operate at standard specifications.
- Derate the output current in applications that:
  - Increase the carrier frequency
  - Have high ambient temperature
  - Install drives side-by-side

Table 1.2 Three-Phase AC 400 V Class

Model	Heavy Duty Rating (HD) [C6-01 = 0] (Default)	
	Maximum Applicable Motor Output kW (HP)	Rated Output Current A
4009	4.0 (5)	9.2
4015	5.5 (10)	14.8
4018	7.5 (10)	18.0
4024	11.0 (15)	24.0
4031	15.0 (20)	31.0
4039	18.5 (25)	39.0
4045	22.0 (30)	45.0

1.2 Features and Advantages of Control Methods

This drive has 2 available control methods from which you can select for different applications.

Table 1.3 Features and Advantages of V/f Control

Control Method Selection	Open Loop V/f Control (V/f)	Notes
Controlled Motor	**Induction Motor**	**-**
Parameter Settings	A1-02 = 0	-
Basic Control	V/f	-
Main Applications	• Lift	-
Maximum Output Frequency	120 Hz	-
Speed Control Range	1:40	This is the range of variable control. When you connect and operate motors in this mode, think about the increase in motor temperature.
Starting Torque	150% / 3 Hz	This is the motor torque that the drive can supply at low speed during start-up and the related output frequency (rotation speed). You must think about drive capacity when a large quantity of torque is necessary at low speed.
Auto-Tuning *1	Rotational and Line-to-Line Resistance (usually not necessary)	Automatically tunes electrical motor parameters.
Torque Limits	No	Controls maximum motor torque to prevent damage to machines and loads.
Automatic Energy-saving Control *1	Yes	Automatically adjusts the voltage that the drive applies to the motor to maximize motor efficiency for small and large loads.

*1 When you can decouple the motor and machine for a test run, use Rotational Auto-Tuning. You must make adjustments to the control in the range where there is no vibration in the machine after Rotational Auto-Tuning.

•

Table 1.4 Features and Advantages of OLV Control

Control Method Selection	Open Loop Vector (OLV)	Notes
Controlled Motor	**Induction Motor**	**-**
Parameter Settings	A1-02 = 2 (Default)	-
Basic Control	Open Loop Current Vector Control	-
Main Applications	• Lift	-
Maximum Output Frequency	120 Hz	-
Speed Control Range	1:100	This is the range of variable control. When you connect and operate motors in this mode, think about the increase in motor temperature.
Starting Torque	150% / 1 Hz *1	This is the motor torque that the drive can supply at low speed during start-up and the related output frequency (rotation speed). You must think about drive capacity when a large quantity of torque is necessary at low speed.
Auto-Tuning *2	Rotational, Stationary, and Line-to-Line Resistance	Automatically tunes electrical motor parameters.
Torque Limits *2	Yes	Controls maximum motor torque to prevent damage to machines and loads.
Automatic Energy-saving Control *2	Yes	Automatically adjusts the voltage that the drive applies to the motor to maximize motor efficiency for small and large loads.

*1 Select the drive capacity accordingly.

*2 When you can decouple the motor and machine for a test run, use Rotational Auto-Tuning. You must make adjustments to the control in the range where there is no vibration in the machine after Rotational Auto-Tuning.

•

2

Mechanical Installation

This chapter gives information about the correct environment and clearances to install the drive.

2.1 Section Safety 24
2.2 Installation Environment 25
2.3 Installation Position and Clearances 26
2.4 Moving the Drive 29
2.5 Drive Models and Drive Watt Loss 30
2.6 Removing/Reattaching Covers 31
2.7 Remove and Reattach the Keypad 33
2.8 Install the Keypad in a Control Panel or Another Device 34

2.1 Section Safety

⚠WARNING

Electrical Shock Hazard

Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive.

If personnel are not approved, it can cause serious injury or death.

Do not modify the drive body or drive circuitry.

Modifications to drive body and circuitry can cause serious injury or death, will cause damage to the drive, and will void the warranty. Yaskawa is not responsible for modifications of the product made by the user.

Fire Hazard

Do not put flammable or combustible materials on top of the drive and do not install the drive near flammable or combustible materials. Attach the drive to metal or other noncombustible material.

Flammable and combustible materials can start a fire and cause serious injury or death.

When you install the drive in an enclosure, use a cooling fan or cooler to decrease the temperature around the drive. Make sure that the intake air temperature to the drive is 50 °C (122 °F) or less for IP20/UL Open Type drives, and 40 °C (104 °F) or less for IP20/UL Type 1 drives.

If the air temperature is too hot, the drive can become too hot and cause a fire and serious injury or death.

⚠CAUTION

Crush Hazard

Tighten terminal cover screws and hold the case safely when you move the drive.

If the drive or covers fall, it can cause moderate injury.

NOTICE

Do not let unwanted objects, for example metal shavings or wire clippings, fall into the drive during drive installation. Put a temporary cover over the drive during installation. Remove the temporary cover before start-up.

Unwanted objects inside of the drive can cause damage to the drive.

Damage to Equipment

When you touch the drive and circuit boards, make sure that you observe correct electrostatic discharge (ESD) procedures.

If you do not follow procedures, it can cause ESD damage to the drive circuitry.

Install vibration-proof rubber on the base of the motor or use the frequency jump function in the drive to prevent specific frequencies that vibrate the motor.

Motor or system resonant vibration can occur in fixed speed machines that are converted to variable speed. Too much vibration can cause damage to equipment.

You can use the drive with an explosion-proof motor, but the drive is not explosion-proof. Install the drive only in the environment shown on the nameplate.

If you install the drive in a dangerous environment, it can cause damage to the drive.

Do not lift the drive with the covers removed.

If the drive does not have covers, you can easily cause damage to the internal parts of the drive.

2.2 Installation Environment

The installation environment is important for the lifespan of the product and to make sure that the drive performance is correct. Make sure that the installation environment agrees with these specifications.

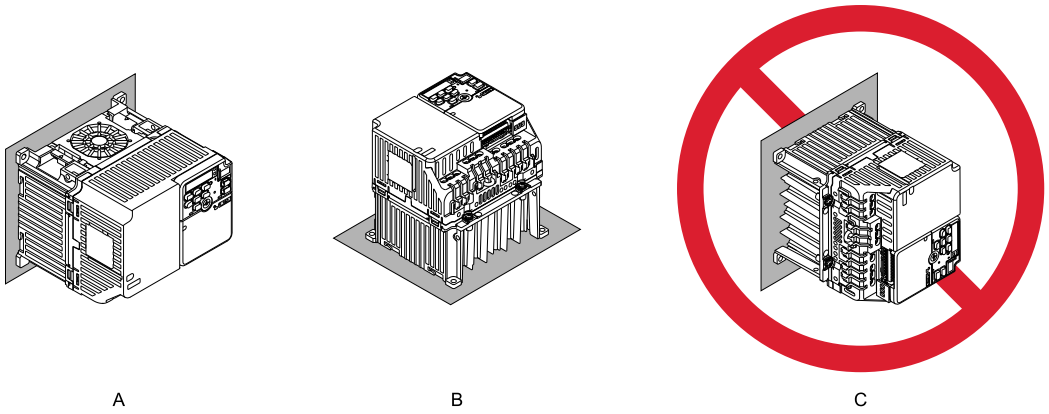
Environment	Conditions
Area of Use	Indoors
Power Supply	Overvoltage Category III (IEC60664)
Ambient Temperature Setting	IP20/UL Open Type: -10 °C to +50 °C (14 °F to 122 °F) IP20/UL Type 1: -10 °C to +40 °C (14 °F to 104 °F) • When you install the drive in an enclosure, use a cooling fan or air conditioner to keep the internal air temperature in the permitted range. • Do not let the drive freeze.
Humidity	95%RH or less Do not let condensation form on the drive.
Storage Temperature	-20 °C to +70 °C (-4 °F to +158 °F) (short-term temperature during transportation)
Surrounding Area	Pollution degree 2 or less (IEC 60664-1) Install the drive in an area without: • Oil mist, corrosive or flammable gas, or dust • Metal powder, oil, water, or other unwanted materials • Radioactive or flammable materials. • Harmful gas or fluids • Salt • Direct sunlight Keep wood and other flammable materials away from the drive.
Altitude	1000 m (3281 ft) Maximum **Note:** Derate the output current by 1% for each 100 m (328 ft) to install the drive in altitudes between 1000 m to 4000 m (3281 ft to 13123 ft). It is not necessary to derate the rated voltage in these conditions: • Installing the drive at 2000 m (6562 ft) or lower • Installing the drive between 2000 m to 4000 m (6562 ft to 13123 ft) and grounding the neutral point on the power supply. Contact Yaskawa or your nearest sales representative when not grounding the neutral point.
Vibration	• 10 Hz to 20 Hz: 1 G (9.8 m/s^2, 32.15 ft/s^2) • 20 Hz to 55 Hz: 0.6 G (5.9 m/s^2, 19.36 ft/s^2)
Installation Orientation	Install the drive vertically or horizontally for sufficient airflow to cool the drive. Refer to the drive Technical Manual for more information.

NOTICE: *Do not put drive peripheral devices, transformers, or other electronics near the drive. Shield the drive from electrical interference if components must be near the drive. Components near the drive can cause incorrect drive operation from electrical interference.*

NOTICE: *Do not let unwanted objects, for example metal shavings or wire clippings, fall into the drive during drive installation. Put a temporary cover over the drive during installation. Remove the temporary cover before start-up. Unwanted objects inside of the drive can cause damage to the drive.*

2.3 Installation Position and Clearances

Install the drive as shown in Figure 2.1 for sufficient airflow to cool the drive.

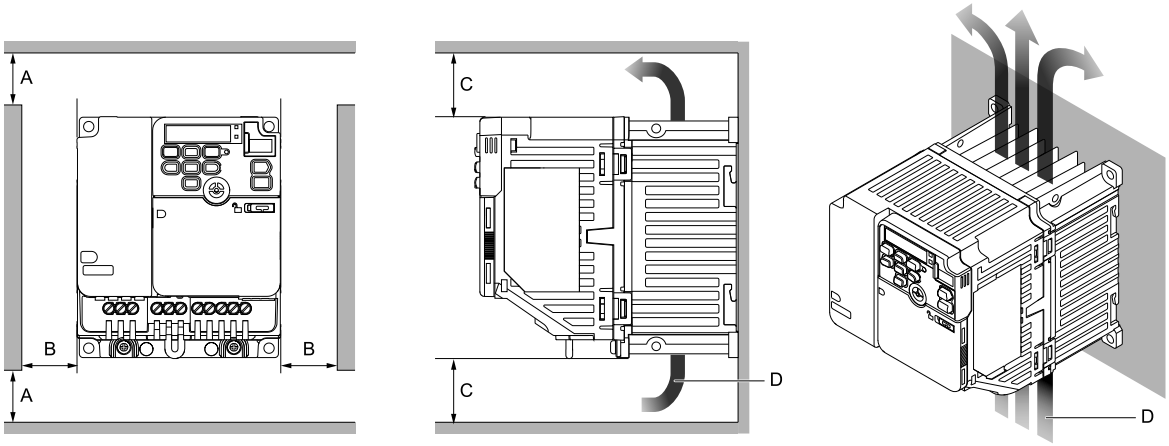


A - Vertical installation
B - Horizontal installation
C - Rotated installation

Figure 2.1 Installation Orientation

◆ Install Single Drive

Use the clearances specified in Figure 2.2 to install the drive. Make sure that there is sufficient space for wiring and airflow.

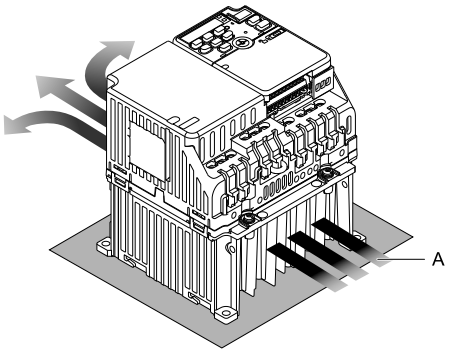


A - 50 mm (2 in) minimum
B - 30 mm (1.18 in) minimum on each side
C - 100 mm (3.94 in) minimum above and below
D - Airflow direction

Figure 2.2 Installation Clearances for One Drive

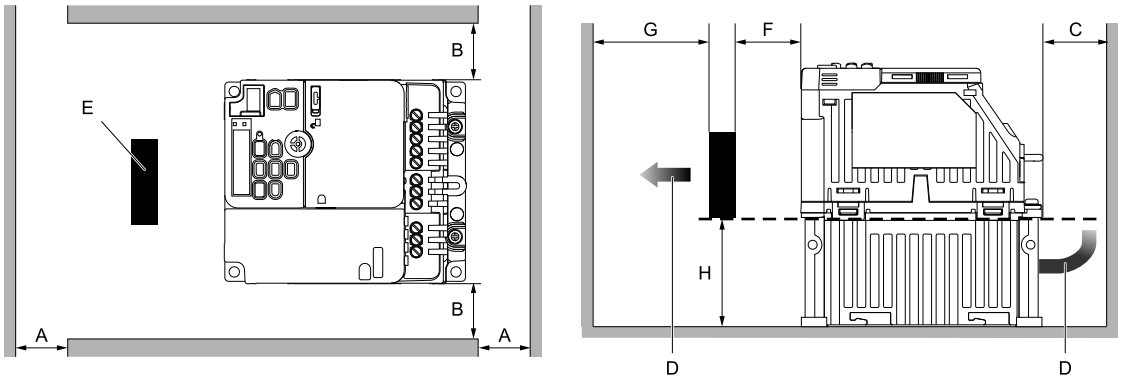
◆ Install the Drive Horizontally

When you install drives horizontally, set *L8-12 = 40 [Ambient Temperature Setting = 40 °C]* and *L8-35 = 1 [Installation Method Selection = Side-by-Side Mounting]*. Use the clearances specified in Figure 2.4 and Figure 2.5 to install the drive. Make sure that there is sufficient space for wiring and airflow. To install the drive model 4009 horizontally, install an external cooling fan. Refer to Table 2.1 for more information about the external cooling fan.



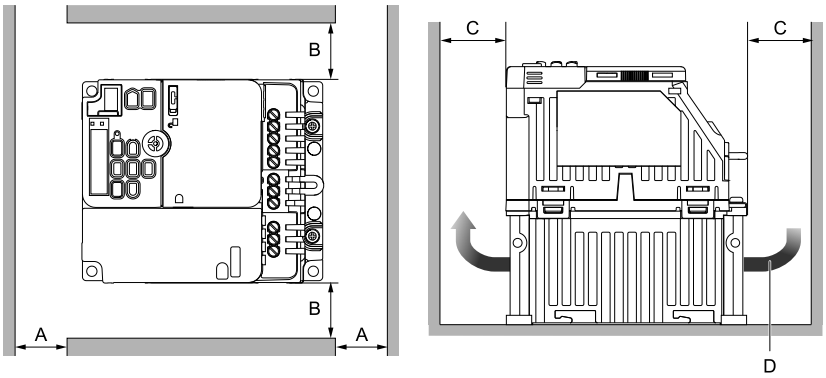
A - Airflow direction

Figure 2.3 Airflow Direction of Horizontal installation



A - 50 mm (2 in) minimum
B - 30 mm (1.18 in) minimum on each side
C - 100 mm (3.94 in) minimum below
D - Airflow direction
E - External cooling fan
F - 30 mm (1.18 in) between the drive and the external cooling fan
G - 120 mm (4.72 in) minimum between the external cooling fan and the enclosure panel
H - Heatsink height

Figure 2.4 Installation Clearances for Horizontal Installation: 4009



A - 50 mm (2 in) minimum
B - 30 mm (1.18 in) minimum on each side
C - 100 mm (3.94 in) minimum above and below
D - Airflow direction

Figure 2.5 Installation Clearances for Horizontal Installation: 4015 - 4045

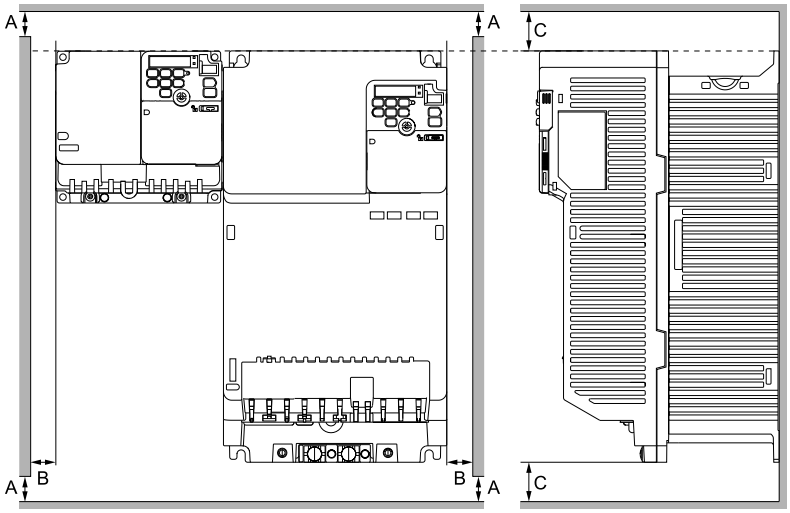
Table 2.1 Specifications of External Cooling Fan for Horizontal (Floor) Installation

Model	Airflow (m^3/min)	Static Pressure (Pa)
4009	1.11 minimum	244 minimum
4015 - 4045	External cooling fan is not necessary	

◆ Install Drives Side-by-Side

When you install drives side-by-side, set to *L8-35 = 1 [Installation Method Selection = Side-by-Side Mounting]*.

Refer to *Derating Depending on Ambient Temperature on page 203* and set derating depending on ambient temperature.



A - 50 mm (1.97 in) minimum
B - 30 mm (1.18 in) minimum on each side
C - 100 mm (3.94 in) minimum above and below

Figure 2.6 Installation Clearances for More than One Drive (Side-by-Side)

Note:
Align the tops of drives that have different dimensions to help when you replace cooling fans.

2.4 Moving the Drive

Obey local laws and regulations when moving and installing this product.

CAUTION! *Crush Hazard. Do not hold the drive by the keypad or front cover. Tighten the screws correctly when you move the drive. If the drive or covers fall, it can cause moderate injury.*

2.5 Drive Models and Drive Watt Loss

◆ Drive Watt Loss (without Built-in EMC Filter)

■ Three-Phase 400 V Class

Table 2.2 Drive Watt Loss (HD, Fc = 2 kHz)

Model	Rated Output Current A	Carrier Frequency kHz	Interior Unit Loss W	Cooling Fin Loss W	Total Loss W
4009A	9.2	2	27	60	87
4015A	14.8	2	48	126	174
4018A	18	2	53	152	205
4024A	24	2	68	191	259
4031A	31	2	81	256	337
4039A	39	2	109	338	447
4045A	45	2	114	328	442

Table 2.3 Drive Watt Loss (HD, Fc = Default Setting)

Model	Rated Output Current A	Carrier Frequency kHz	Interior Unit Loss W	Cooling Fin Loss W	Total Loss W
4009A	9.2	8	32	85	117
4015A	14.8	8	55	166	221
4018A	18	8	61	200	261
4024A	24	8	79	255	334
4031A	31	8	95	338	433
4039A	39	8	127	442	569
4045A	45	8	135	446	581

◆ Drive Watt Loss (with Built-in EMC Filter)

■ Three-Phase 400 V Class

Table 2.4 Drive Watt Loss (HD, Fc = 2 kHz)

Model	Rated Output Current A	Carrier Frequency kHz	Interior Unit Loss W	Cooling Fin Loss W	Total Loss W
4009E	9.2	2	29	60	89
4015E	14.8	2	52	126	178
4018E	18	2	57	152	209
4024E	24	2	73	191	264
4031E	31	2	89	256	345
4039E	39	2	119	338	457
4045E	45	2	128	328	456

Table 2.5 Drive Watt Loss (HD, Fc = Default Setting)

Model	Rated Output Current A	Carrier Frequency kHz	Interior Unit Loss W	Cooling Fin Loss W	Total Loss W
4009E	9.2	8	34	85	119
4015E	14.8	8	59	166	225
4018E	18	8	65	200	265
4024E	24	8	84	255	339
4031E	31	8	103	338	441
4039E	39	8	137	442	579
4045E	45	8	149	446	595

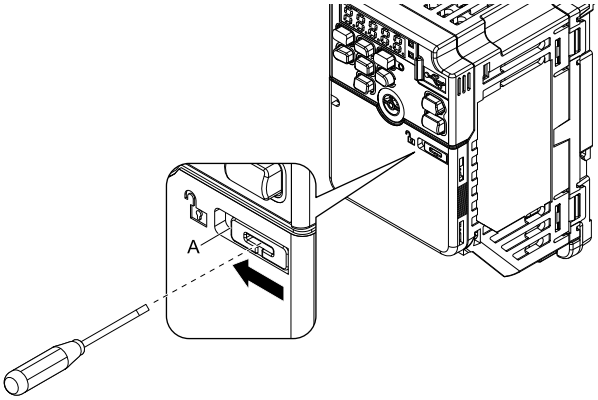
2.6 Removing/Reattaching Covers

DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

◆ Remove the Front Cover

1. Use a slotted screwdriver to unlock the front cover of the drive.

   Use a slotted screwdriver with a tip width of 2.5 mm (0.1 in) or less and a thickness of 0.4 mm (0.02 in) or less.



A - Front cover lock

Figure 2.7 Unlocking

2. Pull down, then pull away from the drive to remove the front cover.

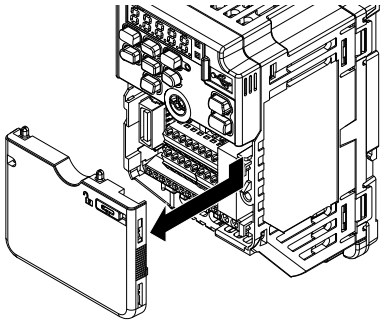


Figure 2.8 Remove the Front Cover

◆ Reattach the Front Cover

1. Reverse the steps to reattach the cover.

Note:
Make sure that you do not pinch wires or signal lines between the front cover and the drive before you reattach the cover.

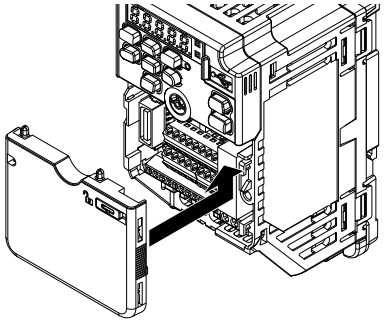
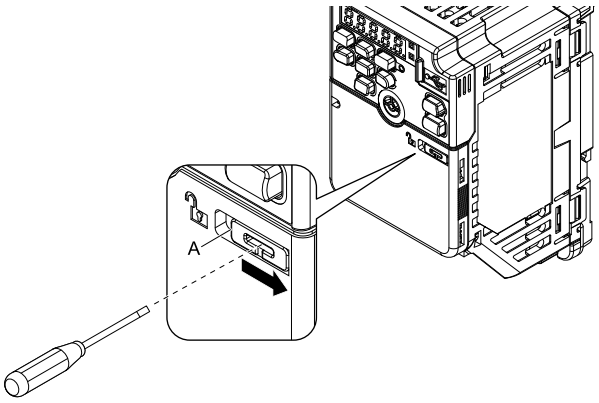


Figure 2.9 Reattach the Front Cover

2. Use a slotted screwdriver to lock the front cover of the drive.

   Use a slotted screwdriver with a tip width of 2.5 mm (0.1 in) or less and a thickness of 0.4 mm (0.02 in) or less.



A - Front cover lock

Figure 2.10 Locking the Front Cover

2.7 Remove and Reattach the Keypad

◆ Remove the Keypad

Remove the front cover.

Push on the tab on the right side of the keypad, then pull the keypad forward to remove it from the drive.

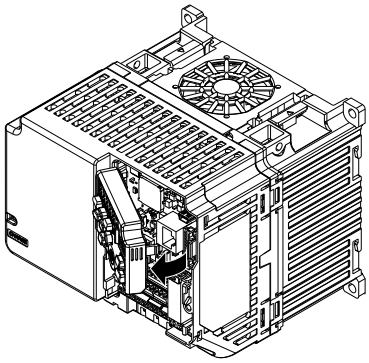


Figure 2.11 Remove the Keypad

◆ Reattach the Keypad

Push in the keypad from the front until the hooks click into place.

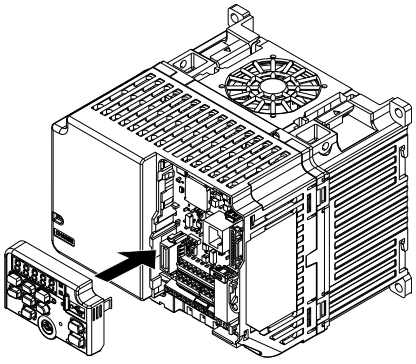


Figure 2.12 Reattach the Keypad

Attach the front cover.

2.8 Install the Keypad in a Control Panel or Another Device

◆ Operate the Keypad from a Remote Location

You can remove the keypad from the drive and connect it to a remote control extension cable 3 m (9.8 ft) long to make operation easier when you cannot access the drive. It is not necessary to open or close the panel door to operate a drive that is in a control panel. To order optional accessories, contact Yaskawa or your nearest sales representative.

Name	Option Model	Intended Use
Keypad Remote Cable	WV001: 1 m (3.3 ft) WV003: 3 m (9.8 ft)	To connect the keypad and drive. This option is an RJ-45, 8-pin straight-through UTP CAT5e cable.
Installation Support Set A	900-192-933-001	To attach the keypad to the control panel. This option uses screws.
Installation Support Set B	900-192-933-002	To attach the keypad to the control panel. This option uses nut clamps. Use this option when weld studs are located in the control panel.

3

Electrical Installation

This chapter gives how to wire the control circuit terminals, motor, and power supply of the drive.

3.1 Section Safety 36
3.2 Electrical Installation 38
3.3 Main Circuit Wiring 41
3.4 Main Circuit Terminal Block Wiring Procedure 48
3.5 Control Circuit Wiring 52
3.6 Control I/O Connections 60
3.7 Connect the Drive to a PC 63
3.8 Braking Resistor Installation 64
3.9 Drive Wiring Protection 66
3.10 Dynamic Braking Option, Motor Protection 67
3.11 Improve the Power Factor 69
3.12 Prevent Switching Surge 70
3.13 Decrease Noise 71
3.14 Protect the Drive during Failures 73
3.15 Wiring Checklist 74
3.16 Motor Application Precautions 75

3.1 Section Safety

⚠DANGER

Electrical Shock Hazard

Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe.

If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.

⚠WARNING

Electrical Shock Hazard

Do not operate the drive when covers are missing. Replace covers and shields before you operate the drive. Use the drive only as specified by the instructions.

Some figures in this section include drives without covers or safety shields to more clearly show the inside of the drive. If covers or safety shields are missing from the drive, it can cause serious injury or death.

Ground the neutral point on the power supply of drive models BxxxE, 2xxxE, and 4xxxE to comply with the EMC Directive before you turn on the EMC filter or if there is high resistance grounding.

If you turn ON the EMC filter, but you do not ground the neutral point, it can cause serious injury or death.

Make sure that the protective ground wire complies with technical standards and local safety regulations. The EN 61800-5-1:2007 standard specifies that you must wire the power supply to automatically de-energize when the protective ground wire disconnects. If you turn on the internal EMC filter, the leakage current of the drive will be more than 3.5 mA. Use the closed-loop crimp terminal to connect a protective ground wire that has a minimum cross-sectional area of 10 mm² (copper wire).

If you do not obey the standards and regulations, it can cause serious injury or death.

When there is a DC component in the protective earthing conductor, the drive can cause a residual current. When a residual current operated protective or monitoring device prevents direct or indirect contact, always use a type B Residual Current Monitor/Residual Current Device (RCM/RCD) as specified by IEC/EN 60755.

If you do not use the correct RCM/RCD, it can cause serious injury or death.

Do not wear loose clothing or jewelry when you do work on the drive. Tighten loose clothing and remove all metal objects, for example watches or rings.

Loose clothing can catch on the drive and jewelry can conduct electricity and cause serious injury or death.

Do not remove covers or touch circuit boards while the drive is energized.

If you touch the internal components of an energized drive, it can cause serious injury or death.

Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive.

If personnel are not approved, it can cause serious injury or death.

Do not modify the drive body or drive circuitry.

Modifications to drive body and circuitry can cause serious injury or death, will cause damage to the drive, and will void the warranty. Yaskawa is not responsible for modifications of the product made by the user.

Fire Hazard

Tighten all terminal screws to the correct tightening torque.

Connections that are too loose or too tight can cause incorrect operation and damage to the drive. Incorrect connections can also cause death or serious injury from fire.

Tighten screws at an angle in the specified range shown in this manual.

If you tighten the screws at an angle not in the specified range, you can have loose connections that can cause damage to the terminal block or start a fire and cause serious injury or death.

⚠WARNING

Damage to Equipment

Do not apply incorrect voltage to the main circuit of the drive. Operate the drive in the specified range of the input voltage on the drive nameplate.

Voltages that are higher than the permitted nameplate tolerance can cause damage to the drive.

Fire Hazard

When you install a dynamic braking option, wire the components as specified by the wiring diagrams.

Incorrect wiring can cause damage to braking components or serious injury or death.

NOTICE

Do not let unwanted objects, for example metal shavings or wire clippings, fall into the drive during drive installation. Put a temporary cover over the drive during installation. Remove the temporary cover before start-up.

Unwanted objects inside of the drive can cause damage to the drive.

Damage to Equipment

When you touch the drive and circuit boards, make sure that you observe correct electrostatic discharge (ESD) procedures.

If you do not follow procedures, it can cause ESD damage to the drive circuitry.

Select a motor that is compatible with the load torque and speed range. When 100% continuous torque is necessary at low speed, use an inverter-duty motor or vector-duty motor. When you use a standard fan-cooled motor, decrease the motor torque in the low-speed range.

If you operate a standard fan-cooled motor at low speed and high torque, it will decrease the cooling effects and can cause heat damage.

Obey the speed range specification of the motor as specified by the manufacturer. When you must operate the motor outside of its specifications, contact the motor manufacturer.

If you continuously operate oil-lubricated motors outside of the manufacturer specifications, it can cause damage to the motor bearings.

When the input voltage is 440 V or higher or the wiring distance is longer than 100 m (328 ft), make sure that the motor insulation voltage is sufficient or use an inverter-duty motor or vector-duty motor with reinforced insulation.

Motor winding and insulation failure can occur.

Before you connect a dynamic braking option to the drive, make sure that qualified personnel read and obey the Braking Unit and Braking Resistor Unit Installation Manual (TOBPC72060001).

If you do not read and obey the manual or if personnel are not qualified, it can cause damage to the drive and braking circuit.

Make sure that all connections are correct after you install the drive and connect peripheral devices.

Incorrect connections can cause damage to the drive.

Note:

- Torque characteristics are different than when you operate the motor directly from line power. Make sure that you understand the load torque characteristics for the application.
- The rated input current of submersible motors is higher than the rated input current of standard motors. Carefully select the correct drive capacity. When the distance between the motor and drive is long, use a wire that can connect the motor to the drive without a reduction in motor torque.
- Do not use unshielded wire for control wiring. Use shielded, twisted-pair wires and ground the shield to the ground terminal of the drive. Unshielded wire can cause electrical interference and unsatisfactory system performance.

3.2 Electrical Installation

DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

WARNING! *Electrical Shock Hazard. De-energize the drive and wait 5 minutes minimum until the Charge LED turns off. Remove the front cover and terminal cover to do work on wiring, circuit boards, and other parts. Use terminals for their correct function only. Incorrect wiring, incorrect ground connections, and incorrect repair of protective covers can cause death or serious injury.*

WARNING! *Electrical Shock Hazard. Correctly ground the drive before you turn on the EMC filter switch. If you touch electrical equipment that is not grounded, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Use the terminals for the drive only for their intended purpose. Refer to the technical manual for more information about the I/O terminals. Wiring and grounding incorrectly or modifying the cover may damage the equipment or cause injury.*

◆ Standard Connection Diagram

Wire the drive as specified by Figure 3.1.

WARNING! *Sudden Movement Hazard. Set the MFDI parameters before you close control circuit switches. Incorrect Run/Stop circuit sequence settings can cause serious injury or death from moving equipment.*

WARNING! *Sudden Movement Hazard. Correctly wire the start/stop and safety circuits before you energize the drive. If you momentarily close a digital input terminal, it can start a drive that is programmed for 3-Wire control and cause serious injury or death from moving equipment.*

WARNING! *Sudden Movement Hazard. When you use a 3-Wire sequence, set A1-03 = 3330 [Initialize Parameters = 3-Wire Initialization] and make sure that b1-17 = 0 [Run Command at Power Up = Disregard Existing RUN Command] (default). If you do not correctly set the drive parameters for 3-Wire operation before you energize the drive, the motor can suddenly rotate when you energize the drive.*

WARNING! *Sudden Movement Hazard. Check the I/O signals and the external sequences for the drive before you set the Application Preset function. When you set the Application Preset function (A1-06 ≠ 0), it changes the I/O terminal functions for the drive and it can cause equipment to operate unusually. This can cause serious injury or death.*

NOTICE: *Fire Hazard. Install sufficient branch circuit short circuit protection as specified by applicable codes and this manual. The drive is suitable for circuits that supply not more than 31,000 RMS symmetrical amperes, 240 Vac maximum (200 V Class), 480 Vac maximum (400 V Class). Incorrect branch circuit short circuit protection can cause serious injury or death.*

NOTICE: *When the input voltage is 440 V or higher or the wiring distance is longer than 100 m (328 ft), make sure that the motor insulation voltage is sufficient or use an inverter-duty motor or vector-duty motor with reinforced insulation. Motor winding and insulation failure can occur.*

Note:

Do not connect the AC control circuit ground to the drive enclosure. Failure to obey can cause incorrect control circuit operation.

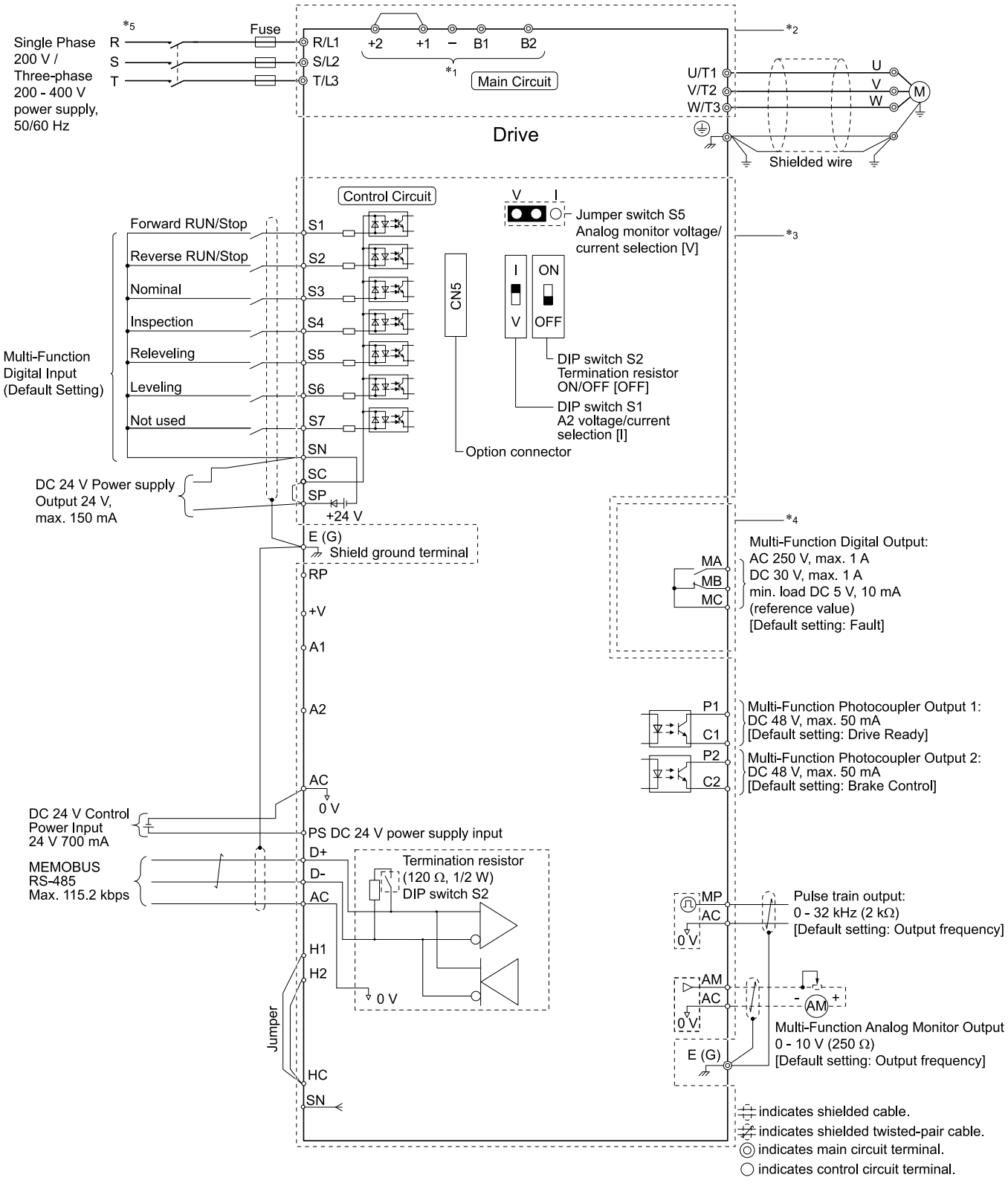


Figure 3.1 Standard Drive Connection Diagram

*1 Set the wiring sequence to de-energize the drive with the MFDO. If the drive outputs a fault during fault restart when you use the fault restart function, set *L5-02 = 1 [Fault Contact at Restart Select = Always Active]* to de-energize the drive. Be careful when you use a cut-off sequence. The default setting for *L5-02* is *0 [Active Only when Not Restarting]*.

*2 When you install a DC reactor, you must remove the jumper between terminals +1 and +2.

*3 When you use a regenerative converter or regenerative unit, set *L8-55 = 0 [Internal DB TransistorProtection = Disable]*. If *L8-55 = 1 [Protection Enabled]*, the drive will detect *rF [Braking Resistor Fault]*.

*4 When you use a regenerative converter, regenerative unit, braking resistor, or braking resistor unit, set *L3-04 = 0 [Stall Prevention during Decel = Disabled]*. If *L3-04 = 1 [General Purpose]*, the drive could possibly not stop in the specified deceleration time.

*5 When you use an ERF-type braking resistor, set *L8-01 = 1 [3% ERF DB Resistor Protection = Enabled]* and set a wiring sequence to de-energize the drive with the MFDO.

*6 Cooling fan wiring is not necessary for self-cooling motors.

*7 Connect peripheral options to terminals -, +1, +2, B1, and B2.

WARNING! *Fire Hazard. Only connect factory-recommended devices or circuits to drive terminals B1, B2, -, +1, and +2. Do not connect an AC power supply lines to these terminals. Incorrect wiring can cause damage to the drive and serious injury or death from fire.*

*8 Connect a 24 V power supply to terminals PS-AC to operate the control circuit while the main circuit power supply is OFF.

*9 To set the MFDI power supply (Sinking/Sourcing Mode or internal/external power supply), install or remove a jumper between terminals SC-SP or SC-SN depending on the application.

NOTICE: *Damage to Equipment. Do not close the circuit between terminals SP-SN. If you close the circuits between terminals SC-SP and terminals SC-SN at the same time, it will cause damage to the drive.*

- Sinking Mode, Internal power supply: Install the jumper to close the circuit between terminals SC-SP.

  NOTICE: *Damage to Equipment. Do not close the circuit between terminals SC-SN. If you close the circuits between terminals SC-SP and terminals SC-SN at the same time, it will cause damage to the drive.*

- Sourcing Mode, Internal power supply: Install the jumper to close the circuit between terminals SC-SN.

  NOTICE: *Damage to Equipment. Do not close the circuit between terminals SC-SP. If you close the circuits between terminals SC-SP and terminals SC-SN at the same time, it will cause damage to the drive.*

- External power supply: Remove the jumper from the MFDI terminals. It is not necessary to close the circuit between terminals SC-SP and terminals SC-SN.

*10 The maximum output current capacity for terminal +V on the control circuit is 20 mA.

NOTICE: *Damage to Equipment. Do not install a jumper between terminals +V and AC. A closed circuit between these terminals will cause damage to the drive.*

*11 DIP switch S1 sets terminal A2 for voltage or current input. The default setting for S1 is current input (“I” side).

*12 Do not ground the control circuit terminals AC or connect them to the drive chassis.

NOTICE: *Do not ground the AC control circuit terminals and only connect the AC terminals according to the product instructions. If you connect the AC terminals incorrectly, it can cause damage to the drive.*

*13 Connect the positive lead from an external 24 Vdc power supply to terminal PS and the negative lead to terminal AC.

NOTICE: *Connect terminals PS and AC correctly for the 24 V power supply. If you connect the wires to the incorrect terminals, it will cause damage to the drive.*

*14 Set DIP switch S2 to “ON” to enable the termination resistor in the last drive in a MEMOBUS/Modbus network.

*15 Use only Sourcing Mode for Safe Disable input.

*16 Use multi-function analog monitor outputs with analog frequency meters, ammeters, voltmeters, and wattmeters. Do not use monitor outputs with feedback-type signal devices.

*17 Jumper S5 sets terminal AM for voltage or current output. The default setting for S5 is voltage output (“V” side).

*18 Disconnect the wire jumpers between H1 and HC and H2 and HC to use the Safe Disable input.

3.3 Main Circuit Wiring

This section gives information about the functions, specifications, and procedures necessary to safely and correctly wire the main circuit in the drive.

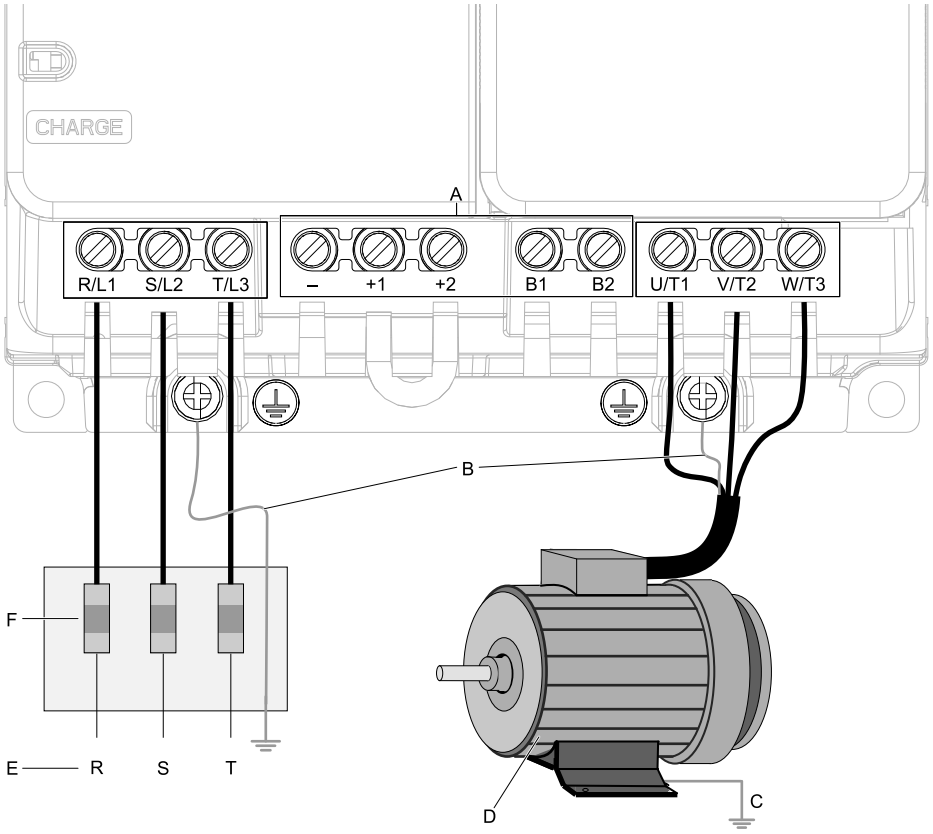
NOTICE: *Damage to Equipment. Do not energize and de-energize the drive more frequently than one time each 30 minutes. If you frequently energize and de-energize the drive, it can cause drive failure.*

Note:

Soldered wire connections can become loose over time and cause unsatisfactory drive performance.

◆ Motor and Main Circuit Connections

WARNING! *Electrical Shock Hazard. Do not connect terminals R/L1, S/L2, T/L3, L/L1, N/L2, U/T1, V/T2, W/T3, -, +1, +2, B1, or B2 to the ground terminal. If you connect these terminals to earth ground, it can cause damage to the drive or serious injury or death.*



Note:

The locations of terminals are different for different drive models.

A - DC bus terminal
B - Connect to the drive ground terminal.
C - Ground the motor case.
D - Three-Phase Motor
E - Use terminals R/L1, S/L2, and T/L3 for three-phase power supply input. Use terminals L/L1 and N/L2 for single-phase power supply input.
F - Input Protection (Fuses or Circuit Breakers)

Figure 3.2 Wiring the Main Circuit and Motor

◆ Configuration of Main Circuit Terminal Block

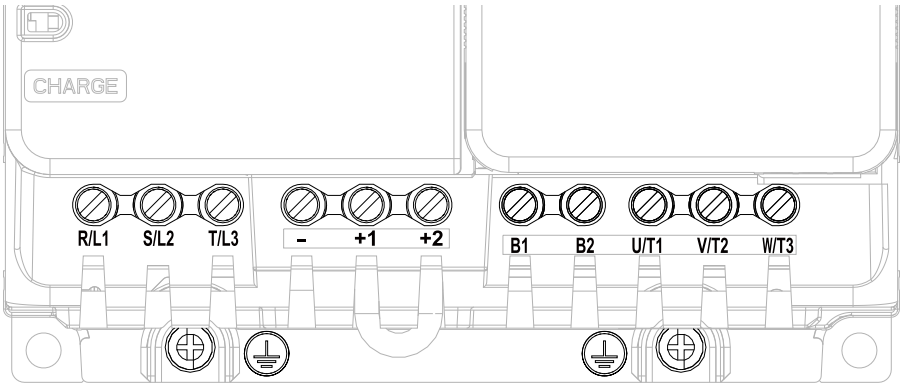


Figure 3.3 Configuration of Main Circuit Terminal Block Models 4009 (Three-Phase, Without a Built-in EMC Filter)

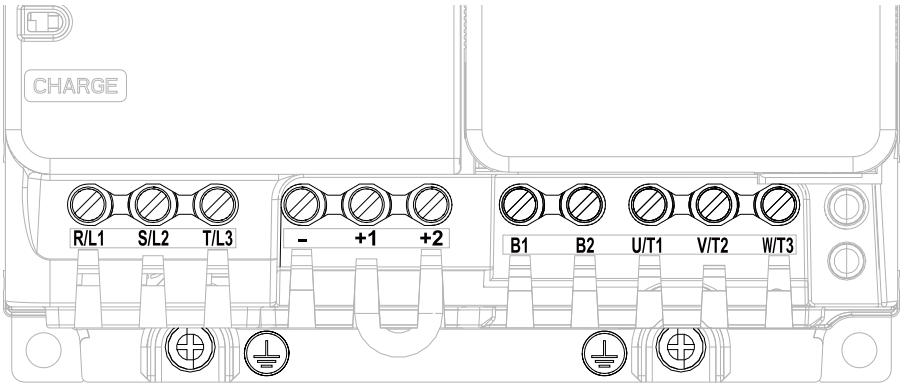


Figure 3.4 Configuration of Main Circuit Terminal Block Models 4009 (Three-Phase, With a Built-in EMC Filter)

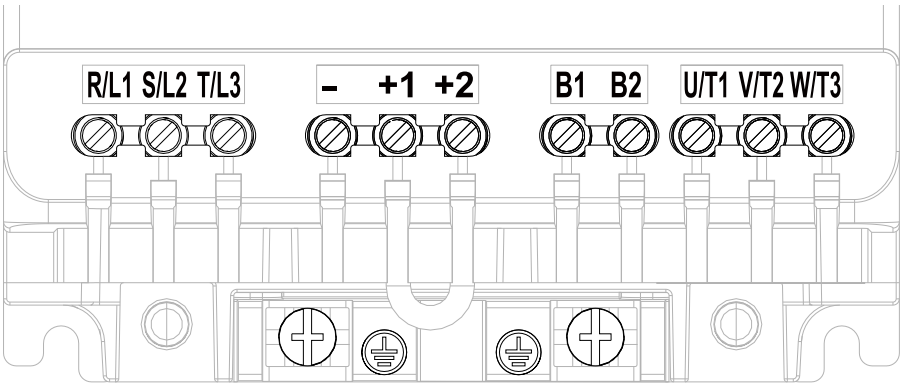


Figure 3.5 Configuration of Main Circuit Terminal Block Models 4015, 4018 (Three-Phase, Without a Built-in EMC Filter)

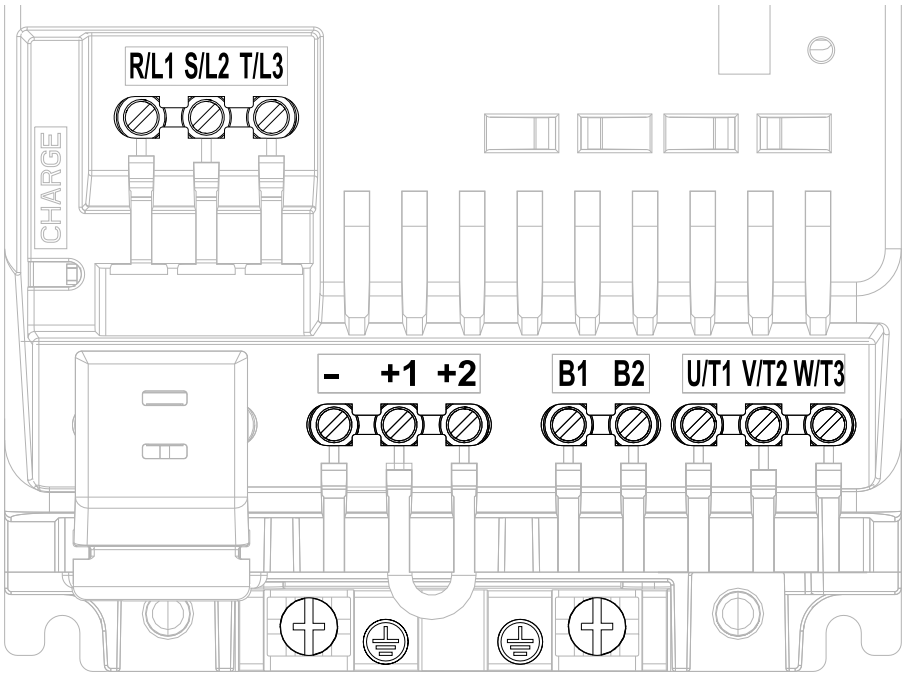


Figure 3.6 Configuration of Main Circuit Terminal Block Models 4015, 4018 (Three-Phase, With a Built-in EMC Filter)

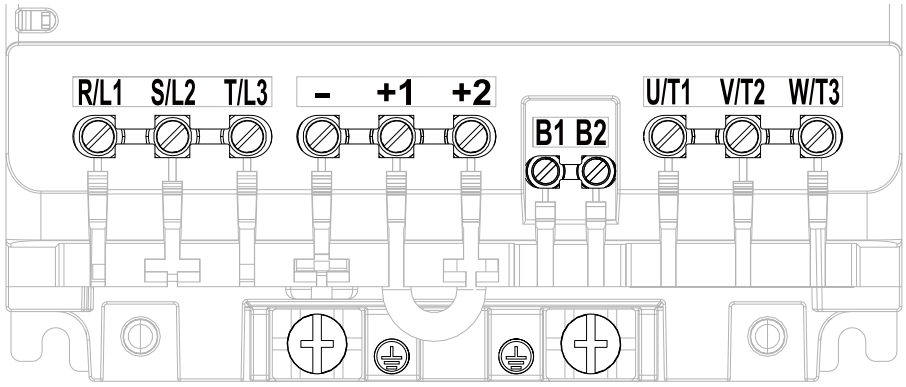


Figure 3.7 Configuration of Main Circuit Terminal Block Models 4024, 4031 (Three-Phase, Without a Built-in EMC Filter)

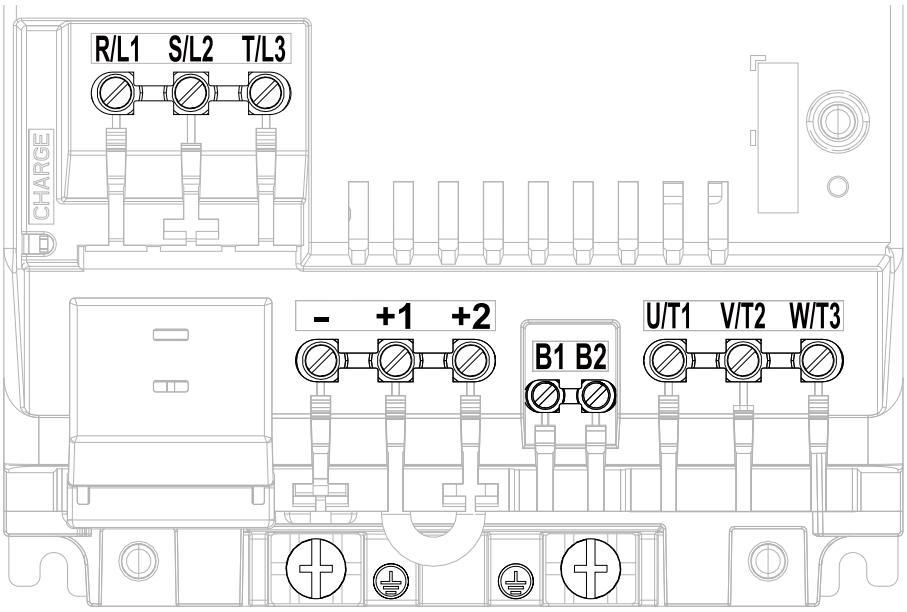


Figure 3.8 Configuration of Main Circuit Terminal Block Models 4024, 4031 (Three-Phase, With a Built-in EMC Filter)

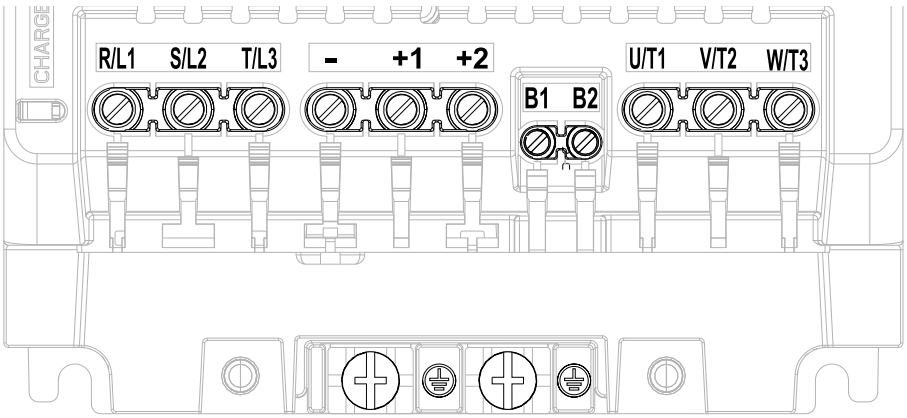


Figure 3.9 Configuration of Main Circuit Terminal Block Models 4039, 4045 (Three-Phase, Without a Built-in EMC Filter)

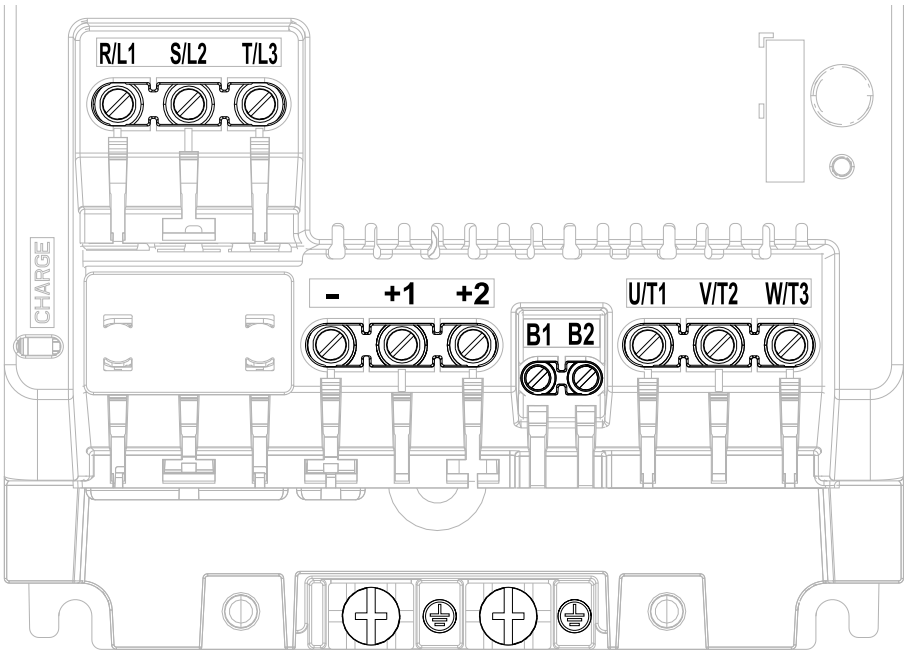


Figure 3.10 Configuration of Main Circuit Terminal Block Models 4039, 4045 (Three-Phase, With a Built-in EMC Filter)

◆ Main Circuit Terminal Functions

Table 3.1 Main Circuit Terminal Functions

Terminal	Name	Function
Model	4009 - 4045	
R/L1	Main circuit power supply input	To connect a commercial power supply.
S/L2		
T/L3		
L/L1	-	
N/L2		
U/T1	Drive output	To connect a motor.
V/T2		
W/T3		
+1	DC reactor connection	Remove the jumper between terminals +1 and +2 to connect a DC reactor.
+2		
B1	Braking resistor connection	To connect a braking resistor or braking resistor unit.
B2		
⏚	Ground Wiring	To ground the drive. • 400 V: C class grounding (ground to 10 Ω or less)

◆ Wire Selection

Select the correct wires for main circuit wiring.

Refer to *Main Circuit Wire Gauges and Tightening Torques on page 123* for wire gauges and tightening torques as specified by European standards.

◆ Main Circuit Terminal and Motor Wiring

This section outlines the various steps, precautions, and checkpoints for wiring the main circuit terminals and motor terminals.

WARNING! *Fire Hazard. Do not connect main power supply wiring to drive motor terminals U/T1, V/T2, and W/T3. Connect main power supply wiring to main circuit input terminals R/L1, S/L2, and T/L3. Incorrect wiring can cause serious injury or death from fire.*

WARNING! *Sudden Movement Hazard. Make sure that you align the phase order for the drive and motor when you connect the motor to drive output terminals U/T1, V/T2, and W/T3. If the phase order is incorrect, it can cause the motor to run in reverse. If the motor accidentally runs in reverse, it can cause serious injury or death.*

NOTICE: *Do not connect phase-advancing capacitors, LC/RC noise filters, or leakage breakers (RCM/RCD) to the motor circuit. If you connect these devices to the output circuits, it can cause damage to the drive and connected equipment.*

■ Cable Length Between Drive and Motor

When the wiring between the drive and the motor is too long, voltage drop along the motor cable can decrease motor torque, usually at low frequency output. If you connect motors in parallel with long motor cable, this is also a problem. Drive output current increases when the leakage current from the cable increases. An increase in leakage current can cause overcurrent and decrease the precision of the current detection.

Use the values in Table 3.2 to adjust the drive carrier frequency. For systems that have 100 m (328 ft) or longer motor wiring, if you use metal conduits or isolated cables for each phase, it will increase stray capacitance.

Table 3.2 Carrier Frequency against Cable Length Between Drive and Motor

Wiring Distance Between the Drive and Motor	50 m (164 ft) Maximum	100 m (328 ft) Maximum	More than 100 m (328 ft)
Carrier Frequency	15 kHz or less	5 kHz or less	2 kHz or less

Note:
- To set the carrier frequency in a drive that is operating more than one motor, calculate the cable length as the total distance of wiring to all connected motors.
- If the length of the wire between the drive and an induction motor is longer than 100 m (328 ft), set *A1-02 = 0 [V/f]*.
- The maximum cable length between the drive and a PM motor is 100 m (328 ft).

■ Ground Wiring

Follow the precautions to wire the ground for one drive or a series of drives.

WARNING! *Electrical Shock Hazard. Make sure that the protective ground wire complies with technical standards and local safety regulations. The EN 61800-5-1:2007 standard specifies that you must wire the power supply to automatically de-energize when the protective ground wire disconnects. If you turn on the internal EMC filter, the leakage current of the drive will be more than 3.5 mA. Use the closed-loop crimp terminal to connect a protective ground wire that has a minimum cross-sectional area of 10 mm² (copper wire). If you do not obey the standards and regulations, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Ground the neutral point on the power supply of drive models 4xxxE to comply with the EMC Directive before you turn on the EMC filter or if there is high resistance grounding. If you turn ON the EMC filter, but you do not ground the neutral point, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Use a ground wire that complies with technical standards on electrical equipment and use the minimum length of ground wire. Incorrect equipment grounding can cause serious injury or death from dangerous electrical potentials on the equipment chassis.*

WARNING! *Electrical Shock Hazard.*
Correctly ground the ground terminals. Obey federal and local electrical wiring codes for correct grounding methods. The maximum grounding resistance is
- *400 V class: ground to 10 Ω or less*

If you touch electrical equipment that is not grounded, it can cause serious injury or death.

Note:
- Only use the drive grounding wire to ground the drive. Do not share the ground wire with other devices such as welding machines or large-current electrical equipment. Incorrect equipment grounding can cause drive or equipment malfunction from electrical interference.
- To connect more than one drive to the same grounding circuit, follow the instructions in the instruction manual. Incorrect equipment grounding can cause drive or equipment malfunction from electrical interference.

■ Wiring the Main Circuit Terminal Block

WARNING! *Electrical Shock Hazard. Before you wire the main circuit terminals, make sure that MCCB and MC are OFF. If you touch electrical equipment when MCCB and MC are ON, it can cause serious injury or death.*

■ Main Circuit Configuration

The figures in this section show the different schematics of the drive main circuit. The connections change when the drive capacity changes. The DC power supply for the main circuit also supplies power to the control circuit.

Note:

Drive models 4009A to 4045A do not have a built-in EMC filter.

WARNING! *Fire Hazard. Do not connect a braking resistor to terminals +1 or -. Use terminals B1 and B2 for the braking resistor connections. If you connect a braking resistor to the incorrect terminals, it can cause damage to the drive and braking circuit and serious injury or death.*

NOTICE: *Do not use the negative DC bus terminal "-" as a ground terminal. This terminal is at high DC voltage potential. Incorrect wiring connections can cause damage to the drive.*

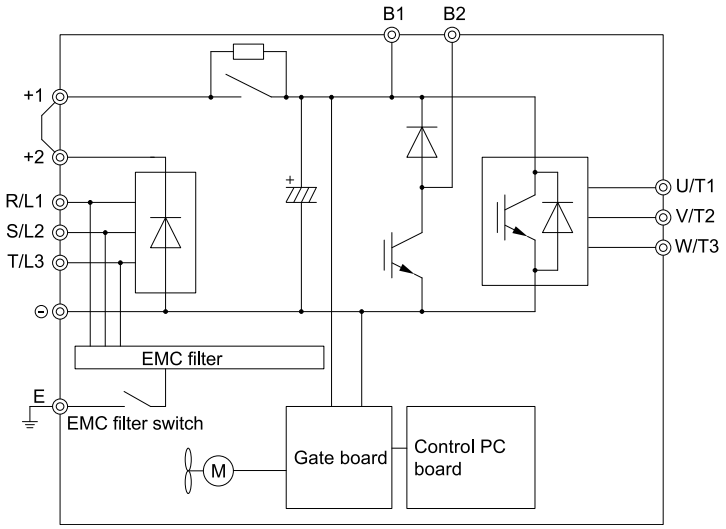


Figure 3.11 Drive Main Circuit Configuration for Models 4009 - 4045

3.4 Main Circuit Terminal Block Wiring Procedure

DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

◆ Wire to the Main Circuit Terminal Block

Wire to the main circuit terminal block correctly as specified by the instructions in the manual.

Read these instructions before wiring the terminal block.

■ Notes on Wiring the Main Circuit Terminal Block

Read these notes before you wire the main circuit terminal block.

- Use UL-Listed, vinyl-coated insulated copper wires for operation with a continuous maximum permitted temperature of 75 °C at 600 V.
- Remove all unwanted objects that are near the terminal block connections.
- Remove the insulation from the connection wires to the wire stripping lengths shown in the manual.
- Do not use bent or crushed wires. Remove the damaged end of the wire before you use it. Incorrect connections can cause death or serious injury from fire.
- Do not solder stranded wire. Soldered wire connections can become loose over time and cause unsatisfactory drive performance.
- If you use stranded wire, make sure that all of the wire strands are in the connection. Also, do not twist the stranded wire too much. Incorrect connections can cause death or serious injury from fire.
- Put the wire all the way into the terminal block. Remove the insulation from the wire to the recommended wire stripping length to fit the wire with insulation in the plastic housing.
- Use a torque driver, torque ratchet, or torque wrench for the screws. A slotted driver or a hex tool will be necessary to wire the screw clamp terminal. Use applicable tools as specified by the recommended conditions in the product manual.
- If you use power tools to tighten the terminal screws, use a low speed setting (300 to 400 r/min). Failure to obey can cause damage to the terminal screws.
- Wire gauges on existing drive models to be replaced may not match wire gauge ranges on new drives. Refer to the drive manuals for correct wire sizes.
- Do not tighten the terminal screws at an angle of 5 degrees or more. Failure to obey can cause damage to the terminal screws.
  If you damage a terminal screw, contact Yaskawa or your nearest sales representative.

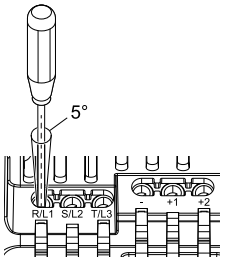


Figure 3.12 Permitted Angle

- Put the bit all the way into the hex socket to tighten the hex socket cap screw.
- When you tighten slotted screws, hold the straight-edge screwdriver perpendicularly to the screw. Make sure that you align the end of the straight-edge screwdriver with the screw groove.

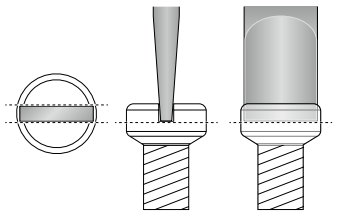
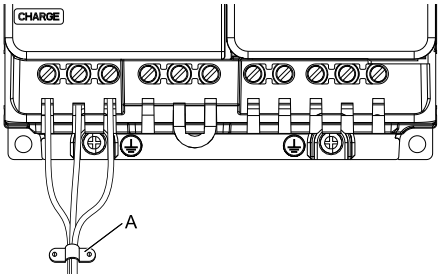


Figure 3.13 Tightening Slotted Screws

- After you connect the wires to the terminal block, lightly pull on the wires to make sure that they do not come out of the terminals.
- Do not let strain on the wiring cause damage. Use a strain relief near the wiring to release the tension. Refer to Figure 3.14 for an example.



A - Cable clamp

Figure 3.14 Strain Relief Example

Table 3.3 Recommended Wiring Tools

Screw Size	Screw Shape	Wire Gauge	Adapter	Bit Manufacturer, Model	Torque Driver Model (Tightening Torque)	Torque Wrench (Tightening Torque)
M3	⊖	-	Bit	PHOENIX CONTACT SF-BIT-SL 0,5X3,0-70	TSD-M 1,2NM (0.3 - 1.2 N·m (2.7 - 10.6 in·lb))	-
M4	⊖	-	Bit	PHOENIX CONTACT SF-BIT-SL 1,0X4,0-70	TSD-M 3NM (1.2 - 3.0 N·m (10.6 - 26.6 in·lb))	-
M5 *1	⊖	≤ 25 mm² (AWG 10)	Bit	PHOENIX CONTACT SF-BIT-SL 1,2X6,5-70	TSD-M 3NM (1.2 - 3.0 N·m (10.6 - 26.6 in·lb))	-
		≥ 30 mm² (AWG 8)			-	4.1 - 4.5 N·m (36.3 - 39.8 in·lb) *2
M6	⑤ (WAF: 5 mm)	-	Bit	PHOENIX CONTACT SF-BIT-HEX 5-50	-	5 - 9 N·m (44.3 - 79.7 in·lb) *2

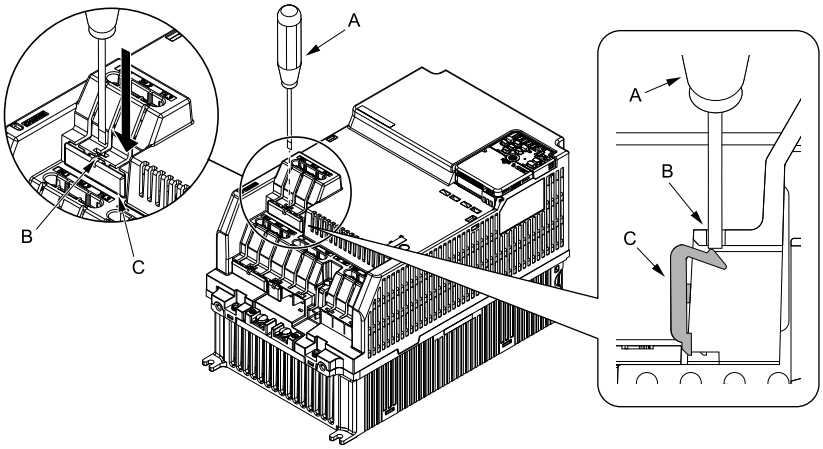
*1 Sselect the correct tools for the wire gauge.
*2 Use 6.35 mm (0.25 in) bit socket holder.

■ Remove IP20 Terminal Protective Cover

These drives have IP20 terminal protective covers. Remove the covers for the application.

Model	Terminal R/L1, S/L2, T/L3	Terminal -
4024, 4031	-	x
4039, 4045	x	x

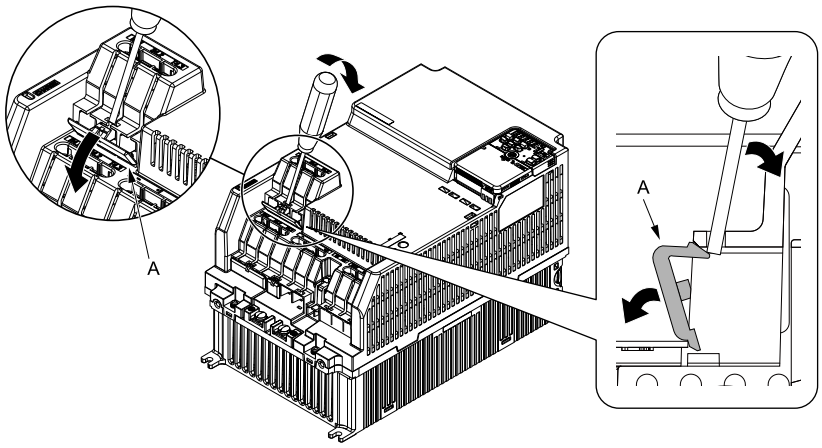
1. Put a slotted screwdriver blade into the slit to push the hook of the IP20 terminal protective cover.



A - Slotted screwdriver
B - Slit
C - IP20 terminal protective cover

Figure 3.15 Put the Screwdriver Blade into the Slit

2. Push up the screwdriver to release the IP20 terminal protective cover.



A - IP20 terminal protective cover

Figure 3.16 Release IP20 terminal protective cover

3. Remove IP20 terminal protective cover.

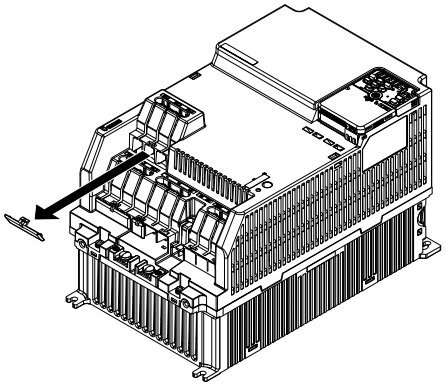


Figure 3.17 Remove IP20 Terminal Protective Cover

■ Main Circuit Terminal Block Wiring Procedure

When terminals R/L1, S/L2, T/L3, and terminal - have IP20 terminal protective covers, remove the cover on the terminal where you will wire.

1. Put wires with prepared ends into the main circuit terminal block.

   Look through the opening in the drive case to make sure that you correctly installed the wires into the terminal block.

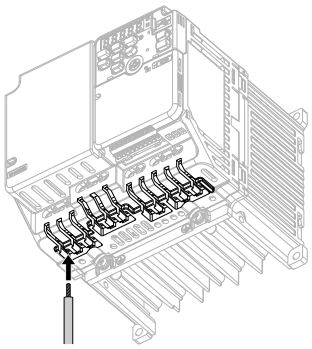


Figure 3.18 Install the Electrical Wire

Note:
There is a jumper between terminals +1 and +2. Remove the jumper, then wire to terminals +1 and +2.

2. Tighten the screws to the specified torque.

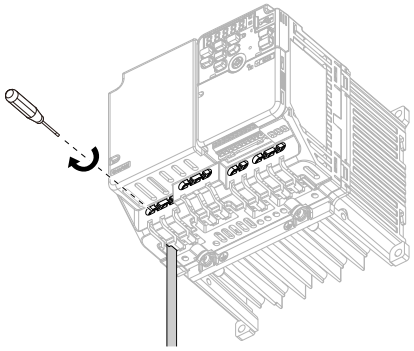


Figure 3.19 Tighten Terminal Block Screws

3.5 Control Circuit Wiring

This section gives information about how to correctly wire the control circuit.

◆ Control Circuit Connection Diagram

Wire the drive control circuit as shown in Figure 3.20.

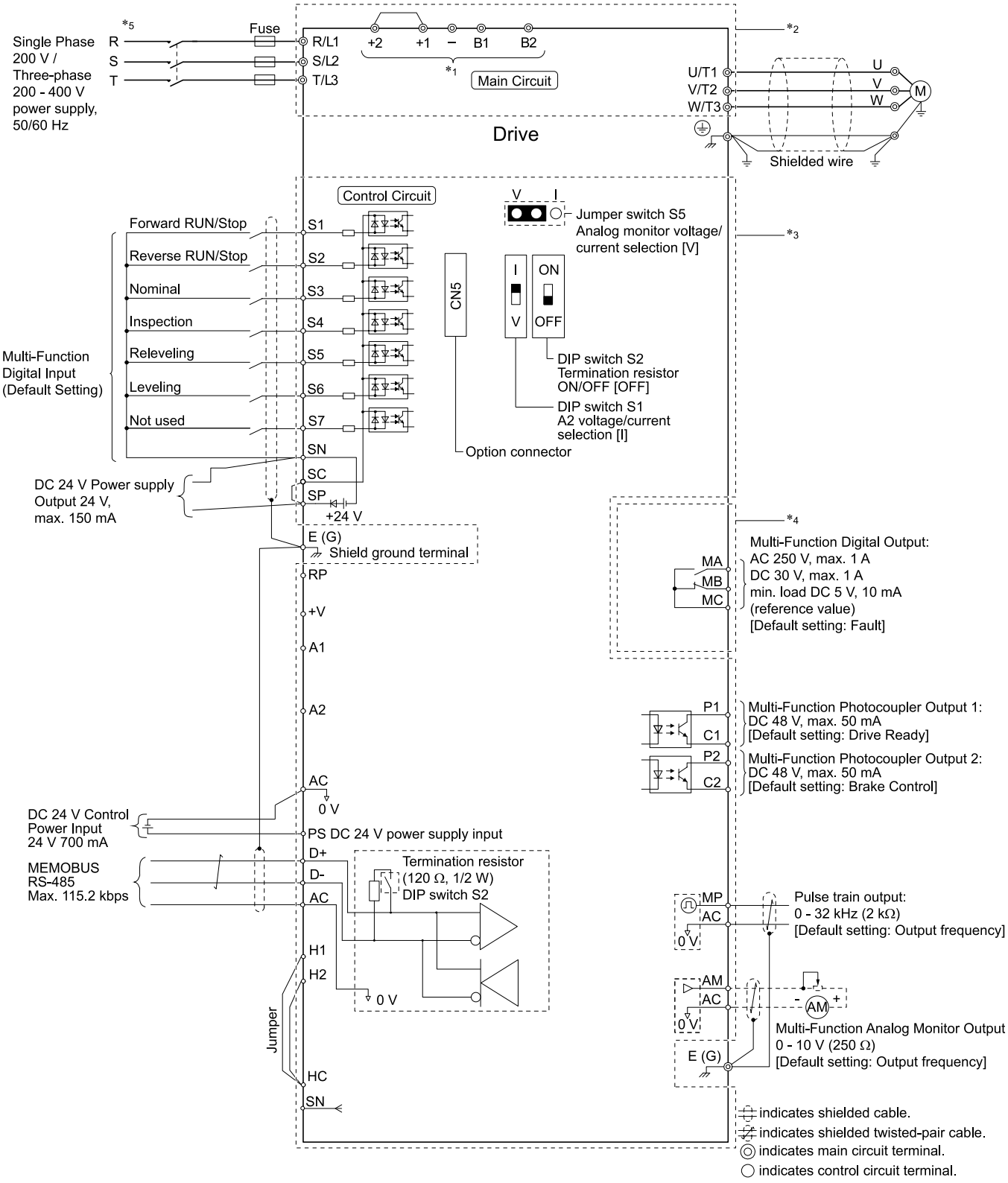


Figure 3.20 Control Circuit Connection Diagram

*1 Connect a 24 V power supply to terminals PS-AC to operate the control circuit while the main circuit power supply is OFF.

*2 To set the MFDI power supply (Sinking/Sourcing Mode or internal/external power supply), install or remove a jumper between terminals SC-SP or SC-SN depending on the application.

NOTICE: *Damage to Equipment. Do not close the circuit between terminals SP-SN. If you close the circuits between terminals SC-SP and terminals SC-SN at the same time, it will cause damage to the drive.*

- Sinking Mode, Internal power supply: Install the jumper to close the circuit between terminals SC-SP.

  NOTICE: *Damage to Equipment. Do not close the circuit between terminals SC-SN. If you close the circuits between terminals SC-SP and terminals SC-SN at the same time, it will cause damage to the drive.*

- Sourcing Mode, Internal power supply: Install the jumper to close the circuit between terminals SC-SN.

  NOTICE: *Damage to Equipment. Do not close the circuit between terminals SC-SP. If you close the circuits between terminals SC-SP and terminals SC-SN at the same time, it will cause damage to the drive.*

- External power supply: Remove the jumper from the MFDI terminals. It is not necessary to close the circuit between terminals SC-SP and terminals SC-SN.

*3 The maximum output current capacity for terminal +V on the control circuit is 20 mA.

NOTICE: *Damage to Equipment. Do not install a jumper between terminals +V and AC. A closed circuit between these terminals will cause damage to the drive.*

*4 DIP switch S1 sets terminal A2 for voltage or current input. The default setting for S1 is current input (“I” side).

*5 Do not ground the control circuit terminals AC or connect them to the drive chassis.

NOTICE: *Do not ground the AC control circuit terminals and only connect the AC terminals according to the product instructions. If you connect the AC terminals incorrectly, it can cause damage to the drive.*

*6 Do not connect terminals PS and AC inversely. Failure to obey will cause damage to the drive.

*7 Set DIP switch S2 to the ON position to enable the termination resistor in the last drive when you use MEMOBUS/Modbus communications.

*8 To use the internal power supply with the Safe Disable input, use sourcing mode.

*9 Use multi-function analog monitor outputs with analog frequency meters, ammeters, voltmeters, and wattmeters. Do not use monitor outputs with feedback-type signal devices.

*10 Jumper S5 sets terminal AM for voltage or current output. The default setting for S5 is voltage output (“V” side).

*11 Disconnect the wire jumpers between H1 and HC and H2 and HC to use the Safe Disable input.

◆ Control Circuit Terminal Block Functions

Hx-xx parameters set functions for the multi-function input and output terminals.

WARNING! *Sudden Movement Hazard. Correctly wire and test all control circuits to make sure that the control circuits operate correctly. If you use a drive that has incorrect control circuit wiring or operation, it can cause death or serious injury.*

WARNING! *Sudden Movement Hazard. Check the I/O signals and the external sequences for the drive before you set the Application Preset function. When you set the Application Preset function (A1-06 ≠ 0), it changes the I/O terminal functions for the drive and it can cause equipment to operate unusually. This can cause serious injury or death.*

NOTICE: *Damage to Equipment. Do not energize and de-energize the drive more frequently than one time each 30 minutes. If you frequently energize and de-energize the drive, it can cause drive failure.*

■ Input Terminals

Refer to Table 3.4 for a list of input terminals and functions.

Table 3.4 Multi-function Input Terminals

Type	Terminal	Name (Default)	Function (Signal Level)
Digital Inputs	S1	Up Command (Closed: Up, Open: Stop)	• Photocoupler • 24 V, 6 mA
	S2	Down Command (Closed: Down, Open: Stop)	
	S3	Multi-function input 1 (Nominal Speed)	
	S4	Multi-function input 2 (Inspection Operation)	
	S5	Multi-function input 3 (Releveling Speed)	
	S6	Multi-function input 4 (Leveling Speed)	
	S7	(Not used)	
	SC	MFDI selection common	MFDI power supply, 24 V (maximum 150 mA)
Safe Disable Input	H1	Safe Disable input 1	• 24 V, 6 mA • ON: Normal operation • OFF: Coasting motor • Internal impedance 4.7 kΩ • OFF Minimum OFF time of 3 ms.
	H2	Safe Disable input 2	
	HC	Safe Disable function common	Safe Disable function common
Master Frequency Reference	RP	Master frequency reference pulse train input (Master frequency reference)	• Response frequency: 0 to 32 kHz • H level duty: 30% to 70% • H level voltage: 3.5 V to 13.2 V • L level voltage: 0.0 V to 0.8 V • Input impedance: 3 kΩ
	AC	Frequency reference common	0 V

■ Output Terminals

Refer to Table 3.5 and Table 3.6 for a list of Output terminals and functions.

Table 3.5 Control Circuit Output Terminals

Type	Terminal	Name (Default)	Function (Signal Level)
Digital Outputs	MA	N.O. output (Fault)	• Relay output • 30 Vdc, 10 mA to 1 A • 250 Vac, 10 mA to 1 A • Minimum load: 5 V, 10 mA (Reference value)
	MB	N.C. output (Fault)	
	MC	Digital output common	
Multi-function Photocoupler Outputs	P1	• P1-PC Multi-function photocoupler output 1 (Drive Ready) • P2-PC Multi-function photocoupler output 2 (Brake Control)	• Photocoupler output • 48 V, 2 mA to 50 mA
	P2		
	PC		

Table 3.6 Control Circuit Monitor Output Terminals

Type	Terminal	Name (Default)	Function (Signal Level)
Monitor Output	AM	Analog monitor output (Output speed)	0 V to 10 V/0% to 100%
	AC	Monitor common	0 V

■ External Power Supply Input Terminals

Refer to Table 3.7 for a list of the functions of the external power supply input terminals.

Table 3.7 External Power Supply Input Terminals

Type	Terminal	Name (Default)	Function
External Power Supply Input Terminals	PS	External 24 V power supply input	Supplies backup power to the drive control circuit, keypad, and option board. 21.6 VDC to 26.4 VDC, 700 mA
	AC	External 24 V power supply ground	0 V

Alarm Display When You Use External 24 V Power Supply

When you use an external 24 V power supply, the drive detects an alarm as shown in Table 3.8 if you set *o2-23 [External 24V Powerloss Detection]* and *o2-26 [Alarm Display at Ext. 24V Power]* for the main circuit power supply. Set the alarm display as necessary.

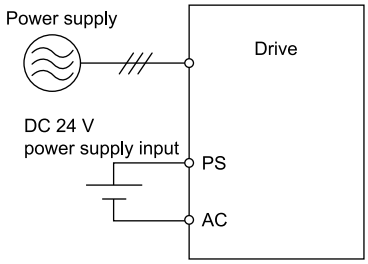


Table 3.8 Power Supply and Alarm Display

Main Circuit Power Supply	External 24 V Power Supply	o2-23 [External 24V Powerloss Detection]	o2-26 [Alarm Display at Ext. 24V Power]	Alarm Display
ON	ON	-	-	-
ON	OFF	0 [Disabled]	-	-
		1 [Enabled]	-	*L24v [Loss of External Power 24 Supply]*
OFF	ON	-	0 [Disabled]	"Ready" LED light flashes quickly
		-	1 [Enabled]	*EP24v [External Power 24V Supply]*

■ Serial Communication Terminals

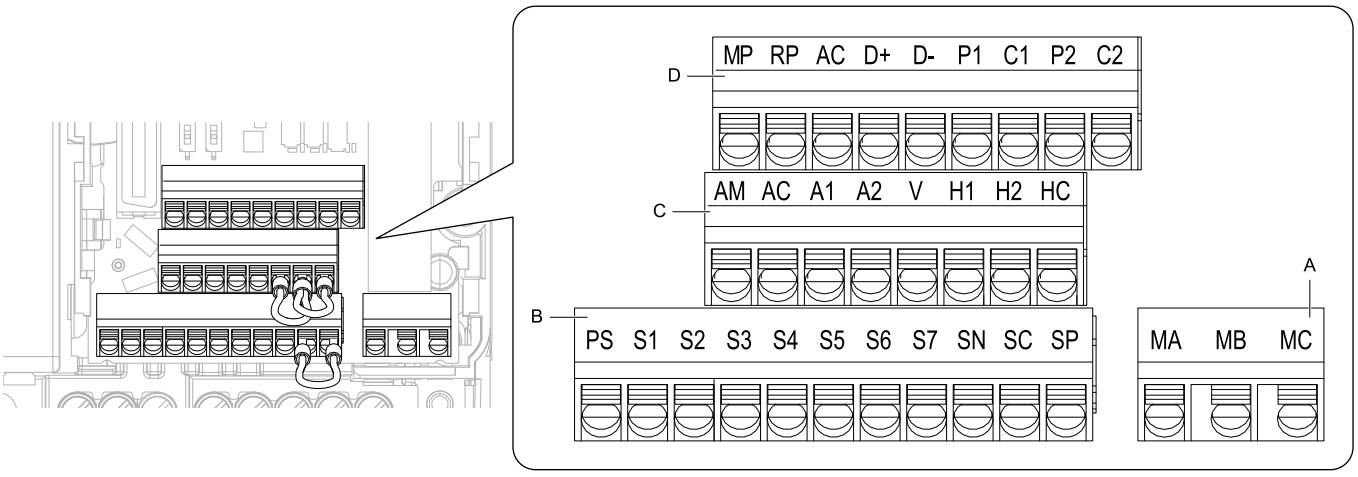
Refer to Table 3.9 for a list of serial communication terminals and functions.

Table 3.9 Serial Communication Terminals

Type	Terminal	Terminal Name	Function (Signal Level)	
Modbus Communication	D+	Communication input/output (+)	MEMOBUS/Modbus communications Use an RS-485 cable to connect the drive. **Note:** Set DIP switch S2 to ON to enable the termination resistor in the last drive in a MEMOBUS/Modbus network.	• RS-485 • MEMOBUS/Modbus communication protocol • Maximum 115.2 kbps
	D-	Communication output (-)		
	AC	Shield ground	0 V	

◆ Control Circuit Terminal Configuration

The control circuit terminals are in the positions shown in Figure 3.21.



A - Terminal block (TB2)
B - Terminal block (TB1-1)
C - Terminal block (TB1-2)
D - Terminal block (TB1-3)

Figure 3.21 Control Circuit Terminal Arrangement

■ Control Circuit Wire Gauges and Tightening Torques

Use the tables in this section to select the correct wires. Use shielded wire to wire the control circuit terminal block. Use crimp ferrules on the wire ends to make the wiring procedure easier and more reliable.

Table 3.10 Control Circuit Wire Gauges and Tightening Torques

Terminal Block	Terminal	Bare Wire		Crimp Ferrule	
		Recomm. Gauge mm² (AWG)	Applicable Gauge mm² (AWG)	Recomm. Gauge mm² (AWG)	Applicable Gauge mm² (AWG)
TB1-1	PS, S1 - S7, SN, SC, SP	0.75 (18)	• Stranded wire 0.25 - 1.0 (24 - 17) • Solid wire 0.25 - 1.5 (24 - 16)	0.5 (20)	0.25 - 0.5 (24 - 20)
TB1-2	AM, AC, A1, A2, +V, H1, H2, HC				
TB1-3	MP, RP, AC, D+, D-, P1, C1, P2, C2				
TB2	MA, MB, MC				

Crimp Ferrules

Attach an insulated sleeve when you use crimp ferrules. Refer to Table 3.11 for the recommended external dimensions and model numbers of crimp ferrules.

Use the CRIMPFOX 6, a crimping tool made by PHOENIX CONTACT.

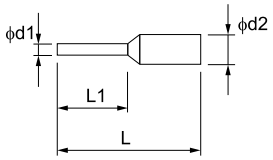


Figure 3.22 External Dimensions of Crimp Ferrules

Table 3.11 Crimp Ferrule Models and Sizes

Wire Gauge mm² (AWG)	Model	L (mm)	L1 (mm)	φd1 (mm)	φd2 (mm)
0.25 (24)	AI 0.25-6 YE AI 0.25-6 BU	10.5	6.0	0.8	2.0
0.34 (22)	AI 0.34-6 TQ	10.5	6.0	0.8	2.0
0.5 (20)	AI 0.5-6 WH AI 0.5-6 OG	12.0	6.0	1.1	2.5
0.75 (18)	AI 0.75-6 GY AI 0.75-6 WH	12.0	6.0	1.3	2.8
1.0 (17)	AI 1-6 RD AI 1-6 YE	12.0	6.0	1.5	3.0

◆ Wiring the Control Circuit Terminal

WARNING! *Electrical Shock Hazard. Do not remove covers or touch circuit boards while the drive is energized. If you touch the internal components of an energized drive, it can cause serious injury or death.*

NOTICE: *Do not let wire shields touch other signal lines or equipment. Insulate the wire shields with electrical tape or shrink tubing. If you do not insulate the wire shields, it can cause a short circuit and damage the drive.*

Note:
- Use a Class 2 power supply to connect external power to the control terminals. If the power supply for peripheral devices is incorrect, it can cause a decrease in drive performance.
- Connect the shield of shielded cable to the applicable ground terminal. Incorrect equipment grounding can cause drive or equipment malfunction from electrical interference.
- Isolate wiring for contact output terminals MA, MB, MC, P1, C1, P2, and C2 from other control circuit wiring. Incorrect wiring procedures can cause the drive and connected equipment to malfunction and cause the drive to trip.
- Isolate control circuit wiring from main circuit wiring (terminals R/L1, S/L2, T/L3, L/L1, N/L2, B1, B2, U/T1, V/T2, W/T3, -, +1, +2) and other high-power wiring. If control circuit wiring is adjacent to main circuit wiring, it can cause incorrect operation of the drive and equipment from electrical interference.

Wire the grounding terminal and main circuit terminals, then wire the control circuit terminals.

1. Remove the front cover from the drive.

   You must remove the keypad to move Jumper S5.

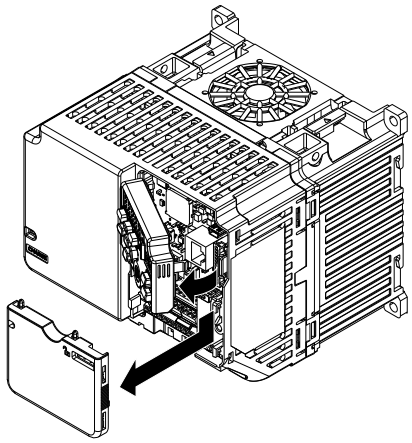


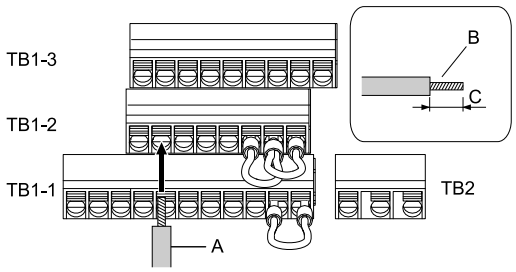
Figure 3.23 Remove the Front Cover

2. Refer to the figure and wire the control circuit.

   Use a slotted screwdriver with a blade width of 2.5 mm (0.1 in) or less and thickness of 0.4 mm (0.01 in) or less.

   WARNING! *Fire Hazard. Tighten all terminal screws to the correct tightening torque. Connections that are too loose or too tight can cause incorrect operation and damage to the drive. Incorrect connections can also cause death or serious injury from fire.*

Note:
- Use shielded, twisted-pair wires and ground the shield to the ground terminal of the drive. Incorrect equipment grounding can cause drive or equipment malfunction from electrical interference.
- Do not use control circuit wiring that is longer than 50 m (164 ft) to supply the analog frequency reference from a remote source. If the control circuit wiring is too long, it can cause unsatisfactory system performance.



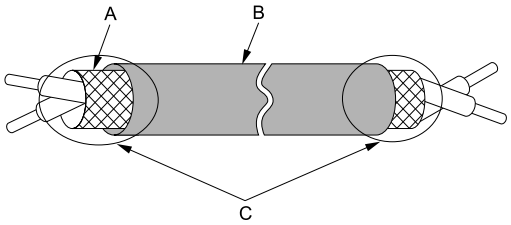
A - Wire with a crimp ferrule attached, or unsoldered wire with the core wires lightly twisted

B - Pull back the shielding and lightly twist the end with your fingers to keep the ends from fraying.

C - When you do not use crimp ferrules, remove approximately 5.5 mm (0.21 in) of the covering at the end of the wire.

Figure 3.24 Wiring Procedure for the Control Circuit

Note:
- It is easier to wire TB1-1 first, then TB1-2, then TB1-3.
- Do not solder the core wire. Soldered wiring connections can become loose and cause the drive to malfunction.
- Refer to Figure 3.25 for information to prepare terminal ends of the shielded wire.
- Prepare the wire ends of shielded twisted-pair wires as shown in Figure 3.25 to use an analog reference from an external frequency setting potentiometer to set the frequency. Connect the shield to terminal E (G) of the drive.



A - Connect the shield to terminal E (G) of the drive.
B - Sheath
C - Insulate with electrical tape or shrink tubing.

Figure 3.25 Prepare the Ends of Shielded Wire

3. Attach the front cover.

   If you moved Jumper S5, attach the keypad before you attach the front cover.

   If you did not move Jumper S5, attach the front cover.

   Make sure that you do not pinch wires or signal lines between the front cover and the drive before you reattach the cover.

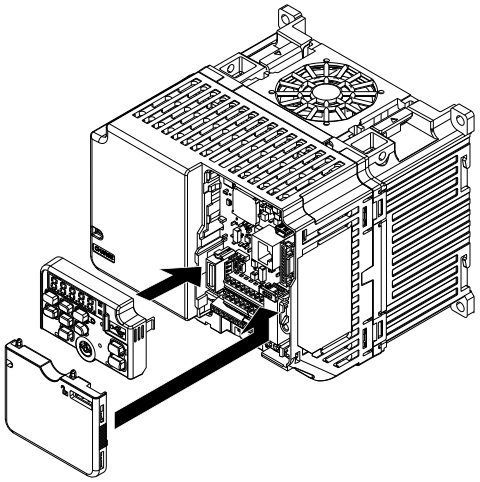


Figure 3.26 Reattach the Front Cover

◆ Switches and Jumpers on the Terminal Board

The terminal board has switches to adapt the drive I/Os to the external control signals as shown in Figure 3.27. Set the switches to select the functions for each terminal.

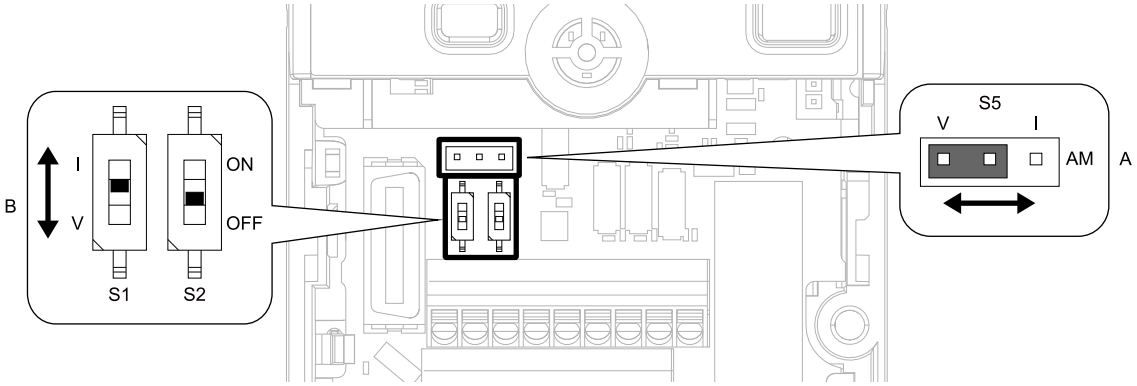


Figure 3.27 Locations of Switches

Table 3.12 I/O Terminals and Switches Functions

Position	Switch	Terminal	Function	Default
A	Jumper switch S5	AM	Sets the output method for terminal AM (voltage or current).	V (voltage output)
B	DIP switch S1	A2	Sets the input method for terminal A2 (voltage or current).	I (current input)
	DIP switch S2	-	Enables and disables the MEMOBUS/Modbus communications termination resistor.	OFF

3.6 Control I/O Connections

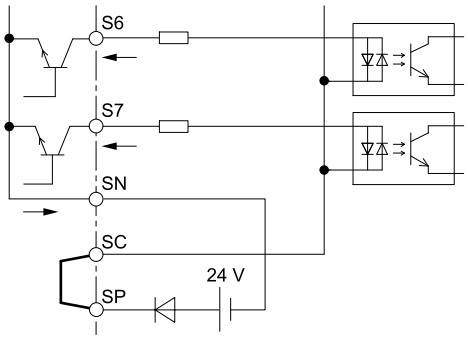
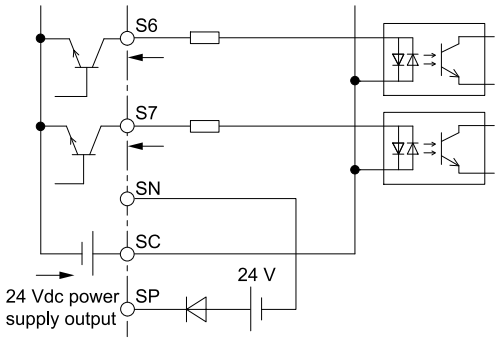
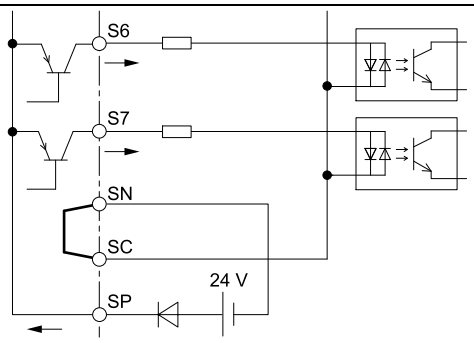
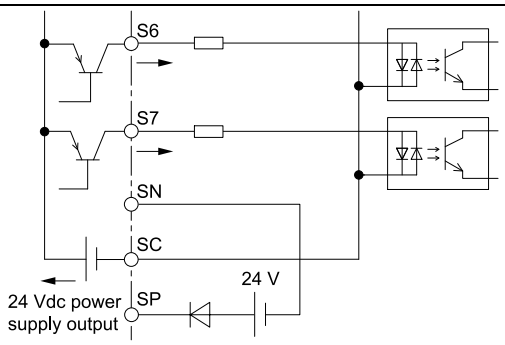
This section gives information about the settings for the listed control circuit I/O signals.

- MFDI (terminals S1 to S7)
- MFAI (terminal A2)
- MFAO (terminal AM)
- MEMOBUS/Modbus communications (terminals D+, D-, AC)

◆ Set Sinking Mode/Sourcing Mode

Close the circuit between terminals SC-SP and SC-SN to set the sinking mode/sourcing mode and the internal/external power supply for the MFDI terminals. The default setting for the drive is internal power supply sinking mode.

NOTICE: *Damage to Equipment. Do not close the circuit between terminals SP-SN. If you close the circuits between terminals SC-SP and terminals SC-SN at the same time, it will cause damage to the drive.*

Mode	Internal Power Supply (Terminal SN-SP)	External 24 V power supply
Sinking Mode (NPN)		
Sourcing Mode (PNP)		

◆ Set the Input Signal for the MFAI Terminal A2

Use terminal A2 to input a voltage or a current signal. Set the signal type as shown in Table 3.13.

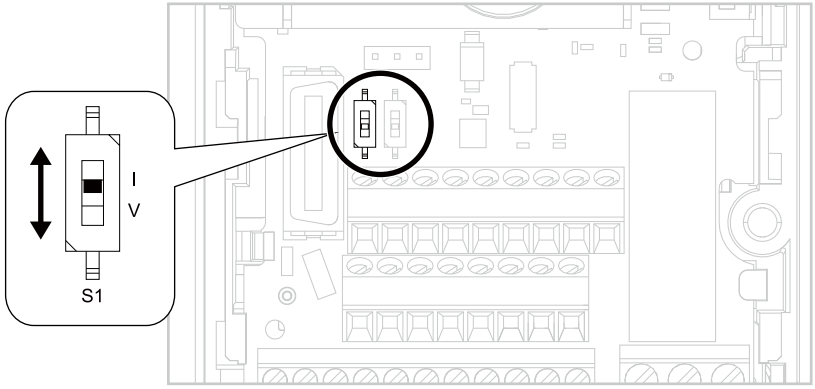


Figure 3.28 Location of DIP Switch S1

Table 3.13 MFAI Terminal A2 Signal Settings

Terminal	Input Signal	DIP Switch Settings		Parameter	
		Switch	Setting	No.	Signal Level
A2	Current input	S1	I (Default)	H3-09	2: 4 mA to 20 mA/0% to 100% (input impedance: 250 Ω) 3: 0 mA to 20 mA/0% to 100% (input impedance: 250 Ω)
	Voltage input		V		0: 0 V to 10 V/0% to 100% (with zero limit) (input impedance: 20 kΩ) 1: 0 V to 10 V/0% to 100% (without zero limit) (input impedance: 20 kΩ)

Note:
Use tweezers or a jig with a tip width of approximately 0.8 mm (0.03 in) to set DIP switches.

◆ Set the Output Signal for the MFAO Terminal AM

Set the signal type for terminal AM to voltage or current output. Use jumper S5 and *H4-07 [Terminal AM Signal Level Select]* to set the signal type.

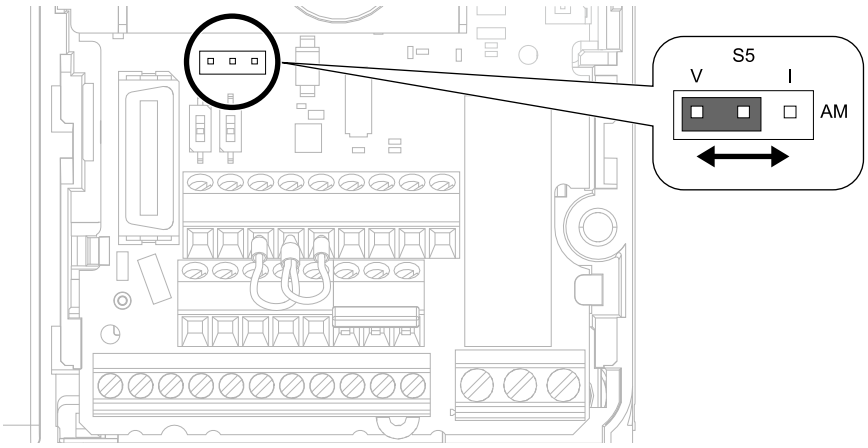


Figure 3.29 Location of Jumper Switch S5

Terminal	Types of Output Signals	Jumper S5	Parameter	
			No.	Signal Level
AM	Voltage output (Default)	V I AM	H4-07	0: 0 V to 10 V
	Current output	V I AM		2: 4 mA to 20 mA

◆ Switch ON Termination Resistor for MEMOBUS/Modbus Communications

When the drive is the last slave in a MEMOBUS/Modbus communications, set DIP switch S2 to the ON position. This drive has a built-in termination resistor for the RS-485 interface.

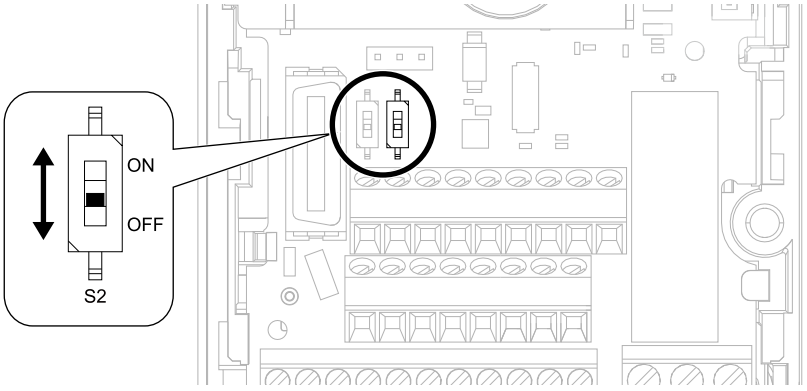


Figure 3.30 Location of DIP Switch S2

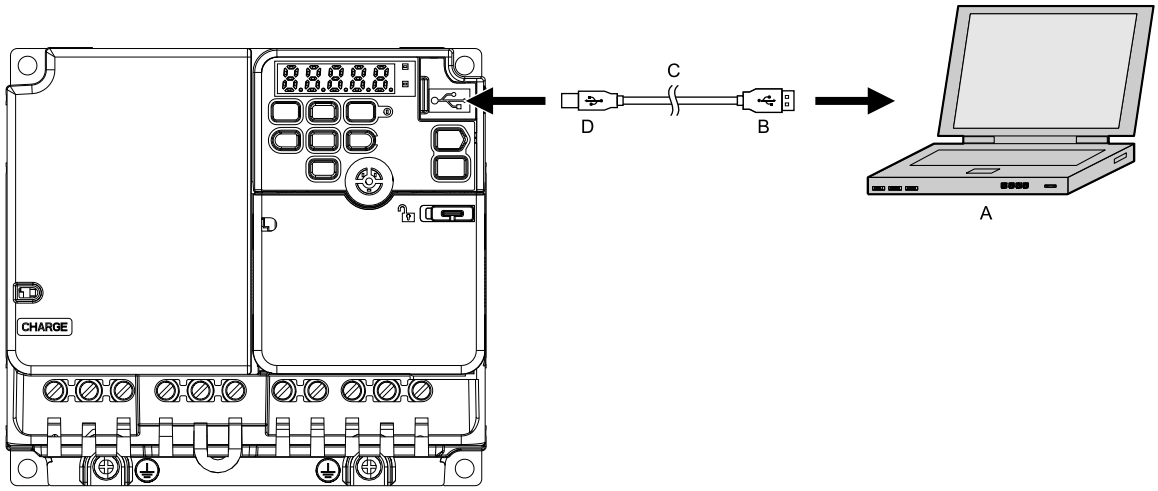
Table 3.14 MEMOBUS/Modbus Communications Termination Resistor Setting

DIP switch S2	Description
ON	The built-in termination resistor is ON.
OFF (Default)	The built-in termination resistor is OFF.

3.7 Connect the Drive to a PC

The drive has a mini-B type USB port.

You can use a USB cable (USB 2.0, type: A - mini-B) to connect the drive to a type-A USB port on a PC. After you connect the drive to the PC, you can use Yaskawa DriveWizard software to monitor drive performance and manage parameter settings.



A - PC
B - Type-A connector
C - USB 2.0, type A - mini-B cable
D - Mini-B type connector

Figure 3.31 Connect to a PC (USB)

Yaskawa recommends that you use a USB cable with connectors connected with shielded wires.

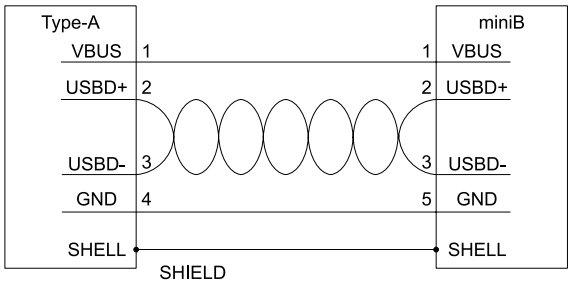


Figure 3.32 Recommended USB Cable

3.8 Braking Resistor Installation

A braking resistor or braking resistor unit (dynamic braking option) helps stop the motor quickly and smoothly when there is high load inertia. If you try to decelerate a motor in less time than usual for a coast to stop, the motor will rotate faster than the synchronous speed that aligns with the set frequency. This will cause the motor to become an induction generator. The inertia energy of the motor and regenerate to the drive and charge the drive DC bus capacitor and increase the voltage. If the voltage is more than the overvoltage level, an *ov [Overvoltage]* will occur. To prevent these overvoltage faults, a dynamic braking option is necessary.

WARNING!
Set L3-04 = 0 [Stall Prevention during Decel = Disabled] when you operate the drive with:
- *a regenerative converter*
- *regenerative unit*
- *braking resistor*
- *braking resistor unit.*

If you set the parameter incorrectly, the drive can decelerate for too long and cause serious injury or death.

NOTICE: *Damage to Equipment. Before you connect a dynamic braking option to the drive, make sure that qualified personnel read and obey the Braking Unit and Braking Resistor Unit Installation Manual (TOBPC72060001). If you do not read and obey the manual or if personnel are not qualified, it can cause damage to the drive and braking circuit.*

Note:
- Select the correct braking circuit size to dissipate the power that is necessary to decelerate the load in the correct time. Before you run the drive, make sure that the braking circuit can dissipate the energy for the set deceleration time.
- To install a dynamic braking option, set *L8-01 = 0 [3% ERF DB Resistor Protection = Disabled]*.

WARNING! *Fire Hazard. Do not connect a braking resistor to terminals +1 or -. Use terminals B1 and B2 for the braking resistor connections. If you connect a braking resistor to the incorrect terminals, it can cause damage to the drive and braking circuit and serious injury or death.*

NOTICE: *Connect braking resistors to the drive as shown in the connection diagram examples. If you wire the braking circuits incorrectly, it can cause damage to the drive or equipment.*

To connect a Yaskawa ERF series braking resistor to the drive, set *L8-01 = 1 [Enabled]*.

To use a non-ERF-type braking resistor, connect a thermal overload relay between the drive and the braking resistor and set a circuit to turn OFF the drive power at the trip contacts of the thermal overload relay.

◆ Install a Braking Resistor: ERF-Type

Connect the braking resistor to drive models 4009 as shown in Figure 3.33.

When you use a braking resistor, set *L8-01 = 1 [3% ERF DB Resistor Protection = Enabled]* and set one of the MFDO parameters *H2-01 to H2-03 = D [MFDO Function Selection = Braking Resistor Fault]*. Use a sequence to turn OFF the power with a MFDO.

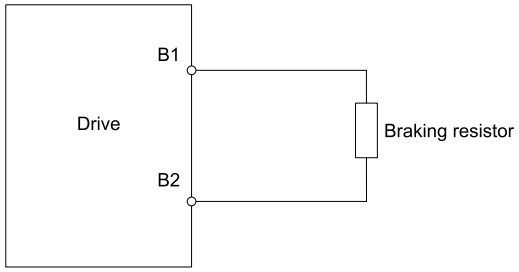


Figure 3.33 Install an ERF-Type Braking Resistor

◆ Install a Braking Resistor Unit: LKEB-Type

Connect the braking resistor unit as shown in Figure 3.34. To install a braking resistor unit, set *L8-01 = 0 [3% ERF DB Resistor Protection = Disabled]*.

This product has a built-in braking transistor.

To prevent overheating the braking resistor unit, set a sequence to de-energize the drive at the trip contacts of the thermal overload relay.

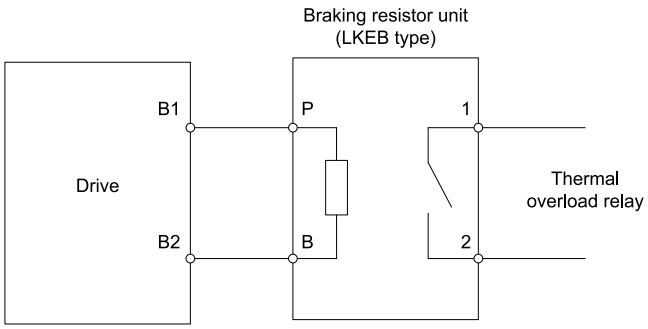


Figure 3.34 Install a Braking Resistor Unit: LKEB-Type

◆ Dynamic Braking Option Overload Protection

To prevent overheating the dynamic braking option, set a sequence to de-energize the drive at the trip contacts of the thermal overload relay.

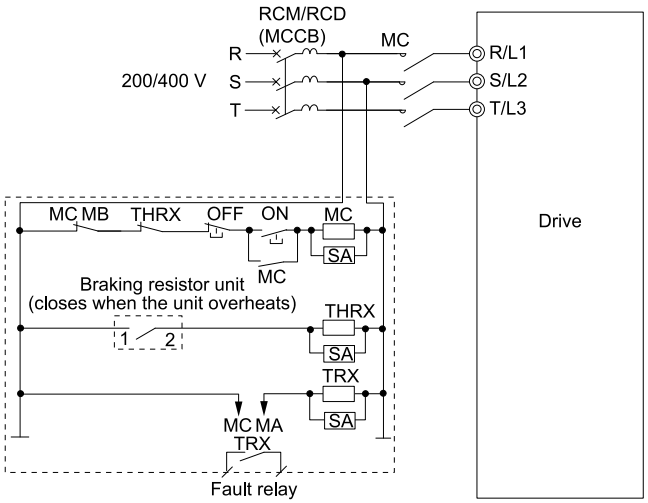


Figure 3.35 Power Supply Interrupt for Overheat Protection Example

WARNING! *Fire Hazard. When you use a braking unit, use a thermal relay on the braking resistors and set a fault contact output for the braking resistor unit to disconnect drive main power through an input contactor. Incorrect braking circuit protection can cause the resistors to become too hot and cause serious injury or death.*

3.9 Drive Wiring Protection

◆ Installing a Residual Current Monitoring/Detection (RCM/RCD)

When the drive output switches at high speeds, it causes high frequency leakage current. To prevent electrical shock and fires caused by ground fault protection that is not sufficient, install an RCM/RCD.

Use a high frequency RCM/RCD at the power input side of the drive and make sure that each drive has a minimum cumulative sensitivity amperage of 30 mA. The specialized breaker detects only the leakage current from frequency bands that are dangerous to humans.

If a device does not have protection against high frequencies, high frequency leakage currents can cause the device to malfunction. If you have a malfunction on a device that is not protected, decrease the carrier frequency of the drive, switch to a better breaker, or use an RCM/RCD with a minimum cumulative sensitivity amperage of 200 mA for each drive.

These conditions can have an effect on leakage current:

- Drive capacity
- Carrier frequency
- Wiring distance and types of motor cables
- EMI/RFI filter

To prevent damage and injury to personnel and drives, use a high-frequency RCM/RCD that is rated for AC and DC power supplies.

Note:

Yaskawa recommends these RCMs/RCDs, which are designed to operate with high frequencies:
- Mitsubishi Electric Corporation, NV series
- Schneider Electric, NS series

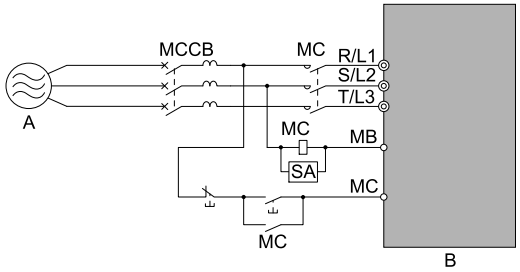
You can use a molded-case circuit breaker (MCCB) as a replacement for an RCM/RCD that is upstream in the power supply system.

◆ Installing a Molded-Case Circuit Breaker (MCCB) or Residual Current Monitor/Device (RCM/RCD)

Install a molded-case circuit breaker (MCCB) or a residual current monitor/device (RCM/RCD) for line protection between the power supply and main circuit power supply input terminals R/L1, S/L2, and T/L3. The MCCB or RCM/RCD gives overload protection and also prevent damage to the main circuit and the devices that are wired to the main circuit.

Use the information in this section to select the correct MCCB or RCM/RCD and to safely connect the device.

- The capacity of the MCCB or RCM/RCD must be 1.5 to 2 times the rated output current of the drive. Use an MCCB or RCM/RCD as an alternative to overheat protection (150% for one minute at the rated output current) to prevent drive faults.
- When you connect more than one drive or the drive and other device to an MCCB or RCM/RCD, refer to Figure 3.36, use a magnetic contactor (MC), and set a sequence that de-energizes the drive when it outputs errors.



A - Power supply **B - Drive**

Figure 3.36 Connect an MCCB

WARNING! *Electrical Shock Hazard. Use an MCCB, RCM/RCD, or Magnetic Contactor (MC) to de-energize the drive before you wire the main circuit terminal. If the main circuit terminal is energized during wiring, it will cause serious injury or death.*

3.10 Dynamic Braking Option, Motor Protection

◆ Install an Electromagnetic Contactor (MC) at the Input Side of the Drive

You can use an MC as an alternative to a molded case circuit breaker (MCCB) when:

- The protective functions of the drive have been triggered
- An emergency stop occurred, and the sequence de-energizes the drive.

If an MC on the input side of the drive (primary side) stops the drive, regenerative braking will not operate, and the drive will coast to stop.

NOTICE: *When you connect electromagnetic switches or magnetic contactors to the output motor circuits, make sure that you sequence them correctly. If the output motor circuit sequence is incorrect, it can cause damage to the drive.*

NOTICE: *Damage to Equipment. Do not energize and de-energize the drive more frequently than one time each 30 minutes. If you frequently energize and de-energize the drive, it can cause drive failure.*

Note:
- When machinery must not restart after recovery from a momentary power loss that occurred during run, install an MC at the input side of the drive and set a sequence that does not automatically set the Run command to ON after recovery of power.
- When it is necessary to stop momentary power loss, for example to maintain a circuit that has momentary power loss, use a delayed-release MC.
- Use an MC (magnetic contactor) to make sure that you can fully remove power to the drive when necessary. Wire the MC to open when a fault output terminal is triggered.

■ Protect the Braking Resistor/Braking Resistor Unit

Use an MC on the input side (primary side) to prevent damage to the braking resistor/braking resistor unit.

WARNING! *Fire Hazard. When you use a braking unit, use a thermal relay on the braking resistors and set a fault contact output for the braking resistor unit to disconnect drive main power through an input contactor. Incorrect braking circuit protection can cause the resistors to become too hot and cause serious injury or death.*

◆ Installing a Thermal Overload Relay on the Drive Output

A thermal overload relay disconnects the power line to the motor during a motor overload condition to prevent damage to the motor.

Install a thermal overload relay between the drive and motor in these conditions:

- When you operate more than one motor with one drive
- When you operate the motor directly from the power line with a power line bypass

When you operate one motor with one drive, it is not necessary to install a thermal overload relay. The drive has electronic motor overload protection in the drive software.

Note:
- When you install a thermal overload relay, set parameter *L1-01 = 0 [Motor Overload (oL1) Protection = Disabled]*.
- Set up a sequence that will trip an external fault (coast to stop) for the contacts of the thermal overload relay.

■ General Precautions When Using Thermal Overload Relays

When you use a motor thermal overload relay on the drive output to prevent nuisance trips and overheating of the motor at low speeds, be sure to think about these application precautions:

- Operation of a low speed motor
- When you operate more than one motor with one drive
- Length of the motor cables
- Nuisance tripping because of high drive carrier frequency

Operation of a Low Speed Motor

Usually, you use thermal overload relays on general-purpose motors (standard motors). When a drive drives a general-purpose motor, the motor current is approximately 5% to 10% more than with a commercial power supply. When a motor with a shaft-driven fan operates at low speeds, the cooling capacity decreases. This can cause the motor to overheat when the load current is in the motor rated value. Enable the electronic thermal protection in the drive when possible to prevent this problem.

The electronic thermal overload function uses the relation between the speed and heat characteristics in the variable speed control range to simulate the cooling ability of general-purpose motors and forced-vented motors to prevent damage to the motor.

Length of the Motor Cables

If you use long motor cables with a high carrier frequency, the increased leakage current can cause nuisance tripping of the thermal relay. To prevent this, decrease the carrier frequency or increase the tripping level of the thermal overload relay.

Nuisance Tripping Because of High Drive Carrier Frequency

High carrier frequency PWM drives make current waveforms that can increase the temperature in overload relays. It may be necessary to increase the trip level setting when encountering nuisance triggering of the relay.

WARNING! *Fire Hazard. Before you increase the detection level of the thermal relay, make sure that a secondary problem is not the cause of the overload. Make sure that you know the local codes for electrical wiring, then adjust the electrothermal settings. Incorrect thermal relay adjustment and incorrect wiring can cause serious injury or death.*

3.11 Improve the Power Factor

◆ Connect an AC Reactor or a DC Reactor

AC reactors and DC reactors decrease surges in current and improve the power factor on the input side of the drive.

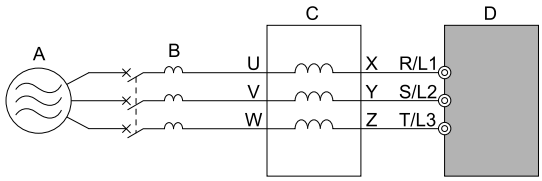
Connect an AC reactor or a DC reactor to the input side (primary side) in the these conditions:

- To decrease harmonic current or improve the power factor of the power supply
- When there is switching of phase advancing capacitor
- With a large capacity power supply transformer (600 kVA or more).

Note:

- You can use an AC reactor and DC reactor together.
- When you connect a thyristor converter (for example, a DC drive) to the same power supply system, use an AC reactor.
- The main circuit terminal block for the drive and the terminal blocks for the AC and DC reactors come in different shapes. Correctly prepare the ends of the wiring.
- Ground the AC and DC reactors (option) on the back of the mounting base. Remove all paint from the mounting surface of the control panel.

■ Connect an AC Reactor

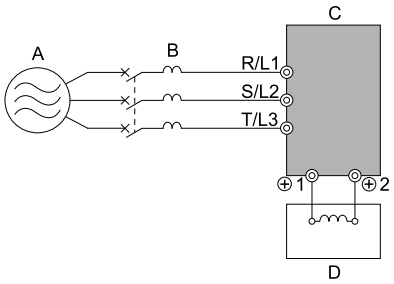


A - Power Supply
B - MCCB
C - AC reactor
D - Drive

Figure 3.37 AC Reactor Connection Example

■ Connect a DC Reactor

When you install a DC reactor, remove the jumper between terminals +1 and +2. If you will not use a DC reactor, do not remove the jumper. Refer to Figure 3.38 for an example of how to wire the DC reactor.



A - Power Supply
B - MCCB
C - Drive
D - DC reactor

Figure 3.38 DC Reactor Connection Example

Note:

You cannot connect a DC reactor to drive models B001 to B018.

For single-phase drives, an AC reactor has the same effect as a DC reactor. To improve the power factor on single-phase drives, install an AC reactor on the drive input side.

3.12 Prevent Switching Surge

◆ Connect a Surge Protective Device

A surge protective device decreases the surge voltage generated when you switch an inductive load near the drive. Inductive loads include:

- Magnetic contactors
- Electromagnetic relays
- Magnetic valves
- Solenoids
- Magnetic brakes.

Always use a surge protective device or diode with inductive loads.

Note:
Do not connect a surge protective device to the drive output side.

3.13 Decrease Noise

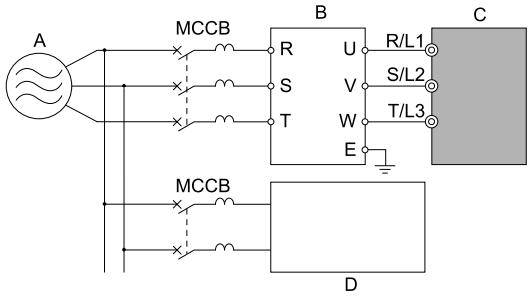
Note:

The main circuit terminal block for the drive and the terminal blocks for the AC and DC reactors come in different shapes. Correctly prepare the ends of the wiring.

◆ Connect a Noise Filter to the Input Side (Primary Side)

High-speed switching makes noise in the drive output. This noise flows from the drive to the power supply, and can have an effect on other equipment. Install a noise filter to the input side of the drive to decrease the quantity of noise that flows to the power supply. A noise filter will also stop noise from entering the drive from the power supply.

- Use a noise filter specially designed for drives.
- Install the noise filter as close as possible to the drive.



A - Power supply
B - Input side (primary side) noise filter
C - Drive
D - Other controller

Figure 3.39 Example of Connecting the Noise Filter on the Input Side (Primary Side)

Note:

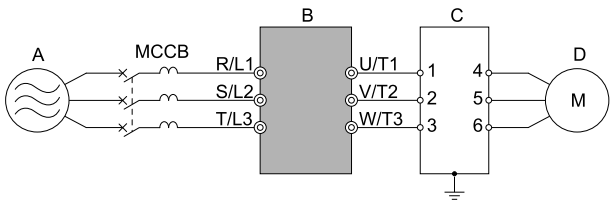
Drive models LA50CxxxxExx have built-in input noise filters. When you use these drives, make sure that the filter specifications align with the application and applicable regulations. Refer to *European Standards on page 114* for more information.

◆ Connect a Noise Filter to the Output Side (Secondary Side)

A noise filter on the output side of the drive decreases inductive noise and radio frequency interference.

Figure 3.40 shows an example of noise filter wiring.

NOTICE: *Do not connect phase-advancing capacitors, LC/RC noise filters, or leakage breakers (RCM/RCD) to the motor circuit. If you connect these devices to the output circuits, it can cause damage to the drive and connected equipment.*



A - Power supply
B - Drive
C - Noise filter on output side (secondary side)
D - Motor

Figure 3.40 Example of Connecting the Noise Filter on the Output Side (Secondary Side)

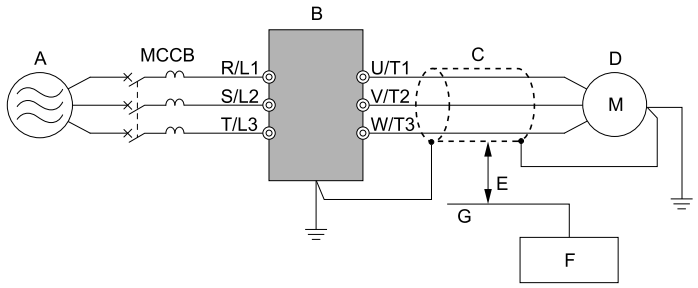
Note:

Glossary

- Radio frequency interference:
  Electromagnetic waves radiated from the drive and cables make noise through the full radio bandwidth that can have an effect on nearby devices.
- Inductive noise:
  The noise from electromagnetic induction can have an effect on the signal line and can cause the controller to malfunction.

■ Prevent Inductive Noise

In addition to installing a noise filter, you can also run all wiring through a grounded metal conduit to decrease inductive noise occurring at the output side. Put the cables a minimum of 30 cm (11.8 in) away from the signal line to prevent induced noise. Ground the cables to metal conduits.



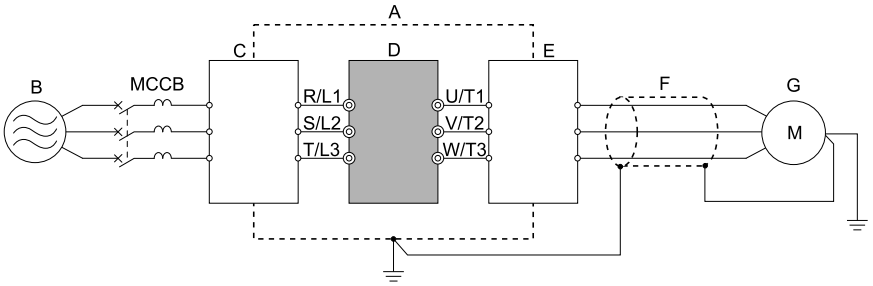
A - Power supply
B - Drive
C - Shielded motor cable
D - Motor
E - Minimum of 30 cm (11.8 in) apart
F - Controller
G - Signal line

Figure 3.41 Prevent Inductive Noise

■ Decrease Radio Frequency Interference

The drive, input lines, and output lines generate radio frequency interference. Use noise filters on input and output sides and install the drive in a steel box to decrease radio frequency interference.

Note:
Keep the cable between the drive and motor as short as possible.



A - Steel box
B - Power supply
C - Noise filter
D - Drive
E - Noise filter
F - Shielded motor cable
G - Motor

Figure 3.42 Decrease Radio Frequency Interference

3.14 Protect the Drive during Failures

◆ Factory-Recommended Branch Circuit Protection for UL Listing

Use branch circuit protection to protect against short circuits and to maintain compliance with UL61800-5-1. Yaskawa recommends connecting semiconductor protection fuses on the input side for branch circuit protection.

WARNING! *Electrical Shock Hazard. After the drive blows a fuse or trips an RCM/RCD, do not immediately energize the drive or operate peripheral devices. Wait for the time specified on the warning label at a minimum and make sure that all indicators are OFF. Then check the wiring and peripheral device ratings to find the cause of the problem. If you do not know the cause of the problem, contact Yaskawa before you energize the drive or peripheral devices. If you do not fix the problem before you operate the drive or peripheral devices, it can cause serious injury or death.*

- 400 V class
  Use the fuses specified in this document to prepare the drive for use on a circuit that supplies not more than 31,000 RMS and not more than 480 Vac when there is a short circuit in the power supply.

The built-in short circuit protection of the drive does not provide branch circuit protection. The user must provide branch circuit protection as specified by the National Electric Code (NEC), the Canadian Electric Code, Part I (CEC), and local codes.

■ Factory Recommended Fuses Three-Phase 400 V Class

Table 3.15 Factory-Recommended Branch Circuit Protection: Three-Phase 400 V Class

Drive Model	Maximum Applicable Motor Output kW (HP)	Time Delay Fuse	Semiconductor Protection Fuse Rated Current Manufacturer: EATON/Bussmann	
	HD	Class J, CC, and T Fuse Rated Current A	Model	Fuse Rated Current A
4009	4.0 (5)	20	FWH-90B	90
4015	5.5 (10)	30	FWH-80B	80
4018	7.5 (10)	40	FWH-100B	100
4024	11.0 (15)	50	FWH-125B	125
4031	15.0 (20)	60	FWH-175B	175
4039	18.5 (25)	70	FWH-200B	200
4045	22.0 (30)	100	FWH-200B	200

3.15 Wiring Checklist

Wire the drive, examine these items, then do a test run.

Table 3.16 Power Supply Voltage

Checked	No.	Item to Check
	1	The power supply voltage must be in the input voltage specification range of the drive.

Table 3.17 Main Circuit Wiring

Checked	No.	Item to Check
	1	• Put the power supply through a molded-case circuit breaker (MCCB) before it gets to the drive input. • Connect an applicable MCCB.
	2	Correctly wire the power supply to drive terminals R/L1, S/L2, and T/L3, or L/L1 and N/L2.
	3	Correctly wire the drive and motor together. The motor lines and drive output terminals U/T1, V/T2, and W/T3 must align to make the correct phase order. **Note:** If the phase order is incorrect, the drive will rotate in the opposite direction.
	4	Use 600 V heat resistant indoor PVC wire for the power supply and motor lines. **Note:** Wire gauge recommendations assume use of 600 V class 2 heat-resistant indoor PVC wire.
	5	Use the correct wire gauges for the main circuit. **Note:** • When the wiring distance between the drive and the motor is long, use this formula for the voltage drop in the wire: Motor rated voltage (V) × 0.02 ≥ √3 × wire resistance (Ω/km) × wiring distance (m) × motor rated current (A) × 10^{-3} • When the cable between the drive and motor is longer than 50 m (164 ft), use parameter *C6-02 [Carrier Frequency Selection]* to decrease the carrier frequency.
	6	Correctly ground the drive.
	7	Tighten the main circuit and grounding terminal screws of the drive to a correct tightening torque.
	8	When you use a braking resistor or a braking resistor unit, install an electromagnetic contactor (MC). Correctly install the resistor and make sure that overload protection uses the MC to shut off the power supply.
	9	Make sure you did not install phase advancing capacitors, input noise filters, or ELCBs, GFCIs, RCM/RCDs on the output side of the drive.

Table 3.18 Control Circuit Wiring

Checked	No.	Item to Check
	1	Use twisted-pair cables for all drive control circuit wiring.
	2	Ground the shields of shielded wiring to terminal E (G).
	3	For 3-Wire sequence, set parameters for MFDI terminals and wire control circuits.
	4	Correctly install any options.
	5	Examine the drive for other wiring errors. Only use a multimeter to check wiring.
	6	Tighten the control circuit terminal screws of the drive to a correct tightening torque.
	7	Pick up all wire clippings.
	8	Make sure that none of the wires on the terminal block touch other terminals or connections.
	9	Make sure that you isolate the control circuit wiring from main circuit wiring in the control panel or in a duct.
	10	Make sure that control circuit wiring is not longer than 50 m (164 ft).
	11	Make sure that Safe Disable input wiring is not longer than 30 m (98 ft).

3.16 Motor Application Precautions

◆ Precautions for Existing Standard Motors

■ Insulation Withstand Voltage

Consider motor voltage tolerance levels and motor insulation in applications with an input voltage of over 440 V or particularly long wiring distances. Use an insulated drive motor.

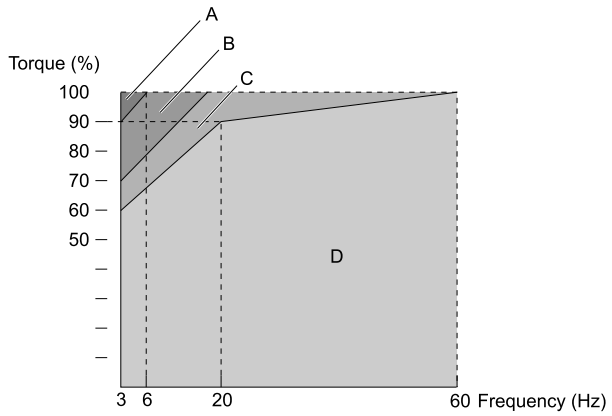
NOTICE: *Use an inverter-duty motor or vector-duty motor with reinforced insulation and windings applicable for use with an AC drive. If the motor does not have the correct insulation, it can cause a short circuit or ground fault from insulation deterioration.*

■ High-Speed Operation

If you operate a motor more than its rated speed, you can have problems with the motor bearing durability and dynamic balance of the machine. Contact the motor or machine manufacturer.

■ Low-Speed Range

When a drive operates a standard motor, it will lose more power compared to operating the motor with a commercial power supply. In the low speed range, the temperature of the motor increases quickly because the motor cannot decrease its temperature when the speed decreases. In these conditions, decrease the load torque of the motor in the low-speed range. Figure 3.43 shows the permitted load characteristics for a Yaskawa standard motor. When 100% continuous torque is necessary at low speeds, use a motor designed to operate with a drive.



A - 25% ED (or 15 min)
B - 40% ED (or 20 min)
C - 60% ED (or 40 min)
D - Continuous operation

Figure 3.43 Permitted Load Characteristics for a Yaskawa Standard Motors

■ Torque Characteristics

When you operate a motor with a drive, the torque characteristics are different than when you operate the motor directly from line power. Make sure that you know about the load torque characteristics for your application.

■ Vibration

Vibrations could occur in the these conditions:

- Resonance with the natural frequency of machinery
  Use caution if you add a variable-speed drive to applications that operate the motor from line power at a constant speed. If resonance occurs, install shock-absorbing rubber around the base of the motor and enable the Jump frequency control.
- The motor is not balanced
  Use caution if the motor speed is more than the rated motor speed.
- Subsynchronous resonance
  Subsynchronous resonance can occur with long motor shafts and in applications such as turbines, blowers, and fans with high inertia loads.

■ Audible Noise

The audible noise of the motor changes when the carrier frequency setting changes. When you use a high carrier frequency, audible noise from the motor is equivalent to the motor noise generated when you operate from line power. If you operate at speeds that are more than the rated rotation speed, the unwanted motor noise increases.

◆ Precautions for Specialized Motors

■ Submersible Motors

The rated current of a submersible motor is more than the rated current of a standard motor. Use a sufficiently large motor cable that will not let voltage drop decrease the maximum torque level.

■ Explosion-Proof Motors

You must test the motor and the drive together for explosion-proof certification. You must also test existing installations of explosion-proof motors. The drive is not designed for explosion-proof areas. Install the drive in a safe location.

The encoder used with pressure-resistant explosion-proof motors is intrinsically safe. When wiring between the drive and encoder, always connect through a specialized pulse coupler.

■ Geared Motors

The continuous speed range is different for different lubricating methods and manufacturers. For oil lubrication, continuous operation in the low-speed range can cause burnout. Contact the manufacturer for more information about applications where operating at more than the rated frequency is necessary.

■ Single-Phase Motors

Variable speed drives are not designed to operate with single-phase motors. The drive is for use with three-phase motors only. If you use capacitors to start the motor, it can cause a high frequency current to flow to the capacitors and can damage the capacitors. A split-phase start or a repulsion start can burn out the starter coils because the internal centrifugal switch is not activated.

■ Motors with Brakes

If you use a drive to operate a motor that has a brake connected to the output side, low voltage levels can cause the brake to possibly not release at start. Use a motor with a brake that has a dedicated source of power for the brake. Connect the brake power supply to the power supply side of the drive. Motors with built-in brakes make noise when operating at low speeds.

◆ Notes on the Power Transmission Mechanism

For power transmission machinery that uses oil to lubricate gearboxes, transmissions, or reduction gears, make sure that you use precaution if you operate the machinery continuously at low speed. Oil does not lubricate the system as well at low speeds. If you operate at frequencies higher than the rated frequency, it can cause problems with the power transmission mechanism. These problems include audible noise, decreased service life, and decreased durability.

4

Startup Procedure and Test Run

4.1 Section Safety 78
4.2 Overview of Keypad Components and Functions 79
4.3 Set up the Drive with User-Parameter Menu 82
4.4 Drive Mode and Programming Mode 84
4.5 Start-up Procedures 88
4.6 Auto-Tuning 91
4.7 Setup Procedure for Elevator Applications 94
4.8 Items to Check before Starting Up the Drive 102
4.9 Keypad Operation 103
4.10 Auto-Tuning 107
4.11 Fine Tuning during Test Runs (Adjust the Control Function) 109

4.1 Section Safety

⚠DANGER

Electrical Shock Hazard

Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe.

If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.

⚠WARNING

Electrical Shock Hazard

Do not operate the drive when covers are missing. Replace covers and shields before you operate the drive. Use the drive only as specified by the instructions.

Some figures in this section include drives without covers or safety shields to more clearly show the inside of the drive. If covers or safety shields are missing from the drive, it can cause serious injury or death.

Do not remove covers or touch circuit boards while the drive is energized.

If you touch the internal components of an energized drive, it can cause serious injury or death.

Sudden Movement Hazard

When you use a mechanical holding brake with the drive in a lifting application, you must close the brake if an input terminal triggers the Baseblock command to stop drive output.

If you enter the baseblock command, the motor will suddenly coast and the load will slip, which can cause serious injury or death.

When you use the drive in a lifting application, you must also install external safety circuitry. The drive does not have protection against accidental load drops in lifting applications. Install electrical and/or mechanical safety circuit mechanisms that are isolated from the drive circuitry.

If you do not use external safety circuitry, the drive could drop the load and cause serious injury or death.

4.2 Overview of Keypad Components and Functions

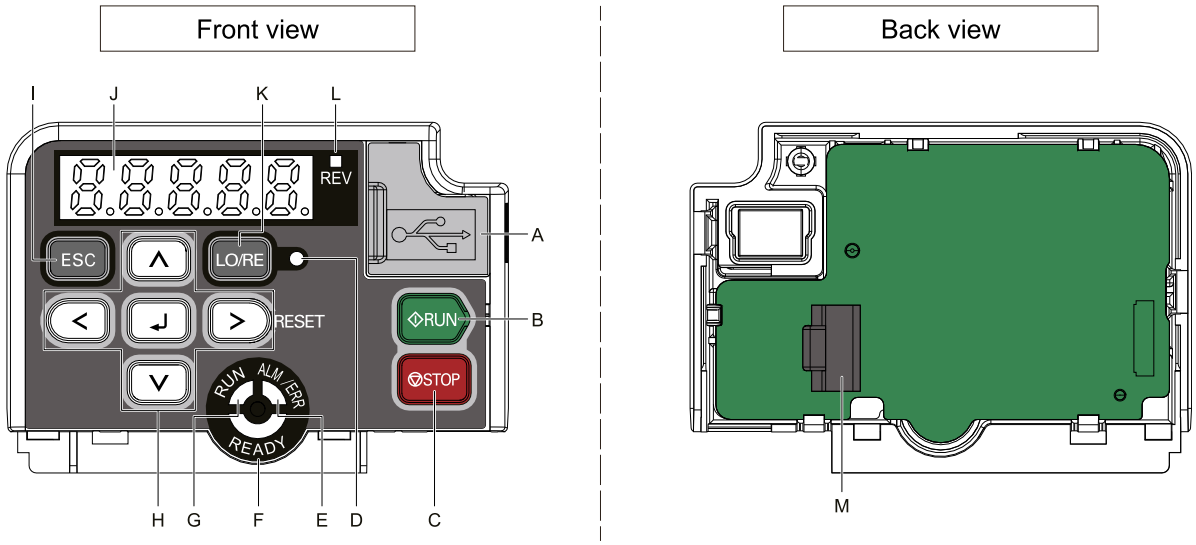


Figure 4.1 Keypad

Table 4.1 Keypad Components and Functions

Symbol	Name	Function
A	USB Terminal Pass-through	Pass-through point to connect a USB cable to the drive to support the DriveWizard PC tool and the DriveWizard Mobile App. Uses a USB cable (USB standard 2.0, type A - mini-B) to connect the drive to a PC, Android smartphone, or tablet.
B	RUN Key	Starts the drive in LOCAL Mode. Starts the operation in Auto-Tuning Mode. **Note:** Before you use the keypad to operate the motor, push LO/RE on the keypad to set the drive to LOCAL Mode.
C	STOP Key	Stops drive operation. **Note:** Uses a stop-priority circuit. Push STOP to stop the motor. This will also apply when a Run command (REMOTE Mode) is active at an external Run command source. To disable STOP priority, set *o2-02 = 0 [STOP Key Function Selection = Disabled]*.
D	LO/RE LED	Illuminated: The keypad controls the Run command (LOCAL Mode). OFF: The control circuit terminal or serial transmission device controls the Run command (REMOTE Mode). **Note:** • LOCAL: Use the keypad to operate the drive. Use the keypad to enter Run/Stop commands and the frequency reference command. • REMOTE: Use the control circuit terminal or serial transmission to operate the drive. Use the Run command source selected in *b1-02*.
E	ALM/ERR LED	Illuminated: The drive detects a fault. OFF: There are no drive faults or alarms. Flashing: • An alarm • Operation Errors • An Auto-Tuning error **Note:** The LED will illuminate to identify a fault if the drive detects a fault and an alarm at the same time.
F	READY LED	Illuminated: The drive is operating or is ready for operation. OFF: • The drive detects a fault. • There is no fault and the drive received a Run command, but the drive cannot run. For example, in Programming Mode. Flashing: The drive is in *STo [Safe Torque OFF]* condition. Flashing quickly: The voltage of the main circuit power supply is not in drive nameplate specifications, and the external 24 V power supply provides the only power to the drive.

Symbol	Name	Function
G	RUN LED	Illuminated: The drive is in normal operation. OFF: The drive is stopped. Flashing: • The drive is decelerating to stop. • The drive received a Run command, but the frequency reference is 0 Hz. Flashing quickly: • When the drive is in LOCAL Mode, the drive received a Run command from the MFDI terminals and is switched to REMOTE Mode. • The drive received a Run command from the MFDI terminals when the drive is not in Drive Mode. • The drive received a Fast Stop command. • The safety function shut off the drive output. • The user pushed STOP on the keypad while the drive is operating in REMOTE Mode.
H	Left Arrow Key	Moves the cursor to the left.
	Up Arrow Key/Down Arrow Key	• Moves to a different screen. • Selects parameter numbers and increments or decrements setting values.
	Right Arrow Key (RESET)	• Moves the cursor to the right. • Resets the drive to clear a fault.
	ENTER Key	• Enters parameter values and settings. • Selects each mode, parameter, and set value.
I	ESC Key	• Goes back to the previous screen. • Push and hold to go back to the frequency reference screen (the initial screen).
J	LED Display	Shows parameters, errors, and other data.
K	LO/RE Selection Key	Switches drive control for the Run command and frequency reference between the keypad (LOCAL) and an external source (REMOTE). **Note:** • The LOCAL/REMOTE Selection Key continuously stays enabled after the drive stops in Drive Mode. If the application must not switch from REMOTE to LOCAL because it will have a negative effect on system performance, set *o2-01 = 0 [LO/RE Key Function Selection = Disabled]* to disable LO/RE. • The drive will not switch between LOCAL and REMOTE when it is receiving a Run command from an external source.
L	REV LED	Illuminated: The drive received a Reverse run command.
N	RJ-45 Connector	Connects to the drive. Use an RJ-45 8-pin straight through UTP CAT5e extension cable to install the keypad in a different location than the drive.

◆ Indicator flashing statuses

Refer to Figure 4.2 for the difference between “flashing” and “flashing quickly”.

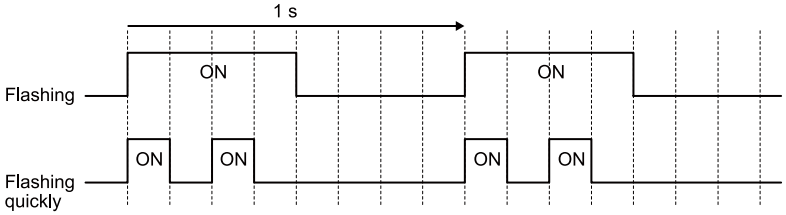


Figure 4.2 About indicator flashing statuses

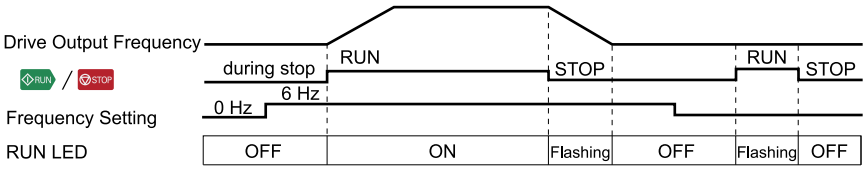


Figure 4.3 Relation between RUN indicator and Drive Operation

◆ Keypad Mode and Menu Displays

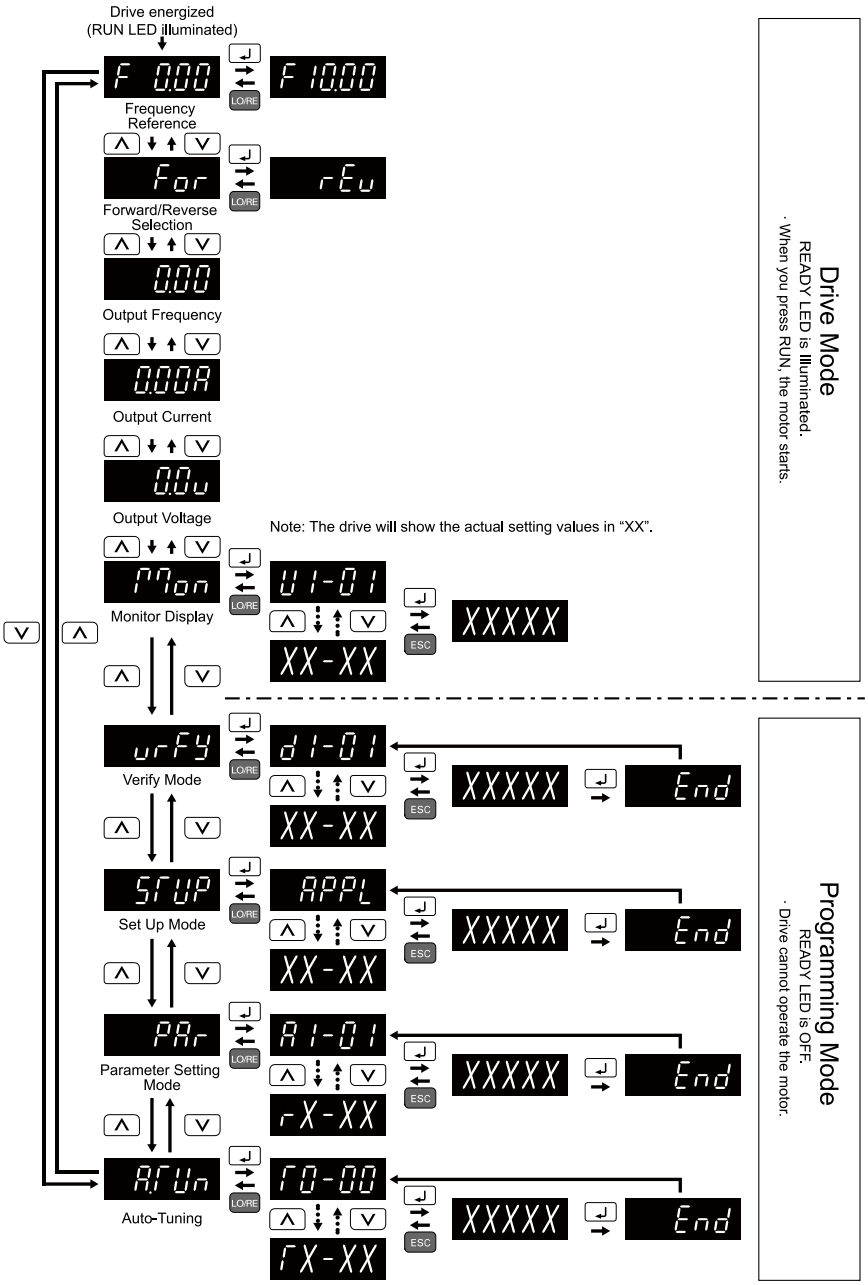


Figure 4.4 Keypad Functions and Display Levels

4.3 Set up the Drive with User-Parameter Menu

Drive parameters are in letter groups from A to U. Setup Mode SrUP contains only the most frequently used parameters to help you set up the drive more easily.

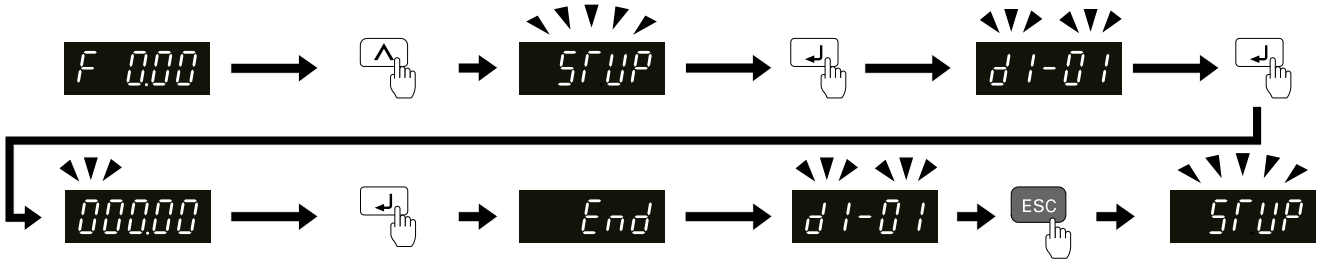


Figure 4.5 Parameters in General-Purpose Setup Mode

Table 4.2 shows the parameters available in Setup Mode. To access parameters not shown in the Setup Mode, use the PAr menu.

Table 4.2 Parameters in General-Purpose Setup Mode

User Parameter	Parameter	Name
A2-01	A1-00	Language Selection
A2-02	A1-02	Control Method Selection
A2-03	d1-18	Speed Reference Selection Mode
A2-04	d1-19	Nominal Speed
A2-05	d1-23	Releveling Speed
A2-06	d1-24	Inspection Operation
A2-07	d1-25	Rescue Operation Speed
A2-08	d1-26	Leveling Speed
A2-09	d1-01	Speed Reference 1
A2-10	d1-02	Speed Reference 2
A2-11	d1-03	Speed Reference 3
A2-12	d1-04	Speed Reference 4
A2-13	d1-05	Speed Reference 5
A2-14	d1-06	Speed Reference 6
A2-15	d1-07	Speed Reference 7
A2-16	d1-08	Speed Reference 8
A2-17	d1-28	Nominal/Leveling Spd Detection
A2-18	d1-29	Inspection Speed Detection
A2-19	C1-01	Acceleration Time 1
A2-20	C1-02	Deceleration Time 1
A2-21	C2-01	S-Curve @ Start of Acceleration
A2-22	C2-02	S-Curve @ End of Acceleration
A2-23	C2-03	S-Curve @ Start of Deceleration
A2-24	C2-04	S-Curve @ End of Deceleration
A2-25	C2-05	Jerk @ Levelig
A2-26	E2-03	Motor No-Load Current
A2-27	S1-02	DC-Injection @ Start
A2-28	S1-03	DC-Injection @ Stop
A2-29	S1-06	Brake Open Delay
A2-30	S1-07	Brake Close Delay

User Parameter	Parameter	Name
A2-31		User Parameter Selection
A2-32		
A2-33		

4.4 Drive Mode and Programming Mode

The keypad display of this drive has two modes: Drive Mode and Programming Mode.

- Drive Mode
  Use this mode to operate the drive. These operations are available:
  - Monitor operation statuses (for example, output frequency, output current, and output voltage)
  - Set the parameters that you can change while the drive is operating (for example, d1-01 to d1-17). Refer to *Parameter Details on page 249* for more information.
- Programming Mode
  Use this mode to set parameters. These operations are available:
  - Examine and set the parameters that are not at default settings (Verify Mode)
  - See and set the basic parameters necessary for the drive operation (Setup Mode)
  - See and set all parameters (Parameter Setting Mode)
  - Automatically set motor parameters (Auto-Tuning Mode)

Table 4.3 gives information about the functions you can access when you push [∧] / [∨].

Note:
Set *b1-08 [Run Command Select in PRG Mode]* to accept or reject the Run command from an external source when in Programming Mode. Refer to *b1-08: Up/Down Command Select in PRG Mode on page 254* for more information.

Table 4.3 Overview of the Modes

Mode	LED Display	Description	Description	Ref.
Drive Mode (Operation of the motor and monitoring of operation status)	F 0.00	Frequency reference display	You can set and monitor frequency references.	-
	MOn	Monitor display	The keypad shows *Ux-xx [Monitor]*.	-
	0.0U	Output voltage display	You can monitor the output voltage reference. Set *o1-01 [User Monitor Selection]* to change the item shown on this screen.	315
	0.00A	Output current display	You can monitor the output current.	-
	0.00	Output frequency display	You can monitor the frequency output from the drive.	-
	For	Forward/reverse selection	For: Motor rotates in forward direction rEv: Motor rotates in reverse direction **Note:** For applications where the motor must not rotate in reverse direction (for example, for fans and pumps), you can use *b1-04 [Reverse Operation Selection]* to prevent reverse rotation. How to set reverse operation rEv For → [↵] → For → [∧] → rEv → [↵]	-
Programming Mode (Parameter Settings)	A.Tun	Auto-Tuning Mode	The drive automatically calculates and sets the motor parameters.	107
	PAr	Parameter Setting Mode	You can see and set all parameters.	86
	SEtUP	Setup Mode	You can see and set the basic parameters necessary for drive operation.	85
	vrFy	Verify Menu	You can examine and set the parameters that are not at default settings.	86

◆ Drive Mode

These operations are available in Drive Mode:

- Operate and stop the drive
- Show the drive status monitors (for example, frequency reference, output frequency, output current, and output voltage)
- Show the alarm content

- Show the alarm history

Note:

To operate the drive, select Drive Mode. You can switch to other modes when the drive stops, but the drive must be in Drive Mode to start operation.

These steps show how to set the frequency reference source to LOCAL (keypad) and change the frequency reference from 0 Hz to 6 Hz.

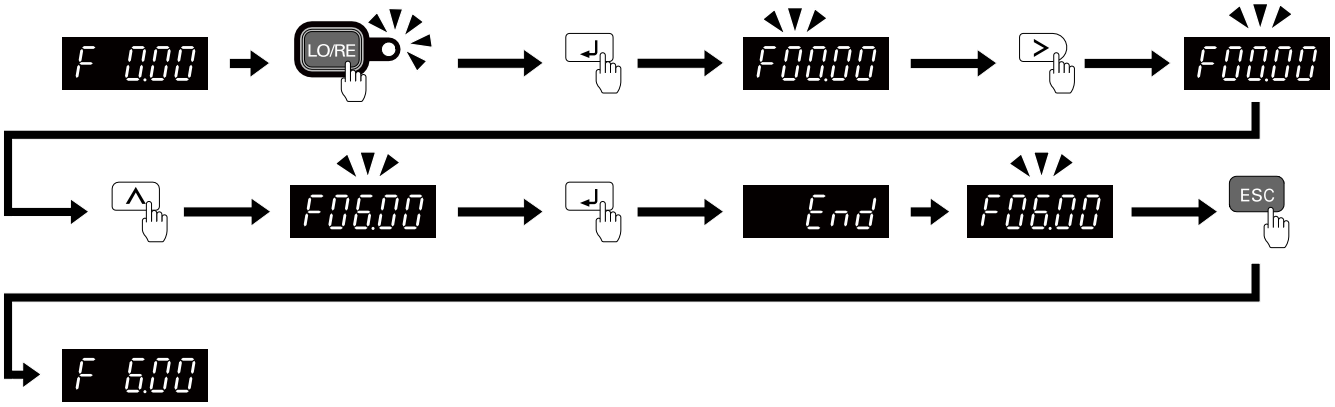


Figure 4.6 Frequency Reference Setting in Drive Mode

Note:

To prevent an incorrect setting, after you enter the frequency reference, you must push the ENTER key to change the frequency reference. Set *o2-05 = 1 [Home Mode Freq Ref Entry Mode = Immediate / MOP-style]* to change the frequency reference value without pushing the ENTER key.

◆ Programming Mode

In Programming Mode, you can set parameters or do Auto-Tuning. This mode has 4 sub-modes for different programming requirements:

- Verify Menu: Use this mode to examine and set the parameters that are not at default settings.
- Setup Mode: Use this mode to see and set the minimum parameters necessary for drive operation. Refer to *Verify and Set the Changed Parameters (Verify Menu) on page 86* for more information.
- Parameter Setting Mode: Use this mode to see and set all parameters.
- Auto-Tuning Mode: Use this mode to automatically set the motor parameters necessary for each control method.

■ Setup Mode

In Setup Mode, you can see and set the minimum parameters necessary for drive operation. Refer to Figure 4.7 for an example.

Note:

1. Refer to *Set up the Drive with User-Parameter Menu on page 82* for more information about Setup mode parameters.
2. Push APPL and ↵ to continue to the application selection screen. When you change the setting, it will optimize the parameter for the application. The default setting is *0 [General-purpose]*.
3. To go back to the initial screen, push ESC. To change another parameter in Setup Mode, push ∧ or ∨.

Change *b1-01 [Frequency Reference Selection 1]* from *1 [Analog Input]* to *0 [Keypad]*.

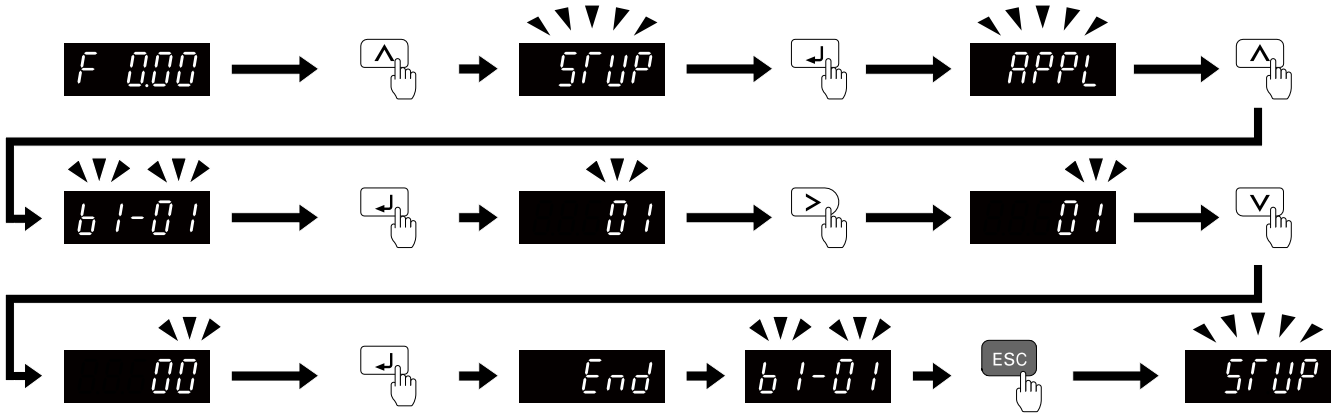


Figure 4.7 Key Operation Examples in Setup Mode

◆ Change Parameter Settings

Show the frequency reference screen in advance.

Note:

Push and hold ESC to go back to the frequency reference screen from any screen.

Use these steps to change *C1-01 [Acceleration Time 1]* from 1.0 s (default) to 2.0 s.

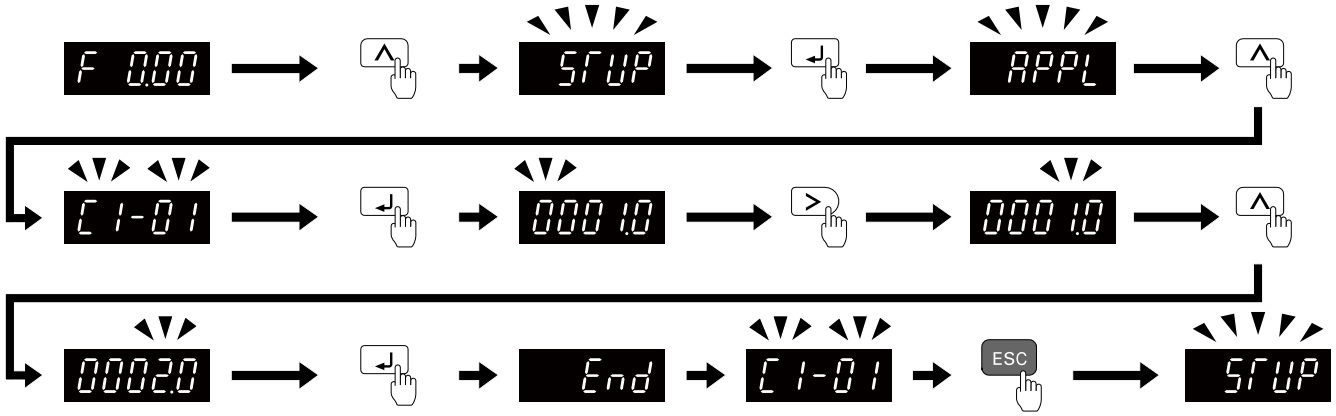


Figure 4.8 Key Operation Examples for Parameter Settings

◆ Verify and Set the Changed Parameters (Verify Menu)

Show the frequency reference screen.

Note:

Push and hold ESC to return to frequency reference screen from any screen.

Use Verify mode to view all parameters that are not at default settings. This is very useful when you replace a drive. When there are no changes to parameter settings, the display shows none. This lets you quickly access and re-edit changed parameters. Figure 4.9 shows the procedure.

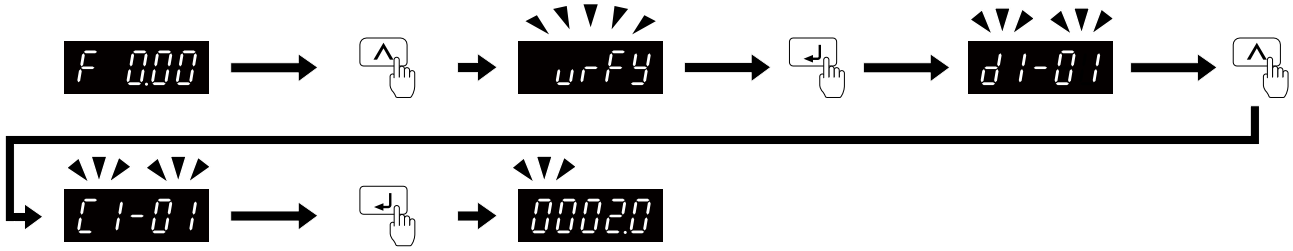


Figure 4.9 Verify and Set the Changed Parameters

◆ How to Switch between LOCAL and REMOTE

LOCAL mode lets you use the keypad to input Run commands. REMOTE mode lets you use other sources than the keypad to input Run commands.

WARNING! *Sudden Movement Hazard. If you change the control source when b1-07 = 1 [LOCAL/REMOTE Run Selection = Accept Existing RUN Command], the drive can start suddenly. Before you change the control source, remove all personnel from the area around the drive, motor, and load. Sudden starts can cause serious injury or death.*

You can use LO/RE or MFDI functions (LOCAL/REMOTE Selection) to switch between LOCAL and REMOTE.

Note:

- LO/RE illuminates while the drive is in LOCAL Mode.
- While you are entering a Run command, you cannot switch between LOCAL and REMOTE.

■ Use the LO/RE Selection Key on the Keypad to Switch between LOCAL and REMOTE

Each time you push LO/RE, the mode switches between LOCAL and REMOTE. The LED illuminates in LOCAL Mode.

Figure 4.10 Use the LO/RE Selection Key to Switch between LOCAL and REMOTE

■ Use MFDI Terminals (S3 to S7) to Switch between LOCAL and REMOTE

When you set *H1-xx = 1 [MFDI Function Selection = LOCAL/REMOTE Selection]*, you can activate/deactivate the terminal to switch between LOCAL and REMOTE. Set *H1-xx = 1* to disable the LO/RE key on the keypad. For information about the MFDI functions, refer to the list in *H1: Digital Inputs on page 274*.

4.5 Start-up Procedures

This section gives the basic steps necessary to start up the drive using only the most basic settings.

Use the flowcharts in this section to find the most applicable start-up method for your application.

◆ Flowchart A: Connect and Run the Motor with Minimal Setting Changes

Flowchart A shows a basic start-up sequence to connect and run a motor with a minimum of setting changes. Settings can change when the application changes.

Use the drive default parameter settings for basic applications where high precision is not necessary.

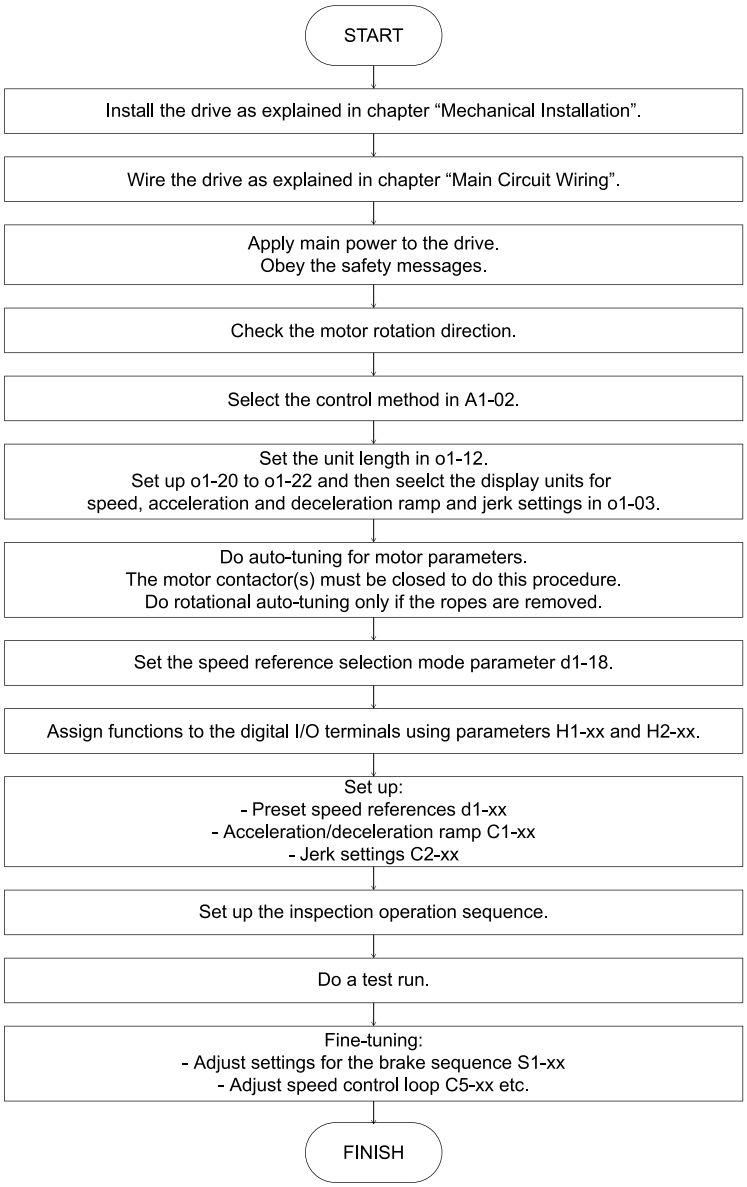


Figure 4.11 Basic Steps before Startup

◆ Power On

Take the following precautions before applying main power to the drive:

WARNING! *Fire Hazard. Verify that the rated voltage of the drive matches the voltage of the incoming power supply before applying power. Failure to comply can result in death or serious injury from fire.*

WARNING! *Fire Hazard. Tighten all terminal screws to the specified tightening torque. Avoid loose electrical connections. Failure to comply can result in death or serious injury from fire.*

WARNING! *Fire Hazard. Connect AC power supply lines to main circuit terminals R/L1, S/L2, and T/L3 (or R/L1 and S/L2 for single-phase power). Do not connect the AC power supply line to the output motor terminals of the drive. Failure to comply can result in death or serious injury from fire.*

WARNING! *Sudden Movement Hazard. Ensure start/stop, I/O and safety circuits are wired properly and in the correct state before energizing or running the drive. Failure to comply can result in death or serious injury from moving equipment.*

WARNING! *Sudden Movement Hazard. Clear personnel, secure equipment and check sequence and safety circuitry before starting the drive. Secure covers, couplings, shaft keys, and machine loads. Ensure start/stop and safety circuits are wired properly and in the correct state. Failure to comply can result in death or serious injury from moving equipment.*

WARNING! *Sudden Movement Hazard. Always check the operation of any emergency circuits after they are wired. Emergency circuits are required to provide safe and quick shutdown of the drive. Do not operate the drive with untested emergency circuits. Failure to comply can result in death or serious injury.*

NOTICE: *Equipment Hazard. The motor may run in reverse if the phase order is backward. Connect motor input terminals U/T1, V/T2, and W/T3 to drive output terminals U/T1,V/T2, and W/T3. The phase order for the drive and motor should match.*

NOTICE: *Equipment Hazard. Check all the wiring including the PG encoder wiring, to ensure that all connections are correct after installing the drive and connecting any other devices. Failure to comply could result in damage to the drive.*

After applying power, the drive mode display should appear and no fault or alarm should be displayed. In the event of a drive fault or error code, refer to *Troubleshooting on page 139*.

◆ Control Method Selection

Select one of the motor control methods after applying power to the drive according to the table below.

Machine Type	Control Method	A1-02 Setting
Induction Motor	V/f Control	0
	Open Loop Vector (OLV) Control	2

◆ Motor Rotation Direction Setup

Check the direction of motor rotation to verify the Up command causes the elevator to move in the upward direction. Perform the following check to confirm proper motor and load direction:

- The drive outputs motor voltage in U/T1-V/T2-W/T3 phase sequence when an Up command is issued. Check the motor rotation with this phase sequence (for most motors clockwise is seen from the shaft side). If motor rotation is incorrect, rewire the drive output to the motor.

DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

◆ Flowchart B: Auto-Tuning for Induction Motors

Flowchart B shows Auto-Tuning for induction motors operating with V/f Control or Open Loop Vector Control. Settings can change when the application changes.

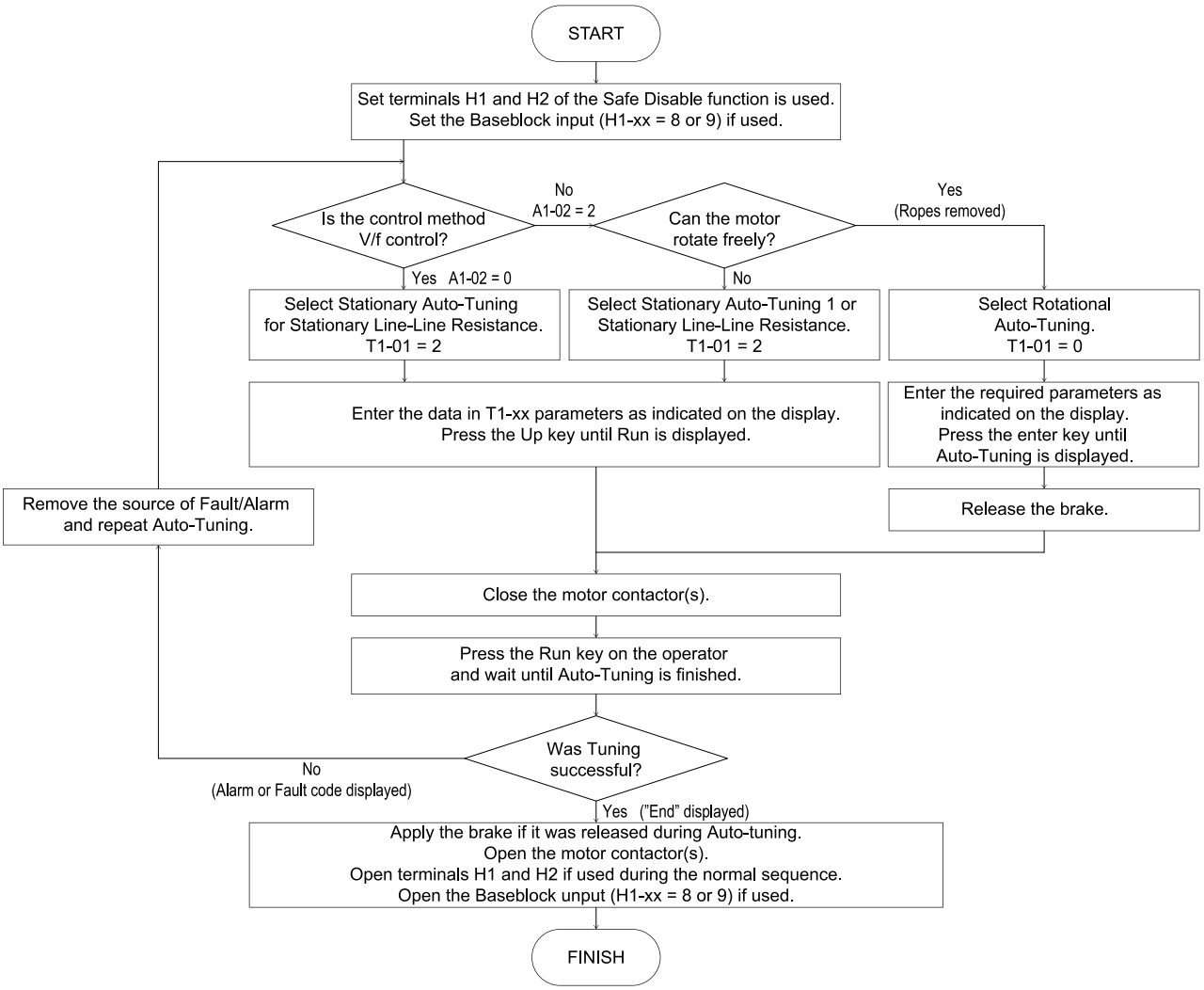


Figure 4.12 Auto-Tuning for Induction Motors

4.6 Auto-Tuning

WARNING! *Sudden Movement Hazard. Ensure the area surrounding the drive motor and load are clear before proceeding with Auto-Tuning. Remove main power from the drive before servicing the drive or motor. Do not touch the motor during Auto-Tuning The drive and motor may start unexpectedly during Auto-Tuning, which could result in death or serious injury. .*

Insufficient torque can cause the elevator car to move in the direction of the load, or cause the motor to behave erratically (reverse operation, stand still, sudden accelerations, etc.).

For more information, refer to the instruction manual included with the motor.

◆ Types of Auto-Tuning

The drive offers two different types of Auto-Tuning for induction motors. The type of Auto-Tuning used differs based on the control mode and other operating conditions. Refer to the tables below to select the type of Auto-Tuning that bests suits the application. For directions for performing Auto-Tuning, refer to *Flowchart A: Connect and Run the Motor with Minimal Setting Changes on page 88*.

The drive will only show Auto-Tuning parameters that are valid for the control mode that has been set in A1-02.

Table 4.4 Status of Input/Output Terminals during Auto-Tuning

Auto-Tuning Type	Parameter	Requirements and Benefits	Control Method (A1-02)	
			V/f (0)	OLV (2)
Rotational Auto-Tuning	T1-01 = 0	• Rotational Auto-Tuning gives the most accurate results, and is recommended if possible. • Motor must run freely or with light load (<30%), i.e. ropes have to be removed.	No	Yes
Stationary Auto-Tuning 1	T1-01 = 1	• A motor test report listing motor data is not available. • Automatically calculates motor parameters needed for vector control. • Use if ropes cannot be removed. Note that the accuracy is lower than with Rotational Auto-Tuning.	No	Yes
Line-to-Line Resistance	T1-01 = 2	• Used when the motor cable exceeds 50 m. • Used in V/f control if drive and motor capacities differ. • Perform when the ropes cannot be removed from the installation for rotational auto-tuning.	Yes	Yes

The following table lists the data that must be entered for Auto-Tuning. Make sure this data is available before starting Auto-Tuning. The necessary information is usually listed on the motor nameplate or in the motor test report provided by the motor manufacturer.

Table 4.5 Status of Input/Output Terminals during Auto-Tuning

Input Value	Input Parameter	Unit	Tuning Type (T1-01)	
			Rotational (0)	Stationary for Line-to-Line Resistance (2)
Motor Rated Power	T1-02	kW	Yes	Yes
Motor Rated Voltage	T1-03	Vac	Yes	No
Motor Rated Current	T1-04	A	Yes	Yes
Motor Rated Frequency	T1-05	Hz	Yes	No
Number of Motor Poles	T1-06	-	Yes	No
Motor Rated Speed	T1-07	r/min	Yes	No

The number of motor poles (T1-06) is usually not found on the motor nameplate, but it can be calculated by: $p = (120 \times f) / n_s$. Where **f** is the motor rated frequency and $\mathbf{n_s}$ is the motor synchronous speed.

◆ Before Auto-Tuning the Drive

WARNING! *Sudden Movement Hazard. When performing Rotational Auto-Tuning for motor data or PG encoder offset, always uncouple the motor from the mechanical system (remove ropes from traction sheave). Failure to comply can cause death or serious injury to personnel, and damage to the equipment.*

WARNING! *Electrical Shock Hazard. Do not touch the motor during Auto-Tuning. Lethal voltages may be present on the motor case. Failure to comply can cause death or serious injury.*

WARNING! *Electrical Shock Hazard. Do not touch the motor until Auto-Tuning is completed. When doing Stationary Auto-Tuning, the motor does not rotate, however, power is applied. Failure to comply can result in death or serious injury.*

WARNING! *Sudden Movement Hazard. The holding brake must remain engaged for the entire Stationary Auto-Tuning sequence. Ensure a brake release digital output signal cannot be issued by the drive. Failure to comply can result in death or serious injury.*

WARNING! *Sudden Movement Hazard. Do not release the mechanical brake during Stationary Auto-Tuning. Ensure that the mechanical brake release circuit is not controlled by the drive multi-function digital outputs exclusively. Failure to comply can result in death or serious injury.*

- Rotational Auto-Tuning is the preferred tuning method because it gives more accurate results than Stationary Auto-Tuning. Perform Rotational Auto-Tuning when the motor can be uncoupled from the elevator mechanical system (remove ropes from traction sheave). Perform Stationary Auto-Tuning when the motor and mechanical system cannot be uncoupled.
- Make sure that the mechanical brake remains applied for all Stationary Auto-Tuning methods. Make sure to release the brake for all Rotational Auto-Tuning methods.
- When using a motor contactor, make sure it is closed throughout the Auto-Tuning process.
- The H1 signal must be ON when performing Auto-Tuning.
- A digital input programmed for Baseblock (H1-xx = 8 or 9) must be set so that the drive is not in a baseblock condition.
- Confirm that the motor is mechanically fixed.
- To cancel Auto-Tuning, press the STOP key on the digital operator.
- Auto-Tuning requires the user to input data from the motor nameplate or motor test report. Make sure this data is available before Auto-Tuning the drive.
- For best performance, the drive input supply voltage must be greater than the motor rated voltage.

Improved performance is possible when using a motor with a base voltage that is 40 V lower than the input supply voltage. This is particularly important when operating the motor above 90% of base speed, where high torque precision is required.

WARNING! *Sudden Movement Hazard. Stay clear of the motor during rotational auto-tuning. During automatic starting of equipment, the machine may start moving suddenly. Clear all personnel from the drive, motor, and machine area before applying power. Secure covers, couplings, shaft keys, and machine loads before applying power to the drive. Failure to comply can result in death or serious injury.*

The following table describes digital input and output terminal operation while Auto-Tuning is executed.

Table 4.6 Status of Input/Output Terminals during Auto-Tuning

Auto-Tuning Type	Digital Input	Digital Output
Rotational Auto-Tuning	Digital input functions are disabled.	Functions the same as during normal operation.
Stationary Auto-Tuning 1	Digital input functions are disabled.	Maintains the status at the start of Auto-Tuning.
Line-to-Line Resistance	Digital input functions are disabled.	Maintains the status at the start of Auto-Tuning.

◆ Auto-Tuning Interruption and Fault Codes

If tuning results are abnormal or the STOP key is pressed before completion, Auto-Tuning will be interrupted and a fault code will appear on the LED operator.

◆ Auto-Tuning Operation Example

The following example demonstrates Rotational Auto-Tuning when using OLV (A1-02 = 2).

■ Selecting the Type of Auto-Tuning

1. Turn on the power to the drive.
   The initial display appears.
2. Press [∧] or [∨] until the Auto-Tuning display appears.
3. Press [↵] to start setting parameters.
4. Press [↵] to display the value for T1-01.
5. Press [>] to select the digit to be modified.
6. Press [∨] to select Rotational Auto-Tuning.
7. Press [↵] to save the setting.
   The display returns to the display shown in step 3.

■ Enter Data from the Motor Nameplate

After selecting the type of Auto-Tuning, enter the data required from the motor nameplate.

These instructions continue from Step 8 in *Selecting the Type of Auto-Tuning on page 92*.

1. Press [∧] to access the motor output power parameter T1-02.
2. Press [↵] to view the default setting.
3. Press [>] to select the correct digit and [∧]/[∨] to change the digit value, enter the motor power nameplate data in kW.
4. Press [↵] to save the settings.
   The display returns to the display in step 1.
5. Repeat steps 1 to 4 to set the remaining motor parameters.
   - *T1-03 [Motor Rated Voltage]*
   - *T1-04 [Motor Rated Current]*
   - *T1-05 [Motor Base Frequency]*
   - *T1-06 [Number of Motor Poles]*
   - *T1-07 [Motor Base Speed]*

■ Start Auto-Tuning

After selecting the type of Auto-Tuning, enter the data required from the motor nameplate.

These instructions continue from Step 5 in *Enter Data from the Motor Nameplate on page 92*.

1. Press [∧] to confirm the motor data.
2. Press [RUN] to start Auto-Tuning..
   The drive begins by injecting current into the motor for about 1 min, and then starts to rotate the motor. Auto-Tuning finishes in approximately one to two minutes if no errors occur.

◆ Input Data for Motor Auto-Tuning: T1

The *T1-xx* parameters are used to set the Auto-Tuning input data.

Note:
- Cycling drive power after setting the *T1* parameters will reset these parameters to default values.
- For motors operating in the field weakening range, first perform the Auto-Tuning with the base data. After Auto-Tuning is complete, change *E1-04 [Maximum Output Frequency]* to the desired value.

For more information refer to *T: Auto-Tuning on page 327*.

4.7 Setup Procedure for Elevator Applications

◆ Up and Down Commands And Speed Reference Selection

WARNING! *Sudden Movement Hazard. Remove the Up/Down Command before resetting alarms and faults. Failure to comply can cause death or serious injury.*

WARNING! *Sudden Movement Hazard. Verify the maximum drive output frequency before starting the drive. The drive is capable of running the motor at high speed. Failure to comply can result in death or serious injury.*

CAUTION! *Sudden Movement Hazard. Verify drive parameter b1-03 = 0 [Stopping Method = Ramp to Stop] before starting the drive. Failure to comply can cause the elevator to free-fall when the Up/Down command is removed.*

■ Speed Reference Selection

Parameter *b1-01* determines the source of the speed reference. On the drive the speed reference can only be modified through the digital operator or using digital inputs to switch between reference values. Therefore, this parameter cannot be modified from its default value of *0*.

Setting of b1-01	Reference Source	Speed Reference Input
0 (default)	Digital operator keypad	Set the speed references in the *d1-xx* parameters and use digital inputs to switch between different reference values.

■ Up/Down Command Source Selection

The input source for the Up and Down command can be selected using parameter *b1-02*.

Setting of b1-02	Up/Down Source	Up/Down Command Input
0	Digital operator keypad	RUN and STOP keys on the operator
1 (default)	Digital inputs	Terminal S1: Run in Up direction Terminal S2: Run in Down direction

■ Travel Start

To start the elevator in up or down direction, the following conditions must be fulfilled:

- A speed reference greater than zero must be provided.
- The Safe Disable signal at terminals H1 and H2 must be closed (drive output enabled).
- If a multi-function digital input is programmed for Baseblock (H1-xx = 8 or 9), this input must be set so the drive is not in a baseblock condition.
- An Up or Down signal must be set at the source specified in b1-02.
- If a multifunction input is programmed for output contactor feedback (H1-xx = 56), then the output contactor must be closed.

■ Travel Stop

The drive stops under the following conditions:

- The Up or Down command is removed.
- d1-18 is set to 1 or 2 and the Up/Down or Leveling Speed signal (H1-xx = 53) is removed.
- d1-18 is set to 3 and all speed inputs are removed.
- A fault occurs. The stopping method depends on the specific fault that occurred, in combination with certain parameter settings.
- The Safe Disable input is opened or a Base Block signal is input. In this case, the brake is applied immediately and the drive output shuts off.

◆ Speed Selection Using Digital Inputs

Set parameter b1-01 = 0 to enable the speed selection using the drive digital inputs. Use parameter d1-18 to determine different travel speeds selected by the digital inputs.

Setting of d1-18	Speed Selection
0 (default)	Multi-speed inputs 1, Speed references are set in d1-01 to d1-08.
1	Separate speed inputs, Speed references are set in d1-19 to d1-24 and d1-26, Higher speed has priority.

■ Multi-Speed Inputs 1

When *d1-18 = 0*, multi-function digital inputs are preset as shown below.

Setting of d1-18	Speed Selection	Set Value	
S5	H1-05	3	Multi-Speed Reference 1
S6	H1-06	4	Multi-Speed Reference 2
S7	H1-07	5	Multi-Speed Reference 3

Different speed reference settings can be selected by combining the three digital inputs as shown in the table below.

Parameters d1-19 through d1-26 are displayed only if d1-18 is set to 1 or 2.

Digital Inputs			Selected Speed
Multi-Speed Reference 1	Multi-Speed Reference 2	Multi-Speed Reference 3	d1-18 = 0
0 (Off)	0 (Off)	0 (Off)	Speed Reference 1 (d1-01)
1 (On)	0 (Off)	0 (Off)	Speed Reference 2 (d1-02)
0 (Off)	1 (On)	0 (Off)	Speed Reference 3 (d1-03)
1 (On)	1 (On)	0 (Off)	Speed Reference 4 (d1-04)
0 (Off)	0 (Off)	1 (On)	Speed Reference 5 (d1-05)
1 (On)	0 (Off)	1 (On)	Speed Reference 6 (d1-06)
0 (Off)	1 (On)	1 (On)	Speed Reference 7 (d1-07)
1 (On)	1 (On)	1 (On)	Speed Reference 8 (d1-08)

Setting d1-18 = 0 :

Up to eight speed references can be set using parameters *d1-01 to d1-08*. The drive starts with an Up or Down command, and stops when the Up or Down command is removed. When *d1-18 = 0*, parameters *d1-19* through *d1-23* will not be displayed.

■ Separate Speed Inputs (d1-18 = 1)

Six different speed settings (defined in the parameters d1-19 to d1-24 and d1-26) can be set and selected using four digital inputs. When *d1-18 = 1*, multi-function digital inputs are preset as shown below:

Setting of d1-18	Speed Selection	Set Value	Details
S3	H1-03	50	Nominal speed (d1-19)
S4	H1-04	54	Inspection speed (d1-24)
S5	H1-05	51	Intermediate speed (d1-20)
S6	H1-06	53	Leveling speed (d1-26)

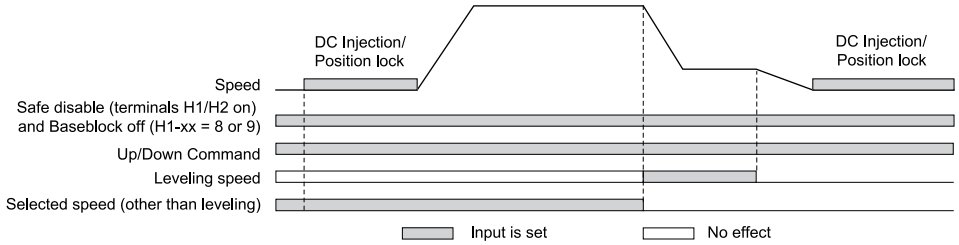
Different speed settings can be selected depending on the assignment of the speed selection digital inputs (H1-xx) as shown in the table below. Parameters d1-19 through d1-26 are displayed only if d1-18 is set to 1 or 2.

Leveling and Nominal Speed assigned (H1-xx = 50 and H1-xx = 53)				Leveling Speed not assigned (H1-xx ≠ 53)			Nominal Speed not assigned (H1-xx ≠ 50)			Selected Speed
50	51	52	53	50	51	52	51	52	53	
1	0	0	A	1	0	0	0	0	0	Nominal Speed (d1-19)
0	1	0	A	0	1	0	1	0	0	Intermediate Speed 1 (d1-20)
1	1	1	A	1	1	1	N/A	N/A	N/A	Intermediate Speed 2 (d1-21)
0	1	1	A	0	1	1	1	1	0	Intermediate Speed 3 (d1-22)
0	0	1	A	0	0	1	0	1	0	Releveling Speed (d1-23)
0	0	0	1	0	0	0	A	A	A	Leveling Speed (d1-26)
0	0	0	0	N/A	N/A	N/A	N/A	N/A	N/A	Zero Speed

0 = Off, *1* = On, *A* = no influence, *N/A* = Not available

Higher Speed has Priority and the Leveling Speed Input is Assigned (d1-18 = 1 and H1-xx = 53) (default) :

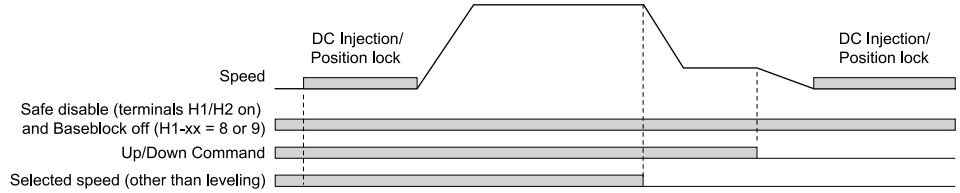
The higher speed has priority over the leveling speed. The leveling signal is disregarded as long as any other speed selection input is active. The drive decelerates to the leveling speed (d1-26) when the selected speed reference signal is removed.



Higher Speed Priority is Selected and the Leveling Speed Input is Not Assigned (d1-18 = 1 and H1-xx ≠ 53) :

The drive decelerates to the leveling speed (d1-26) when the selected speed reference signal is removed.

If no speed reference is selected at start, the drive will trigger an “FrL” fault. Set parameter S6-15 to 0 to disable Speed Reference Missing (FrL) detection. With this setting the drive starts using leveling speed if no other speed reference is selected.



◆ Multi-Function Terminal Setup

■ Multi-Function Digital Inputs (Terminals S3 to S7)

The H1 parameters assign functions to digital input terminals S3 to S7. Refer to *H1: Digital Inputs on page 274*.

■ Multi-Function Digital Outputs

The H2 parameters assign functions to digital output terminals. Refer to *H2: Digital Outputs on page 280*.

■ Multi-Function Analog Output

The H4 parameters assign a function to analog output terminal AM. Select the function for this terminals by entering the last three digits of the desired U monitor. Refer to *U: Monitors on page 238*.

◆ Accel/Decel Ramp and Jerk Settings

Acceleration and deceleration ramps are set using the C1-xx parameters. Use the C2-xx parameters to adjust the jerk at the start of acceleration or deceleration.

The figure below explains how accel/decel ride and jerk settings can be used to adjust the ride profile.

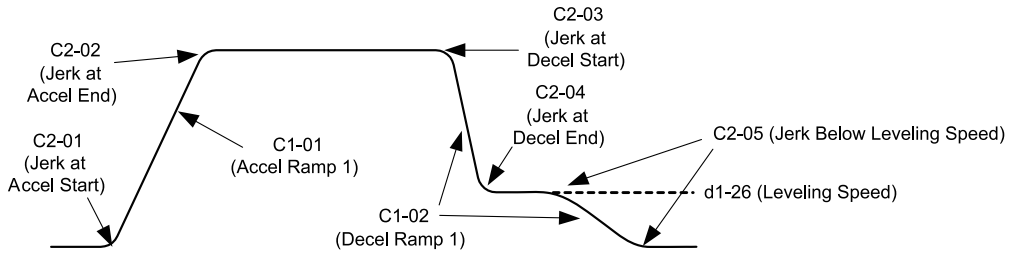


Figure 4.13 Accel/Decel Ramp and Jerk Function

Units used to set the acceleration and deceleration ramp as well as the Jerk function change with the setting of parameter.

◆ Inspection Operation

■ Start Condition in Inspection Operation

Inspection operation is performed when an Up or Down signal is input while one of the following conditions is true:

- Parameter d1-18 is set to 0 and the selected speed is higher than d1-28 but lower than d1-29.
- Parameter d1-18 is set to 1 and a digital input programmed for Inspection Operation Speed *[H1-xx = 54]* is enabled.

Inspection Operation uses the same acceleration characteristics and brake sequence at start as normal operation.

The carrier frequency is set to 2 kHz during Inspection Operation, but can be changed using parameter C6-21.

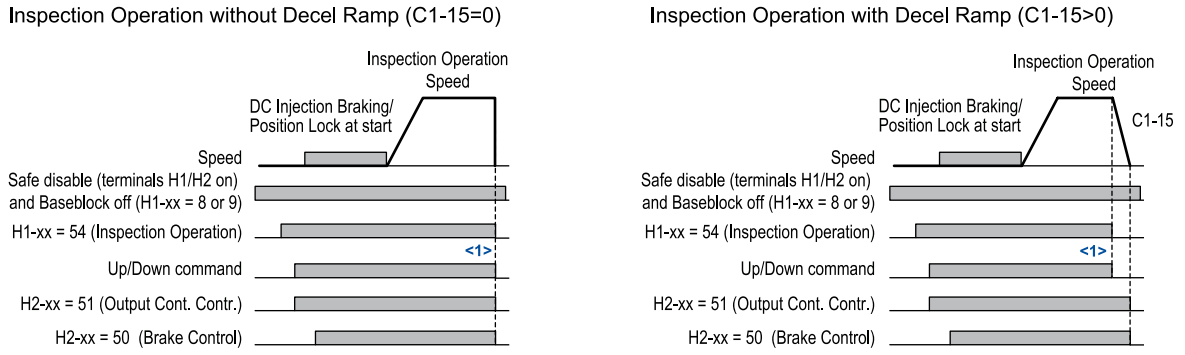
■ Stop Condition in Inspection Operation

To stop the drive during Inspection Operation, either remove the Up or Down command or reset the input terminal for Inspection Operation.

A deceleration ramp can be set for Inspection Operation using parameter C1-15.

- If C1-15 = 0.00, the drive immediately applies the brake, shuts off the drive output, and opens the motor contactor, i.e., the multi-function output terminals set for “Brake Control” *[H2-xx = 50]* and “Output Contactor Control” *[H2-xx = 51]* are cleared.
- If C1-15 > 0.00, the drive decelerates to stop at the rate set to C1-15, then applies the brake, shuts the output off, and opens the motor contactor.

■ Inspection Operation Timing Chart



<1> The drive stops if either the Up/Down command or Inspection Operation signals are removed.

Figure 4.14 Inspection Operation Sequence

◆ Brake Sequence

WARNING! *Sudden Movement Hazard. Be sure to set an acceptable deceleration time in parameter C1-09 [EmergencyStop Ramp] when using the fast-stop feature. Rapid deceleration may cause the drive to fault on an overvoltage condition and lead to an uncontrolled motor state. Failure to comply can result in death or serious injury*

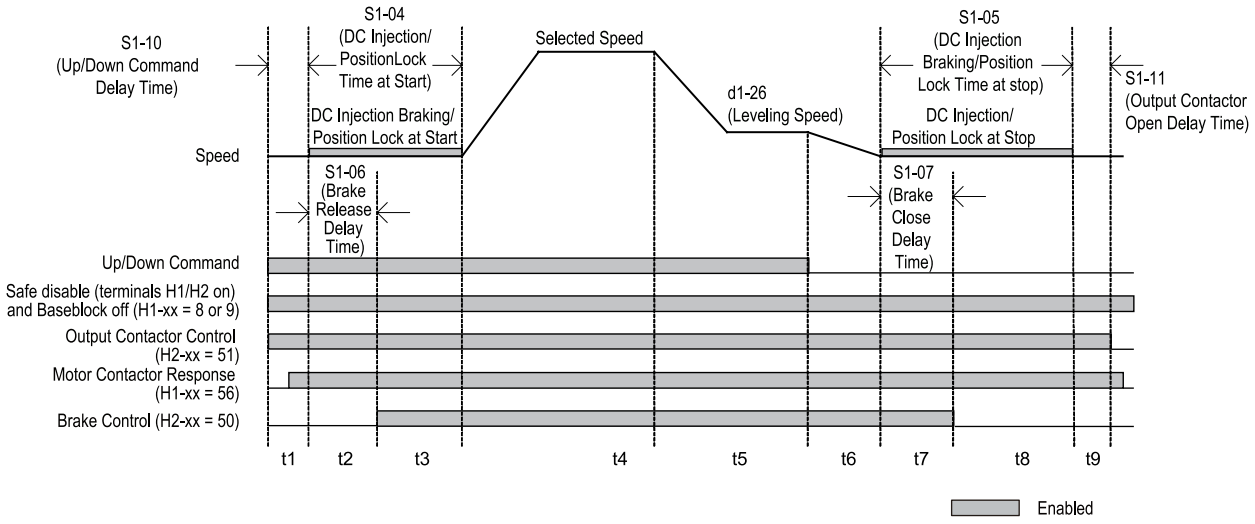


Figure 4.15 Brake Sequence Timing Diagram

Table 4.7 Timing Zones for Brake Sequence Timing Diagram

Timing Zone	Description
t1	Up or Down command is issued.
	Safe Disable terminals H1-H2-HC must be set and Baseblock must be disabled (digital inputs set to H1-xx = 8 or 9).
	Speed reference must be selected by multi-function input terminals.
	Output contactor control signal is set *[H2-xx = 51]* by the drive.
	Drive waits for the "Motor Contactor Feedback" signal *[H1-xx = 56]* to be issued. If the motor contactor feedback is not received within t1, or if the feedback signal is on before the contactor control command has been issued, an SE1 fault is triggered. If the motor contactor feedback signal is not used, then the drive waits for the operation start delay time set in S1-10 to pass, then proceeds to the next step.
t2	After the delay time set in S1-10 has passed, the drive outputs current to the motor. DC Injection Braking or Position Lock begins.
	After the brake release delay time set in S1-06 has passed, the drive sets the "Brake Control" output *[H2-xx = 50]* in order to release the brake.
t3	DC Injection Braking or Position Lock will continue until: • The time S1-04 has elapsed, • Or the time S1-06 has elapsed if S1-06 > S1-04 (this setting should be avoided since the motor could be driven against the applied brake).
t4	The drive accelerates up to the selected speed. The speed is kept constant until the leveling speed is selected.
t5	Leveling speed is selected. The drive decelerates to the leveling speed and maintains that speed until the Up or Down command is removed.
t6	The Up or Down signal is cleared. The drive decelerates to zero speed.
t7	The motor speed reaches the zero speed level (S1-01). DC Injection Braking or Position Lock is then executed for the time set in S1-05.
	After the delay time to apply the brake set in S1-07 has passed, the drive clears the "Brake Control" output *[H2-xx = 50]*. The brake applies.
t8	The drive continues DC Injection or Position Lock until the time S1-05 has passed. When S1-05 has passed the drive output is shut off.
t9	After the delay for the magnetic contactor set in S1-11 has passed, the drive resets the output terminal set for "Output Contactor Control" *[H2-xx = 51]*. The Safe Disable Inputs can be cleared and Baseblock can be enabled.

◆ Rescue Operation

In the event of a power outage, Rescue Operation allows the elevator to travel to the nearest floor by switching to a backup battery or UPS (Uninterruptable Power Supply) for power.

An input terminal set for Rescue Operation (H1-xx = 55) can be used to initiate Rescue Operation. During Rescue Operation, the drive uses the speed reference set in d1-25 to travel to the nearest floor.

NOTICE: *Equipment Hazard. Do not use the Rescue Operation feature for extended periods. Failure to comply can result in drive heat sink overtemperature alarms (oH).*

NOTICE: *Equipment Hazard. When changing parameters while the drive is supplied from the rescue operation power supply, wait at least 5 s after entering parameters before switching off the power supply. Instantly switching off the power can cause parameter settings corruption that can only be resolved by initializing the drive. Failure to comply can cause erroneous drive performance.*

■ Drive Power Supply for Rescue Operation

There are various methods of supplying power to the drive for rescue operation. Independent of the chosen method, the voltage in the DC bus of the drive and the voltage supplied to the drive control circuit must meet the specifications provided in the following table.

The DC bus voltage can either be supplied by a battery connected to the DC bus terminals of the drive or by a UPS connected to drive terminals L1 and L2. The control circuit voltage is supplied directly from the drive's DC bus.

When using a single-phase AC power supply for rescue operation such as a single-phase UPS, the ripple in the DC bus voltage will be higher than with a three-phase or battery supply. Make sure that the DC bus voltage never falls below the minimum value listed in the following table.

Table 4.8 Power Supply Ratings for Rescue Operation

Timing Zone	Description	Control Circuit Voltage
Induction Motor	200 V class drives: 115 to 300 Vdc 400 V class drives: 230 to 600 Vdc	Same as DC Bus Voltage.

■ Parameter Setup

Adjust drive parameters as described below when using Rescue Operation.

- Select the type of Rescue Operation power supply for the drives main circuit in parameter S4-06.
- When using a UPS, set the UPS power value to parameter S4-07. Use parameter S4-08 to decide if the Rescue Operation speed shall be limited automatically depending on the UPS power.
- If deterioration of the battery or UPS should be detected, also set up parameters S4-12 and S4-13. Measure the DC bus voltage during operation with the rescue power supply and set the measured value to parameter S4-12. Set the deterioration detection level to parameter S4-13.
- Set parameters S4-01 to S4-04 to select if light load direction search should be automatically performed when Rescue Operation is started and to configure the light load search function.

■ Wiring Examples

Switching the main power supply to a battery or UPS requires magnetic contactors that must be controlled by an external controller. Wiring methods and the sequence used for the magnetic contactors depend on the application. This instruction manual describes the following configurations:

- • A single-phase, 230 V UPS is used as backup power supply for a 200 V or 400 V class drive.
- • Two separate batteries for the main power and control power supplies. Main power battery voltage is below 250 Vdc for 200 V class drives or 500 Vdc for 400 V class drives.
- • Two separate batteries. One is used for the main power supply, a second battery supplies the controller via an optional 24 V Backup Power Supply Unit.
- • A single battery with minimum 250 Vdc for 200 V class drives or 500 Vdc for 400 V class drives is used for the main and control power supply.

Select the configuration that matches your application. Follow the corresponding instructions for wiring and drive settings. For configurations not covered in the list above, contact your Yaskawa representative or our sales office directly for consultation.

WARNING! *Electrical Shock Hazard. Make sure that the drive and all devices connected to the drive have been shut off prior to performing and type of maintenance or wiring. After shutting off the power, wait for at least the amount of time specified on the drive before touching any components or perform wiring. The internal capacitor remains charged even after the power supply is turned off. Do not connect or disconnect wiring while the power is on. Never remove or install option cards or attempt to replace the cooling fan while the drive is switched on. Failure to comply can result in death or serious injury*

NOTICE: *Be sure to thoroughly read the instructions for wiring and magnetic contactor sequence described in this section before setting up the drive for Rescue Operation. Failure to follow these instructions can damage the drive.*

NOTICE: *Equipment Hazard. Do not use the Rescue Operation feature for extended periods. Failure to comply can result in drive heat sink overtemperature alarms (oH).*

■ Using a Single-Phase 230 Vac UPS (Uninterruptable Power Supply)

Follow the instructions when using a single-phase 230 V UPS for Rescue Operation. A 230 V UPS can be used for both 200 V and 400 V class drives.

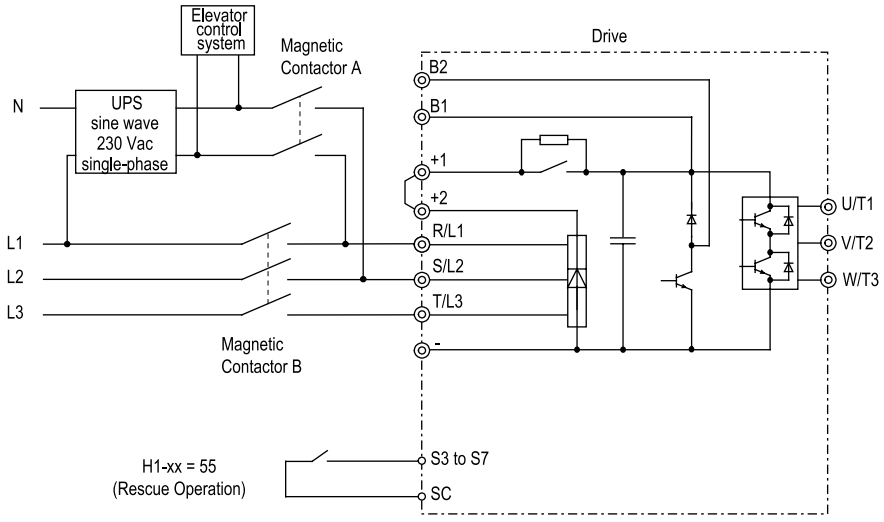


Figure 4.16 Wiring Diagram when Using a UPS

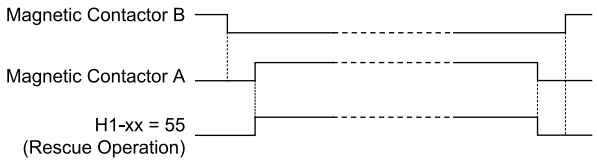


Figure 4.17 Magnetic Contactor Sequence

Operation Sequence

Starting Rescue Operation

1. Open contactor B.
2. Set the input terminal programmed for Rescue Operation *[H1-xx = 55]*.
3. Close contactor A.
4. Set the Up/Down command.

Ending Rescue Operation

1. After the car has stopped, open contactor A.
2. Clear the input terminal set for Rescue Operation *[H1-xx = 55]*.
3. Close contactor B to return to operation with normal power supply.

Application Precautions

The drive may fault on a control power supply fault (Uv2) if the UPS can't provide enough voltage, or if the Light Load Direction Search is not set properly.

■ Using a Single Battery with Minimum 230 Vdc

Follow the instructions when using one battery to supply both, main circuit and controller. The battery voltage must be at least 230 Vdc for 400 V class drives, battery voltage of 250 Vdc is recommended.

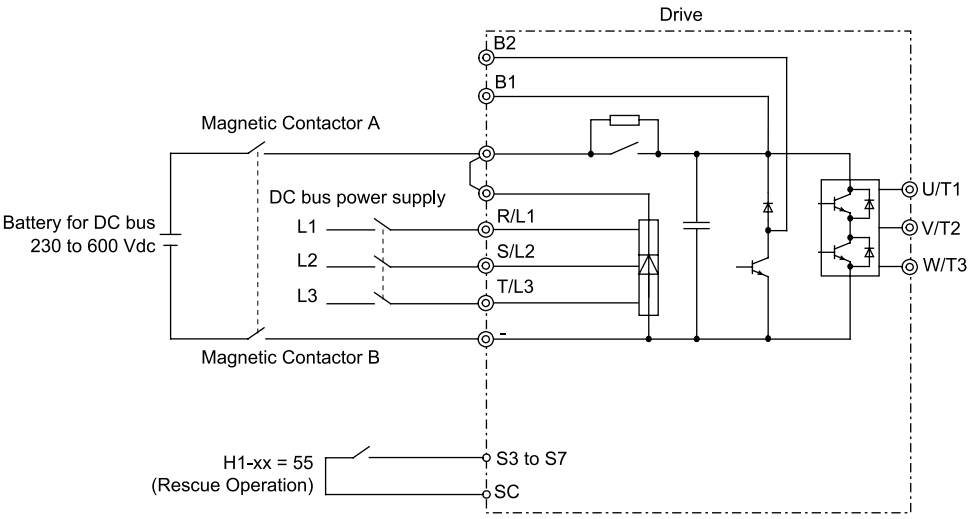


Figure 4.18 Wiring Diagram when Using a UPS

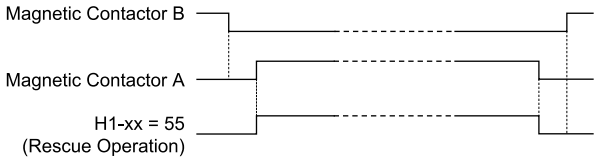


Figure 4.19 Magnetic Contactor Sequence

Operation Sequence

Starting Rescue Operation

1. Open contactor B.
2. Set the input terminal programmed for Rescue Operation *[H1-xx = 55]*.
3. Close contactor A.
4. Set the Up/Down command.

Ending Rescue Operation

1. After the car has stopped, open contactor A.
2. Clear the input terminal set for Rescue Operation *[H1-xx = 55]*.
3. Close contactor B to return to operation with normal power supply.

4.8 Items to Check before Starting Up the Drive

◆ Check before You Energize the Drive

Check the items in Table 4.9 before you energize the drive.

Table 4.9 Items to Check before You Energize the Drive

Items to Check	Description
Input Power Supply Voltage	The voltage of the input power supply must be: Three-phase 400 V class: three-phase 380 Vac to 480 Vac 50/60 Hz, 513 Vdc to 679 Vdc
	Correctly wire power supply input terminals R/L1, S/L2, and T/L3, or L and N.
	Correctly ground the drive and motor.
Connection between Drive Output Terminals and Motor Terminals	Make sure that you connected drive output terminals U/T1, V/T2, and W/T3 in the correct sequence to align with motor terminals U, V, and W and tighten the screws to a correct tightening torque.
Control Circuit Terminal Wiring	Make sure that you connected the drive control circuit terminals in the correct sequence to align with devices and switches and tighten the screws to a correct tightening torque.
Control Circuit Terminal Status	Turn OFF the inputs from all devices and switches connected to the drive control circuit terminals.
Connection between Machinery and Motor	Disengage all couplings and belts that connect the motor and machinery.

◆ Check after You Energize the Drive

Check the items in Table 4.10 after you energize the drive. The keypad display is different depending on drive status.

Table 4.10 Display Status after You Energize the Drive

Status	Display	Description
During Usual Operation		The LED display shows the frequency reference.
When the Drive Detects a Fault		The display is different for different faults. Refer to “Troubleshooting” to remove the cause of the fault. The ALM/ERR LED will illuminate.

4.9 Keypad Operation

◆ Show the Monitor

Show the frequency reference screen.

Note:

Push and hold ESC to return to frequency reference screen from any screen.

Use these steps to access Mon menu and check drive operation status.

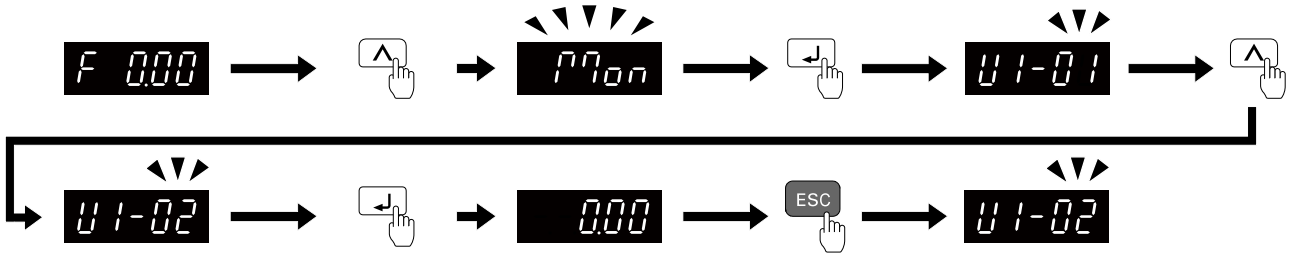


Figure 4.20 How to Monitor the Parameter Setting Values

◆ Check Modified Parameters

Show the frequency reference screen.

Note:

Push and hold ESC to return to frequency reference screen from any screen.

Use Verify mode to view all parameters that are not at default settings. This is very useful when you replace a drive. This lets you quickly access and re-edit changed parameters.

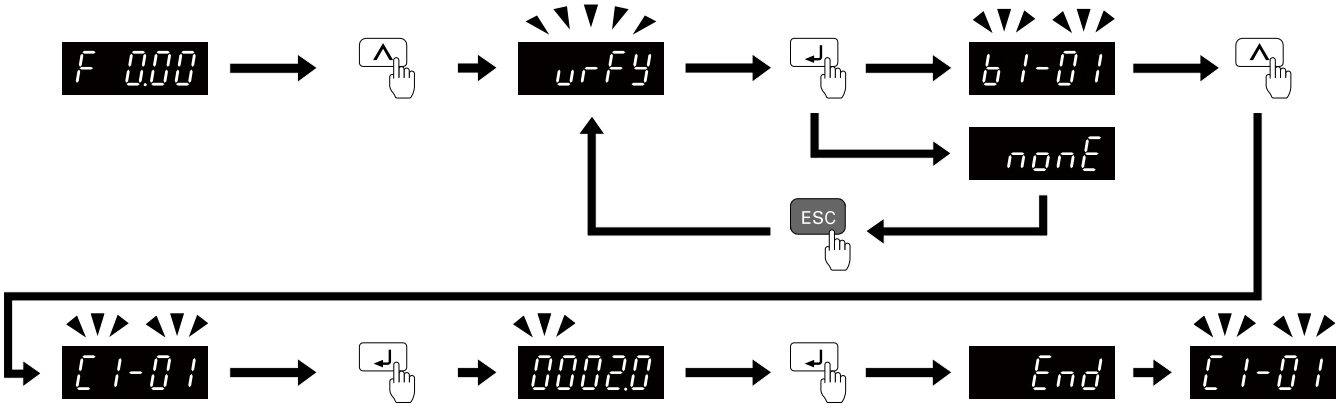


Figure 4.21 How to Examine the Changed Parameters

◆ Set and View Quick Setup Parameters

Show the frequency reference screen.

Note:

Push and hold ESC to return to frequency reference screen from any screen.

The setup mode shows the parameters set in *A2-01 to A2-32 [User Parameter 1 to User Parameter 32]*. This lets you quickly access and change these parameters.

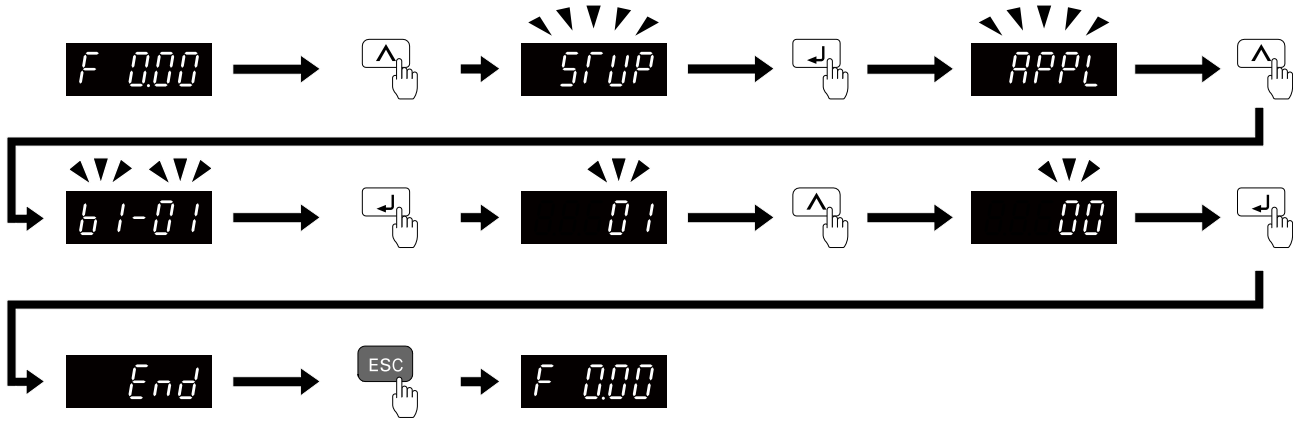


Figure 4.22 Set and View Quick Setup Parameters

Continue to change the parameters or press and hold ESC to go back to the frequency reference screen.

◆ Change Parameter Settings

Show the frequency reference screen in advance.

Note:

Push and hold ESC to go back to the frequency reference screen from any screen.

This example shows how to change *C1-01 [Acceleration Time 1]*. Set the parameter to the necessary value.

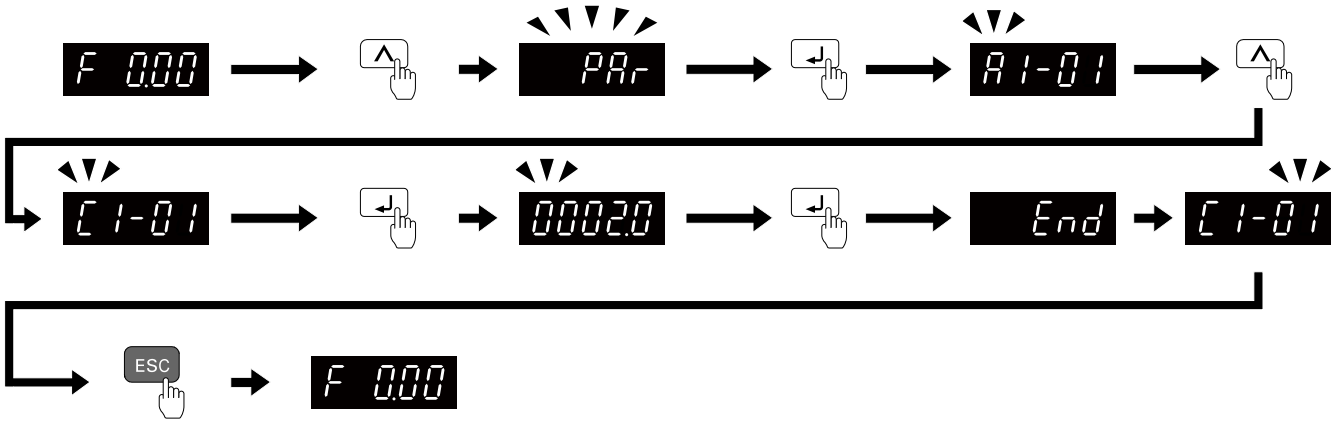


Figure 4.23 How to Change the Parameter Setting

Continue to change parameters or push and hold ESC to go back to the frequency reference screen.

◆ Save a Backup of Parameters

Show the frequency reference screen in advance.

Note:

Push and hold ESC to go back to the frequency reference screen from any screen.

Use these steps to save a backup of the drive parameters to the keypad.

Making backups of the parameter settings can save time when setting parameters after replacing a drive. If you set up more than one drive, you can copy the parameter settings from a drive that completed a test run to the other drives.

Note:

- Make sure that you stop the motor before you back up parameters.
- The drive does not accept Run commands while it is making a backup.
- Set *o3-02 = 0 [Copy Allowed Selection = Disabled]* to protect the parameters saved in the keypad.

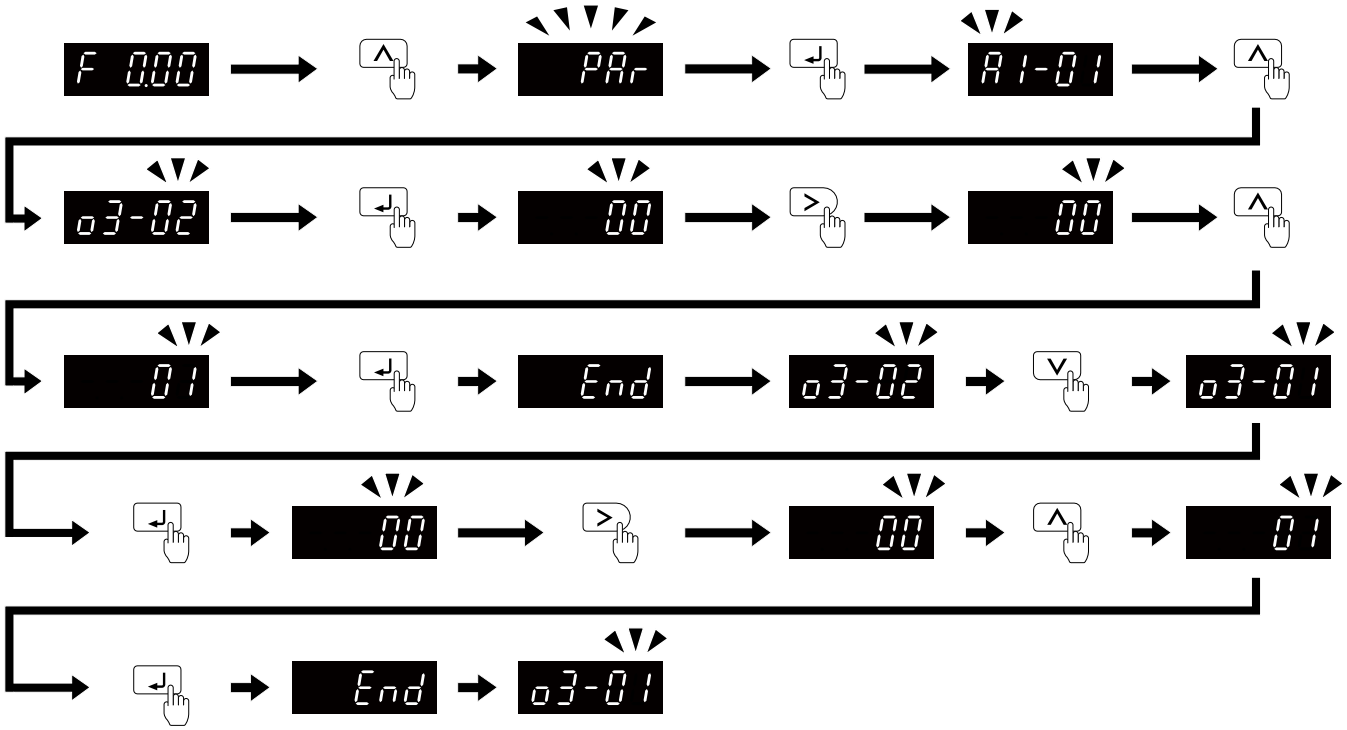


Figure 4.24 How to Save Backed-up Parameters

Push and hold ESC to go back to the frequency reference screen.

◆ Write Backed-up Parameters to the Drive

Show the frequency reference screen in advance.

Note:

Push and hold ESC to go back to the frequency reference screen from any screen.

Use these steps to write the parameters backed up in the keypad into a different drive.

Note:

- Make sure that you stop the drive before you restore the backed-up parameters.
- The drive does not accept Run commands while it is restoring parameters.

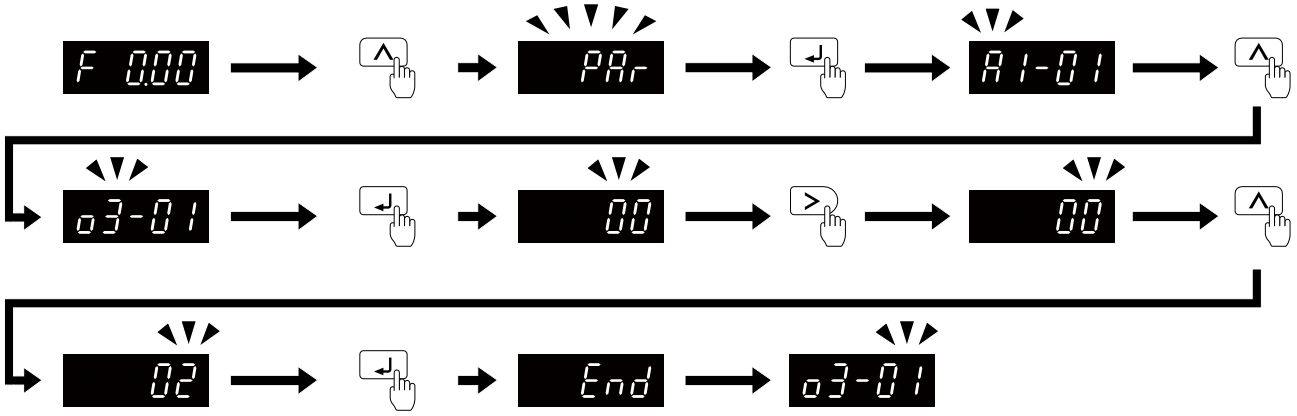


Figure 4.25 Writing backed up parameters

Push and hold ESC to go back to the frequency reference screen.

◆ Verify Keypad Parameters and Drive Parameters

Show the frequency reference screen in advance.

Note:

Push and hold ESC to go back to the frequency reference screen from any screen.

This procedure verifies that the parameter setting values that were backed up in the keypad agree with the parameter setting values in the drive.

Note:
- Make sure that you stop the drive before you examine parameters.
- The drive does not accept Run commands while it is restoring parameters.

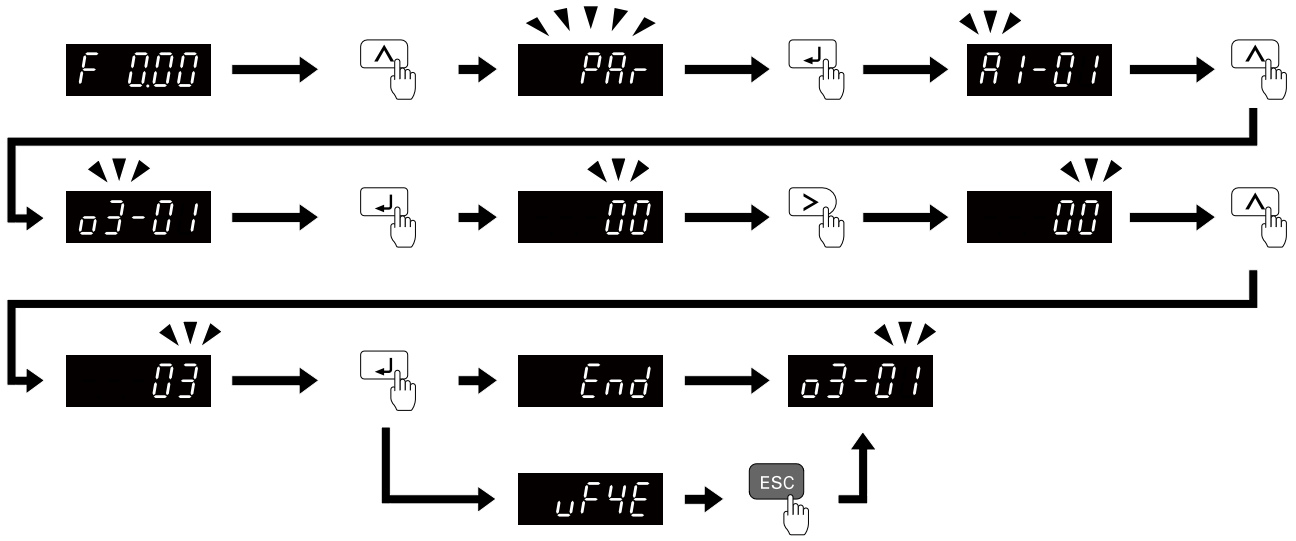


Figure 4.26 Verify Keypad Parameters and Drive Parameters

Push and hold ESC to go back to the frequency reference screen.

◆ Delete Parameters Backed Up to the Keypad

Show the frequency reference screen in advance.

Note:

Push and hold ESC to go back to the frequency reference screen from any screen.

Use these steps to erase the parameters backed up in the keypad.

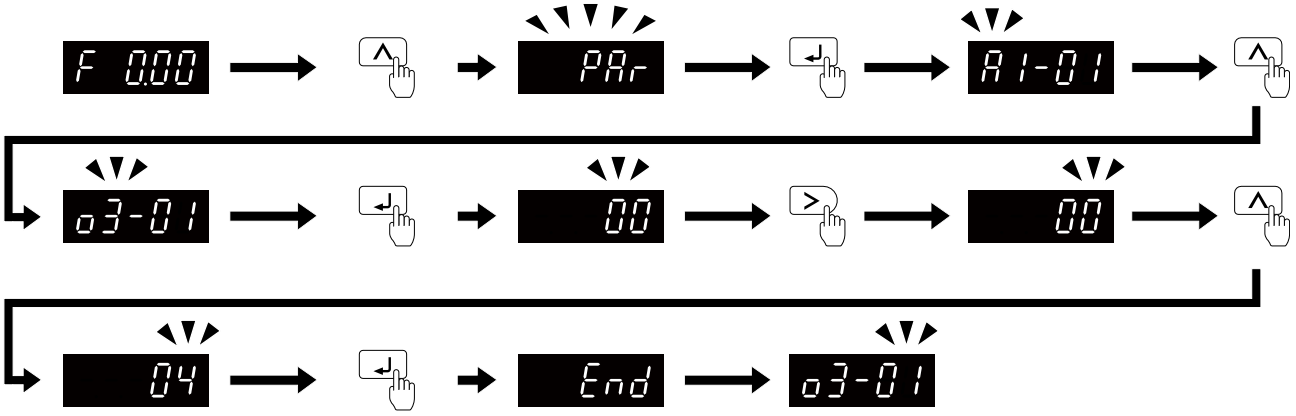


Figure 4.27 How to Erase the Backed-up Parameters

Push and hold ESC to go back to the frequency reference screen.

4.10 Auto-Tuning

Auto-Tuning uses motor characteristics to automatically set drive parameters for vector control. Think about the type of motor, drive control method, and the motor installation environment and select the best Auto-Tuning method.

WARNING! *Injury to Personnel. Rotational Auto-Tuning rotates the motor at 50% or more of the motor rated frequency. Make sure that there are no issues related to safety in the area around the drive and motor. Increased motor frequency can cause serious injury or death.*

◆ Precautions before Auto-Tuning

Examine the topics in this section before you start Auto-Tuning.

■ Prepare for Basic Auto-Tuning

- You must input data from the motor nameplate or motor test report to do Auto-Tuning. Make sure that this data is available before you do Auto-Tuning.
- For best performance, make sure that the drive input supply voltage is equal to or more than the motor rated voltage.

  Note:

  Better performance is possible when you use a motor with a rated voltage that is less than the input supply voltage (by 20 V for 200 V class models or by 40 V for 400 V class models). This is very important when you operate the motor at more than 90% of base speed, where high torque precision is necessary. If the input power supply is equal to the motor rated voltage, the drive output voltage will not be sufficient and performance will decrease.

- Push STOP on the keypad to cancel Auto-Tuning.
- If a Safe Disable input signal is input to the drive during Auto-Tuning, Auto-Tuning measurements will not complete successfully. If this occurs, cancel the Auto-Tuning, then do it again.
- Table 4.11 shows the status of multi-function input/output terminals during Auto-Tuning.

Table 4.11 Status of Input/Output Terminals during Auto-Tuning

Auto-Tuning Type	Mode		Parameter	Multi-Function Input	Multi-Function Output *1
Induction Motor Auto-Tuning	Rotational	Rotational Auto-Tuning	T1-01 = 0	Disabled	Functions the same as during usual operation.
	Stationary	Stationary Auto-Tuning 1	T1-01 = 1	Disabled	
		Line-to-Line Resistance	T1-01 = 2	Disabled	Keeps the status at the start of Auto-Tuning.

*1 When you set a terminal to *H2-xx = E [MFDO Function Selection = Fault]*, it will function the same as during usual operation.

WARNING! *Crush Hazard. Wire a sequence that will not let a multi-function output terminal open the holding brake during Stationary Auto-Tuning. If the holding brake is open during Stationary Auto-Tuning, it can cause serious injury or death.*

WARNING! *Sudden Movement Hazard. Before you do Rotational Auto-Tuning, disconnect the load from the motor. The load can move suddenly and cause serious injury or death.*

WARNING! *Injury to Personnel. Rotational Auto-Tuning rotates the motor at 50% or more of the motor rated frequency. Make sure that there are no issues related to safety in the area around the drive and motor. Increased motor frequency can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. During Auto-Tuning, the motor will receive high voltage when the motor is stopped. Do not touch the motor until Auto-Tuning is complete. If you touch a motor that is energized, it can cause serious injury or death.*

■ Precautions before Rotational Auto-Tuning

WARNING! *Electrical Shock Hazard. During Auto-Tuning, the motor will receive high voltage when the motor is stopped. Do not touch the motor until Auto-Tuning is complete. If you touch a motor that is energized, it can cause serious injury or death.*

- Before you do Rotational Auto-Tuning to prevent drive malfunction, uncouple the motor from the load.
- Make sure that the motor magnetic brake is released.
- Make sure that external force from the machine will not cause the motor to rotate.

■ Precautions before Stationary Auto-Tuning

- Make sure that the motor magnetic brake is not open.
- Make sure that external force from the machine will not cause the motor to rotate.

WARNING! *Electrical Shock Hazard. During Auto-Tuning, the motor will receive high voltage when the motor is stopped. Do not touch the motor until Auto-Tuning is complete. If you touch a motor that is energized, it can cause serious injury or death.*

■ Automatically Set E2-02 [Motor Rated Slip] and E2-03 [Motor No-Load Current]

After Stationary Auto-Tuning is complete, use this procedure to do the operation in test mode:

1. Check the *E2-02* and *E2-03* values on the “Modified Parameters/Fault Log” screen or the “Parameters” screen.
2. Operate the motor in Drive Mode with these conditions:
   - Uncouple the motor from the load
   - Make sure that you connect all wiring between the drive and motor
   - Make sure that a mechanical brake on the motor shaft is not locked
   - Keep a constant speed of 30% of *E1-06 [Base Frequency]* (default value = maximum frequency) or more for 1 second or longer.
3. After the motor stops, examine the values of *E2-02* and *E2-03* again in the Verify Menu or Parameter Setting Mode.
4. Make sure that the input data is correct.
   When the settings in *E2-02* and *E2-03* are different than in step 1, the drive set the values automatically.

4.11 Fine Tuning during Test Runs (Adjust the Control Function)

This section gives information about the adjustment procedures to stop hunting or oscillation errors caused by the control function during a test run. Adjust the applicable parameters as specified by your control method and drive status.

Note:

This section only lists frequently adjusted parameters. If you must adjust parameters that have a higher degree of precision, contact Yaskawa.

◆ V/f Control

Table 4.12 Parameters for Fine Tuning the Drive (*A1-02 = 0 [V/f]*)

Issue	Possible Solutions	Parameter	Default	Recommended Setting
Hunting or oscillation at mid-range speeds (10 Hz to 40 Hz)	• If torque is not sufficient with heavy loads, decrease the setting value. • If hunting or oscillation occur with light loads, increase the setting value. • If hunting occurs with a low-inductance motor, for example a motor with a larger frame size or a high-frequency motor, lower the setting value. Set *n1-01 = 1 [Hunting Prevention Selection = Enabled]*.	n1-02 [Hunting Prevention Gain Setting]	1.00	0.10 - 2.00
• The volume of the motor excitation sound is too high. • Hunting or oscillation at low speeds (10 Hz or lower), or at mid-range speeds (10 Hz to 40 Hz)	• If the volume of the motor excitation sound is too high, increase the carrier frequency. • If hunting or oscillation occur at low or mid-range speeds, decrease the carrier frequency.	C6-02 [Carrier Frequency Selection]	8 *1	1 to upper limit value
• Unsatisfactory motor torque and speed response • Hunting or oscillation	• If torque or speed response are slow, decrease the setting value. • If hunting or oscillation occur, increase the setting value.	C4-02 [Torque Compensation Delay Time]	200 ms *2	100 ms to 1000 ms
• Torque at low speeds (10 Hz or lower) is not sufficient. • Hunting or oscillation	• If torque at low speeds (10 Hz or lower) is not sufficient, increase the setting value. • If hunting or oscillation occur with light loads, decrease the setting value.	C4-01 [Torque Compensation Gain]	1.00	0.50 - 1.50
• If you use the drive with an IE3 high efficiency motor, the current that is more than the motor rated current will flow and trip at overload. Hunting or oscillation.	Decrease the setting value in these conditions: • Drive trips at overload. • Hunting or oscillation.	C4-01 [Torque Compensation Gain]	1.00	0.00 - 1.00
• Torque at low speeds (10 Hz or lower) is not sufficient. • Large initial vibration at start up.	• If torque at low speeds (10 Hz or lower) is not sufficient, increase the setting value. • If there is large initial vibration at start up, decrease the setting value	• E1-08 [Mid Point A Voltage] • E1-10 [Minimum Output Voltage]	• E1-08: 17.2 V *3 • E1-10: 7.4 V *3	Default setting +/- 5 V *4
Speed precision is unsatisfactory.	Set *E2-01 [Motor Rated Current], E2-02 [Motor Rated Slip], and E2-03 [Motor No-Load Current]*, then adjust *C3-01*.	C3-01 [Slip Compensation Gain]	0.0 (no slip compensation)	0.5 - 1.5

*1 The default setting changes when the settings for *o2-04 [Drive Model (KVA) Selection]* change.
*2 The default setting changes when the settings for *A1-02 [Control Method Selection] and o2-04* change.
*3 The default setting changes when the settings for *A1-02 and E1-03 [V/f Pattern Selection]* change.
*4 Recommended settings are for 200 V class drives. Multiply the voltage by 2 for 400 V class drives.

◆ Open Loop Vector Control Method

In Open Loop Vector Control, keep *C4-01 [Torque Compensation Gain]* at its default setting (1.00).

If you cannot get speed precision during regeneration in Open Loop Vector Control, set *C3-04 = 1 [Slip Compensation at Regen = Enabled Above 6 Hz]*.

Table 4.13 Parameters for Fine Tuning the Drive (*A1-02 = 2 [OLV]*)

Issue	Possible Solutions	Parameter	Default	Recommended Setting
• Unsatisfactory motor torque and speed response • Hunting or oscillation at mid-range speeds (10 Hz to 40 Hz)	• To increase the speed of torque or speed response, decrease the setting value in 0.05-unit increments. • If hunting or oscillation occur, decrease the setting value in 0.05-unit increments.	n2-01 [Automatic Freq Regulator Gain]	1.00	0.50 to 2.00
	• To increase the speed of torque or speed response, decrease the setting value in 10 ms increments and examine the response. • If hunting or oscillation occur or if the load inertia is too much, increase the setting value in 50 ms increments and examine the response. **Note:** Make sure that this parameter setting is: *n2-02 ≤ n2-03 [Automatic Freq Regulator Time 2]*. When you adjust *n2-02*, you must also increase the *C4-02 [Torque Compensation Delay Time]* value by the same ratio.	n2-02 [Automatic Freq Regulator Time 1]	50 ms	50 ms to 2000 ms
ov [Overvoltage] occurs when the drive stops accelerating, starts to decelerate, or when there are large changes in the load.	• If *ov* occurs, increase the setting value in 50 ms increments and examine the response. • If the response is not sufficient, decrease the setting value in 10 ms increments and examine the response. **Note:** Make sure that this parameter setting is: *n2-02 [Automatic Freq Regulator Time 1] ≤ n2-03*. When you adjust *n2-03*, you must also increase the *C4-06 [Motor 2 Torque Comp Delay Time]* value by the same ratio.	n2-03 [Automatic Freq Regulator Time 2]	750 ms	750 ms to 2000 ms
	• If *ov* occurs, increase the setting value in 10 ms increments and examine the response. • If the response is not sufficient, decrease the setting value in 2 ms increments and examine the response. **Note:** Make sure that this parameter setting is: *C4-02 [Torque Compensation Delay Time] ≤ C4-06*. When you adjust *C4-06*, you must also increase the *n2-03 [Automatic Freq Regulator Time 2]* value by the same ratio.	C4-06 [Motor 2 Torque Comp Delay Time]	150 ms	150 ms to 750 ms
• Unsatisfactory motor torque and speed response • Hunting or oscillation	• If torque or speed response are slow, decrease the setting value in 2 ms increments. • If hunting or oscillation occur, increase the setting value in 10 ms increments. **Note:** Make sure that this parameter setting is: *C4-02 ≤ C4-06 [Motor 2 Torque Comp Delay Time]*. When you adjust *C4-02*, you must also increase the *n2-02 [Automatic Freq Regulator Time 1]* value by the same ratio.	C4-02 [Torque Compensation Delay Time]	20 ms **1*	20 ms - 100 ms **1*
• Speed response is slow. • Speed is not stable.	• If speed response is slow, decrease the setting value in 10 ms increments. • If speed is not stable, increase the value in 10 ms increments.	C3-02 [Slip Compensation Delay Time]	200 ms **1*	100 ms to 500 ms
Speed precision is unsatisfactory.	• If speed is too slow, increase the setting value in 0.1 unit increments. • If speed is too fast, decrease the setting value in 0.1 unit increments.	C3-01 [Slip Compensation Gain]	1.0 **2*	0.5 to 1.5
• The volume of the motor excitation sound is too high. • Hunting or oscillation at low speeds (10 Hz or lower)	• If the volume of the motor excitation sound is too high, increase the carrier frequency. • If hunting or oscillation occur at low speeds, decrease the carrier frequency.	C6-02 [Carrier Frequency Selection]	8	1 to upper limit value
• Torque at low speeds (10 Hz or lower) is not sufficient. speed response is slow. • Speed response is slow. • Large initial vibration at start up.	• If torque or speed response are slow, increase the setting value. • If there is large initial vibration at start up, decrease the setting value **Note:** If you set the value too high, the drive can output a large torque reference although the load is light.	• E1-08 [Mid Point A Voltage] • E1-10 [Minimum Output Voltage]	• E1-08: 25.0 V • E1-10: 5 V	Default setting +/- 2 V **3*

*1 The default setting changes when the settings for *A1-02 [Control Method Selection]* and *o2-04 [Drive Model (KVA) Selection]* change.
*2 The default setting changes when the settings for *A1-02 [Control Method Selection]* and *E1-03 [V/f Pattern Selection]* change.
*3 Recommended settings are for 200 V class drives. Multiply the voltage by 2 for 400 V class drives.

5

Standards Compliance

This chapter gives information about how to make the machines and devices that use this product comply with European standards and UL standards.

5.1 Section Safety 112
5.2 European Standards 114
5.3 UL Standards 132

5.1 Section Safety

⚠DANGER

Electrical Shock Hazard

Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe.

If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.

⚠WARNING

Electrical Shock Hazard

Do not operate the drive when covers are missing. Replace covers and shields before you operate the drive. Use the drive only as specified by the instructions.

Some figures in this section include drives without covers or safety shields to more clearly show the inside of the drive. If covers or safety shields are missing from the drive, it can cause serious injury or death.

Always ground the motor-side grounding terminal.

If you do not ground the equipment correctly, it can cause serious injury or death if you touch the motor case.

Do not remove covers or touch circuit boards while the drive is energized.

If you touch the internal components of an energized drive, it can cause serious injury or death.

Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive.

If personnel are not approved, it can cause serious injury or death.

Do not wear loose clothing or jewelry when you do work on the drive. Tighten loose clothing and remove all metal objects, for example watches or rings.

Loose clothing can catch on the drive and jewelry can conduct electricity and cause serious injury or death.

Do not modify the drive body or drive circuitry.

Modifications to drive body and circuitry can cause serious injury or death, will cause damage to the drive, and will void the warranty. Yaskawa is not responsible for modifications of the product made by the user.

Fire Hazard

Tighten all terminal screws to the correct tightening torque.

Connections that are too loose or too tight can cause incorrect operation and damage to the drive. Incorrect connections can also cause death or serious injury from fire.

Tighten screws at an angle in the specified range shown in this manual.

If you tighten the screws at an angle not in the specified range, you can have loose connections that can cause damage to the terminal block or start a fire and cause serious injury or death.

Damage to Equipment

Do not apply incorrect voltage to the main circuit of the drive. Operate the drive in the specified range of the input voltage on the drive nameplate.

Voltages that are higher than the permitted nameplate tolerance can cause damage to the drive.

Fire Hazard

Do not put flammable or combustible materials on top of the drive and do not install the drive near flammable or combustible materials. Attach the drive to metal or other noncombustible material.

Flammable and combustible materials can start a fire and cause serious injury or death.

⚠WARNING

Crush Hazard

Wear eye protection when you do work on the drive.

If you do not use correct safety equipment, it can cause serious injury or death.

Electrical Shock Hazard

After the drive blows a fuse or trips an RCM/RCD, do not immediately energize the drive or operate peripheral devices. Wait for the time specified on the warning label at a minimum and make sure that all indicators are OFF. Then check the wiring and peripheral device ratings to find the cause of the problem. If you do not know the cause of the problem, contact Yaskawa before you energize the drive or peripheral devices.

If you do not fix the problem before you operate the drive or peripheral devices, it can cause serious injury or death.

NOTICE

Damage to Equipment

When you touch the drive and circuit boards, make sure that you observe correct electrostatic discharge (ESD) procedures.

If you do not follow procedures, it can cause ESD damage to the drive circuitry.

Do not break the electrical connection between the drive and the motor when the drive is outputting voltage.

Incorrect equipment sequencing can cause damage to the drive.

Before you connect a dynamic braking option to the drive, make sure that qualified personnel read and obey the Braking Unit and Braking Resistor Unit Installation Manual (TOBPC72060001).

If you do not read and obey the manual or if personnel are not qualified, it can cause damage to the drive and braking circuit.

Make sure that all connections are correct after you install the drive and connect peripheral devices.

Incorrect connections can cause damage to the drive.

Note:

Do not use unshielded cable for control wiring. Use shielded, twisted-pair wires and ground the shield to the ground terminal of the drive. Unshielded wire can cause electrical interference and unsatisfactory system performance.

5.2 European Standards

Figure 5.1 CE Mark

The CE Mark identifies that the product meets environmental and safety standards in the European Union. Products manufactured, sold, or imported in the European Union must display the CE Mark.

European Union standards include standards for electrical appliances (Low Voltage Directive), standards for electrical noise (EMC Directive).

This product displays the CE Mark in accordance with the Low Voltage Directive, the EMC Directive.

Table 5.1 Harmonized Standards

European Directive	Harmonized Standard
CE Low Voltage Directive Compliance 2014/35/EU	EN 61800-5-1:2007
EMC Directive 2014/30/EU	EN 61800-3:2004/A1:2012

The customer must display the CE Mark on the final device containing this product. Customers must verify that the final device complies with EU standards.

◆ EU Declaration of Conformity

EU Declaration of Conformity

Original

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Directive of the European Parliament and Council

Low Voltage Directive (LVD)	**: 2014/35/EU**
Electromagnetic Compatibility Directive (EMC)	**: 2014/30/EU**
Restriction of the use of certain Hazardous Substances (RoHS)	**: 2011/65/EU**

Applied harmonized Standards

EN 50581:2012
EN 61800-5-1:2007
EN 61800-3:2004/A1:2012

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11.Dec.2019

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EU Declaration of Conformity

YASKAWA

Translation – German | French | Italian | Spanish | Portugese

EG-Konformitätserklärung | Déclaratione de conformiteé CE
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Directive of the European Parliament and Council
Richtlinie des Europäischen Parlamentes und Rates / Directive du Parlement européen et du Conseil
Direttiva del Parlamento europeo e del Consiglio / Directiva del Parlamento Europeo y del Consejo /
Diretiva do Parlamento Europeu e do Conselho

Low Voltage Directive (LVD) : **2014/35/EU**
Niederspannungsrichtlinie / Directive Basse Tension
Direttiva sulla bassa tensione / Directiva de Baja Tensión / Diretiva "Baixa Tensão"

Electromagnetic Compatibility Directive (EMC) : **2014/30/EU**
EMV-Richtlinie / Directive CEM
Direttiva EMC / Directiva sobre Compatibilidad Electromagnética / Diretiva CEM

Restriction of the use of certain Hazardous Substances (RoHS) : **2011/65/EU**
Beschränkung der Verwendung bestimmter gefährlicher Stoffe in Elektro- und Elektronikgeräten.
Relative à la limitation de l'utilisation de certaines substances dangereuses dans les équipements électriques et électroniques.
Sulla restrizione dell'uso di determinate sostanze pericolose nelle apparecchiature elettriche ed elettroniche.
Sobre restricciones a la utilización de determinadas sustancias peligrosas en aparatos eléctricos y electrónicos.
Relativa à restrição do uso de determinadas substâncias perigosas em equipamentos eléctricos e electrónicos.

Applied harmonized Standards:

EN 61800-5-1:2007

EN 50581:2012
EN 61800-3:2004/A1:2012

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Ort, Datum / Lieu et date / Luogo, data / Lugar, Fecha / Local, data

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EF-overensstemmelseserklæring | EG-försäkran om överensstämmelse
EY-vaatimustenmukaisuusvakuutus | EK atbilstības deklarācija
EÜ vastavusdeklaratsioon

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Eiropas Parlamenta un Padomes Direktīva / Euroopa Parlamendi ja nõukogu direktiiv

Low Voltage Directive (LVD) : 2014/35/EU
Lavspændingsdirektivet / Lågspänningsdirektivet / Pienjännitedirektiivi
Zemsprieguma direktīva / Madalpingedirektiiv

Electromagnetic Compatibility Directive (EMC) : 2014/30/EU
EMC-direktivet / EMC-direktivet / EMC-direktiivi
EMS direktīva / Elektromagnetilise ühilduvuse direktiiv

Restriction of the use of certain hazardous substances (RoHS) : 2011/65/EU
Om begrænsning af anvendelsen af visse farlige stoffer i elektrisk og elektronisk udstyr.
Om begränsning av användning av vissa farliga ämnen i elektrisk och elektronisk utrustning.
Tiettyjen vaarallisten aineiden käytön rajoittamisesta sähkö- ja elektroniikkalaitteissa.
Par dažu bīstamu vielu izmantošanas ierobežošanu elektriskās un elektroniskās iekārtās.
Dėl tam tikrų pavojingų medžiagų naudojimo elektros ir elektroninėje įrangoje apribojimo.

Applied harmonized Standards:

EN 61800-5-1:2007

EN 50581:2012
EN 61800-3:2004/A1:2012

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By, dato / Ort och datum / Paikka, pvm / Vieta, datums / Koht, kuupäev
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Δήλωση Συμμόρφωσης ΕΚ | ЕО-Декларация за съответствие
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Richtlijn van het Europese Parlement en de Europese Raad
Treoir ó Pharlaimint na hEorpa agus ón gComhairle / Οδηγία του Ευρωπαϊκού Κοινοβουλίου και του Συμβουλίου
Директива на Европейския парламент и Съвета / Directiva Parlamentului European şi a Consiliului

Low Voltage Directive (LVD) : 2014/35/EU
Laagspanningsrichtlijn / Treoir maidir le hÍsealvoltas
Οδηγία για τη χαμηλή τάση / Директивата за ниско напрежение
Directive voltaj scăzut

Electromagnetic Compatibility Directive (EMC) : 2014/30/EU
EMC-richtlijn / Treoir maidir le Comhoiriúnacht Leictreamaighnéadach
Οδηγία ηλεκτρομαγνητικής συμβατότητας (EMC) / Директива за електромагнитна съвместимост
Directive CEM

A Restriction of the use of certain hazardous substances (RoHS) : 2011/65/EU
Betreffende beperking van het gebruik van bepaalde gevaarlijke stoffen in elektrische en elektronische apparatuur.
για τον περιορισμό της χρήσης ορισμένων επικίνδυνων ουσιών σε ηλεκτρικό και ηλεκτρονικό εξοπλισμό.
относно ограничението за употребата на определени опасни вещества в електрическото и електронното оборудване.
Privind restricţiile de utilizare a anumitor substanţe periculoase în echipamentele electrice şi electronice.

Applied harmonized Standards:

EN 61800-5-1:2007
EN 50581:2012
EN 61800-3:2004/A1:2012

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Deklaracja zgodności WE | EB atitikties deklaracija
ES Prohlášení o shodě | Vyhlásenie o zhode ES
EK megfelelőségi nyilatkozat

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Prohlašuje na svou výhradní odpovědnost shodu níže uvedených výrobků
potvrdzuje výlučnú zodpovednosť za zhodu pre nasledujúce výrobky
saját kizárólagos felelősségére kijelenti, hogy a következő termékek megfelelnek az alábbiakban megfogalmazott követelményeknek

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Směrnice Evropského parlamentu a Rady / Smernice Európskeho parlamentu a Rady
Az Európai Parlament és az Európai Tanács irányelve

Low Voltage Directive (LVD) : 2014/35/EU
Dyrektywa dot. niskich napięć / Žemos įtampos direktyva
Směrnice o zařízeních nízkého napětí / Smernica o nízkom napätí
Kisfeszültségről szóló irányelv

Electromagnetic Compatibility Directive (EMC) : 2014/30/EU
Dyrektywa EMC / EMS direktyva / Směrnice o elektromagnetické kompatibilitě
Smernica EMC / Elektromágneses összeférhetőségről szóló irányelv

Restriction of the use of certain hazardous substances (RoHS) : 2011/65/EU
W sprawie ograniczenia stosowania niektórych niebezpiecznych substancji w sprzęcie elektrycznym i elektronicznym.
Dėl tam tikrų pavojingų medžiagų naudojimo elektros ir elektroninėje įrangoje apribojimo.
O omezení používání některých nebezpečných látek v elektrických a elektronických zařízeních.
O obmedzení používania určitých nebezpečných látok v elektrických a elektronických zariadeniach.
Egyes veszélyes anyagok elektromos és elektronikus berendezésekben való alkalmazásának korlátozásáról.

Applied harmonized Standards:

EN 61800-5-1:2007

EN 50581:2012
EN 61800-3:2004/A1:2012

Place / Date
Miejscowość, data / Vieta, data / Místo, datum / Miesto, dátum / Hely, dátum
YASKAWA Europe GmbH
Hauptstraße 185
65760 Eschborn

Dec 11th , 2019

General Manager
European Technology Center
Drives and Motion Division
i.V. T. Unger
Tobias Unger

EU Declaration of Conformity

YASKAWA

Translation - Croatian | Slovene | Maltese

EZ Izjava o sukladnosti | Deklaracija o skladnosti ES
Dikjarazzjoni tal-KE dwar il-Konformità

YASKAWA Europe GmbH
Hauptstraße 185
65760 Eschborn

declares under sole responsibility conformity of the following products
pod isključivom odgovornošću izjavljuje sukladnost sljedećih proizvoda
na lastno odgovornost potrjuje skladnost naslednjih izdelkov
tiddikjara taħt ir-responsabbiltà unika tagħha l-konformità tal-prodotti li ġejjin

Model: CIPR-LA50 □ □ □□□ □ □ □ □ - □□□□□□
(Where □ can be a letter from "A" to "Z" or a number from "0" to "9")

Directive of the European Parliament and Council
Direktiva Europskog parlamenta i Vijeća / Direktiva Evropskega parlamenta in Sveta
Eiropas Parlamenta un Padomes Direktīva / Euroopa Parlamendi ja nõukogu direktiiv
Direttiva tal-Parlament Ewropew u tal-Kunsill

Low Voltage Directive (LVD) : 2014/35/EU
Direktiva o niskom naponu / Nizkonapetostna direktiva
Direttiva dwar il-Voltaġġ Baxx

Electromagnetic Compatibility Directive (EMC) : 2014/30/EU
Direktiva o elektromagnetskoj kompatibilnosti (EMC) / EMC direktiva
Direttiva dwar l-EMC

Restriction of the use of certain hazardous substances (RoHS) : 2011/65/EU
O ograničenju uporabe određenih opasnih tvari u električnoj i elektroničkoj opremi.
O omejevanju uporabe nekaterih nevarnih snovi v električni in elektronski opremi.
Dwar ir-restrizzjoni tal-użu ta' ċerti sustanzi perikolużi fit-tagħmir elettriku u elettroniku.

Applied harmonized Standards:

EN 61800-5-1:2007

EN 50581:2012
EN 61800-3:2004/A1:2012

Place / Date
Mjesto, datum / Kraj, datum / Post, Data

YASKAWA Europe GmbH
Hauptstraße 185
65760 Eschborn

Dec 11th , 2019

General Manager
European Technology Center
Drives and Motion Division

Tobias Unger

◆ CE Low Voltage Directive Compliance

It has been confirmed that this product complies with the CE Low Voltage Directive by conducting a test according to EN 61800-5-1:2007.

The following conditions must be satisfied for machines and devices incorporating this product to comply with the CE Low Voltage Directive.

■ Area of Use

Install this product in a location with Overvoltage Category III and pollution degree 2 or less as specified in IEC/CE 60664.

■ Guarding Against Debris

When you install IP20/UL Open type drives, use an enclosure that does not let unwanted material enter the drive from above or below.

■ Wiring Diagram

Refer to Figure 5.2 for an example of a drive that is wired to comply with the CE Low Voltage Directive.

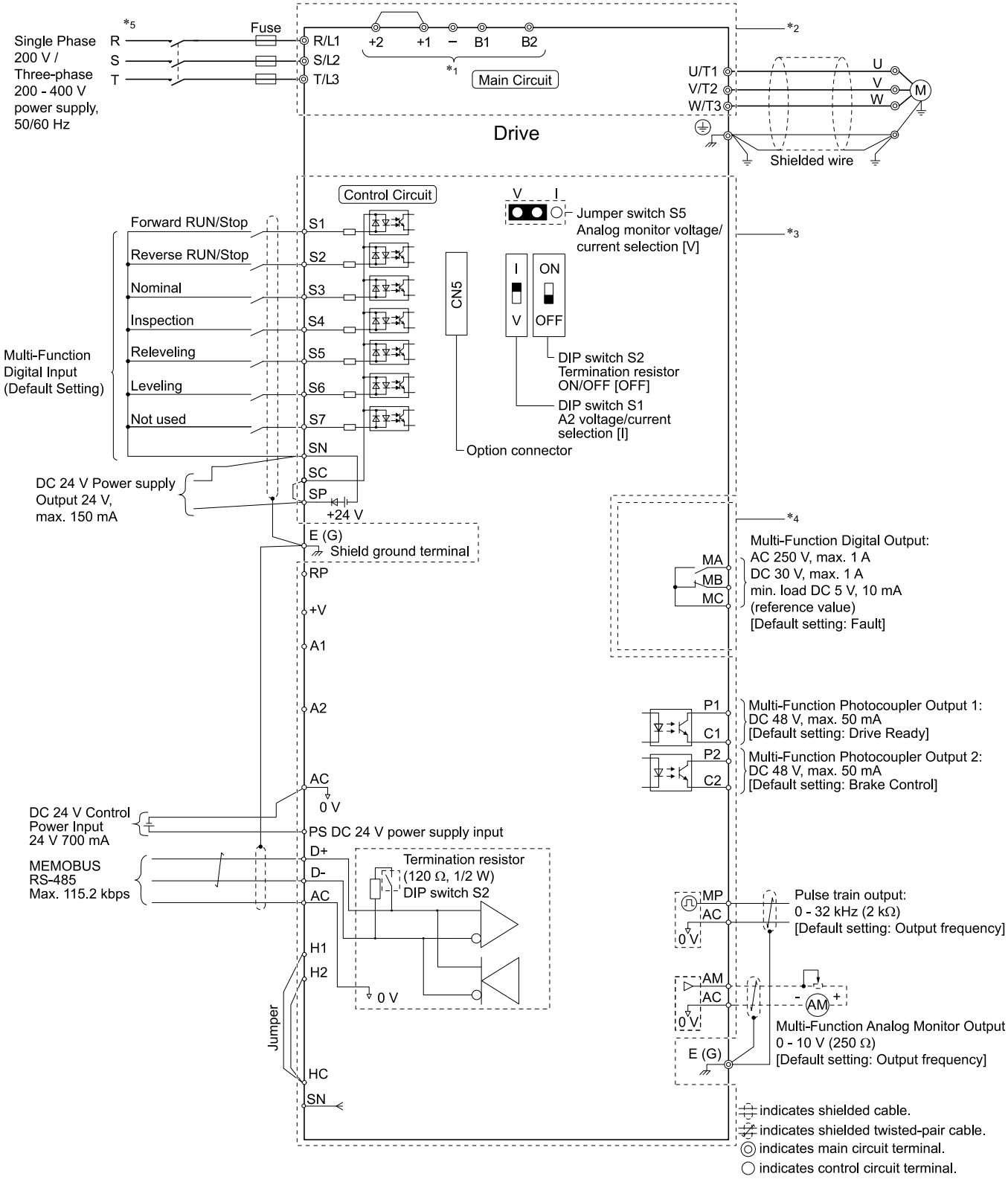


Figure 5.2 Wiring Diagram for CE Low Voltage Directive Compliance

*1 Use terminals B1, B2, -, +1, and +2 to connect options to the drive.

WARNING! *Fire Hazard. Only connect factory-recommended devices or circuits to drive terminals B1, B2, -, +1, and +2. Do not connect an AC power supply lines to these terminals. Incorrect wiring can cause damage to the drive and serious injury or death from fire.*

*2 For circuit protection, the main circuit is separated from the surface case that can touch the main circuit.

*3 The control circuit is a Safety Extra-Low Voltage circuit. Separate this circuit from other circuits with reinforced insulation. Make sure that the Safety Extra-Low Voltage circuit is connected as specified.

*4 Reinforced insulation separates the output terminals from other circuits. Users can also connect circuits that are not Safety Extra-Low Voltage circuits if the drive output is 250 Vac 1 A maximum or 30 Vdc 1 A maximum.

*5 Set *L8-05 = 1 [Output Phase Loss Protect Select = Enabled]* or set the wiring sequence to prevent input phase loss.

■ Main Circuit Wire Gauges and Tightening Torques

WARNING! *Electrical Shock Hazard. Make sure that the protective ground wire complies with technical standards and local safety regulations. The EN 61800-5-1: 2007 standard specifies that users must wire the power supply to automatically turn off when the protective ground wire disconnects. If you turn on the internal EMC filter, the leakage current of the drive will be more than 3.5 mA. You can also connect a protective ground wire that has a minimum cross-sectional area of 10 mm ^{2} (copper wire). If you do not obey the standards and regulations, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Only connect factory-recommended devices or circuits to drive terminals B1, B2, -, +1, and +2. Do not connect AC power to these terminals. Incorrect wiring can cause damage to the drive and serious injury or death from fire.*

Note:

- The recommended wire gauges are based on drive continuous current ratings with 75 °C (167 °F) 600 V class 2 heat-resistant indoor PVC wire. Assume these conditions:
  - Ambient temperature: 40 °C (104 °F) maximum
  - Wiring distance: 100 m (3281 ft) maximum
  - Normal Duty rated current value
- Refer to the instruction manual for each device for recommended wire gauges to connect peripheral devices or options to terminals +1, +2, -, B1, and B2. Contact Yaskawa or your nearest sales representative if the recommended wire gauges for the peripheral devices or options are out of the range of the applicable gauges for the drive.

Wire Selection Precautions

Think about line voltage drop before selecting wire gauges. Select wire gauges that drop the voltage by 2% or less of the rated voltage. Increase the wire gauge and the cable length when the risk of voltage drops increases. Calculate line voltage drop with this formula:

Line voltage drop (V) = $\sqrt{3}$ × wire resistance (Ω/km) × wiring distance (m) × motor rated current (A) × 10^{-3}.

Precautions during Wiring

- Refer to “Yaskawa AC Drive Option Braking Unit, Braking Resistor Unit Instruction Manual (TOBPC72060001)” for information about wire gauges and tightening torques to connect braking resistor units.
- Use terminals +1 and - to connect a regenerative converter or regenerative unit.

WARNING! *Fire Hazard. Do not connect a braking resistor to terminals +1 or -. Use terminals B1 and B2 for the braking resistor connections. If you connect a braking resistor to the incorrect terminals, it can cause damage to the drive and braking circuit and serious injury or death.*

Screw Shape

These tables use icons in Table 5.2 to show the shapes of the screw heads.

Table 5.2 Icons to Identify Screw Shapes

Icon	Screw Shape
	+/-
	Slotted (-)
	Hex socket cap (WAF: 5 mm)

Three-Phase 400 V Class

Model	Terminal	Recomm. Gauge mm²	Applicable Gauge mm²	Wire Stripping Length *1 mm	Terminal Screw		Tightening Torque N·m (in·lb)
					Size	Shape	
4009	R/L1, S/L2, T/L3	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	B1, B2	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	4 *2	2.5 - 6 *2	-	M4	⊕	1.2 - 1.5 (10.6 - 13.3)
4015	R/L1, S/L2, T/L3	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	4	2.5 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	B1, B2	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	4 *2	2.5 - 16	-	M5	⊕	2.0 - 2.5 (17.7 - 22.1)
4018	R/L1, S/L2, T/L3	4	2.5 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	4	2.5 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	4	4 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	B1, B2	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	4 *2	4 - 16	-	M5	⊕	2.0 - 2.5 (17.7 - 22.1)
4024	R/L1, S/L2, T/L3	6	4 - 10	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	6	4 - 10	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	10	2.5 - 16	18	M5	⊖	2.3 - 2.5 (19.8 - 22)
	B1, B2	2.5	2.5 - 4	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	6 *2	6 - 16 *2	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)
4031	R/L1, S/L2, T/L3	10	4 - 16	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	6	2.5 - 10	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	16	4 - 25	18	M5	⊖	2.3 - 2.5 (19.8 - 22)
	B1, B2	4	2.5 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	10	6 - 16	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)

Model	Terminal	Recomm. Gauge mm²	Applicable Gauge mm²	Wire Stripping Length *1 mm	Terminal Screw		Tightening Torque N·m (in·lb)
					Size	Shape	
4039	R/L1, S/L2, T/L3	16	4 - 25	18	M5	⊖	2.3 - 2.5 (19.8 - 22)
	U/T1, V/T2, W/T3	10	4 - 16	18	M5	⊖	2.3 - 2.5 (19.8 - 22)
	-, +1, +2	16	6 - 25	18	M5	⊖	2.3 - 2.5 (19.8 - 22)
	B1, B2	6	4 - 10	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	10	6 - 16	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)
4045	R/L1, S/L2, T/L3	25	6 - 35	18	M5	⊖	• ≤ 25 mm² 2.3 - 2.5 (19.8 - 22) • 35 mm² ≤ 4.1 - 4.5 (36 - 40)
	U/T1, V/T2, W/T3	16	4 - 25	18	M5	⊖	2.3 - 2.5 (19.8 - 22)
	-, +1, +2	25	6 - 35	18	M5	⊖	• ≤ 25 mm² 2.3 - 2.5 (19.8 - 22) • 35 mm² ≤ 4.1 - 4.5 (36 - 40)
	B1, B2	10	2.5 - 16	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	10	6 - 16	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)

*1 Remove insulation from the ends of wires to expose the length of wire shown.
*2 If you turn on the internal EMC filter, the leakage current of the drive will be more than 3.5 mA. Use the closed-loop crimp terminals to connect a protective ground wire that has a minimum cross-sectional area of 10 mm² (copper wire).

■ Connect a Fuse to the Input Side (Primary Side)

The drive circuit protection must comply with EN 61800-5-1:2007 for protection against a short circuit in the internal circuitry. Connect semiconductor fuses on the input side for branch circuit protection.

WARNING! *Electrical Shock Hazard. After the drive blows a fuse or trips an RCM/RCD, do not immediately energize the drive or operate peripheral devices. Wait for the time specified on the warning label at a minimum and make sure that all indicators are OFF. Then check the wiring and peripheral device ratings to find the cause of the problem. If you do not know the cause of the problem, contact Yaskawa before you energize the drive or peripheral devices. If you do not fix the problem before you operate the drive or peripheral devices, it can cause serious injury or death.*

Three-Phase 400 V Class

Table 5.3 Factory-Recommended Branch Circuit Protection: Three-Phase 400 V Class

Drive Model	Semiconductor Protection Fuse Rated Current Manufacturer: EATON/Bussmann
4009	FWH-90B
4015	FWH-80B
4018	FWH-100B
4024	FWH-125B

Drive Model	Semiconductor Protection Fuse Rated Current Manufacturer: EATON/Bussmann
4031	FWH-175B
4039	FWH-200B
4045	FWH-200B

■ CE Standards Compliance for DC Power Supply Input

To comply with CE Standards, install a fuse for the DC power supply input.

Figure 5.3 shows a wiring example for a DC power supply that has two drives connected in parallel.

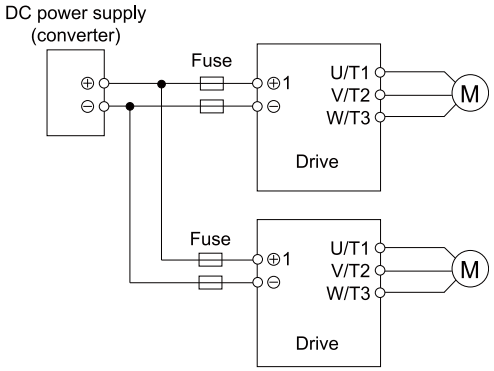


Figure 5.3 Wiring Example for DC Power Supply Input

WARNING! *Electrical Shock Hazard. Do not ground the main circuit bus. Incorrect wiring can cause serious injury or death.*

Note:
- Install a fuse for each drive when operating more than one drive. If one fuse blows, replace all fuses.
- Install the external filter (system) to comply with the EMC Directive.

Refer to Table 5.4 for the recommended fuses.

Table 5.4 Recommended Fuse: Three-Phase 400 V Class

Drive Model	Fuse Manufacturer: Bussmann Model	Drive Model	Fuse Manufacturer: Bussmann Model
4009	FWH-90B	4031	FWH-175B
4015	FWH-80B	4039	FWH-200B
4018	FWH-100B	4045	FWH-200B
4024	FWH-125B		

◆ EMC Directive

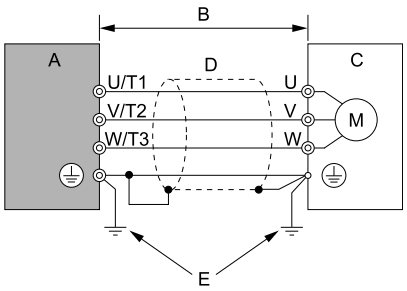
Use drives with built-in EMC filters or install external EMC filters to the drive input side to comply with the EMC Directive.

Drives with built-in EMC filters (models 4xxxE) were tested in accordance with European standard EN 61800-3:2004/A1:2012, and comply with the EMC Directive.

■ Install a Drive to Conform to the EMC Directive

Install drive models 4xxxE with this procedure to comply with the EMC Directive when the drive is a single unit or installed in a larger device.

1. Install the drive on a grounded metal plate.
2. Wire the drive and motor.
3. Ground the wire shielding on the drive side and motor side.



A - Drive
B - Wiring length *1
C - Motor
D - Metal conduit
E - Grounding wire

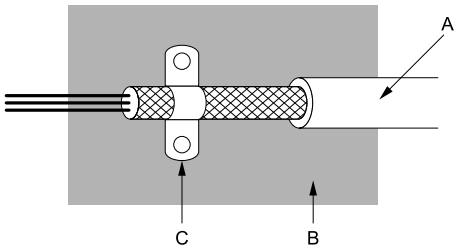
Figure 5.4 Wiring the Drive and Motor

*1 The maximum wiring length between the drive and motor is:
- 4xxxE: 20 m (65.6 ft)

Note:
- Use a braided shield cable for the drive and motor wiring or put the wires through a metal conduit.
- Keep the cable between the drive and motor and the grounding wire as short as possible.

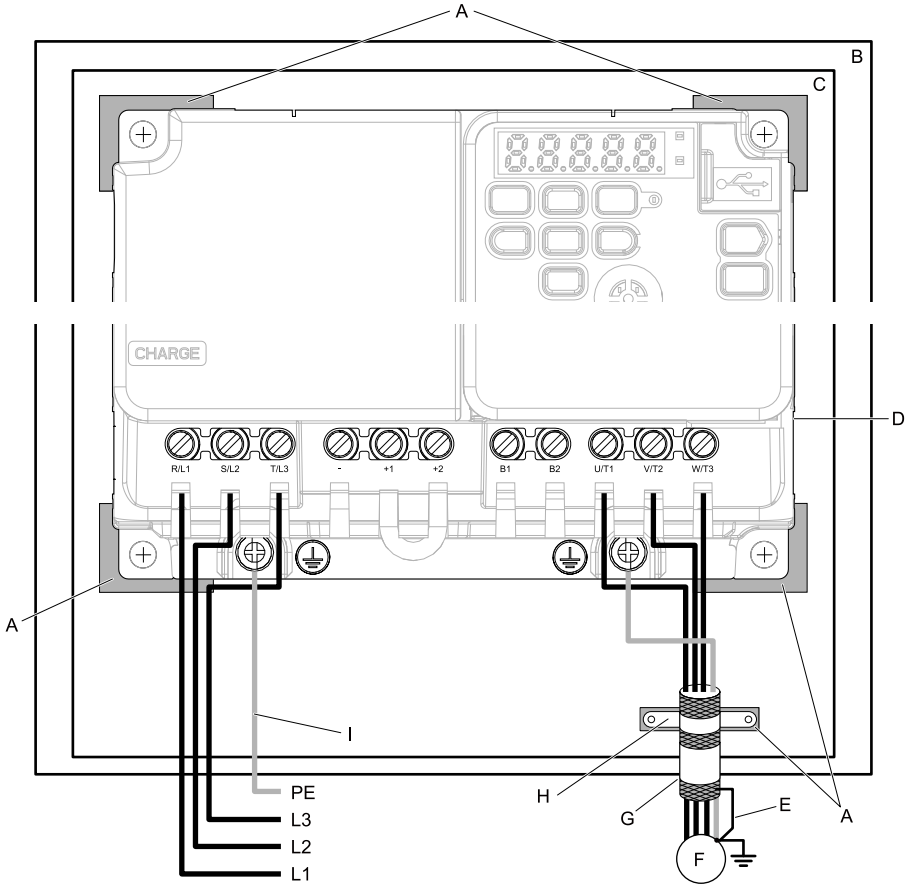
4. Use a cable clamp to ground the motor cable to the metal plate.

Note:
Make sure that the protective ground wire complies with technical specifications or local safety standards.



A - Braided shield cable
B - Metal plate
C - Cable clamp (conductive)

Figure 5.5 Ground the Shield



A - Grounding surface (Remove any paint or sealant.)
B - Enclosure panel
C - Metal plate
D - Drive
E - Shielded wire
F - Motor
G - Motor cable
H - Cable clamp
I - Grounding wire

Figure 5.6 Install a Drive with a Built-in EMC Filter

Ground Wiring

WARNING! *Electrical Shock Hazard. Do not remove covers or touch circuit boards while the drive is energized. If you touch the internal components of an energized drive, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Ground the neutral point on the power supply of drive models BxxxE, 2xxxE, and 4xxxE to comply with the EMC Directive before you turn on the EMC filter or if there is high resistance grounding. If you turn ON the EMC filter, but you do not ground the neutral point, it can cause serious injury or death.*

Enable the Internal EMC Filter

On drive models 4xxxE, move the screw or screws to turn ON and OFF (enable and disable) the EMC filter.

Make sure that the symmetric grounding network is applied, and install the screw or screws in the ON position to enable the built-in EMC filter in compliance with the EMC Directive. The EMC filter switch screw or screws are installed in the OFF position by default.

WARNING! *Electrical Shock Hazard. Disconnect all power to the drive, wait for the time specified on the warning label, and check the drive for dangerous voltages before you remove covers or touch EMC filter screws. If you touch the screws when there are dangerous voltages, it will cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Do not remove covers or touch circuit boards while the drive is energized. If you touch the internal components of an energized drive, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Ground the neutral point on the power supply of drive models BxxxE, 2xxxE, and 4xxxE to comply with the EMC Directive before you turn on the EMC filter or if there is high resistance grounding. If you turn ON the EMC filter, but you do not ground the neutral point, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. Connect the ground cable correctly. If you touch electrical equipment that is not grounded, it can cause serious injury or death.*

NOTICE: *To disable the internal EMC filter, move the screws from ON to OFF and then tighten to the specified torque. If you fully remove the screws or tighten the screws to an incorrect torque, it can cause drive failure.*

NOTICE: *Move the EMC switch screw or screws to the OFF position for networks that are not symmetrically grounded. If the screws are not in the correct position, it can cause damage to the drive.*

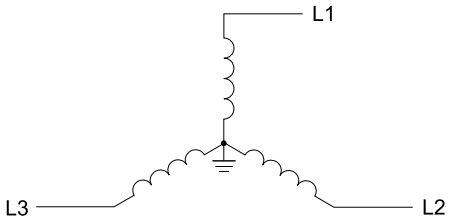
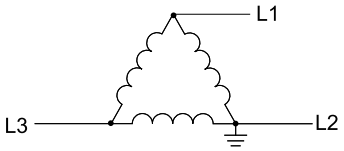
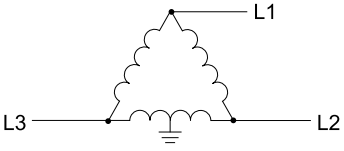


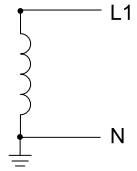
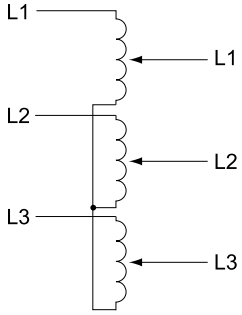
Figure 5.7 Symmetric Grounding

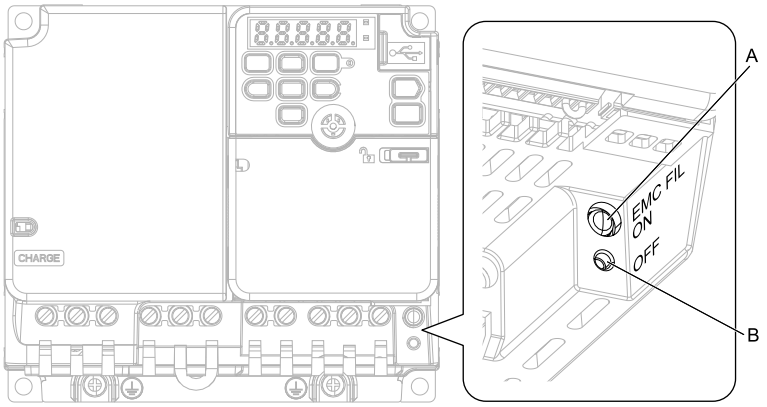
NOTICE: *When you use the drive with a non-grounding, high-resistance grounding, or asymmetric-grounding network, put the EMC Filter screw or screws in the OFF position to disable the built-in EMC filter. Failure to obey the instructions can damage the drive.*

Table 5.5 shows asymmetric grounding networks.

Table 5.5 Asymmetric Grounding

Type of Grounding	Diagram
Grounded at the corner of the delta connection	
Grounded at the middle of the side	

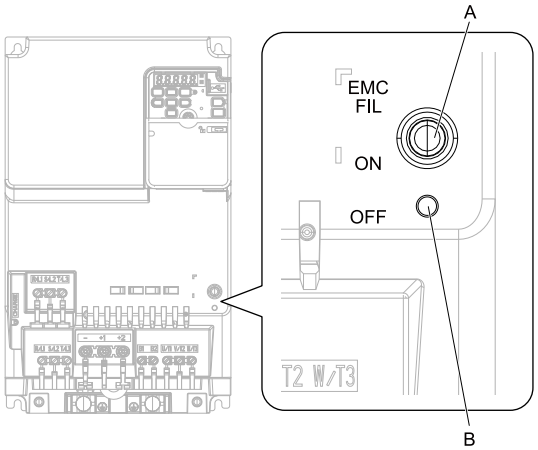
Type of Grounding	Diagram
Single-phase, grounded at the end point	
Three-phase variable transformer without solidly grounded neutral	



A - SW (ON) **B - SW (OFF)**

Screw Size
M3 × 20

Figure 5.8 EMC Filter Switch Location for Models 4009E



A - SW (ON) **B - SW (OFF)**

Screw Size
M4 × 20

Figure 5.9 EMC Filter Switch Location for Models 4015E - 4045E

If you lose an EMC filter switch screw, use the correct replacement screw and install the new screw with the correct tightening torque.

NOTICE: *Only use the screws specified in this manual. If you use screws that are not approved, it can cause damage to the drive.*

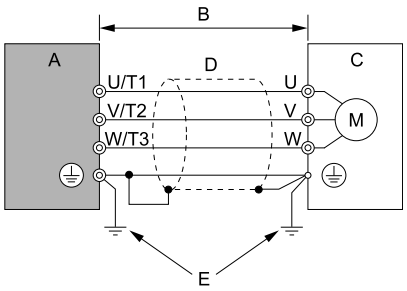
■ Installing the External EMC Noise Filter

Drive models 4xxxA must align with the conditions in this section to comply with EN 61800-3:2004/A1:2012.

Connect an EMC noise filter that complies with European standards as specified by Yaskawa to the input side (primary side). Refer to *External EMC Noise Filter Selection on page 131* to select the correct EMC noise filter.

Use this procedure to install an EMC noise filter to make equipment and devices added to the drive comply with the EMC Directive.

1. Install the drive and EMC noise filter on the same grounded metal plate.
2. Wire the drive and motor.
3. Ground the wire shielding on the drive side and motor side.



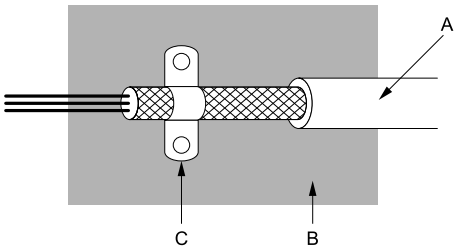
A - Drive
B - 10 m (32.8 ft) maximum
C - Motor
D - Metal conduit
E - Grounding wire

Figure 5.10 Wiring the Drive and Motor

Note:
- Use a braided shield cable for the drive and motor wiring or put the wires through a metal conduit.
- Keep the wire as short as possible. The maximum wiring length between the drive and motor is:
  –4xxxA: 10 m (32.8 ft)
- Keep the grounding wire as short as possible.

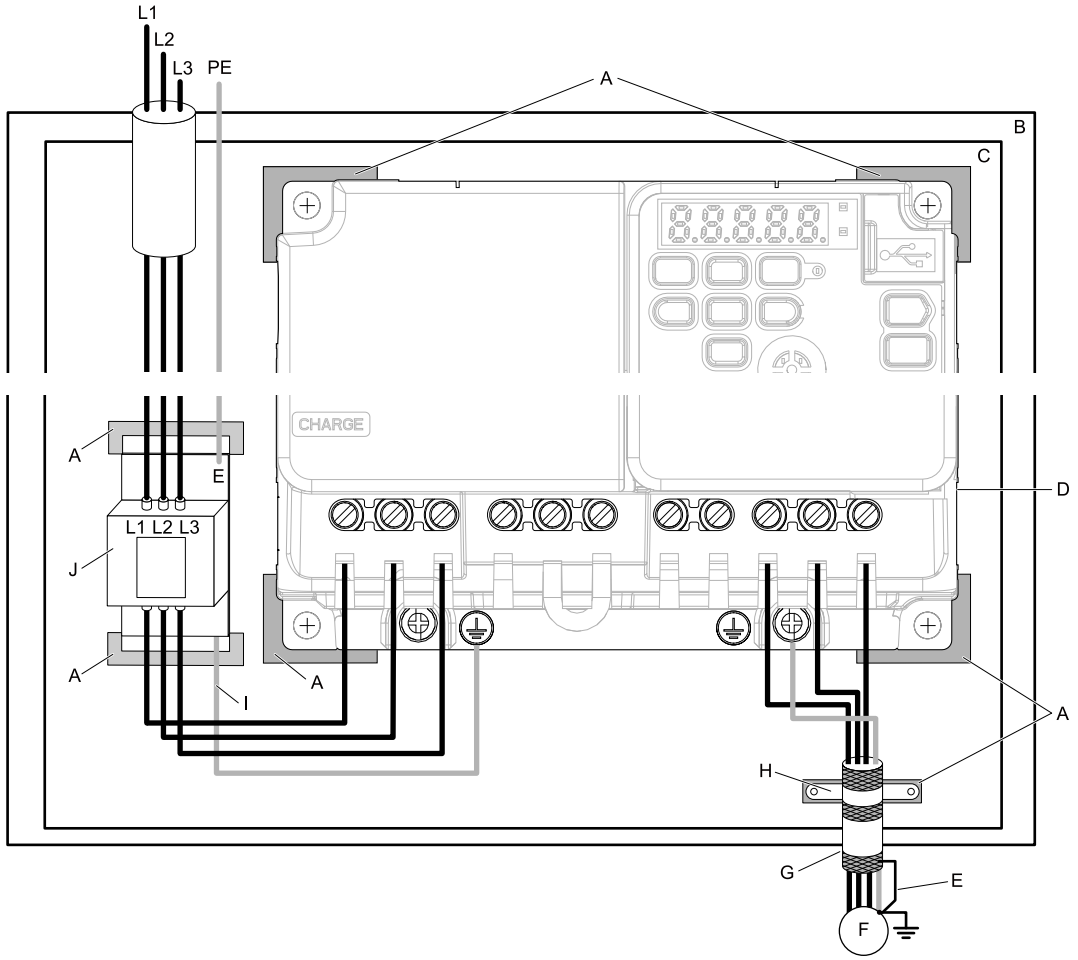
4. Use a cable clamp to ground the motor cable to the metal plate.

Note:
Make sure that the protective ground wire complies with technical specifications or local safety standards.



A - Braided shield cable
B - Metal plate
C - Cable clamp (conductive)

Figure 5.11 Ground the Shield



A - Grounding surface (Remove any paint or sealant.)
B - Enclosure panel
C - Metal plate
D - Drive
E - Ground the shield.
F - Motor
G - Motor cable (Braided shield cable: 10 m (32.8 ft) maximum)
H - Cable clamp
I - Grounding wire
J - EMC noise filter

Figure 5.12 EMC Noise Filter and Drive Installation Procedure

5. Connect the DC reactor to decrease harmonic distortion.

Ground Wiring

WARNING! *Electrical Shock Hazard. Do not remove covers or touch circuit boards while the drive is energized. If you touch the internal components of an energized drive, it can cause serious injury or death.*

External EMC Noise Filter Selection

Table 5.6 External EMC Noise Filter (4xxxA)

Drive model	EMC Noise Filter Model	Quantity	Manufacturer
4009	FS23639-15-07	1	Schaffner
4015	FS5972-35-07 *1	1	Schaffner
4018	FS5972-35-07 *1	1	Schaffner
4024	FS5972-60-07 *1	1	Schaffner
4031	FS5972-60-07 *1	1	Schaffner
4039	RTEN-5100	1	TDK
4045	RTEN-5100	1	TDK

*1 When you install an external EMC noise filter, change the terminals or use the junction terminal.

5.3 UL Standards

Figure 5.13 UL/cUL Mark

The UL/cUL Mark indicates that this product satisfies stringent safety standards. This mark appears on products in the United States and Canada. It shows UL approval, indicating that it has been determined that the product complies with safety standards after undergoing strict inspection and assessment.
You must use UL Listed or UL Recognized parts for all primary components that are built into electrical equipment that has UL approval.

This product has been tested in accordance with UL standard UL61800-5-1, and has been verified to be in compliance with UL standards.

Machines and devices integrated with this product must satisfy the following conditions for compliance with UL standards.

◆ Area of Use

Install this product in a location with Overvoltage Category III and pollution degree 2 or less as specified in UL61800-5-1.

■ Ambient Temperature Setting

Maintain the ambient temperature within the following ranges according to the enclosure type.

- IP20/UL Type 1: -10 °C to +40 °C (14 °F to 104 °F)
- IP20/UL Open Type: -10 °C to +50 °C (14 °F to 122 °F)

◆ Wire the Main Circuit Terminal Block

Wire the main circuit terminal block correctly as specified by the instructions in the manual.

To select the correct wire gauge, refer to *Factory-Recommended Branch Circuit Protection for UL Listing on page 136*.

■ Notes on Wiring the Main Circuit Terminal Block

Read these notes before you wire the main circuit terminal block.

- Use UL-Listed, vinyl-coated insulated copper wires for operation with a continuous maximum permitted temperature of 75 °C at 600 V.
- Remove all unwanted objects that are near the terminal block connections.
- Remove the insulation from the connection wires to the wire stripping lengths shown in the manual.
- Do not use bent or crushed wires. Remove the damaged end of the wire before you use it. Incorrect connections can cause death or serious injury from fire.
- Do not solder stranded wire. Soldered wire connections can become loose over time and cause unsatisfactory drive performance.
- If you use stranded wire, make sure that all of the wire strands are in the connection. Also, do not twist the stranded wire too much. Incorrect connections can cause death or serious injury from fire.
- Put the wire all the way into the terminal block. Remove the insulation from the wire to the recommended wire stripping length to fit the wire with insulation in the plastic housing.
- Use a torque driver, torque ratchet, or torque wrench for the screws. A slotted driver or a hex tool will be necessary to wire the screw clamp terminal. Use applicable tools as specified by the recommended conditions in the product manual.
- If you use power tools to tighten the terminal screws, use a low speed setting (300 to 400 r/min). Failure to obey can cause damage to the terminal screws.
- Wire gauges on existing drive models to be replaced may not match wire gauge ranges on new drives. Refer to the drive manuals for correct wire sizes.
- Do not tighten the terminal screws at an angle of 5 degrees or more. Failure to obey can cause damage to the terminal screws.

If you damage a terminal screw, contact Yaskawa or your nearest sales representative.

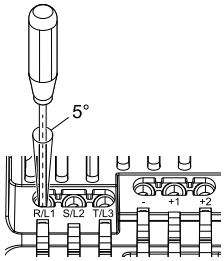


Figure 5.14 Permitted Angle

- Put the bit all the way into the hex socket to tighten the hex socket cap screw.
- When you tighten slotted screws, hold the straight-edge screwdriver perpendicularly to the screw. Make sure that you align the end of the straight-edge screwdriver with the screw groove.

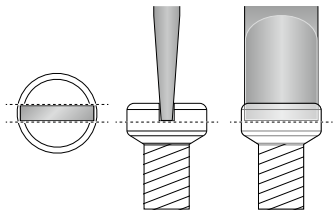
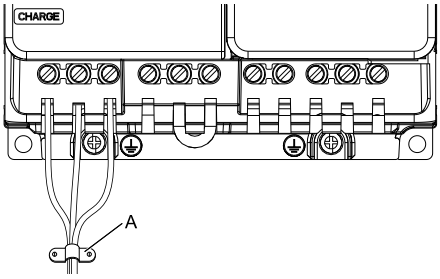


Figure 5.15 Tightening Slotted Screws

- After you connect the wires to the terminal block, lightly pull on the wires to make sure that they do not come out of the terminals.
- Do not let strain on the wiring cause damage. Use a strain relief near the wiring to release the tension. Refer to Figure 5.16 for an example.



A - Cable clamp

Figure 5.16 Strain Relief Example

Table 5.7 Recommended Wiring Tools

Screw Size	Screw Shape	Wire Gauge	Adapter	Bit Manufacturer, Model	Torque Driver Model (Tightening Torque)	Torque Wrench (Tightening Torque)
M3	⊖	-	Bit	PHOENIX CONTACT SF-BIT-SL 0,5X3,0-70	TSD-M 1,2NM (0.3 - 1.2 N·m (2.7 - 10.6 in·lb))	-
M4	⊖	-	Bit	PHOENIX CONTACT SF-BIT-SL 1,0X4,0-70	TSD-M 3NM (1.2 - 3.0 N·m (10.6 - 26.6 in·lb))	-
M5 *1	⊖	≤ 25 mm² (AWG 10)	Bit	PHOENIX CONTACT SF-BIT-SL 1,2X6,5-70	TSD-M 3NM (1.2 - 3.0 N·m (10.6 - 26.6 in·lb))	-
		≥ 30 mm² (AWG 8)			-	4.1 - 4.5 N·m (36.3 - 39.8 in·lb) *2
M6	⑤ (WAF: 5 mm)	-	Bit	PHOENIX CONTACT SF-BIT-HEX 5-50	-	5 - 9 N·m (44.3 - 79.7 in·lb) *2

*1 Sselect the correct tools for the wire gauge.
*2 Use 6.35 mm (0.25 in) bit socket holder.

■ Main Circuit Wire Gauges and Tightening Torques

Comply with local standards for correct wire gauges in the region where you will use the drive.

WARNING! *Electrical Shock Hazard. Only connect factory-recommended devices or circuits to drive terminals B1, B2, -, +1, and +2. Do not connect AC power to these terminals. Incorrect wiring can cause damage to the drive and serious injury or death from fire.*

Note:
- The recommended wire gauges are based on drive continuous current ratings with 75 °C (167 °F) 600 V class 2 heat-resistant indoor PVC wire. Assume these conditions:
  - Ambient temperature: 40 °C (104 °F) maximum
  - Wiring distance: 100 m (3281 ft) maximum
  - Normal Duty rated current value
- Refer to the instruction manual for each device for recommended wire gauges to connect peripheral devices or options to terminals +1, +2, -, B1, and B2. Contact Yaskawa or your nearest sales representative if the recommended wire gauges for the peripheral devices or options are out of the range of the applicable gauges for the drive.

Wire Selection Precautions

Think about line voltage drop before selecting wire gauges. Select wire gauges that drop the voltage by 2% or less of the rated voltage. Increase the wire gauge and the cable length when the risk of voltage drops increases. Calculate line voltage drop with this formula:

Line voltage drop (V) = $\sqrt{3}$ × wire resistance (Ω/km) × wiring distance (m) × motor rated current (A) × 10^{-3}.

Precautions during Wiring

- Refer to “Yaskawa AC Drive Option Braking Unit, Braking Resistor Unit Instruction Manual (TOBPC72060001)” for information about wire gauges and tightening torques to connect braking resistor units.
- Use terminals +1 and - to connect a regenerative converter or regenerative unit.

WARNING! *Fire Hazard. Do not connect a braking resistor to terminals +1 or -. Use terminals B1 and B2 for the braking resistor connections. If you connect a braking resistor to the incorrect terminals, it can cause damage to the drive and braking circuit and serious injury or death.*

Screw Shape

These tables use icons in Table 5.8 to show the shapes of the screw heads.

Table 5.8 Icons to Identify Screw Shapes

Icon	Screw Shape
	+/-
	Slotted (-)
	Hex socket cap (WAF: 5 mm)

Recommended Wire Gauges and Tightening Torques - Three-Phase 400 V Class

Model	Terminal	Recomm. Gauge AWG, kcmil	Applicable Gauge AWG, kcmil	Wire Stripping Length *1 mm	Terminal Screw		Tightening Torque N·m (in·lb)
					Size	Shape	
4009	R/L1, S/L2, T/L3	12	14 - 10	10	M4		1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	14	14 - 12	10	M4		1.5 - 1.7 (13.5 - 15)
	-, +1, +2	10	12 - 8	10	M4		1.5 - 1.7 (13.5 - 15)
	B1, B2	14	14 - 12	10	M4		1.5 - 1.7 (13.5 - 15)
		10 *2	14 - 10 *2	-	M4		1.2 - 1.5 (10.6 - 13.3)
4015	R/L1, S/L2, T/L3	10	12 - 8	10	M4		1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	10	12 - 8	10	M4		1.5 - 1.7 (13.5 - 15)
	-, +1, +2	10	14 - 8	10	M4		1.5 - 1.7 (13.5 - 15)
	B1, B2	14	14 - 12	10	M4		1.5 - 1.7 (13.5 - 15)
		10 *2	14 - 6 *2	-	M5		2.0 - 2.5 (17.7 - 22.1)

Model	Terminal	Recomm. Gauge AWG, kcmil	Applicable Gauge AWG, kcmil	Wire Stripping Length *1 mm	Terminal Screw		Tightening Torque N·m (in·lb)
					Size	Shape	
4018	R/L1, S/L2, T/L3	8	14 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	10	14 - 8	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	8	12 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	B1, B2	12	14 - 10	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	10 *2	10 - 6 *2	-	M5	⊕	2.0 - 2.5 (17.7 - 22.1)
4024	R/L1, S/L2, T/L3	8	12 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	8	12 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	6	12 - 4	18	M5	⊖	• ≤ AWG 10 2.3 - 2.5 (19.8 - 22) • AWG 8 ≤ 4.1 - 4.5 (36 - 40)
	B1, B2	10	12 - 8	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	8	10 - 6	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)
4031	R/L1, S/L2, T/L3	6	12 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	U/T1, V/T2, W/T3	8	12 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	-, +1, +2	4	10 - 2	18	M5	⊖	• ≤ AWG 10 2.3 - 2.5 (19.8 - 22) • AWG 8 ≤ 4.1 - 4.5 (36 - 40)
	B1, B2	10	14 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	6	10 - 6	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)
4039	R/L1, S/L2, T/L3	4	10 - 2	18	M5	⊖	• ≤ AWG 10 2.3 - 2.5 (19.8 - 22) • AWG 8 ≤ 4.1 - 4.5 (36 - 40)
	U/T1, V/T2, W/T3	6	12 - 4	18	M5	⊖	• ≤ AWG 10 2.3 - 2.5 (19.8 - 22) • AWG 8 ≤ 4.1 - 4.5 (36 - 40)
	-, +1, +2	2	8 - 2	18	M5	⊖	4.1 - 4.5 (36 - 40)
	B1, B2	8	12 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	6	10 - 6	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)

Model	Terminal	Recomm. Gauge AWG, kcmil	Applicable Gauge AWG, kcmil	Wire Stripping Length *1 mm	Terminal Screw		Tightening Torque N·m (in·lb)
					Size	Shape	
4045	R/L1, S/L2, T/L3	2	8 - 2	18	M5	⊖	4.1 - 4.5 (36 - 40)
	U/T1, V/T2, W/T3	4	10 - 2	18	M5	⊖	• ≤ AWG 10 2.3 - 2.5 (19.8 - 22) • AWG 8 ≤ 4.1 - 4.5 (36 - 40)
	-, +1, +2	2	6 - 2	18	M5	⊖	4.1 - 4.5 (36 - 40)
	B1, B2	8	12 - 6	10	M4	⊖	1.5 - 1.7 (13.5 - 15)
	⏚	6	10 - 6	-	M6	⊕	5.4 - 6.0 (47.8 - 53.1)

*1 Remove insulation from the ends of wires to expose the length of wire shown.
*2 If you turn on the internal EMC filter, the leakage current of the drive will be more than 3.5 mA. Use the closed-loop crimp terminals to connect a protective ground wire that has a minimum cross-sectional area of 10 mm^2 (copper wire).

■ Factory-Recommended Branch Circuit Protection for UL Listing

Use branch circuit protection to protect against short circuits and to maintain compliance with UL61800-5-1. Yaskawa recommends connecting semiconductor protection fuses on the input side for branch circuit protection.

WARNING! *Electrical Shock Hazard. After the drive blows a fuse or trips an RCM/RCD, do not immediately energize the drive or operate peripheral devices. Wait for the time specified on the warning label at a minimum and make sure that all indicators are OFF. Then check the wiring and peripheral device ratings to find the cause of the problem. If you do not know the cause of the problem, contact Yaskawa before you energize the drive or peripheral devices. If you do not fix the problem before you operate the drive or peripheral devices, it can cause serious injury or death.*

- 400 V class
  Use the fuses specified in this document to prepare the drive for use on a circuit that supplies not more than 31,000 RMS and not more than 480 Vac when there is a short circuit in the power supply.

The built-in short circuit protection of the drive does not provide branch circuit protection. The user must provide branch circuit protection as specified by the National Electric Code (NEC), the Canadian Electric Code, Part I (CEC), and local codes.

Factory Recommended Fuses Three-Phase 400 V Class

Table 5.9 Factory-Recommended Branch Circuit Protection: Three-Phase 400 V Class

Drive Model	Maximum Applicable Motor Output kW (HP)	Time Delay Fuse	Semiconductor Protection Fuse Rated Current Manufacturer: EATON/Bussmann	
	HD	Class J, CC, and T Fuse Rated Current A	Model	Fuse Rated Current A
4009	4.0 (5)	20	FWH-90B	90
4015	5.5 (10)	30	FWH-80B	80
4018	7.5 (10)	40	FWH-100B	100
4024	11.0 (15)	50	FWH-125B	125
4031	15.0 (20)	60	FWH-175B	175
4039	18.5 (25)	70	FWH-200B	200
4045	22.0 (30)	100	FWH-200B	200

◆ Low Voltage Wiring for Control Circuit Terminals

You must provide low voltage wiring as specified by the National Electric Code (NEC), the Canadian Electric Code, Part I (CEC), and local codes. Yaskawa recommends the NEC class 1 circuit conductor. Use the UL approved class 2 power supply for external power supply.

Table 5.10 Control Circuit Terminal Power Supplies

Input/Output	Terminals	Power Supply Specifications
Digital input	S1 to S7, SN, SC, SP	Uses the LVLC power supply in the drive. Use the UL Listed class 2 power supply for external power supply.
Analog input	A1, A2, AC ,+V	Uses the LVLC power supply in the drive. Use the UL Listed class 2 power supply for external power supply.
Analog output	AM, AC	Uses the LVLC power supply in the drive.
Pulse train output	MP, AC	Uses the LVLC power supply in the drive. Use the UL Listed class 2 power supply for external power supply.
Pulse train input	RP, AC	Uses the LVLC power supply in the drive. Use the UL Listed class 2 power supply for external power supply.
Safe disable input	H1, H2, HC	Uses the LVLC power supply in the drive. Use the UL Listed class 2 power supply for external power supply.
Serial communication input/output	D+, D-, AC	Uses the LVLC power supply in the drive. Use the UL Listed class 2 power supply for external power supply.
24 V external power supply	PS, AC	Use the UL Listed class 2 power supply.

◆ Drive Motor Overload and Overheat Protection

The drive motor overload and overheat protection function complies with the National Electric Code (NEC) and the Canadian Electric Code, Part I (CEC).

Set the Motor Rated Current and *L1-01: Motor Overload (oL1) Protection on page 293*, and *L1-02: Motor Overload Protection Time on page 295* correctly to enable motor overload and overheat protection.

Refer to the control method and set the motor rated current with *E2-01: Motor Rated Current (FLA) on page 271*.

6

Troubleshooting

6.1 Section Safety 140
6.2 Types of Faults, Minor Faults, Alarms, and Errors 142
6.3 List of Fault, Minor Fault, Alarm, and Error Codes 143
6.4 Fault 146
6.5 Minor Faults/Alarms 158
6.6 Parameter Setting Errors 166
6.7 Auto-Tuning Errors 168
6.8 Backup Function Operating Mode Display and Errors 171
6.9 Diagnosing and Resetting Faults 172
6.10 Troubleshooting Without Fault Display 173

6.1 Section Safety

DANGER

Electrical Shock Hazard

Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe.

If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.

WARNING

Electrical Shock Hazard

Do not operate the drive when covers are missing. Replace covers and shields before you operate the drive. Use the drive only as specified by the instructions.

Some figures in this section include drives without covers or safety shields to more clearly show the inside of the drive. If covers or safety shields are missing from the drive, it can cause serious injury or death.

Always ground the motor-side grounding terminal.

If you do not ground the equipment correctly, it can cause serious injury or death if you touch the motor case.

After the drive blows a fuse or trips an RCM/RCD, do not immediately energize the drive or operate peripheral devices. Wait for the time specified on the warning label at a minimum and make sure that all indicators are OFF. Then check the wiring and peripheral device ratings to find the cause of the problem. If you do not know the cause of the problem, contact Yaskawa before you energize the drive or peripheral devices.

If you do not fix the problem before you operate the drive or peripheral devices, it can cause serious injury or death.

Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive.

If personnel are not approved, it can cause serious injury or death.

Do not wear loose clothing or jewelry when you do work on the drive. Tighten loose clothing and remove all metal objects, for example watches or rings.

Loose clothing can catch on the drive and jewelry can conduct electricity and cause serious injury or death.

Do not remove covers or touch circuit boards while the drive is energized.

If you touch the internal components of an energized drive, it can cause serious injury or death.

Do not modify the drive body or drive circuitry.

Modifications to drive body and circuitry can cause serious injury or death, will cause damage to the drive, and will void the warranty. Yaskawa is not responsible for modifications of the product made by the user.

Fire Hazard

Tighten all terminal screws to the correct tightening torque.

Connections that are too loose or too tight can cause incorrect operation and damage to the drive. Incorrect connections can also cause death or serious injury from fire.

Tighten screws at an angle in the specified range shown in this manual.

If you tighten the screws at an angle not in the specified range, you can have loose connections that can cause damage to the terminal block or start a fire and cause serious injury or death.

Damage to Equipment

Do not apply incorrect voltage to the main circuit of the drive. Operate the drive in the specified range of the input voltage on the drive nameplate.

Voltages that are higher than the permitted nameplate tolerance can cause damage to the drive.

⚠WARNING

Fire Hazard

Do not put flammable or combustible materials on top of the drive and do not install the drive near flammable or combustible materials. Attach the drive to metal or other noncombustible material.

Flammable and combustible materials can start a fire and cause serious injury or death.

Crush Hazard

Wear eye protection when you do work on the drive.

If you do not use correct safety equipment, it can cause serious injury or death.

Use a crane or hoist to move large drives when necessary.

If you try to move a large drive without a crane or hoist, it can cause serious injury or death.

NOTICE

Damage to Equipment

When you touch the drive and circuit boards, make sure that you observe correct electrostatic discharge (ESD) procedures.

If you do not follow procedures, it can cause ESD damage to the drive circuitry.

Do not break the electrical connection between the drive and the motor when the drive is outputting voltage.

Incorrect equipment sequencing can cause damage to the drive.

Make sure that all connections are correct after you install the drive and connect peripheral devices.

Incorrect connections can cause damage to the drive.

Note:

Do not use unshielded wire for control wiring. Use shielded, twisted-pair wires and ground the shield to the ground terminal of the drive. Unshielded wire can cause electrical interference and unsatisfactory system performance.

6.2 Types of Faults, Minor Faults, Alarms, and Errors

If the drive or motor do not operate correctly, check the drive keypad for a code or message.

If problems occur that are not identified in this manual, contact the nearest Yaskawa representative with this information:

- Drive model
- Drive software version
- Date of purchase
- Description of the problem (for example failure conditions and modified parameters)

Table 6.1 contains descriptions of the different types of faults, minor faults, alarms, and errors that can occur during drive operation.

Contact Yaskawa if there is damage to the drive. Contact information is on the back cover of the manual.

Table 6.1 Types of Faults, Minor Faults, Alarms, and Errors

Type	Drive Response
Fault	When the drive detects a fault, it will cause these conditions: • The keypad shows the fault code, and ALM/ERR stays illuminated. • The drive shuts off output, and the motor coasts to a stop. Some faults let the user select a motor stopping method. • The terminal set to *H2-01 to H2-03 = E [MFDO Function Selection = Fault]* will activate. The drive will not operate until you clear the fault with a Fault Reset and the drive goes back to usual status.
Minor Faults/Alarms	When the drive detects a minor fault or an alarm, it will cause these conditions: • The keypad shows the alarm code, and ALM/ERR flashes. • The drive will continue to operate the motor. Some alarms let you select a motor stopping method. • If the drive detects a minor fault, the terminal set to *H2-01 to H2-03 = 10 [Alarm]* will activate If you do not set parameters *H2-01 to H2-03*, the drive will not trigger MFDO terminals if it detects a minor fault. • The drive will not output a minor fault signal if it detects an alarm. It is not necessary to do Fault Reset.
Operation Errors	An error occurs when parameter settings do not agree or a parameter combination is incorrect. The drive will not operate until you set the parameters correctly. When the drive detects an operation error, these conditions will result: • The keypad shows the error code. • Multi-function outputs do not output an alarm signal. Find the parameters that caused the error and correct the settings.
Auto-Tuning Error	An error occurs during Auto-Tuning. When the drive detects an operation error, it will cause these conditions: • The keypad shows the error code. • Multi-function outputs do not output an alarm signal. • The motor coasts to stop. Remove the cause of the error and do Auto-Tuning again.
Copy Function Error	An error occurs when you use the keypad for a backup, restore, or verify operation. When the drive detects a copy function error, it will cause these conditions: • The keypad shows the error code. • Multi-function outputs do not output an alarm signal. Push a key on the keypad to clear the error. Remove the cause of the error and try the backup, restore, or verify operation again.

6.3 List of Fault, Minor Fault, Alarm, and Error Codes

Table 6.2 shows the possible fault, minor fault, alarm, and error codes.

The display codes are in alphabetical order. Search the table for the code shown on the keypad, and identify its causes and possible solutions.

Note:

The number in parentheses adjacent to the code in the table identifies the fault code or minor fault code (hex. number) that was read during MEMOBUS/Modbus communications.

Example: AEr (0032)

Table 6.2 List of Fault, Minor Fault, Alarm, and Error Codes

Display (Hex.)	Name	ALM LED	Type	Ref.
bAT (0085)	Keypad Battery Low Voltage	Flashing	Alarm	158
bAT (0402)	Keypad Battery Low Voltage	Illuminated	Fault	146
bb (0008)	Baseblock	Flashing	Alarm	158
bCE (008A)	Bluetooth Communication Error	Flashing	Alarm	158
bCE (0416)	Bluetooth Communication Fault	Illuminated	Fault	146
boL (0045)	Braking Transistor Overload	Flashing	Alarm	158
boL (004F)	BrakingTransistor Overload Fault	Illuminated	Fault	146
bUS (0015)	Option Communication Error	Flashing	Alarm	158
bUS (0022)	Option Communication Error	Illuminated	Fault	146
bUSy	Busy	-	Not an alarm.	158
CALL (001D)	Serial Comm Transmission Error	Flashing	Alarm	159
CE (0014)	Modbus Communication Error	Flashing	Alarm	159
CE (0021)	Modbus Communication Error	Illuminated	Fault	146
CF (0025)	Control Fault	Illuminated	Fault	147
CoF (0046)	Current Offset Fault	Illuminated	Fault	147
CPEr	Control Mode Mismatch	-	Backup Function Runtime Error	171
CPF00	Control Circuit Error	Illuminated	Fault	147
CPF01	Control Circuit Error	Illuminated	Fault	147
CPF02, CPF03 (0083, 0084)	Control Circuit Error	Illuminated	Fault	148, 148
CPF06 (0087)	EEPROM Memory Data Error	Illuminated	Fault	148
CPF08 (0089)	Terminal Board Connection Error	Illuminated	Fault	148
CPF11 - CPF14 (008C - 008F)	Control Circuit Error	Illuminated	Fault	148 - 148
CPF16 - CPF24 (0091 - 0099)	Control Circuit Error	Illuminated	Fault	148 - 149
CPF38 (00A7)	EEPROM Memory Data Error	Illuminated	Fault	149
CPyE	Error Writing Data	-	Backup Function Runtime Error	171
CrST	Remove RUN Command to Reset	Flashing	Not an alarm.	159
CSEr	Control Mode Mismatch	-	Backup Function Runtime Error	171
dFPS	Drive Model Mismatch	-	Backup Function Runtime Error	171
dv7 (005B)	Polarity Judge Timeout	Illuminated	Fault	149
EF (0007)	FWD/REV Run Command Input Error	Flashing	Alarm	159
EF1 (0042)	External Fault (Terminal S1)	Illuminated	Fault	149
EF1 (0039)	External Fault (Terminal S1)	Flashing	Alarm	159
EF2 (003A)	External Fault (Terminal S2)	Flashing	Alarm	160
EF2 (0043)	External Fault (Terminal S2)	Illuminated	Fault	149
EF3 (0009)	External Fault (Terminal S3)	Flashing	Alarm	160
EF3 (0011)	External Fault (Terminal S3)	Illuminated	Fault	150
EF4 (000A)	External Fault (Terminal S4)	Flashing	Alarm	160

Display (Hex.)	Name	ALM LED	Type	Ref.
EF4 (0012)	External Fault (Terminal S4)	Illuminated	Fault	150
EF5 (000B)	External Fault (Terminal S5)	Flashing	Alarm	160
EF5 (0013)	External Fault (Terminal S5)	Illuminated	Fault	150
EF6 (000C)	External Fault (Terminal S6)	Flashing	Alarm	160
EF6 (0014)	External Fault (Terminal S6)	Illuminated	Fault	150
EF7 (000D)	External Fault (Terminal S7)	Flashing	Alarm	160
EF7 (0015)	External Fault (Terminal S7)	Illuminated	Fault	150
End1	Excessive Rated Voltage Setting	Flashing	Auto-Tuning Error	168
End2	Iron Core Saturation Coefficient	Flashing	Auto-Tuning Error	168
End3	Rated Current Setting Alarm	Flashing	Auto-Tuning Error	168
End4	Adjusted Slip Calculation Error	Flashing	Auto-Tuning Error	168
End5	Resistance Tuning Error	Flashing	Auto-Tuning Error	168
End6	Leakage Inductance Alarm	Flashing	Auto-Tuning Error	168
End7	No-Load Current Alarm	Flashing	Auto-Tuning Error	168
EP24v (0081)	External Power 24V Supply	Flashing	Alarm	161
Er-01	Motor Data Error	Flashing	Auto-Tuning Error	168
Er-02	Drive in an Alarm State	Flashing	Auto-Tuning Error	169
Er-03	STOP Button was Pressed	Flashing	Auto-Tuning Error	169
Er-04	Resistance Tuning Error	Flashing	Auto-Tuning Error	169
Er-05	No-Load Current Alarm	Flashing	Auto-Tuning Error	169
Er-08	Rated Slip Error	Flashing	Auto-Tuning Error	169
Er-09	Acceleration Error	Flashing	Auto-Tuning Error	170
Er-11	Motor Speed Error	Flashing	Auto-Tuning Error	170
Er-12	Current Detection Error	Flashing	Auto-Tuning Error	170
Er-13	Leakage Inductance Alarm	Flashing	Auto-Tuning Error	170
Er-15	Torque Saturation Error	Flashing	Auto-Tuning Error	170
Err (001F)	EEPROM Write Error	Illuminated	Fault	150
GF (0006)	Ground Fault	Illuminated	Fault	151
HCA (0034)	High Current Alarm	Flashing	Alarm	161
iFEr	Communication Err	-	Backup Function Runtime Error	171
L24v (0021)	Loss of External Power 24 Supply	Flashing	Alarm	161
LF (001C)	Output Phase Loss	Illuminated	Fault	151
LF2 (0036)	Output Current Imbalance	Illuminated	Fault	151
LoG	Log Com Error	Flashing	Alarm	161
LT-1 (0035)	Cooling Fan Maintenance Time	Flashing	Alarm	161
LT-2 (0036)	Capacitor Maintenance Time	Flashing	Alarm	161
LT-3 (0043)	SoftChargeBypassRelay MainteTime	Flashing	Alarm	161
LT-4 (0044)	IGBT Maintenance Time (50%)	Flashing	Alarm	162
ndAT	Model,VolClass,Capacity Mismatch	-	Backup Function Runtime Error	171
oC (0007)	Overcurrent	Illuminated	Fault	151
oH (0003)	Heatsink Overheat	Flashing	Alarm	162
oH (0009)	Heatsink Overheat	Illuminated	Fault	152
oH1 (000A)	Heatsink Overheat	Illuminated	Fault	152
oH2 (0004)	External Overheat (H1-XX=B)	Flashing	Alarm	162
oH3 (0022)	Motor Overheat (PTC Input)	Flashing	Alarm	162
oL1 (000B)	Motor Overload	Illuminated	Fault	152

Display (Hex.)	Name	ALM LED	Type	Ref.
oL2 (000C)	Drive Overload	Illuminated	Fault	153
oL3 (0005)	Overtorque 1	Flashing	Alarm	162
oL3 (000D)	Overtorque Detection 1	Illuminated	Fault	154
oL4 (0006)	Overtorque 2	Flashing	Alarm	163
oL4 (000E)	Overtorque Detection 2	Illuminated	Fault	154
oPE01	Drive Capacity Setting Fault	Flashing	Parameter Setting Error	166
oPE02	Parameter Range Setting Error	Flashing	Parameter Setting Error	166
oPE03	Multi-Function Input Setting Err	Flashing	Parameter Setting Error	166
oPE08	Parameter Selection Error	Flashing	Parameter Setting Error	167
oPE10	V/f Data Setting Error	Flashing	Parameter Setting Error	167
oPE33	Digital Output Selection Error	Flashing	Parameter Setting Error	167
oPr (001E)	Keypad Connection Fault	Illuminated	Fault	154
oS (0010)	Overspeed	Flashing	Alarm	163
oS (0018)	Overspeed	Illuminated	Fault	163
ov (0002)	Overvoltage	Flashing	Alarm	163
ov (0008)	Overvoltage	Illuminated	Fault	154
ovEr	Too Many Parameters Changed	-	Not an alarm.	163
PASS	Modbus Communication Test	Flashing	Not an alarm.	163
PF (0047)	Input Phase Loss	Flashing	Alarm	163
PF (001B)	Input Phase Loss	Illuminated	Fault	155
rdEr	Error Reading Data	-	Backup Function Runtime Error	171
rF (004E)	Braking Resistor Fault	Illuminated	Fault	155
rH (0010)	Braking Resistor Overheat	Illuminated	Fault	155
rr (000F)	Dynamic Braking Transistor Fault	Illuminated	Fault	156
SC (0005)	Short Circuit/IGBT Failure	Illuminated	Fault	156
SCF (040F)	Safety Circuit Fault	Illuminated	Fault	156
SE (0020)	Modbus Test Mode Error	Flashing	Alarm	164
SToF (003B)	Safe Torque OFF	Flashing	Alarm	164
TiM (0089)	Keypad Time Not Set	Flashing	Alarm	164
TiM (0401)	Keypad Time Not Set	Illuminated	Fault	156
TrPC (0042)	IGBT Maintenance Time (90%)	Flashing	Alarm	164
UL3 (001E)	Undertorque Detection 1	Flashing	Alarm	164
UL3 (0029)	Undertorque Detection 1	Illuminated	Fault	156
UL4 (001F)	Undertorque Detection 2	Flashing	Alarm	164
UL4 (002A)	Undertorque Detection 2	Illuminated	Fault	156
Uv (0001)	DC Bus Undervoltage	Flashing	Alarm	164
Uv1 (0002)	DC Bus Undervoltage	Illuminated	Fault	156
Uv2 (0003)	Control Power Undervoltage	Illuminated	Fault	157
Uv3 (0004)	Soft Charge Answerback Fault	Illuminated	Fault	157
vAEr	Voltage Class, Capacity Mismatch	-	Backup Function Runtime Error	171
vFyE	Parameters do not Match	-	Backup Function Runtime Error	171

6.4 Fault

This section gives information about some of the causes and possible solutions of faults. You must use the Fault Reset operation to remove the fault before you can operate the drive. Use the information in this table to remove the cause of the fault.

Code	Name	Causes	Possible Solutions
bAT	Keypad Battery Low Voltage	The keypad battery voltage is low.	Replace the keypad battery.
Note: Use *o4-24 [bAT Detection Selection]* to enable/disable *bAT* detection.			
Code	**Name**	**Causes**	**Possible Solutions**
bCE	Bluetooth Communication Fault	The smartphone or tablet with DriveWizard Mobile installed is too far from the keypad.	Use the smartphone or tablet 10 m (32.8 ft) or nearer to the keypad. **Note:** *bCE* can occur when the smartphone or tablet is 10 m (32.8 ft) or nearer to the keypad depending on the specifications of the smartphone or tablet.
		Radio waves from a different device are causing interference with communications between the smartphone or tablet and keypad.	Make sure that no device around the keypad uses the same radio bandwidth (2400 MHz to 2480 MHz), and prevent radio interference.
Note: • The drive detects this error when you use the Bluetooth LCD keypad and operate the drive with a smartphone or tablet. • Do a Fault Reset to clear the fault. • Set the stopping method for this fault in *o2-27 [bCE Detection Selection]*.			
Code	**Name**	**Causes**	**Possible Solutions**
boL	BrakingTransistor Overload Fault	The duty cycle of the braking transistor is high (the regeneration power or repetition frequency is high).	• Install a regenerative converter. • Increase the deceleration time.
		You enabled the protective function for the braking transistor when you have a regenerative converter.	Set *L8-55 = 0 [Internal DB TransistorProtection Selection = Disable]*.
		The braking transistor in the drive is broken.	Replace the drive.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
bUS	Option Communication Error	The drive did not receive a signal from the controller.	Correct wiring errors.
		The communications cable wiring is incorrect.	
		There is a short-circuit in the communications cable or the communications cable is not connected.	• Repair short circuits and connect cables. • Replace the defective communications cable.
		Electrical interference caused a communication data error.	• Examine the control circuit lines, main circuit lines, and ground wiring, and decrease the effects of electrical interference. • Make sure that a magnetic contactor is not the source of the electrical interference, then use a Surge Protective Device if necessary. • Use only the recommended cables or other shielded line. Ground the shield on the controller side or the drive input power side. • Separate the communication wiring from drive power lines, and install a noise filter to the input side of the power supply for communication. • Decrease the effects of electrical interference from the controller.
		The option is incorrectly installed to the drive.	Correctly install the option to the drive.
		The option is damaged.	If the fault continues and the wiring is correct, replace the option.
Note: • The drive detects this error if the Run command or frequency reference is assigned to the option card. • Do a Fault Reset to clear the fault. • If the drive detects this error, the drive will operate the motor as specified by the stopping method set in *F6-01 [Communication Error Selection]*.			
Code	**Name**	**Causes**	**Possible Solutions**
CE	Modbus Communication Error	The communications cable wiring is incorrect.	Correct wiring errors.
		There is a short circuit in the communications cable or the communications cable is not connected.	• Repair short circuits and connect cables. • Replace the defective communications cable.

Code	Name	Causes	Possible Solutions
		Electrical interference caused a communication data error.	• Examine the control circuit lines, main circuit lines, and ground wiring, and decrease the effects of electrical interference. • Make sure that a magnetic contactor is not the source of the electrical interference, then use a Surge Protective Device if necessary. • Use only the recommended cables or other shielded line. Ground the shield on the controller side or the drive input power side. • Separate the communication wiring from drive power lines, and install a noise filter to the input side of the power supply for communication. • Decrease the effects of electrical interference from the controller.

Note:
• The drive detects this error if it does not correctly receive control data for the *CE* detection time set to *H5-09 [CE Detection Time]*.
• Do a Fault Reset to clear the fault.
• If the drive detects this error, the drive will operate the motor as specified by the stopping method set in *H5-04 [Communication Error Stop Method]*.

Code	Name	Causes	Possible Solutions
CF	Control Fault	Motor parameters are set incorrectly	Correctly set the motor parameters and do Auto-Tuning again.
		The torque limit setting is too low.	Adjust *L7-01 to L7-04 [Torque Limit]*.
		The load inertia is too large.	• Adjust *C1-02, C1-04, C1-06, and C1-08 [Deceleration Times]*. • Set the frequency reference to the minimum output frequency, and stop the Run command when the drive stops deceleration.
		The drive is trying to ramp to stop a machine that cannot do ramp to stop or on a machine for which deceleration is not necessary.	Correctly set *b1-03 [Stopping Method Selection]*.
		The motor and drive are connected incorrectly.	Correct wiring errors.
		Line-to-line Resistance Tuning is not done.	Do Stationary Auto-Tuning for Line-to-Line Resistance.
		The drive received a Run command while the motor was coasting.	• Examine the sequence and input the Run command after the motor fully stops. • Set *b3-01 = 1 [Speed Search at Start Selection = Enabled]*.

Note:
• The drive detects this error if the torque reference is more than the torque limit for 3 seconds or longer while the drive ramps to stop.
• Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
CoF	Current Offset Fault	The drive starts operation while the induced voltage stays in the motor (during coasting to a stop or after fast deceleration).	• Make a sequence that does not restart operation when induced voltage stays in the motor. • Set *b3-01 = 1 [Speed Search at Start Selection = Enabled]*. • Use *Speed Search from Fmax or Fref [H1-xx = 61, 62]* to do a speed search through one of the external terminals. **Note:** When controlling the PM motor, External Speed Search commands 1 and 2 operate the same.
		A drive hardware problem occurred.	Replace the drive.

Note:
• The drive detects this error if the current offset value is more than the permitted setting range while the drive automatically adjusts the current offset.
• Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
CPF00	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• Do a Fault Reset to clear the fault.
• Fault trace is not available for these faults.

Code	Name	Causes	Possible Solutions
CPF01	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• Do a Fault Reset to clear the fault.
• Fault trace is not available for these faults.

Code	Name	Causes	Possible Solutions
CPF02	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF03	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF00, CPF01, CPF02, CPF03, CPF08, CPF11 - CPF14, CPF16 - CPF24, CPF38	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF06	Control Circuit Error (EEPROM memory Data Error)	The drive power supply was de-energized while a communication option entered a parameter Write command.	Set *A1-03 = 2220, 3330 [Initialize Parameters = 2-Wire Initialization, 3-Wire Initialization]* and initialize the drive.
		An EEPROM peripheral circuit error occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about how to replace the control board, contact Yaskawa or your nearest sales representative.
Note: • The drive detects this error if there is an error in the data written to the drive EEPROM. • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF08	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF11	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF14	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF16	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			

Code	Name	Causes	Possible Solutions
CPF24	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF38	Control Circuit Error	A drive hardware problem occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF06	Control Circuit Error (EEPROM memory Data Error)	The drive power supply was de-energized while a communication option entered a parameter Write command.	Set *A1-03 = 2220, 3330 [Initialize Parameters = 2-Wire Initialization, 3-Wire Initialization]* and initialize the drive.
		An EEPROM peripheral circuit error occurred.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about how to replace the control board, contact Yaskawa or your nearest sales representative.
Note: • The drive detects this error if there is an error in the data written to the drive EEPROM. • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
CPF25	Terminal Board not Connected	The terminal board is not correctly connected to the drive.	1. De-energize the drive. 2. Correctly connect the terminal board to the drive. 3. Re-energize the drive.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
dCE1	Communication Error1	A drive hardware problem occurred temporarily due to noise.	• Remove the cause of the noise. • If the fault stays, replace the control board or the drive.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
dCE2	Communication Error2	A drive hardware problem occurred temporarily due to noise.	• Remove the cause of the noise. • If the fault stays, replace the control board or the drive.
Note: • Do a Fault Reset to clear the fault. • Fault trace is not available for these faults.			
Code	**Name**	**Causes**	**Possible Solutions**
dv7	Polarity Judge Timeout	There is a disconnection in the motor coil winding.	Measure the motor line-to-line resistance and replace the motor if a coil is disconnected.
		The screws on the drive output terminals are loose.	Tighten the terminal screws to the correct tightening torque.
Note: • The drive detects this error if it cannot detect polarity in a pre-set length of time. • Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
EF1	External Fault (Terminal S1)	MFDI terminal S1 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S1.
		External Fault [H1-01 = 20 to 2B] is set to MFDI terminal S1, but the terminal is not in use.	Correctly set the MFDI.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
EF2	External Fault (Terminal S2)	MFDI terminal S2 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S2.

Code	Name	Causes	Possible Solutions
		External Fault [H1-02 = 20 to 2B] is set to MFDI terminal S2, but the terminal is not in use.	Correctly set the MFDI.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
EF3	External Fault (Terminal S3)	MFDI terminal S3 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S3.
		External Fault [H1-03 = 20 to 2B] is set to MFDI terminal S3, but the terminal is not in use.	Correctly set the MFDI.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
EF4	External Fault (Terminal S4)	MFDI terminal S4 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S4.
		External Fault [H1-04 = 20 to 2B] is set to MFDI terminal S4, but the terminal is not in use.	Correctly set the MFDI.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
EF5	External Fault (Terminal S5)	MFDI terminal S5 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S5.
		External Fault [H1-05 = 20 to 2B] is set to MFDI terminal S5, but the terminal is not in use.	Correctly set the MFDI.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
EF6	External Fault (Terminal S6)	MFDI terminal S6 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S6.
		External Fault [H1-06 = 20 to 2B] is set to MFDI terminal S6, but the terminal is not in use.	Correctly set the MFDI.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
EF7	External Fault (Terminal S7)	MFDI terminal S7 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S7.
		External Fault [H1-07 = 20 to 2B] is set to MFDI terminal S7, but the terminal is not in use.	Correctly set the MFDI.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
Err	EEPROM Write Error	There was a problem with the EEPROM hardware.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. Contact Yaskawa or your nearest sales representative to replace the board.
		Electrical interference corrupted the data while it was writing to the EEPROM of the drive.	• Push ENTER Key. • Set the parameters again.
Note: Do a Fault Reset to clear the fault.			
Code	**Name**	**Causes**	**Possible Solutions**
FRL	Speed Reference Missing	No speed reference is selected at start.	Select speed reference before start.

Code	Name	Causes	Possible Solutions
GF	Ground Fault	Overheating caused damage to the motor or the motor insulation is not satisfactory.	Measure the motor insulation resistance, and replace the motor if there is electrical conduction or unserviceable insulation.
		The motor main circuit cable is contacting ground to make a short circuit.	• Examine the motor main circuit cable for damage, and repair short circuits. • Measure the resistance between the motor main circuit cable and the ground terminal. If there is electrical conduction, replace the cable.
		An increase in the stray capacitance of the cable and the ground terminal caused an increase in the leakage current.	• If the wiring length of the cable is more than 100 m, decrease the carrier frequency. • Decrease the stray capacitance.
		There was a problem with the drive hardware.	Replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• The drive detects this fault if a current short to ground was more than 50% of rated current on the output side of the drive.
• Do a Fault Reset to clear the fault.
• *L5-08 [Fault Reset Enable Select Grp2]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
LF	Output Phase Loss	The motor main circuit cable is disconnected.	Connect motor main circuit cable wiring. Correct wiring errors in the main circuit drive input power.
		There is a disconnection in the motor coil winding.	If a coil is disconnected, measure the motor Line-to-Line Resistance and replace the motor.
		The screws on the drive output terminals are loose.	Tighten the terminal screws to the correct tightening torque.
		The rated output current of the motor is less than 5% of the drive rated current.	Examine the drive capacity or the motor output to be applied.
		You are trying to use a single-phase motor.	The drive cannot operate a single-phase motor.
		The output transistor in the drive is damaged.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• The drive detects this fault if phase loss occurs on the output side of the drive.
• Do a Fault Reset to clear the fault.
• Set *L8-07 [Output Phase Loss Protection Sel]* to enable and disable *LF* detection.

Code	Name	Causes	Possible Solutions
LF2	Output Current Imbalance	Phase loss occurred in the wiring on the output side of the drive.	Examine for wiring errors or disconnected wires on the output side of the drive, and repair problems.
		The output terminal screws of the drive are loose.	Tighten the terminal screws to the correct tightening torque.
		There is not balance between the three phases of the PM motor impedance.	• Measure the Line-to-Line Resistance for each motor phase and make sure that resistance is equal in the three phases, and that all wires are connected correctly. • Replace the motor.
		The drive output circuit is broken.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• The drive detects this fault if there is not balance between the three phases of the output current from the PM motor.
• Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
oC	Overcurrent	The load is too heavy.	• Measure the current flowing into the motor. • Replace the drive with a larger capacity model if the current value is more than the drive rated current. • Decrease the load or replace with a larger drive to prevent sudden changes in the current level.
		Overheating caused damage to the motor or the motor insulation is not satisfactory.	Measure the motor insulation resistance, and replace the motor if there is electrical conduction or unserviceable insulation.
		The motor main circuit cable is contacting ground to make a short circuit.	• Examine the motor main circuit cable for damage, and repair short circuits. • Measure the resistance between the motor main circuit cable and the ground terminal. If there is electrical conduction, replace the cable.
		A short circuit or ground fault on the drive output side caused damage to the output transistor of the drive.	• Make sure that there is not a short circuit in terminal B1 and terminals U/T1, V/T2, and W/T3. Make sure that there is not a short circuit in terminals - and terminals U/T1, V/T2, and W/T3. • If there is a short circuit, contact Yaskawa or your nearest sales representative.

Code	Name	Causes	Possible Solutions
		The acceleration time is too short.	• Calculate the torque necessary during acceleration related to the load inertia and the specified acceleration time. • Increase the values set in *C1-01, C1-03, C1-05, or C1-07 [Acceleration Times]* to get the necessary torque. • Increase the values set in *C2-01 to C2-04 [S-Curve Characteristics]* to get the necessary torque. • Replace the drive with a larger capacity model.
		The drive is trying to operate a specialized motor or a motor that is larger than the maximum applicable motor output of the drive.	• Examine the motor nameplate, the motor, and the drive to make sure that the drive rated current is larger than the motor rated current. • Replace the drive with a larger capacity model.
		A magnetic contactor was switched at the output.	Set the operation sequence to not turn ON or OFF the magnetic contactor while the drive is outputting voltage.
		The V/f pattern settings are incorrect.	• Examine the ratios between the V/f pattern frequency and voltage. Decrease the voltage if it is too high compared to the frequency. • Adjust *E1-04 to E1-10 [V/f Pattern Parameters]*. For motor 2, adjust *E3-04 to E3-10*.
		The torque compensation gain is too large.	Decrease the value set in *C4-01 [Torque Compensation Gain]* to make sure that the motor does not stall.
		Electrical interference caused a problem.	Examine the control circuit lines, main circuit lines, and ground wiring, and decrease the effects of electrical interference.
		The control method is set incorrectly for the motor.	Set *A1-02 [Control Method Selection]* correctly.
		The motor main circuit cable is too long.	• Replace the drive with a larger capacity model. • Decrease *C6-02 [Carrier Frequency]*. Or set *C6-02 = B*.

Note:
• This fault occurs if the drive sensors detect a drive output current more than the specified overcurrent detection level.

Code	Name	Causes	Possible Solutions
oH	Heatsink Overheat	The ambient temperature is high and the heatsink temperature of the drive is more than the value set in *L8-02 [Overheat Alarm Level]*.	• Measure the ambient temperature. • Increase the airflow in the control panel. • Install a cooling device (cooling fan or air conditioner) to lower the ambient temperature. • Remove objects near the drive that are producing too much heat.
		The load is too heavy.	• Measure the output current. • Decrease the load. • Decrease the value set in *C6-02 [Carrier Frequency Selection]*.
		The internal cooling fan of the drive stopped.	1. Use the procedures in this manual to replace the cooling fan. 2. Set *o4-03 = 0 [Fan Operation Time Setting = 0 h]*.

Note:
• The drive detects this fault if the heatsink temperature of the drive is more than the value set in *L8-02*.
• Do a Fault Reset to clear the fault.
• If the drive detects this fault, it will operate the motor as specified by the Stopping Method set in *L8-03 [Overheat Pre-Alarm Selection]*.

Code	Name	Causes	Possible Solutions
oH1	Heatsink Overheat	The ambient temperature is high and the heatsink temperature of the drive is more than the *oH1* detection level.	• Measure the ambient temperature. • Increase the airflow in the control panel. • Install a cooling device (cooling fan or air conditioner) to lower the ambient temperature. • Remove objects near the drive that are producing too much heat.
		The load is too heavy.	• Measure the output current. • Decrease the load. • Decrease the value set in *C6-02 [Carrier Frequency Selection]*.

Note:
• The drive detects this fault if the heatsink temperature of the drive is more than the *oH1* detection level. *o2-04 [Drive Model (KVA) Selection]* determines the *oH1* detection level.
• Do a Fault Reset to clear the fault.
• *L5-08 [Fault Reset Enable Select Grp2]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
oL1	Motor Overload	The load is too large.	Decrease the load. **Note:** Reset *oL1* when *U4-16 [Motor oL1 Level]* < 100.
		The acceleration/deceleration times or cycle times are too short.	• Examine the acceleration/deceleration times and the motor start/stop frequencies (cycle times). • Increase the values set in *C1-01 to C1-08 [Acceleration/Deceleration Times]*.

Code	Name	Causes	Possible Solutions
		Overload occurred while running at low speed.	• Decrease the load when running at low speed. • Increase the motor speed. • If the motor is run frequently at low speeds, replace the motor with a larger motor or use a drive-dedicated motor. **Note:** For general-purpose motors, overload can occur while running at low speed when operating at below the rated current.
		L1-01 [Motor Overload (oL1) Protection] is set incorrectly.	Set *L1-01* in as specified by the motor qualities for a drive-dedicated motor.
		The V/f pattern does not fit the motor qualities.	• Examine the ratios between the V/f pattern frequency and voltage. Decrease the voltage if it is too high compared to the frequency. • Adjust *E1-04 to E1-10 [V/f Pattern Parameters]*. For motor 2, adjust *E3-04 to E3-10*. Decrease the values set in *E1-08 [Mid Point A Voltage]* and *E1-10 [Minimum Output Voltage]*. **Note:** If the values set in *E1-08* and *E1-10* are too low, the overload tolerance will decrease at low speeds.
		E1-06 [Base Frequency] is set incorrectly.	Set *E1-06* to the rated frequency shown on the motor nameplate.
		One drive is operating more than one motor.	Set *L1-01 = 0 [Motor Overload (oL1) Protection = Disabled]*, connect thermal overload relay to each motor to prevent damage to the motor.
		The electronic thermal protector qualities and the motor overload properties do not match.	• Examine the motor qualities and set *L1-01 [Motor Overload (oL1) Protection]* correctly. • Connect a thermal overload relay to the motor.
		The electronic thermal protector is operating at an incorrect level.	Set *E2-01 [Motor Rated Current (FLA)]* correctly to the value specified by the motor nameplate.
		Phase loss in the input power supply is causing the output current to change.	Make sure that there is no phase loss, and repair problems.
		The motor main circuit cable is too long.	• Replace the drive with a larger capacity model. • Decrease *C6-02 [Carrier Frequency]*. Or set *C6-02 = B*.

Note:
• The drive detects this fault if the electronic thermal protector of the drive started the motor overload protection.
• Do a Fault Reset to clear the fault.
• *L5-07 [Fault Reset Enable Select Grp1]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
oL2	Drive Overload	The load is too large.	Decrease the load.
		The acceleration/deceleration times or cycle times are too short.	• Examine the acceleration/deceleration times and the motor start/stop frequencies (cycle times). • Increase the values set in *C1-01 to C1-08 [Acceleration/ Deceleration Times]*.
		The V/f pattern does not fit the motor qualities.	• Examine the ratios between the V/f pattern frequency and voltage. Decrease the voltage if it is too high compared to the frequency. • Adjust *E1-04 to E1-10 [V/f Pattern Parameters]*. Decrease the values set in *E1-08 [Mid Point A Voltage]* and *E1-10 [Minimum Output Voltage]*. For motor 2, adjust *E3-04 to E3-10*. **Note:** If the values set in *E1-08* and *E1-10* are too low, the overload tolerance will decrease at low speeds.
		The drive capacity is too small.	Replace the drive with a larger capacity model.
		Overload occurred while running at low speed.	• Decrease the load when running at low speed. • Replace the drive with a larger capacity model. • Decrease the value set in *C6-02 [Carrier Frequency Selection]*.
		The torque compensation gain is too large.	Decrease the value set in *C4-01 [Torque Compensation Gain]* to make sure that the motor does not stall.
		Phase loss in the input power supply is causing the output current to change.	• Correct errors with the wiring for main circuit drive input power. • Make sure that there is no phase loss, and repair problems.

Note:
• The drive detects this fault if the electronic thermal protector of the drive started the drive overload protection.
• Do a Fault Reset to clear the fault.
• *L5-07 [Fault Reset Enable Select Grp1]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
oL3	Overtorque Detection 1	A fault occurred on the machine. Example: The machine is locked.	Examine the machine and remove the cause of the fault.
		The parameters are incorrect for the load.	Adjust *L6-02 [Torque Detection Level 1]* and *L6-03 [Torque Detection Time 1]* settings.

Note:
• The drive detects this fault if the drive output current is more than the level set in *L6-02* for longer than *L6-03*.
• Do a Fault Reset to clear the fault.
• If the drive detects this fault, it will operate the motor as specified by the Stopping Method set in *L6-01 [Torque Detection Selection 1]*.
• *L5-07 [Fault Reset Enable Select Grp1]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
oL4	Overtorque Detection 2	A fault occurred on the machine. Example: The machine is locked.	Examine the machine and remove the cause of the fault.
		The parameters are incorrect for the load.	Adjust *L6-05 [Torque Detection Level 2]* and *L6-06 [Torque Detection Time 2]* settings.

Note:
• The drive detects this fault if the drive output current is more than the level set in *L6-05* for longer than *L6-06*.
• Do a Fault Reset to clear the fault.
• If the drive detects this fault, it will operate the motor as specified by the Stopping Method set in *L6-04 [Torque Detection Selection 2]*.
• *L5-07 [Fault Reset Enable Select Grp1]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
oL5	Mechanical Weakening Detection 1	The drive detected overtorque as specified by the conditions for mechanical weakening detection set in *L6-08 [Mechanical Fatigue Detect Select]*.	Do a deterioration diagnostic test on the machine side.

Note:
• Do a Fault Reset to clear the fault.
• If the drive detects this fault, it will operate the motor as specified by the Stopping Method set in *L6-08*.

Code	Name	Causes	Possible Solutions
oPr	Keypad Connection Fault	The keypad is not securely connected to the connector on the drive.	Examine the connection between the keypad and the drive.
		The connection cable between the drive and the keypad is disconnected.	• Remove the keypad and then reconnect it. • Replace the cable if damaged.

Note:
• The drive detects this fault if these conditions are correct:
– *o2-06 = 1 [Keypad Disconnect Detection = Enabled]*.
– *b1-02 = 0 [Run Command Selection 1 = Keypad]*, or the drive is operating in LOCAL Mode with the keypad.
• Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
ov	Overvoltage	The deceleration time is too short and too much regenerative energy is flowing back into the drive.	• Increase the values set in *C1-02, C1-04, C1-06, or C1-08 [Deceleration Times]*. • Connect a dynamic braking option to the drive. • Perform Deceleration Rate Tuning.
		The acceleration time is too short.	• Make sure that sudden drive acceleration does not cause the fault. • Increase the values set in *C1-01, C1-03, C1-05, or C1-07 [Acceleration Times]*. • Increase the value set in *C2-02 [S-Curve Time @ End of Accel]*. • Set *L3-11 = 1 [Overvoltage Suppression Select = Enabled]*.
		The braking load is too large.	Connect a dynamic braking option to the drive.
		There are surge voltages in the input power supply	Connect a DC reactor to the drive. **Note:** If you turn the phase advancing capacitors ON and OFF and use thyristor converters in the same power supply system, there can be surge voltages that irregularly increase the input voltage.
		The drive output cable or motor is shorted to ground (the current short to ground is charging the main circuit capacitor of the drive through the power supply).	1. Examine the motor main circuit cable, terminals, and motor terminal box, and then remove ground faults. 2. Re-energize the drive.
		The power supply voltage is too high.	Decrease the power supply voltage to match the drive rated voltage.
		The braking resistor or braking resistor unit wiring is incorrect.	Correct wiring errors in the connection to the braking resistor or braking resistor unit.
		Electrical interference caused a drive malfunction.	• Examine the control circuit lines, main circuit lines, and ground wiring, and decrease the effects of electrical interference. • Make sure that a magnetic contactor is not the source of the electrical interference, then use a Surge Protective Device if necessary.

Code	Name	Causes	Possible Solutions
		The Short Circuit Braking function used in OLV/PM control method.	Connect a braking resistor to the drive.
		There is motor hunting.	• Set *n1-01 = 1 [Hunting Prevention Selection = Enabled]* and adjust *n1-02 [Hunting Prevention Gain Setting]*. • Adjust *n2-02 [Automatic Freq Regulator Time 1]* and *n2-03 [Automatic Freq Regulator Time 2]*.

Note:
• The drive detects this error if the DC bus voltage is more than the *ov* detection level while the drive is running.
• The *ov* detection level is approximately 410 V with 200 V class drives. The detection level is approximately 820 V for 400 V class drives.
• Do a Fault Reset to clear the fault.
• *L5-08 [Fault Reset Enable Select Grp2]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
PF	Input Phase Loss	There is a phase loss in the drive input power.	Correct errors with the wiring for main circuit drive input power.
		There is loose wiring in the drive input power terminals.	Tighten the terminal screws to the correct tightening torque.
		The drive input power voltage is changing too much.	• Examine the input power for problems. • Make the drive input power stable. • If the input power supply is good, examine the magnetic contactor on the main circuit side for problems.
		There is unsatisfactory balance between voltage phases.	• Examine the input power for problems. • Make the drive input power stable. • Set *L8-05 = 0 [Input Phase Loss Protection Sel = Disabled]*.
		The main circuit capacitors have become unserviceable.	• Examine the capacitor maintenance time in monitor *U4-05 [CapacitorMaintenance]*. If *U4-05* is more than 90%, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative. • If drive input power is correct and the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• The drive detects this error if the DC bus voltage changes irregularly without regeneration.
• Do a Fault Reset to clear the fault.
• Use *L8-05* to enable and disable *PF* detection.

Code	Name	Causes	Possible Solutions
PGo	Encoder (PG) Feedback Loss	The holding brake is stopping the motor.	Release the holding brake.

Note:
• Do a Fault Reset to clear the fault.
• If the drive detects this error, it will operate the motor as specified by the Stopping Method set in *F1-02 [PG Open Circuit Detection Select]*.

Code	Name	Causes	Possible Solutions
rF	Braking Resistor Fault	The resistance of the dynamic braking option connected to the drive is too low.	Use a dynamic braking option that fits the model and duty rating of the drive.
		A regenerative converter or regenerative unit is connected to the drive.	Set *L8-55 = 0 [Internal DB TransistorProtection = Disable]*.

Note:
Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
rH	Braking Resistor Overheat	The deceleration time is too short and excessive regenerative energy is flowing back into the drive.	• Check the load level, deceleration time, and speed. • Decrease the load. • Increase the values set in *C1-02, C1-04, C1-06, or C1-08 [Deceleration Times]*. • Use a dynamic braking option that lets you use more power.
		The duty cycle is too high.	Examine the duty cycle. **Note:** When *L8-01 = 1 [3% ERF DB Resistor Protection = Enabled]*, the maximum braking duty cycle is 3%.
		The braking load is too heavy.	• Calculate the braking load and braking power again, and decrease the braking load. • Use a braking resistor that improves braking power.
		The braking resistor is not sufficient.	Use the braking resistor specifications to select a sufficient braking resistor.

Note:
• The drive detects this error if the braking resistor overheat protective function is active.
• The magnitude of the braking load causes the braking resistor overheat alarm, NOT the surface temperature. If the duty cycle is higher than the braking resistor rating, the drive will show the alarm.
• Do a Fault Reset to clear the fault.
• Parameter *L8-01* enables and disables *rH* detection.

Code	Name	Causes	Possible Solutions
rr	Dynamic Braking Transistor Fault	The drive control circuit is damaged.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
		There is a malfunction in the internal braking transistor of the drive.	

Note:
Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
SC	Short Circuit/IGBT Failure	Overheating caused damage to the motor or the motor insulation is not satisfactory.	Measure the motor insulation resistance, and replace the motor if there is electrical conduction or unserviceable insulation.
		The motor main circuit cable is contacting ground to make a short circuit.	• Examine the motor main circuit cable for damage, and repair short circuits. • Measure the resistance between the motor main circuit cable and the ground terminal. If there is electrical conduction, replace the cable.
		A short circuit or ground fault on the drive output side caused damage to the output transistor of the drive.	• Make sure that there is not a short circuit in terminal B1 and terminals U/T1, V/T2, and W/T3. Make sure that there is not a short circuit in terminals - and terminals U/T1, V/T2, and W/T3. • If there is a short circuit, contact Yaskawa or your nearest sales representative.
		When *A1-02 = 5, 6 [Control Method Selection = OLV/PM or AOLV/PM]*, the output current is more than the value set in *L8-27 [Overcurrent Detection Gain]*.	Set *L8-27* correctly.

Note:
• The drive detects this error if there is a short circuit or ground fault on the drive output side, or an IGBT failure.
• Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
SCF	Safety Circuit Fault	The safety circuit is broken.	Replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
Do a Fault Reset to clear the fault.

Code	Name	Causes	Possible Solutions
TiM	Keypad Time Not Set	There is a battery in the keypad, but the date and time are not set.	Use the keypad to set the date and time.

Note:
• Do a Fault Reset to clear the fault.
• Parameter *o4-24 [bAT Detection Selection]* enables and disables *TiM* detection.

Code	Name	Causes	Possible Solutions
UL3	Undertorque Detection 1	A fault occurred on the machine. Example: There is a broken pulley belt.	Examine the machine and remove the cause of the fault.
		The parameters are incorrect for the load.	Adjust *L6-02 [Torque Detection Level 1]* and *L6-03 [Torque Detection Time 1]* settings.

Note:
• The drive detects this error if the drive output current is less than the level set in *L6-02* for longer than *L6-03*.
• Do a Fault Reset to clear the fault.
• If the drive detects this error, it will operate the motor as specified by the Stopping Method set in *L6-01 [Torque Detection Selection 1]*.

Code	Name	Causes	Possible Solutions
UL4	Undertorque Detection 2	A fault occurred on the machine. Example: There is a broken pulley belt.	Examine the machine and remove the cause of the fault.
		The parameters are incorrect for the load.	Adjust *L6-05 [Torque Detection Level 2]* and *L6-06 [Torque Detection Time 2]* settings.

Note:
• The drive detects this error if the drive output current is less than the level set in *L6-05* for longer than *L6-06*.
• Do a Fault Reset to clear the fault.
• If the drive detects this error, it will operate the motor as specified by the Stopping Method set in *L6-04 [Torque Detection Selection 2]*.

Code	Name	Causes	Possible Solutions
Uv1	DC Bus Undervoltage	There is a phase loss in the drive input power.	Correct errors with the wiring for main circuit drive input power.
		There is loose wiring in the drive input power terminals.	Tighten the terminal screws to the correct tightening torque.
		The drive input power voltage is changing too much.	• Examine the input power for problems. • Make the drive input power stable. • If the input power supply is good, examine the magnetic contactor on the main circuit side for problems.
		There was a loss of power.	Use a better power supply.

Code	Name	Causes	Possible Solutions
		The main circuit capacitors have become unserviceable.	Examine the capacitor maintenance time in monitor *U4-05 [CapacitorMaintenance]*. If *U4-05* is more than 90%, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
		The relay or contactor on the soft-charge bypass relay is damaged.	*U4-06 [PreChargeRelayMainte]* shows the performance life of the soft-charge bypass relay. If *U4-06* is more than 90%, replace the board or the drive. For information about replacing the board, contact Yaskawa or your nearest sales representative.

Note:
- The drive detects this error if the DC bus voltage decreases below the level set in *L2-05 [Undervoltage Detection Lvl (Uv1)]* while the drive is running.
- The *Uv1* detection level is approximately 190 V for a 200 V class drives. The detection level is approximately 380 V for 400 V class drives. The detection level is approximately 350 V when *E1-01 [Input AC Supply Voltage]* < 400.
- Do a Fault Reset to clear the fault.
- Fault trace is not available for this fault.
- *L5-08 [Fault Reset Enable Select Grp2]* disables the Auto Restart function.

Code	Name	Causes	Possible Solutions
Uv2	Control Power Undervoltage	The value set in *L2-02 [Power Loss Ride Through Time]* increased and the momentary power loss recovery unit is not connected to the drive.	Connect the momentary power loss recovery unit to the drive.
		There was a problem with the drive hardware.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
- The drive detects this error if the control power supply voltage decreases.
- Do a Fault Reset to clear the fault.
- Fault trace is not available for this fault.

Code	Name	Causes	Possible Solutions
Uv3	Soft Charge Answerback Fault	The relay or contactor on the soft-charge bypass relay is damaged.	• Re-energize the drive. • If the fault stays, replace the control board or the drive. • Check monitor *U4-06 [PreChargeRelayMainte]* shows the performance life of the soft-charge bypass relay. If *U4-06* is more than 90%, replace the board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
- Do a Fault Reset to clear the fault.
- Fault trace is not available for these faults.

6.5 Minor Faults/Alarms

This section gives information about the causes and possible solutions when a minor fault or alarm occurs. Use the information in this table to remove the cause of the minor fault or alarm.

Code	Name	Causes	Possible Solutions
bAT	Keypad Battery Low Voltage	The keypad battery voltage is low.	Replace the keypad battery.
Note: • If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate. • *Parameter o4-24 [bAT Detection Selection]* enables and disables *bAT* detection.			
Code	**Name**	**Causes**	**Possible Solutions**
bb	Baseblock	An external baseblock command was entered through MFDI terminal S1 to S7, and the drive output stopped as shown by an external baseblock command.	Examine the external sequence and timing of the baseblock command input.
Note: The drive will not output a minor fault signal for this alarm.			
Code	**Name**	**Causes**	**Possible Solutions**
bCE	Bluetooth Communication Error	The smartphone or tablet with DriveWizard Mobile installed is too far from the keypad.	Use the smartphone or tablet 10 m (32.8 ft) or nearer to the keypad. **Note:** *bCE* can occur when the smartphone or tablet is 10 m or nearer to the keypad depending on the specifications of the smartphone or tablet.
		Radio waves from a different device are causing interference with the communication between the smartphone or tablet and keypad.	Make sure that no device around the keypad uses the same radio bandwidth (2400 MHz to 2480 MHz), and prevent radio interference.
Note: • The drive detects this error when you operate the drive with a smartphone or tablet with a Bluetooth LCD Keypad. • If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate. • Parameter *o2-27 [bCE Detection selection]* enables and disables *bCE* detection.			
Code	**Name**	**Causes**	**Possible Solutions**
boL	Braking Transistor Overload	The duty cycle of the braking transistor is high (the regeneration power or repetition frequency is high).	• Install a regenerative converter. • Increase the deceleration time.
		You enabled the protective function for the braking transistor when you have a regenerative converter.	Set *L8-55 = 0 [Internal DB TransistorProtection Selection = Disable]*.
		The braking transistor in the drive is broken.	Replace the drive.
Note: If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.			
Code	**Name**	**Causes**	**Possible Solutions**
bUS	Option Communication Error	The communications cable wiring is incorrect.	Correct wiring errors.
		There is a short-circuit in the communications cable or the communications cable is not connected.	• Repair short circuits and connect cables. • Replace the defective communications cable.
		Electrical interference caused a communication data error.	• Examine the control circuit lines, main circuit lines, and ground wiring, and decrease the effects of electrical interference. • Make sure that a magnetic contactor is not the source of the electrical interference, then use a Surge Protective Device if necessary. • Use only the recommended cables or other shielded line. Ground the shield on the controller side or the drive input power side. • Separate the communication wiring from drive power lines, and install a noise filter to the input side of the power supply for communication. • Decrease the effects of electrical interference from the controller.
		The option card is incorrectly installed to the drive.	Correctly install the option card to the drive.
		The option card is damaged.	If the alarm continues and the wiring is correct, replace the option card.
Note: • The drive detects this error if the Run command or frequency reference is assigned to the option card. • If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate. • If the drive detects this error, it will operate the motor as specified by the stopping method set in *F6-01 [Communication Error Selection]*.			
Code	**Name**	**Causes**	**Possible Solutions**
bUSy	Busy	You set the drive to use MEMOBUS/Modbus communications to change parameters, but you used the keypad to change parameters.	Use MEMOBUS/Modbus communications to enter the enter command, then use the keypad to change the parameter.
		You tried to change a parameter while the drive was changing setting.	Wait until the process is complete.

Code	Name	Causes	Possible Solutions
CALL	Serial Comm Transmission Error	The communications cable wiring is incorrect.	Correct wiring errors.
		There is a short-circuit in the communications cable or the communications cable is not connected.	• Repair the short-circuited or disconnected portion of the cable. • Replace the defective communications cable.
		A programming error occurred on the controller side.	Examine communications at start-up and correct programming errors.
		The communications circuitry is damaged.	• Do a self-diagnostics check. • If the problem continues, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
		The termination resistor setting for MEMOBUS/ Modbus communications is incorrect.	On the last drive in a MEMOBUS/Modbus network, set DIP switch S2 to the ON position to enable the termination resistor.

Note:
• The drive detects this error if it does not correctly receive control data from the controller when energizing the drive.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

Code	Name	Causes	Possible Solutions
CE	Modbus Communication Error	The communications cable wiring is incorrect.	Correct wiring errors.
		There is a short-circuit in the communications cable or the communications cable is not connected.	• Repair short circuits and connect cables. • Replace the defective communications cable.
		Electrical interference caused a communication data error.	• Examine the control circuit lines, main circuit lines, and ground wiring, and decrease the effects of electrical interference. • Make sure that a magnetic contactor is not the source of the electrical interference, then use a Surge Protective Device if necessary. • Use only the recommended cables or other shielded line. Ground the shield on the controller side or the drive input power side. • Separate the communication wiring from drive power lines, and install a noise filter to the input side of the power supply for communication. • Decrease the effects of electrical interference from the controller.
		The communication protocol is not compatible.	• Examine the values set in *H5-xx*. • Examine the settings on the controller side and correct the difference in communication conditions.
		The value set in *H5-09 [CE Detection Time]* is too small for the communications cycle.	• Change the controller software settings. • Increase the value set in *H5-09*.
		The controller software or hardware is causing a communication problem.	Examine the controller and remove the cause of the problem.

Note:
• The drive detects this error if it does not correctly receive control data for the *CE* detection time set to *H5-09*.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• If the drive detects this error, it will operate the motor as specified by the stopping method set in *H5-04 [Communication Error Stop Method]*.

Code	Name	Causes	Possible Solutions
CrST	Cannot Reset	The drive received a fault reset command when a Run command was active.	Turn off the Run command then de-energize and re-energize the drive.

Code	Name	Causes	Possible Solutions
EF	FWD/REV Run Command Input Error	A forward command and a reverse command were input at the same time for longer than 0.5 s.	Examine the forward and reverse command sequence and correct the problem.

Note:
• If the drive detects *EF*, the motor will ramp to stop.
• If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will be ON.

Code	Name	Causes	Possible Solutions
EF0	Option Card External Fault	The communication option card received an external fault from the controller.	1. Find the device that caused the external fault and remove the caus. 2. Clear the external fault input from the controller.
		A programming error occurred on the controller side.	Examine the operation of the controller program.

Note:
• The drive detects this error if the alarm function on the external device side is operating.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• If the drive detects this error, it will operate the motor as specified by the stopping method set in *F6-03 [Comm External Fault (EF0) Select]*.

Code	Name	Causes	Possible Solutions
EF1	External Fault (Terminal S1)	MFDI terminal S1 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S1.

Code	Name	Causes	Possible Solutions
		External Fault [H1-01 = 2C to 2F] is set to MFDI terminal S1, but the terminal is not in use.	Correctly set the MFDI.
Note: If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate.			
Code	**Name**	**Causes**	**Possible Solutions**
EF2	External Fault (Terminal S2)	MFDI terminal S2 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S2.
		External Fault [H1-02 = 2C to 2F] is set to MFDI terminal S2, but the terminal is not in use.	Correctly set the MFDI.
Note: If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate.			
Code	**Name**	**Causes**	**Possible Solutions**
EF3	External Fault (Terminal S3)	MFDI terminal S3 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S3.
		External Fault [H1-03 = 2C to 2F] is set to MFDI terminal S3, but the terminal is not in use.	Correctly set the MFDI.
Note: If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate.			
Code	**Name**	**Causes**	**Possible Solutions**
EF4	External Fault (Terminal S4)	MFDI terminal S4 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S4.
		External Fault [H1-04 = 2C to 2F] is set to MFDI terminal S4, but the terminal is not in use.	Correctly set the MFDI.
Note: If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate.			
Code	**Name**	**Causes**	**Possible Solutions**
EF5	External Fault (Terminal S5)	MFDI terminal S5 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S5.
		External Fault [H1-05 = 2C to 2F] is set to MFDI terminal S5, but the terminal is not in use.	Correctly set the MFDI.
Note: If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate.			
Code	**Name**	**Causes**	**Possible Solutions**
EF6	External Fault (Terminal S6)	MFDI terminal S6 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S6.
		External Fault [H1-06 = 2C to 2F] is set to MFDI terminal S6, but the terminal is not in use.	Correctly set the MFDI.
Note: If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate.			
Code	**Name**	**Causes**	**Possible Solutions**
EF7	External Fault (Terminal S7)	MFDI terminal S7 caused an external fault through an external device.	1. Find the device that caused the external fault and remove the cause. 2. Clear the external fault input in the MFDI.
		The wiring is incorrect.	Correctly connect the signal line to MFDI terminal S7.
		External Fault [H1-07 = 2C to 2F] is set to MFDI terminal S7, but the terminal is not in use.	Correctly set the MFDI.
Note: If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will activate.			

Code	Name	Causes	Possible Solutions
EP24v	External Power 24V Supply	The voltage of the main circuit power supply decreased, and the 24 V power supply is supplying power to the drive.	• Examine the main circuit power supply. • Turn ON the main circuit power supply to run the drive.

Note:
• Set *o2-26 [Ext. Power 24V Supply Display]* to enable or disable *EP24v* detection.
• The drive will not output an alarm signal for this alarm.

Code	Name	Causes	Possible Solutions
HCA	High Current Alarm	The load is too heavy.	• Decrease the load for applications with repetitive starts and stops. • Replace the drive with a larger capacity model.
		The acceleration time is too short.	• Calculate the torque necessary during acceleration related to the load inertia and the specified acceleration time. • Increase the values set in *C1-01, C1-03, C1-05, or C1-07 [Acceleration Times]* until you get the necessary torque. • Increase the values set in *C2-01 to C2-04 [S-Curve Characteristics]* until you get the necessary torque. • Replace the drive with a larger capacity model.
		The drive is trying to operate a specialized motor or a motor that is larger than the maximum applicable motor output of the drive.	• Examine the motor nameplate, the motor, and the drive to make sure that the drive rated current is larger than the motor rated current. • Replace the drive with a larger capacity model.
		The current level temporarily increased because of speed search after a momentary power loss or while trying to Auto Restart.	If speed search or Auto Restart cause an increase in current, the drive can temporarily show this alarm. The time that the drive shows the alarm is short. No more steps are necessary to clear the alarm.

Note:
• The drive detects this error if the drive output current is more than the overcurrent alarm level (150% of the rated current).
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

Code	Name	Causes	Possible Solutions
L24v	Loss of External Power 24 Supply	The voltage of the backup 24 V power supply has decreased. The main circuit power supply is operating correctly.	• Examine the external 24 V power supply for disconnected wires and wiring errors and repair the problems. • Examine the external 24 V power supply for problems.

Note:
• Set *o2-23 [External 24V Powerloss Detection]* to enable or disable *L24v* detection.
• The drive will not output an alarm signal for this alarm.

Code	Name	Causes	Possible Solutions
LoG	Log Com Error	There is not a micro SD in the keypad.	Put a micro SD card in the keypad.
		• The drive is connected to USB. • The number of log communication files is more than 1000. • The micro SD card does not have available memory space. • The line number data in a log communication file was changed. • A communication error between the keypad and drive occurred during a log communication.	Set *o5-01 = 0 [Log Start/Stop Selection = OFF]*.

Note:
If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 6A [MFDO Function Select = Data Logger Error]* will be ON.

Code	Name	Causes	Possible Solutions
LT-1	Cooling Fan Maintenance Time	The cooling fan is at 90% of its expected performance life.	1. Use the procedures in this manual to replace the cooling fan. 2. Set *o4-03 = 0 [Fan Operation Time Setting = 0 h]* to reset the cooling fan operation time.

Note:
When the estimated performance life is expired, the terminal assigned to *H2-01 to H2-03 = 2F [MFDO Function Select = Maintenance Notification]* will be ON.

Code	Name	Causes	Possible Solutions
LT-2	Capacitor Maintenance Time	The capacitors for the main circuit and control circuit are at 90% of expected performance life.	Replace the board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
When the estimated performance life is expired, the terminal assigned to *H2-01 to H2-03 = 2F [MFDO Function Select = Maintenance Notification]* will be ON.

Code	Name	Causes	Possible Solutions
LT-3	SoftChargeBypassRelay MainteTime	The soft charge bypass relay is at 90% of its expected performance life.	Replace the board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
When the estimated performance life is expired, the terminal assigned to *H2-01 to H2-03 = 2F [MFDO Function Select = Maintenance Notification]* will be ON.

Code	Name	Causes	Possible Solutions
LT-4	IGBT Maintenance Time (50%)	The IGBT is at 50% of its expected performance life.	Check the load, carrier frequency, and output frequency.

Note:
When the estimated performance life is expired, the terminal assigned to *H2-01 to H2-03 = 2F [MFDO Function Select = Maintenance Notification]* will be ON.

Code	Name	Causes	Possible Solutions
oH	Heatsink Overheat	The ambient temperature is high and the heatsink temperature is more than the *L8-02 [Overheat Alarm Level]*.	• Measure the ambient temperature. • Increase the airflow around the drive. • Install a cooling device (cooling fan or air conditioner) to lower the ambient temperature. • Remove objects near the drive that are producing too much heat.
		There is not sufficient airflow around the drive.	• Give the drive the correct installation space as shown in the manual. • Make sure that there is sufficient circulation around the control panel. • Examine the drive for dust or other unwanted materials that could clog the cooling fan. • Remove unwanted materials that prevent air circulation.
		The internal cooling fan or fans have stopped.	1. Use the procedures in this manual to replace the cooling fan. 2. Set *o4-03 = 0 [Fan Operation Time Setting = 0 h]*.

Note:
• The drive detects this error if the heatsink temperature of the drive is more than *L8-02*.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• If the drive detects this error, it will operate the motor as specified by the stopping method set in *L8-03 [Overheat Pre-Alarm Selection]*.

Code	Name	Causes	Possible Solutions
oH2	External Overheat (H1-XX=B)	An external device sent an *oH2*.	1. Find the external device that output the overheat alarm. 2. Remove the cause of the problem. 3. Clear the *Overheat Alarm (oH2) [H1-xx = B]* set to MFDI terminals S1 to S7.

Note:
If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

Code	Name	Causes	Possible Solutions
oH3	Motor Overheat (PTC Input)	The thermistor wiring that detects motor temperature is defective.	Correct wiring errors.
		A fault occurred on the machine. Example: The machine is locked.	Examine the machine and remove the cause of the fault
		The motor has overheated.	• Check the load level, acceleration/deceleration time, and motor start/stop frequency (cycle time). • Decrease the load. • Increase the values set in *C1-01 to C1-08 [Acceleration/ Deceleration Times]*. • Set *E2-01 [Motor Rated Current (FLA)]* correctly to the value specified by the motor nameplate. • Make sure that the motor cooling system is operating correctly, and repair or replace it if it is damaged. • Adjust *E1-04 to E1-10 [V/f Pattern Parameters]*. For motor 2, adjust *E3-04 to E3-10*. Decrease the values set in *E1-08 [Mid Point A Voltage]* and *E1-10 [Minimum Output Voltage]*. **Note:** If the values set in *E1-08* and *E1-10* are too low, the overload tolerance will decrease at low speeds.

Note:
• The drive detects this fault if the motor overheat signal input from analog input terminal A1 or A2 is more than the fault detection level.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• If the drive detects this error, it will operate the motor as specified by the stopping method set in *L1-03 [Motor Thermistor oH Alarm Select]*.

Code	Name	Causes	Possible Solutions
oL3	Overtorque 1	A fault occurred on the machine. Example: The machine is locked.	Examine the machine and remove the cause of the fault
		The parameters are incorrect for the load.	Adjust *L6-02 [Torque Detection Level 1]* and *L6-03 [Torque Detection Time 1]*.

Note:
• The drive detects this fault if the drive output current is more than the level set in *L6-02* for longer than *L6-03*.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• Set the conditions that trigger the minor fault using *L6-01 [Torque Detection Selection 1]*.

Code	Name	Causes	Possible Solutions
oL4	Overtorque 2	A fault occurred on the machine. Example: The machine is locked.	Examine the machine and remove the cause of the fault
		The parameters are incorrect for the load.	Adjust *L6-05 [Torque Detection Level 2]* and *L6-06 [Torque Detection Time 2]*.

Note:
• The drive detects this error if the drive output current is more than the level set in *L6-05* for longer than *L6-06*.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• Set the conditions that trigger the minor fault using *L6-04 [Torque Detection Selection 2]*.

Code	Name	Causes	Possible Solutions
oS	Overspeed	There is overshoot.	• Decrease *C5-01 [ASR Proportional Gain 1]* and increase *C5-02 [ASR Integral Time 1]*. • Adjust the pulse train gain with *H6-02 to H6-05 [Pulse Train Input Setting Parameters]*.
		There is an incorrect number of PG pulses set in the drive.	Set *H6-02 [Terminal RP Frequency Scaling]* to the pulse train frequency during 100% reference (maximum motor rotation speed).
		The *oS* detection level is set incorrectly.	Adjust *F1-08 [Overspeed Detection Level]* and *F1-09 [Overspeed Detection Delay Time]*.

Note:
• The drive detects this error if the motor speed is more than the value set in *F1-08* for longer than *F1-09*.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• If the drive detects this error, it will operate the motor as specified by the stopping method set in *F1-03 [Overspeed Detection Selection]*.

Code	Name	Causes	Possible Solutions
ov	Overvoltage	There are surge voltages in the input power supply.	Connect a DC reactor to the drive. **Note:** If you turn the phase advancing capacitors ON and OFF and use thyristor converters in the same power supply system, there can be surge voltages that irregularly increase the input voltage.
		The drive output cable or motor is shorted to ground (the current short to ground is charging the main circuit capacitor of the drive through the power supply).	1. Examine the motor main circuit cable, terminals, and motor terminal box, and then remove ground faults. 2. Re-energize the drive.
		The power supply voltage is too high.	Decrease the power supply voltage to match the drive rated voltage.
		Electrical interference caused a drive malfunction.	• Examine the control circuit lines, main circuit lines, and ground wiring, and decrease the effects of electrical interference. • Make sure that a magnetic contactor is not the source of the electrical interference, then use a Surge Protective Device if necessary. • Set *L5-01 ≠ 0 [Number of Auto-Restart Attempts ≠ 0 times]*.

Note:
• The drive detects this error if the DC bus voltage is more than the *ov* detection level when the Run command has not been input (while the drive is stopped).
• The *ov* detection level is approximately 410 V with 200 V class drives. The detection level is approximately 820 V for 400 V class drives.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

Code	Name	Causes	Possible Solutions
ovEr	Too Many Parameters Changed	You tried to change more than 150 parameters.	Make sure that parameters that do not have an effect on drive operation are at their default settings. **Note:** • You can change 150 parameters maximum. • If you change parameters that have dependencies, the drive can detect *ovEr* when the number of changed parameters is fewer than 150.

Code	Name	Causes	Possible Solutions
PASS	Modbus Communication Test	The MEMOBUS/Modbus communications test is complete.	The *PASS* display will turn off after communications test mode is cleared.

Code	Name	Causes	Possible Solutions
PF	Input Phase Loss	There is a phase loss in the drive input power.	Correct all wiring errors with the main circuit power supply.
		Loose wiring in the input power terminals.	Tighten the screws to the correct tightening torque.
		The drive input power voltage is changing too much.	• Examine the supply voltage for problems. • Make the drive input power stable.
		Unsatisfactory balance between voltage phases.	• Examine the supply voltage for problems. • Make the drive input power stable. • If the supply voltage is good, examine the magnetic contactor on the main circuit side for problems.

Code	Name	Causes	Possible Solutions
		The main circuit capacitors have become unserviceable.	• Examine the capacitor maintenance time in monitor *U4-05 [CapacitorMaintenance]*. • If *U4-05* is more than 90%, replace the capacitor. Contact Yaskawa or your nearest sales representative for more information.
			• Examine the supply voltage for problems. • Re-energize the drive. • If the alarm stays, replace the circuit board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• The drive detects this error if the DC bus voltage changes irregularly without regeneration.
• If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will be ON.
• Use *L8-05 [Input Phase Loss Protection Sel]* to enable and disable *PF* detection.

Code	Name	Causes	Possible Solutions
SE	Modbus Test Mode Error	MEMOBUS/Modbus communications self-diagnostics *[H1-xx = 67]* was done while the drive was running.	Stop the drive and do MEMOBUS/Modbus communications self-diagnostics.

Note:
If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

Code	Name	Causes	Possible Solutions
SToF	Safe Torque OFF Failure	One of the two terminals H1-HC and H2-HC received the Safe Disable input signal.	• Make sure that the Safe Disable signal is input from an external source to terminals H1-HC or H2-HC. • When the Safe Disable function is not in use, connect terminals H1-HC and H2-HC.
		The Safe Disable input signal is wired incorrectly.	
		There is internal damage to one Safe Disable channel.	Replace the board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
If the drive detects this error, the terminal assigned to *H2-01 to H2-03 = 10 [MFDO Function Select = Alarm]* will be ON.

Code	Name	Causes	Possible Solutions
TiM	Keypad Time Not Set	There is a battery in the keypad, but the date and time are not set.	Set the date and time with the keypad.

Note:
• Parameter *o4-24 [bAT Detection Selection]* enables and disables *TiM* detection.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

Code	Name	Causes	Possible Solutions
TrPC	IGBT Maintenance Time (90%)	The IGBT is at 90% of its expected performance life.	Replace the IGBT or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

Code	Name	Causes	Possible Solutions
UL3	Undertorque Detection 1	A fault occurred on the machine. Example: There is a broken pulley belt.	Examine the machine and remove the cause of the fault
		The parameters are incorrect for the load.	Adjust *L6-02 [Torque Detection Level 1]* and *L6-03 [Torque Detection Time 1]*.

Note:
• The drive detects this error if the drive output current is less than the level set in *L6-02* for longer than *L6-03*.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• Set the conditions that trigger the minor fault using *L6-01 [Torque Detection Selection 1]*.

Code	Name	Causes	Possible Solutions
UL4	Undertorque Detection 2	A fault occurred on the machine. Example: There is a broken pulley belt.	Examine the machine and remove the cause of the fault
		The parameters are incorrect for the load.	Adjust *L6-05 [Torque Detection Level 2]* and *L6-06 [Torque Detection Time 2]*.

Note:
• The drive detects this error if the drive output current is less than the level set in *L6-05* for longer than *L6-06*.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.
• Set the conditions that trigger the minor fault using *L6-04 [Torque Detection Selection 2]*.

Code	Name	Causes	Possible Solutions
Uv	DC Bus Undervoltage	The drive input power voltage is changing too much.	• Use a better power supply voltage to align with the drive rated voltage. • Make the drive input power stable. • If there is not a fault with the input power supply, examine the magnetic contactor on the main circuit side for faults.
		A phase loss occurred in the drive input power.	Correct errors with the wiring for main circuit drive input power.

Code	Name	Causes	Possible Solutions
		There is loose wiring in the drive input power terminals.	Examine for loose screws and tighten them as specified by the tightening torque values in the manual.
		There was a loss of power.	Use a better power supply.
		The main circuit capacitors have deteriorated.	Examine the capacitor maintenance time in monitor *U4-05 [CapacitorMaintenance]*. If *U4-05* is more than 90%, replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
		The drive input power transformer is too small and voltage drops when the power is switched on.	• Check for an alarm when a molded-case circuit breaker, Leakage Breaker (ELCB, GFCI, or RCM/RCD) (with overcurrent protective function), or magnetic contactor is ON. • Check the capacity of the drive power supply transformer.
		Air inside the drive is too hot.	Measure the ambient temperature of the drive.
		The Charge LED is broken.	Replace the board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.

Note:
• The drive detects this error if one of these conditions is correct when the Run command has not been input (while the drive is stopped).
– The DC bus voltage < *L2-05 [Undervoltage Detection Lvl (Uv1)]*.
– The Contactor that prevents inrush current in the drive was opened.
– There is low voltage in the control drive input power.
• If the drive detects this error, the terminal set to *H2-01 to H2-03 = 10 [MFDO Function Selection = Alarm]* will activate.

6.6 Parameter Setting Errors

Parameter setting errors occur when multiple parameter settings do not agree, or when parameter setting values are not correct. Refer to the table in this section, examine the parameter setting that caused the error, and remove the cause of the error. You must first correct the parameter setting errors before you can operate the drive. The drive will not send notification signals for the faults and alarms when these parameter setting errors occur.

Code	Name	Causes	Possible Solutions
oPE01	Drive Capacity Setting Error	The value set in *o2-04 [Drive Model (KVA) Selection]* does not agree with the drive model.	Set *o2-04* to the correct value.
Code	**Name**	**Causes**	**Possible Solutions**
oPE02	Parameter Range Setting Error	Parameter settings are not in the applicable setting range.	1. Push ↵ to show *U1-18 [oPE Fault Parameter]*, and find parameters that are not in the applicable setting range. 2. Correct the parameter settings. **Note:** • If more than one error occurs at the same time, other *oPExx* errors have priority over *oPE02*.
		Set *E2-01 ≤ E2-03 [Motor Rated Current (FLA) ≤ Motor No-Load Current]*.	Make sure that *E2-01 > E2-03*. **Note:** If it is necessary to set *E2-01 < E2-03*, first lower the value set in *E2-03*, and then set *E2-01*.
Code	**Name**	**Causes**	**Possible Solutions**
oPE03	Multi-Function Input Setting Err	The settings for these parameters do not agree: • *H1-01 to H1-07 [Terminals S1 to S7 Function Selection]* • *H7-01 to H7-04 [Virtual Multi-Function Inputs 1 to 4]*	Correct the parameter settings.
		The settings for MFDIs overlap. **Note:** This does not include *H1-xx = 20 to 2F [MFDI Function Select = External Fault]* and *[Reserved]*.	Set the parameters correctly to prevent MFDI function overlap.
		You did not set these pairs of MFDI functions to Digital Inputs (*H1-xx* and *H7-01 to H7-04*) at the same time: • Setting values *10 [Up Command]* and *11 [Down Command]* • Setting values *75 [Up 2 Command]* and *76 [Down 2 Command]* • Setting values *42 [Run Command (2-Wire Sequence 2)]* and *43 [FWD/REV (2-Wire Sequence 2)]*	Set the MFDI pairs.
		You set a minimum of two of these MFDI combinations to Digital Inputs (*H1-xx* and *H7-01 to H7-04*) at the same time: • Setting values *10 [Up Command]* and *11 [Down Command]* • Setting values *75 [Up 2 Command]* and *76 [Down 2 Command]* • Setting value *A [Accel/Decel Ramp Hold]* • Setting value *1E [Reference Sample Hold]* • Setting values *44 to 46 [Add Offset Frequency 1 to 3 (d7-01 to d7-03)]*	Remove the function settings that are not in use.
		You set these commands in Digital Inputs (*H1-xx* and *H7-01 to H7-04*) at the same time: • Setting values *61 [Speed Search from Fmax]* and *62 [Speed Search from Fref]* • Setting values *16 [Motor 2 Selection]* and *1A [Accel/Decel Time Selection 2]* • Setting values *40, 41 [Forward RUN (2-Wire), Reverse RUN (2-Wire)]* and *42, 43 [Run Command (2-Wire Sequence 2), FWD/REV (2-Wire Sequence 2)]*	Remove the function settings that are not in use.
		Settings for N.C. and N.O. input *[H1-xx]* for these functions were selected at the same time: • Setting value *15 [Fast Stop (N.O.)]* • Setting value *17 [Fast Stop (N.C.)]*	Remove one of the function settings.
		You entered these settings while *H1-xx = 2 [External Reference 1/2 Selection]*: • *b1-15 = 1 [Analog Input]* • *H3-02 ≠ 0 [Terminal A1 Function Selection ≠ Frequency Reference]* or *H3-10 ≠ 0 [Terminal A2 Function Selection ≠ Frequency Reference]*	Set *H3-02 = 0* or *H3-10 = 0*.

Code	Name	Causes	Possible Solutions
		These parameters are set at the same time: • *H1-xx ≠ 6A [Drive Enable]* • *H2-xx = 38 [Drive Enabled]*	Correct the parameter settings.
		These parameters are set at the same time: • *H6-01 ≠ 3 [PG Speed Feedback (V/F Control)]* • *H1-xx = 7E [Reverse Rotation Identifier]*	Correct the parameter settings.
		These parameters are set at the same time: • *H1-xx = 75/76 [Up 2 /Down 2 Command]* • *H3-01, H3-09 = 1 [Terminal A1, A2 Signal Level Select = 0 to +10V(Without Limit)]*	Remove one of the function settings.
		• Multistep frequency reference [D1-18 = 0] and H1-xx = 50/150, 51/151, 52/152 or 53/153 are selected. • Speed reference by digital input [D1-18 = 1] and multi-step speed references 3/103, 4/104 or 5/105 are selected.	Check if contradictory settings have been assigned to the multi-function terminals at the same time. Correct setting errors.
Code	**Name**	**Causes**	**Possible Solutions**
oPE08	Parameter Selection Error	A function was set that is not compatible with the control method selected in *A1-02 [Control Method Selection]*.	1. Push ENTER Key to show *U1-18 [oPE Fault Parameter]*, and find parameters that are not in the applicable setting range. 2. Correct the parameter settings. **Note:** If more than one error occurs at the same time, other *oPExx* errors have priority over *oPE02*.
		When *A1-02 = 2 [OLV]*, you used these parameter settings: • *n2-02 > n2-03 [Automatic Freq Regulator Time 1 > Automatic Freq Regulator Time 2]* • *C4-02 > C4-06 [Torque Compensation Delay Time > Motor 2 Torque Comp Delay Time]*	• Set *n2-02 < n2-03*. • Set *C4-02 < C4-06*.
Code	**Name**	**Causes**	**Possible Solutions**
oPE10	V/f Data Setting Error	The parameters that set the V/f pattern do not satisfy these conditions: • For motor 1: *E1-09 ≤ E1-07 < E1-06 ≤ E1-11 ≤ E1-04 [Minimum Output Frequency ≤ Mid Point A Frequency < Base Frequency ≤ Mid Point B Frequency ≤ Maximum Output Frequency]* • For motor 2: *E3-09 ≤ E3-07 < E3-06 ≤ E3-11 ≤ E3-04 [Minimum Output Frequency ≤ Mid Point A Frequency < Base Frequency ≤ Mid Point B Frequency ≤ Maximum Output Frequency]*	Set the parameters correctly to satisfy the conditions.
Code	**Name**	**Causes**	**Possible Solutions**
oPE33	Digital Output Selection Error	These two parameters are set at the same time: • *H2-60 ≠ F [Term MA,MB,MC Secondary Function ≠ Not Used]* • *H2-01 = 1xx [Term MA,MB,MC Function Selection = Inverse output of xx]*	Clear the *H2-01 to H2-03 = 1xx [Inverse output of xx]* settings. **Note:** If you use the function to output logical calculation results (*H2-60, H2-63, H2-66 ≠ F*), you cannot set *H2-01 to H2-03 = 1xx*.
		These two parameters are set at the same time: • *H2-63 ≠ F [Terminal P1 Secondary Function ≠ Not Used]* • *H2-02 = 1xx [Term P1 Function Selection = Inverse output of xx]*	
		These two parameters are set at the same time: • *H2-66 ≠ F [Terminal P2 Secondary Function ≠ Not Used]* • *H2-03 = 1xx [Term P2 Function Selection = Inverse output of xx]*	
		These parameter pairs are set incorrectly: • *H2-21 [Comparator 1 Lower Limit] > H2-22 [Comparator 1 Upper Limit]* • *H2-27 [Comparator 2 Lower Limit] > H2-28 [Comparator 2 Upper Limit]*	• Set parameters *H2-21 ≤ H2-22*. • Set parameters *H2-27 ≤ H2-28*.

6.7 Auto-Tuning Errors

This table gives information about errors detected during Auto-Tuning. If the drive detects an Auto-Tuning error, the keypad will show the error and the motor will coast to stop. The drive will not send notification signals for faults and alarms when Auto-Tuning errors occur.

Two types of Auto-Tuning errors are: *Endx* and *Erx*. *Endx* identifies that Auto-Tuning has successfully completed with calculation errors. Find and repair the cause of the error and do Auto-Tuning again, or set the motor parameters manually. You can use the drive in the application if you cannot find the cause of the *Endx* error.

Erx identifies that Auto-Tuning was not successful. Find and repair the cause of the error and do Auto-Tuning again.

Code	Name	Causes	Possible Solutions
End1	Excessive Rated Voltage Setting	The torque reference was more than 20% during Auto-Tuning or the no-load current that was measured after Auto-Tuning is more than 80%.	• Make sure that the input motor nameplate data is correct. • Do Auto-Tuning again and correctly set the motor nameplate data. • If you can uncouple the motor and load, remove the motor from the machine and do Rotational Auto-Tuning again. • If you cannot uncouple the motor and load, use the results from Auto-Tuning.
Code	**Name**	**Causes**	**Possible Solutions**
End2	Iron Core Saturation Coefficient	The motor nameplate data entered during Auto-Tuning is incorrect.	• Make sure that the input motor nameplate data is correct. • Do Auto-Tuning again and correctly set the motor nameplate data.
		Auto-Tuning results were not in the applicable parameter setting range, and *E2-07* or *E2-08 [Motor Saturation Coefficient 2]* have temporary values.	• Examine and repair damaged motor wiring. • If you can uncouple the motor and load, remove the motor from the machine and do Rotational Auto-Tuning again.
Code	**Name**	**Causes**	**Possible Solutions**
End3	Rated Current Setting Alarm	The rated current value is incorrect.	Do Auto-Tuning again and set the correct rated current shown on the motor nameplate.
Code	**Name**	**Causes**	**Possible Solutions**
End4	Adjusted Slip Calculation Error	The Auto-Tuning results were not in the applicable parameter setting range.	• Make sure the input motor nameplate data is correct. • Do Rotational Auto-Tuning again and correctly set the motor nameplate data. • If you cannot uncouple the motor and load, do Stationary Auto-Tuning 2.
		The motor rated slip that was measured after Stationary Auto-Tuning was 0.2 Hz or lower.	
		The motor rated slip that was measured after compensation with *E2-08 [Motor Saturation Coefficient 2]* is not in the applicable range.	
		The secondary resistor measurement results were not in the applicable range.	
Code	**Name**	**Causes**	**Possible Solutions**
End5	Resistance Tuning Error	The Auto-Tuning results of the Line-to-Line Resistance were not in the applicable range.	• Make sure that the input motor nameplate data is correct. • Examine and repair damaged motor wiring.
Code	**Name**	**Causes**	**Possible Solutions**
End6	Leakage Inductance Alarm	The Auto-Tuning results were not in the applicable parameter setting range.	Make sure that the input motor nameplate data is correct, and do Auto-Tuning again.
		A1-02 [Control Method Selection] setting is not applicable.	• Examine the value set in *A1-02*. • Make sure that the input motor nameplate data is correct, and do Auto-Tuning again.
Code	**Name**	**Causes**	**Possible Solutions**
End7	No-Load Current Alarm	The Auto-Tuning results of the motor no-load current value were not in the applicable range.	Examine and repair damaged motor wiring.
		Auto-Tuning results were less than 5% of the motor rated current.	Make sure that the input motor nameplate data is correct, and do Auto-Tuning again.
Code	**Name**	**Causes**	**Possible Solutions**
Er-01	Motor Data Error	The motor nameplate data entered during Auto-Tuning is incorrect.	• Make sure that the motor nameplate data is correct. • Do Auto-Tuning again and correctly set the motor nameplate data.
		The combination of the motor rated power and motor rated current do not match.	• Examine the combination of drive capacity and motor output. • Do Auto-Tuning again, and correctly set the motor rated power and motor rated current.

Code	Name	Causes	Possible Solutions
		The combination of the motor rated current that was entered during Auto-Tuning and *E2-03 [Motor No-Load Current]* do not match.	• Examine the motor rated current and the no-load current. • Set *E2-03* correctly. • Do Auto-Tuning again, and correctly set the motor rated current.
		The combination of the setting values of Motor Base Frequency and Motor Base Speed do not match.	Do Auto-Tuning again, and correctly set the Motor Base Frequency and Motor Base Speed.
Code	**Name**	**Causes**	**Possible Solutions**
Er-02	Drive in an Alarm State	The motor nameplate data entered during Auto-Tuning is incorrect.	• Make sure that the motor nameplate data entered in Auto-Tuning is correct. • Do Auto-Tuning again and correctly set the motor nameplate data.
		You did Auto-Tuning while the drive had a minor fault or alarm.	Clear the minor fault or alarm and do Auto-Tuning again.
		There is a defective motor cable or cable connection.	Examine and repair motor wiring.
		The load is too large.	• Decrease the load. • Examine the machine area to see if, for example, the motor shaft is locked.
		The drive detected a minor fault during Auto-Tuning.	1. Stop Auto-Tuning. 2. Examine the minor fault code and remove the cause of the problem. 3. Do Auto-Tuning again.
Code	**Name**	**Causes**	**Possible Solutions**
Er-03	STOP Button was Pressed	During Auto-Tuning, STOP was pushed.	Auto-Tuning did not complete correctly. Do Auto-Tuning again.
Code	**Name**	**Causes**	**Possible Solutions**
Er-04	Line-to-Line Resistance Error	The Auto-Tuning results were not in the applicable parameter setting range.	• Examine and repair motor wiring. • Disconnect the machine from the motor and do Rotational Auto-Tuning again.
		Auto-Tuning did not complete in a pre-set length of time.	
		There is a defective motor cable or cable connection.	
		The motor nameplate data entered during Auto-Tuning is incorrect.	• Make sure that the input motor nameplate data is correct. • Do Auto-Tuning again and correctly set the motor nameplate data.
Code	**Name**	**Causes**	**Possible Solutions**
Er-05	No-Load Current Error	The Auto-Tuning results were not in the applicable parameter setting range.	• Examine and repair motor wiring. • Disconnect the machine from the motor and do Rotational Auto-Tuning again.
		Auto-Tuning did not complete in a pre-set length of time.	
		The motor nameplate data entered during Auto-Tuning is incorrect.	• Make sure that the input motor nameplate data is correct. • Do Auto-Tuning again and correctly set the motor nameplate data.
		Rotational Auto-Tuning was done with a load that was more than 30% of the rating connected to the motor.	• Disconnect the machine from the motor and do Rotational Auto-Tuning again. • If you cannot uncouple the motor and load, make sure that the load is less than 30% of the motor rating. If a mechanical brake is installed in the motor, release the brake during Rotational Auto-Tuning.
Code	**Name**	**Causes**	**Possible Solutions**
Er-08	Rated Slip Error	The motor nameplate data entered during Auto-Tuning is incorrect.	• Make sure that the input motor nameplate data is correct. • Do Auto-Tuning again and correctly set the motor nameplate data.
		Auto-Tuning did not complete in a pre-set length of time.	• Examine and repair the motor wiring. • If the motor and machine are connected during Rotational Auto-Tuning, decouple the motor from the machinery.
		The Auto-Tuning results were not in the applicable parameter setting range.	
		Rotational Auto-Tuning was done with a load that was more than 30% of the rating connected to the motor.	• Disconnect the machine from the motor and do Rotational Auto-Tuning again. • If you cannot uncouple the motor and load, make sure that the load is less than 30% of the motor rating. If a mechanical brake is installed in the motor, release the brake during Rotational Auto-Tuning.

Code	Name	Causes	Possible Solutions
Er-09	Acceleration Error	The motor did not accelerate for the specified acceleration time.	1. Increase the value set in *C1-01 [Acceleration Time 1]*. 2. Disconnect the machine from the motor and do Rotational Auto-Tuning again.
		The value of *L7-01 or L7-02 [Forward/Reverse Torque Limit]* is small.	Increase the value set in *L7-01 or L7-02*.
		Rotational Auto-Tuning was done with a load that was more than 30% of the rating connected to the motor.	• Disconnect the machine from the motor and do Rotational Auto-Tuning again. • If you cannot uncouple the motor and load, make sure that the load is less than 30% of the motor rating. If a mechanical brake is installed in the motor, release the brake during Rotational Auto-Tuning.
Code	**Name**	**Causes**	**Possible Solutions**
Er-11	Motor Speed Error	The torque reference during acceleration is too high (100%).	• Increase the value set in *C1-01 [Acceleration Time 1]*. • Disconnect the machine from the motor and do Rotational Auto-Tuning again.
Code	**Name**	**Causes**	**Possible Solutions**
Er-12	Current Detection Error	There is a phase loss in the drive input power. (U/T1, V/T2, W/T3)	Examine and repair motor wiring.
		The current exceeded the current rating of the drive.	• Check the motor wiring for any short circuits between the wires. • Check and turn ON any magnetic contactors used between motors. • Replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
		The output current is too low.	
		You tried Auto-Tuning without a motor connected to the drive.	Connect the motor and do Auto-Tuning.
		There was a current detection signal error.	Replace the control board or the drive. For information about replacing the control board, contact Yaskawa or your nearest sales representative.
Code	**Name**	**Causes**	**Possible Solutions**
Er-13	Leakage Inductance Alarm	The motor rated current value is incorrect.	Correctly set the rated current indicated on the motor nameplate and perform Auto-Tuning again.
		The drive could not complete tuning for leakage inductance in fewer than 300 seconds.	Examine and repair motor wiring.
Code	**Name**	**Causes**	**Possible Solutions**
Er-15	Torque Saturation Error	During Inertia Tuning, the output torque was more than the value set in *L7-01 to L7-04 [Torque Limit]*.	• Increase the value set in *L7-01 to L7-04 [Torque Limit]* as much as possible. • Decrease the values set for the frequency and amplitude of the test signals used when doing inertia tuning. First, decrease the test signal amplitude, and then do Inertia Tuning. If the error continues, decrease the test signal frequency and do Inertia Tuning again.

6.8 Backup Function Operating Mode Display and Errors

◆ Operating Mode Display

When you use the LCD keypad to do the backup function, the keypad shows the running operation on the LCD display. These indicators do not show that an error has occurred.

Keypad Display	Name	Display	Status
Drive and Keypad mismatch. Should the parameters be restored?	Detection of inconsistency between the drive and keypad	Normally displayed	The drive detected the connection of a keypad from a different drive. Select [Yes] to copy parameters backed up in the keypad to the connected drive.
Restore Restore from keypad	Restoring parameters	Flashing	The parameters stored in the keypad have been restored to the drive.
End	Backup/restore/verify operation ended normally	Normally displayed	The parameter backup, restore, or verify operation ended normally.
Backup Backup from Drive	Backing up parameters	Flashing	The parameters stored in the drive are being backed up to the keypad.
Verify Keypad & Drive	Verifying parameters	Flashing	The parameter settings stored in the keypad and the parameter settings in the drive match or are being compared.

◆ Backup Function Runtime Errors

When an error occurs, the keypad shows a code to identify the error.

The table in this section show the error codes. If there are errors, refer to these tables:

Note:

Push any key on the keypad to clear an error.

Code	Name	Causes	Possible Solutions
CPEr	Control Mode Mismatch	The keypad setting and drive setting for *A1-02 [Control Method Selection]* do not agree.	1. Set *A1-02* on the drive to the same value that is on the keypad. 2. Restore the parameters.
Code	**Name**	**Causes**	**Possible Solutions**
CPyE	Error Writing Data	Parameter restore did not end correctly.	Restore the parameters.
Code	**Name**	**Causes**	**Possible Solutions**
CSEr	Control Mode Mismatch	The keypad is broken.	Replace the keypad.
Code	**Name**	**Causes**	**Possible Solutions**
dFPS	Drive Model Mismatch	You tried to restore parameters to a different drive model than the one that you backed up.	1. Examine the drive model that you used to back up the parameters. 2. Restore the parameters.
Code	**Name**	**Causes**	**Possible Solutions**
iFEr	Keypad Communication Error	There was a communications error between the keypad and the drive.	Examine the connector or cable connection.
Code	**Name**	**Causes**	**Possible Solutions**
ndAT	Error Received Data	The parameter settings for model and specifications (power supply voltage and capacity) are different between the keypad and the drive.	1. Make sure that drive model and the value set in *o2-04 [Drive Model (KVA) Selection]* agree. 2. Restore the parameters.
		The parameters are not stored in the keypad.	1. Connect a keypad that has the correct parameters. 2. Restore the parameters.
Code	**Name**	**Causes**	**Possible Solutions**
rdEr	Error Reading Data	You tried to back up the data when *o3-02 = 0 [Copy Allowed Selection = Disabled]*.	Set *o3-02 = 1 [Enabled]* and back up again.
Code	**Name**	**Causes**	**Possible Solutions**
vAEr	Voltage Class, Capacity Mismatch	The power supply specifications or drive capacity parameter settings are different between the keypad and the drive.	1. Make sure that drive model and the value set in *o2-04 [Drive Model (KVA) Selection]* agree. 2. Restore the parameters.
Code	**Name**	**Causes**	**Possible Solutions**
vFyE	Parameters do not Match	The parameters that are backed up in the keypad and the parameters in the drive are not the same.	1. Restore or backup the parameter again. 2. Verify the parameters.

6.9 Diagnosing and Resetting Faults

When a fault occurs and the drive stops, do the procedures in this section to remove the cause of the fault, then re-energize the drive.

◆ Fault and Power Loss Occur at the Same Time

WARNING! *Crush Hazard. Wear eye protection when you do work on the drive. If you do not use correct safety equipment, it can cause serious injury or death.*

WARNING! *Electrical Shock Hazard. After the drive blows a fuse or trips an RCM/RCD, do not immediately energize the drive or operate peripheral devices. Wait for the time specified on the warning label at a minimum and make sure that all indicators are OFF. Then check the wiring and peripheral device ratings to find the cause of the problem. If you do not know the cause of the problem, contact Yaskawa before you energize the drive or peripheral devices. If you do not fix the problem before you operate the drive or peripheral devices, it can cause serious injury or death.*

1. Supply power to the control circuit from the external 24 V input.
2. Use monitor parameters *U2-xx [Fault Trace]* to show the fault code and data about the operating status of the drive immediately before the fault occurred.
3. Use the information in the Troubleshooting tables to remove the fault.

Note:

1. To find the faults that were triggered, check the fault history in *U2-02 [Previous Fault]*. To find information about drive status (such as frequency, current, and voltage) when the faults were triggered, check *U2-03 to U2-20*.
2. If the fault display stays after you re-energize the drive, remove the cause of the fault and reset.

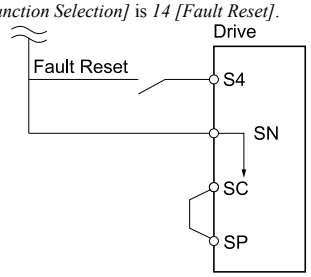
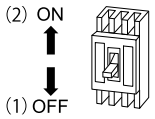
◆ Fault Occurs Without Power Loss

1. Examine the fault code shown on the keypad.
2. Use the information in the Troubleshooting tables to remove the fault.
3. Do a fault reset.

◆ Fault Reset Procedure

If a fault occurs, you must remove the cause of the fault and re-energize the drive. Table 6.3 lists the different methods to reset the drive after a fault.

Table 6.3 Fault Reset Methods

Methods	Description
Method 1	While the keypad is showing the fault or alarm code, push RESET on the keypad.
Method 2	Switch ON the MFDI terminal set to *H1-xx = 14 [MFDI Function Select = Fault Reset]*. **Note:** The default setting for *H1-04 [Terminal S4 Function Selection]* is *14 [Fault Reset]*. 
Method 3	1. De-energize the drive main circuit power supply. 2. Energize the drive again after the keypad display goes out. 

Note:

If the drive receives a Run command from a communication option or control circuit terminal, the drive will not reset the fault. Turn the Run command OFF to reset the fault. If you do a fault reset when the drive has a Run command, the keypad will show minor fault *CrST [Remove RUN Command to Reset]*.

6.10 Troubleshooting Without Fault Display

If the drive or motor operate incorrectly, but the keypad does not show a fault or error code, refer to the items this section.

- Motor hunting and oscillation
- Unsatisfactory motor torque
- Unsatisfactory speed precision
- Unsatisfactory motor torque and speed response
- Motor noise

◆ The Parameter Settings Will Not Change

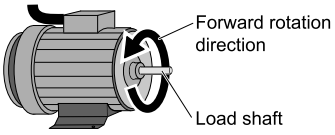
Causes	Possible Solutions
The drive is operating the motor (the drive is in Drive Mode).	Stop the drive and change to Programming Mode.
Parameter *A1-01 = 0 [Access Level Selection = Operation Only]*.	Set *A1-01 = 2 [Access Level Selection = Advanced Level]* or *A1-01 = 3 [Expert Level]*.
Parameter *H1-xx = 1B [MFDI Function Select = Programming Lockout]*.	Activate the terminals to which *H1-xx = 1B* is set, and then change the parameters.
You entered an incorrect password in *A1-04 [Password]*.	• Enter the correct password to *A1-04* again. • If you forgot the password, set the password again with *A1-04* and *A1-05 [Password Setting]*. **Note:** If you set the password, you cannot change these parameters until the password aligns: *• A1-01 [Access Level Selection]* *• A1-02 [Control Method Selection]* *• A1-03 [Initialize Parameters]* *• A1-06 [Application Preset]* *• A1-07 [DriveWorksEZ Function Selection]* *• A2-01 to A2-32 [User Parameter 1 to User Parameter 32]*
The drive detected *Uv [Undervoltage]*.	• View *U1-07 [DC Bus Voltage]* to see the power supply voltage. • Examine the main circuit wiring.

◆ The Motor Does Not Rotate after You Enter a Run Command

Causes	Possible Solutions
The drive is not in Drive Mode.	1. Make sure that the READY LED on the keypad is ON. 2. If the READY LED is OFF, push and hold the ESC Key to go back to the frequency reference screen (the initial screen).
The drive stopped and you pushed LO/RE to transfer the Run command source to the keypad.	Do one of these two: • Push LO/RE. • Re-energize the drive. **Note:** When LO/RE must not change the Run command source, set *o2-01 = 0 [LO/RE Key Function Selection = Disabled]*.
Auto-Tuning completed.	Push and hold the ESC Key to go back to the frequency reference screen (the initial screen). **Note:** When Auto-Tuning completes, the drive changes to Programming Mode. The drive will not accept a Run command unless the drive is in Drive Mode.
The drive received a Fast Stop command.	Turn off the fast stop input signal.
The settings for the source that supplies the Run command are incorrect.	Set *b1-02 [Run Command Selection 1]* correctly.
The frequency reference source is not set correctly.	Set *b1-01 [Frequency Reference Selection 1]* correctly.
There is defective wiring in the control circuit terminals.	• Correctly wire the drive control circuit terminals. • View *U1-10 [Input Terminal Status]* for input terminal status.
The settings for voltage input and current input of the master frequency reference are incorrect.	Examine these analog input terminal signal level settings: • Terminal A1: *H3-01 [Terminal A1 Signal Level Select]* • Terminal A2: DIP switch S1 and *H3-09 [Terminal A2 Signal Level Select]*
The selection for the sinking/sourcing mode and the internal/external power supply is incorrect.	• For sinking mode, close the circuit between terminals SC-SP with a wire jumper. • For sourcing mode, close the circuit between terminals SC-SN with a wire jumper. • For external power supply, remove the wire jumper.
The frequency reference is too low.	• View *U1-01 [Freq Reference]*. • Increase the frequency reference to a value higher than *E1-09 [Minimum Output Frequency]*.

Causes	Possible Solutions
The MFAI setting is incorrect.	• Make sure that the functions set to the MFAI are correct. The frequency reference is 0 when *H3-02, H3-10 = 1 [MFAI Function Select = Frequency Gain]* and voltage (current) is not input. • Use *U1-13, U1-14 [Terminal A1, A2 Input Voltage]* to make sure that the analog input values set to terminals A1 and A2 are applicable.
You pushed STOP.	Turn the Run command OFF then ON from an external input. **Note:** When you push STOP during operation, the drive will ramp to stop. Set *o2-02 = 0 [STOP Key Function Selection = Disabled]* to disable the STOP function.
The 2-wire sequence and 3-wire sequence are not set correctly.	• Set one of the parameters *H1-03 to H1-07 [Terminals S3 to S7 Function Select]* to *0 [3-Wire Sequence]* to enable the 3-wire sequence. • If a 2-wire sequence is necessary, make sure that *H1-03 to H1-07 ≠ 0*.

◆ The Motor Rotates in the Opposite Direction from the Up/Down Command

Causes	Possible Solutions
The phase wiring between the drive and motor is incorrect.	• Examine the wiring between the drive and motor. • Connect drive output terminals U/T1, V/T2, and W/T3 in the correct sequence to agree with motor terminals U, V, and W. • Switch two motor cables U, V, and W to reverse motor direction.
Drive control circuit terminals for the Up and Down commands are switched.	• Connect drive output terminals U/T1, V/T2, and W/T3 in the correct sequence to agree with motor terminals U, V, and W. • Switch two motor cables U, V, and W to reverse motor direction.  **Figure 6.1 Forward Rotating Motor** **Note:** • For Yaskawa motors, the forward direction is counterclockwise when looking from the motor shaft side. • Refer to the motor specifications, and make sure that the forward rotation direction is correct for the application. The forward rotation direction of motors can be different for different motor manufacturers and types.
The signal connections for forward run and reverse run on the drive control circuit terminals and control panel side are incorrect.	Correctly wire the control circuit.

◆ The Motor Rotates in Only One Direction

Causes	Possible Solutions
The drive will not let the motor rotate in reverse.	Set *b1-04 = 0 [Reverse Operation Selection = Reverse Enabled]*.
The drive did not receive a Reverse run signal and 3-Wire sequence is selected.	Activate the terminals to which *H1-xx = 0 [3-Wire Sequence]* is set, and then enable reverse operation.

◆ The Motor Is Too Hot

Causes	Possible Solutions
The load is too heavy.	• Decrease the load. • Increase the acceleration and deceleration times. • Examine the values set in *L1-01 [Motor Overload (oL1) Protection]*, *L1-02 [Motor Overload Protection Time]*, and *E2-01 [Motor Rated Current (FLA)]*. • Use a larger motor. **Note:** The motor also has a short-term overload rating. Examine this rating carefully before setting drive parameters.
The motor is running continuously at a very low speed.	• Change the run speed. • Use a drive-dedicated motor.
The drive is operating in a vector control mode, but Auto-Tuning has not been done.	• Do Auto-Tuning. • Calculate motor parameter and set motor parameters. • Set *A1-02 = 0 [Control Method Selection = V/f Control]*.

Causes	Possible Solutions
The voltage insulation between motor phases is not sufficient.	• Use a motor with a voltage tolerance that is higher than the maximum voltage surge. • Use a drive-dedicated motor that is rated for use with AC drives for applications that use a motor on drives rated higher than 400 V class. • Install an AC reactor on the output side of the drive and set *C6-02 = 1 [Carrier Frequency Selection = 2.0 kHz]*. **Note:** When the motor is connected to the drive output terminals U/T1, V/T2, and W/T3, surges occur between the drive switching and the motor coils. These surges can be three times the drive input power supply voltage (600 V for a 200 V class drive, 1200 V for a 400 V class drive).
The air around the motor is too hot.	• Measure the ambient temperature. • Decrease the temperature in the area until it is in the specified temperature range.
The motor fan stopped or is clogged.	• Clean the motor fan. • Make the drive environment better.

◆ The Correct Auto-Tuning Mode Is Not Available

Causes	Possible Solutions
The desired Auto-Tuning mode is not available for the selected control mode.	Change the motor control method with parameter *A1-02 [Control Method Selection]*.

◆ The Motor Stalls during Acceleration or Accel/Decel Time Is Too Long

Causes	Possible Solutions
The drive and motor system are at the torque limit or current suppression will not let the drive accelerate.	• Decrease the load. • Use a larger motor. **Note:** Although the drive has a Stall Prevention function and a Torque Compensation Limit function, if you try to accelerate too fast or try to drive a load that is too large, it can be too much for the limits of the motor.
Torque limit is set incorrectly.	Set the torque limit correctly.
The acceleration time setting is too short.	Examine the values set in *C1-01, C1-03, C1-05, or C1-07 [Acceleration Time]* and set them to applicable values.
The load is too large.	• Increase the acceleration time. • Examine the mechanical brake and make sure that it is fully releasing. • Decrease the load to make sure that the output current stays less than the motor rated current. • Use a larger motor. **Note:** • In extruder and mixer applications, the load can increase as the temperature decreases. • Although the drive has a Stall Prevention function and a Torque Compensation Limit function, if you try to accelerate too fast or try to drive a load that is too large, it can be too much for the limits of the motor.
The motor characteristics and drive parameter settings are not compatible.	• Set the correct V/f pattern to agree with the characteristics of the motor. • Examine the V/f pattern set in *E1-03 [V/f Pattern Selection]*. • Do Rotational Auto-Tuning.
The drive is operating in vector control mode, but you did not complete Auto-Tuning.	• Do Auto-Tuning. • Calculate motor data and reset motor parameters. • Set *A1-02 = 0 [Control Method Selection = V/f Control]*.
The Stall Prevention level during acceleration setting is too low.	Increase the value set in *L3-02 [Stall Prevent Level during Accel]*. **Note:** If the *L3-02* value is too low, the acceleration time can be unsatisfactorily long.
The Stall Prevention level during run setting is too low.	Increase the value set in *L3-06 [Stall Prevent Level during Run]*. **Note:** If the *L3-06* value is too low, speed will decrease before the drive outputs torque.
The drive is at the limit of the V/f motor control method.	• When the motor cable is longer than 50 m (164 ft), do Auto-Tuning for line-to-line resistance. • Set the V/f pattern to "High Starting Torque". • Use a Vector Control method. **Note:** V/f control method does not supply high torque at low speeds.

◆ There Is Too Much Motor Oscillation and the Rotation Is Irregular

Causes	Possible Solutions
Unsatisfactory balance of motor phases.	• Make sure that the drive input power voltage supplies stable power. • Set *L8-05 = 0 [Input Phase Loss Protect Select = Disabled]*.
The motor is hunting.	• Set *n1-01 = 1 [Hunting Prevention Selection = Enabled]*. • Increase the value of *n2-01 [SpdFeedbackDetectCtr (AFR) Gain]* or *n2-02 [SpdFeedbackDetCtr(AFR)TimeConst1]*.

◆ Deceleration Takes Longer Than Expected When Dynamic Braking Is Enabled

Causes	Possible Solutions
The stall prevention during deceleration setting is incorrect.	• Examine the setting for *L3-04 [Decel Stall Prevention Selection]*. • When the drive has a dynamic braking option installed, set *L3-04 = 0 [Disabled]*. • If the drive detects *ov [Overvoltage]*, set *L3-04 = 3 [General Purpose w/ DB resistor]*.
The deceleration time setting is too long.	Set *C1-02, C1-04, C1-06, or C1-08 [Deceleration Times]* to applicable values.
The motor torque is not sufficient.	Use a larger motor. **Note:** If these items are correct, the demand on the motor is more than the motor capacity: • Parameter settings are correct. • The drive does not detect *ov [Overvoltage]*.

The drive and motor system reached the torque limit.	• Examine the values set in *L7-01 to L7-04 [Torque Limit]* and increase them if necessary.
	Note:
	If the torque limit is enabled, deceleration time can increase because the drive cannot output more torque than the limit.
	• If *H3-02, H3-10 = 10, 11, 12, 15 [MFAI Function Select = Torque Limit]*, examine the settings for the MFAIs.
	  • Examine the values set in *H3-02 and H3-10*.
	  • Use *U1-13, U1-14 [Terminal A1, A2 Input Voltage]* to make sure that the analog input values set to terminals A1 and A2 are applicable.

The load is more than the internal torque limit as specified by the drive rated current.	Replace the drive with a larger capacity model.

◆ There Is Audible Noise from the Drive or Motor Cables When the Drive Is Energized

Causes	Possible Solutions
The relay switching in the drive is making too much noise.	• Use *C6-02 [Carrier Frequency Selection]* to decrease the carrier frequency. • Connect a noise filter to the input side of the drive power supply. • Connect a noise filter to the output side of the drive. • Isolate the control circuit wiring from the main circuit wiring. • Use a metal cable gland to wire the drive. • Shield the periphery of the drive with metal. • Make sure that the drive and motor are grounded correctly. • Make sure that ground faults have not occurred in the wiring or motor.

◆ Residual Current Monitoring/Detection (RCM/RCD) Trips During Run

Causes	Possible Solutions
There is too much leakage current from the drive.	• Increase the RCM/RCD sensitivity or use RCM/RCD with a higher threshold. • Use *C6-02 [Carrier Frequency Selection]* to decrease the carrier frequency. • Decrease the length of the cable used between the drive and the motor. • Install a noise filter or AC reactor on the output side of the drive. Set *C6-02 = 1 [2.0 kHz]* when connecting an AC reactor. • Disable the internal EMC filter.

◆ Motor Rotation Causes Unexpected Audible Noise from Connected Machinery

Causes	Possible Solutions
The carrier frequency and the resonant frequency of the connected machinery are the same.	• Adjust *C6-02 to C6-05 [Carrier Frequency]*. • Set *C6-02 = 1 to 6 [Carrier Frequency Selection = Frequency other than Swing PWM]*. **Note:** If *C6-02 = 7 to A [Carrier Frequency Selection = Swing PWM]*, the drive will not know if the noise comes from the drive or the machine.
The drive output frequency and the resonant frequency of the connected machinery are the same.	• Adjust *d3-01 to d3-04 [Jump Frequency]*. • Put the motor on a rubber pad to decrease vibration.

◆ Motor Rotation Causes Oscillation or Hunting

Causes	Possible Solutions
The frequency reference is assigned to an external source, and there is electrical interference in the signal.	Make sure that electrical interference does not have an effect on the signal lines. • Isolate control circuit wiring from main circuit wiring. • Use twisted-pair cables or shielded wiring for the control circuit.
The cable between the drive and motor is too long.	• Do Auto-Tuning. • Make the wiring as short as possible.

◆ The Starting Torque Is Not Sufficient

Causes	Possible Solutions
Auto-Tuning has not been done in vector control method.	Do Auto-Tuning.
The control method was changed after doing Auto-Tuning.	Do Auto-Tuning again.
Stationary Auto-Tuning for Line-to-Line Resistance was done.	Do Rotational Auto-Tuning.

◆ The Output Frequency Is Lower Than the Frequency Reference

Causes	Possible Solutions
A large load triggered Stall Prevention function during acceleration.	• Decrease the load. • Adjust *L3-02 [Stall Prevent Level during Accel]*.
L3-01 = 3 [Stall Prevent Select duringAccel = ILim Mode] has been set.	1. Check whether the V/f pattern and motor parameter settings are appropriate, and set them correctly. 2. If this does not solve the problem, and it is not necessary to limit the current level of stall during acceleration, adjust *L3-02*. 3. If this does not solve the problem, set *L3-01 = 1 [Enabled]*.
The motor is rotating at this speed: *S1-02 [DC Injection/Zero SpeedThreshold]* ≤ Motor Speed < *E1-09 [Minimum Output Frequency]*	Set *E1-09* < *S1-02*.

◆ The Motor Will Not Restart after a Loss of Power

Causes	Possible Solutions
The drive did not receive a Run command after applying power.	• Examine the sequence and wiring that enters the Run command. • Set up a relay to make sure that the Run command stays enabled during a loss of power.
For applications that use 3-wire sequence, the momentary power loss continued for a long time, and the relay that keeps the Run command has been switched off.	Examine the wiring and circuitry for the relay that keeps the Run command enabled during the momentary power loss ride-thru time.

7

Periodic Inspection and Maintenance

This chapter gives information about how to examine and maintain drives in use, how to replace cooling fans and other parts, and how to store drives.

7.1 Section Safety 180
7.2 Inspection 182
7.3 Maintenance 184
7.4 Replace Cooling Fans 186
7.5 Replace the Drive 191
7.6 Storage Guidelines 193

7.1 Section Safety

⚠DANGER

Electrical Shock Hazard

Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe.

If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.

Disconnect all power to the drive and wait for the time specified on the warning label before you remove covers. Check the drive for dangerous voltages before servicing or repair work.

If you do work on the drive when it is energized and there is no cover over the electronic circuits, it will cause serious injury or death from electrical shock. The drive has internal capacitors that stay charged after you de-energize the drive.

⚠WARNING

Electrical Shock Hazard

The motor will run after you de-energize the drive. PM motors can generate induced voltage to the terminal of the motor after you de-energize the drive.

If you touch a motor that is moving or energized, it can cause serious injury or death.

Do not operate the drive when covers are missing. Replace covers and shields before you operate the drive. Use the drive only as specified by the instructions.

Some figures in this section include drives without covers or safety shields to more clearly show the inside of the drive. If covers or safety shields are missing from the drive, it can cause serious injury or death.

Always ground the motor-side grounding terminal.

If you do not ground the equipment correctly, it can cause serious injury or death if you touch the motor case.

Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive.

If personnel are not approved, it can cause serious injury or death.

Do not wear loose clothing or jewelry when you do work on the drive. Tighten loose clothing and remove all metal objects, for example watches or rings.

Loose clothing can catch on the drive and jewelry can conduct electricity and cause serious injury or death.

Fire Hazard

Tighten all terminal screws to the correct tightening torque.

Connections that are too loose or too tight can cause incorrect operation and damage to the drive. Incorrect connections can also cause death or serious injury from fire.

Damage to Equipment

Do not apply incorrect voltage to the main circuit of the drive. Operate the drive in the specified range of the input voltage on the drive nameplate.

Voltages that are higher than the permitted nameplate tolerance can cause damage to the drive.

Fire Hazard

Do not put flammable or combustible materials on top of the drive and do not install the drive near flammable or combustible materials. Attach the drive to metal or other noncombustible material.

Flammable and combustible materials can start a fire and cause serious injury or death.

⚠WARNING

Electrical Shock Hazard

Do not modify the drive body or drive circuitry.

Modifications to drive body and circuitry can cause serious injury or death, will cause damage to the drive, and will void the warranty. Yaskawa is not responsible for modifications of the product made by the user.

Sudden Movement Hazard

Make sure that you align the phase order for the drive and motor when you connect the motor to drive output terminals U/T1, V/T2, and W/T3.

If the phase order is incorrect, it can cause the motor to run in reverse. If the motor accidentally runs in reverse, it can cause serious injury or death.

⚠CAUTION

Burn Hazard

Do not touch a hot drive heatsink. De-energize the drive, wait for a minimum of 15 minutes, then make sure that the heatsink is cool before you replace the cooling fans.

If you touch a hot drive heatsink, it can burn you.

NOTICE

Damage to Equipment

When you touch the drive and circuit boards, make sure that you observe correct electrostatic discharge (ESD) procedures.

If you do not follow procedures, it can cause ESD damage to the drive circuitry.

Use the instructions in this manual to replace the cooling fans. When you do maintenance on the fans, replace all the fans to increase product life.

If you install the fans incorrectly, it can cause damage to the drive.

Make sure that all connections are correct after you install the drive and connect peripheral devices.

Incorrect connections can cause damage to the drive.

Do not energize and de-energize the drive more frequently than one time each 30 minutes.

If you frequently energize and de-energize the drive, it can cause drive failure.

Do not operate a drive or connected equipment that has damaged or missing parts.

You can cause damage to the drive and connected equipment.

Note:

Do not use unshielded cable for control wiring. Use shielded, twisted-pair wires and ground the shield to the ground terminal of the drive. Unshielded wire can cause electrical interference and unsatisfactory system performance.

7.2 Inspection

Power electronics have limited life and can show changes in performance and deterioration of performance after years of use in usual conditions. To help prevent these problems, it is important to do preventive maintenance and regular inspection, and replace parts on the drive.

Drives contain different types of power electronics, for example power transistors, semiconductors, capacitors, resistors, fans, and relays. The electronics in the drive are necessary for correct motor control.

Follow the inspection lists in this chapter as a part of a regular maintenance program.

Note:

Examine the drive one time each year at a minimum.

The operating conditions, environmental conditions, and use conditions will have an effect on the examination frequency for connected equipment.

Examine the drive more frequently if you use the drive in bad conditions or in these conditions:

- High ambient temperatures
- Frequent starting and stopping
- Changes in the AC power supply or load
- Too much vibration or shock loading
- Dust, metal dust, salt, sulfuric acid, or chlorine atmospheres
- Unsatisfactory storage conditions.

◆ Recommended Daily Inspection

Examine the items in Table 7.1 each day to make sure that the components do not wear out or fail. Make a copy of this checklist and put a check mark in the “Checked” column after each inspection.

Table 7.1 Daily Inspection Checklist

Inspection Area	Inspection Points	Corrective Action	Checked
Motor	Examine for unusual oscillation or noise coming from the motor.	• Check the load coupling. • Measure motor vibration. • Tighten all loose components.	
Cooling System	Examine for unusual heat from the drive or motor and visible discoloration.	• Check for a load that is too heavy. • Tighten loose screws. • Check for a dirty heatsink or motor. • Measure the ambient temperature.	
	Examine the cooling fans.	• Check for a clogged or dirty fan. • Use the performance life monitor to check for correct fan operation.	
Surrounding Environment	Make sure that the installation environment is applicable.	Remove the source of contamination or correct unsatisfactory environment.	
Load	Make sure that the drive output current is not more than the motor or drive rating for an extended period of time.	• Check for a load that is too heavy. • Check the correct motor parameter settings.	
Power Supply Voltage	Examine main power supply and control voltages.	• Correct the voltage or power supply to agree with nameplate specifications. • Verify all main circuit phases.	

◆ Recommended Periodic Inspection

Examine the items in Table 7.2 to Table 7.6 one time each year at a minimum. The operating conditions, environmental conditions, and use conditions will have an effect on the examination frequency for connected equipment. You must use your experience with the application to select the correct inspection frequency for each drive installation. Periodic inspections will help to prevent performance deterioration and product failure. Make a copy of this checklist and put a check mark in the “Checked” column after each inspection.

DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

Table 7.2 Main Circuit Periodic Inspection Checklist

Inspection Area	Inspection Points	Corrective Action	Checked
General	• Examine equipment for discoloration from too much heat or deterioration. • Examine for damaged parts.	• Replace damaged components as necessary. • The drive does not have many serviceable parts and it could be necessary to replace the drive.	
	Examine for dirt, unwanted particles, or dust on components.	• Examine enclosure door seal. • Use a vacuum cleaner to remove unwanted particles and dust without touching the components. • If you cannot remove unwanted particles and dust with a vacuum cleaner, replace the components.	
Conductors and Wiring	• Examine wiring and connections for discoloration or damage. Examine wiring and connections for discoloration from too much heat. • Examine wire insulation and shielding for discoloration and wear.	Repair or replace damaged wiring.	
Terminal Block	Examine terminals for stripped, damaged, or loose connections.	• Tighten loose screws. • Replace damaged screws.	
Electromagnetic Contactors and Relays	• Examine contactors and relays for too much noise during operation. • Examine coils for signs of too much heat, such as melted or broken insulation.	• Check coil voltage for overvoltage or undervoltage conditions. • Replace broken relays, contactors, or circuit boards that you can remove.	
Dynamic Braking Option	Examine the insulation for discoloration from too much heat.	If there is discoloration in the option, check to make sure that there is not damage to the wiring. A small quantity of discoloration is not a problem.	
Electrolytic Capacitor	• Examine for leaks, discoloration, or cracks. • Examine if the cap has come off, if there is swelling, or if there are leaks from broken sides.	The drive does not have many serviceable parts and it could be necessary to replace the drive.	
Diodes, IGBT (Power Transistor)	Examine for dust or other unwanted material collected on the surface.	Use a vacuum cleaner to remove unwanted particles and dust without touching the components.	

Table 7.3 Motor Periodic Inspection Checklist

Inspection Area	Inspection Points	Corrective Action	Checked
Operation Check	Check for increased vibration or unusual noise.	Stop the motor and contact approved maintenance personnel as necessary.	

Table 7.4 Control Circuit Periodic Inspection Checklist

Inspection Area	Inspection Points	Corrective Action	Checked
General	• Examine terminals for stripped, damaged, or loose connections. • Make sure that all terminals have been correctly tightened.	• Tighten loose screws. • Replace damaged screws or terminals. • If terminals are integral to a circuit board, it could be necessary to replace the control board or the drive.	
Circuit Boards	• Check for odor, discoloration, or rust. • Make sure that all connections are correctly fastened. • Make sure that the surface of the circuit board does not have dust or oil mist.	• Tighten loose connections. • Use a vacuum cleaner to remove unwanted particles and dust without touching the components. • If you cannot remove unwanted particles and dust with a vacuum cleaner, replace the components. • Do not use solvents to clean the board. • The drive does not have many serviceable parts and it could be necessary to replace the drive.	

Table 7.5 Cooling System Periodic Inspection Checklist

Inspection Area	Inspection Points	Corrective Action	Checked
Cooling Fan	• Check for unusual oscillation or unusual noise. • Check for damaged or missing fan blades.	Clean or replace the fans as necessary.	
Heatsink	• Examine for dust or other unwanted material collected on the surface. • Examine for dirt.	Use a vacuum cleaner to remove unwanted particles and dust without touching the components.	
Air Duct	Examine air intake, exhaust openings and make sure that there are no unwanted materials on the surface.	Clear blockages and clean air duct as necessary.	

Table 7.6 Keypad Periodic Inspection Checklist

Inspection Area	Inspection Points	Corrective Action	Checked
General	• Make sure that the keypad shows the data correctly. • Examine for dust or other unwanted material that collected on components in the area.	• If you have problems with the display or the keys, contact Yaskawa or your nearest sales representative. • Clean the keypad.	

7.3 Maintenance

The drive Maintenance Monitors keep track of component wear and tell the user when the end of the estimated performance life is approaching. The Maintenance Monitors prevent the need to shut down the full system for unexpected problems. Users can set alarm notifications for the maintenance periods for these drive components:

- Cooling fan
- Electrolytic capacitor
- Soft charge bypass relay
- IGBT

Contact Yaskawa or your nearest sales representative for more information about part replacement.

◆ Replaceable Parts

You can replace these parts of the drive:

- Cooling fan

If there is a failure in the main circuit, replace the drive.

If the drive is in the warranty period, contact Yaskawa or your nearest sales representative before you replace parts. Yaskawa reserves the right to replace or repair the drive as specified by the Yaskawa warranty policy.

DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

◆ Part Replacement Guidelines

Table 7.7 shows the standard replacement period for replacement parts. When you replace these parts, make sure that you use Yaskawa replacement parts for the applicable model and design revision number of your drive.

Table 7.7 Standard Replacement Period

Parts	Standard Replacement Period
Cooling fan	10 years
Electrolytic capacitor *1	10 years

*1 If there is damage to parts that you cannot repair or replace, replace the drive.

Note:

Performance life estimate is based on these use conditions. These conditions are provided for the purpose of replacing parts to maintain performance. Some parts may require more frequent replacement due to poor environments or rigorous use. Operating conditions for performance life estimate: Ambient temperature: Yearly average of 40 °C (IP20/UL Open Type), Load factor: 80%, Operating rate: 24 hours a day

◆ Monitors that Display the Lifespan of Drive Components

The drive keypad shows percentage values for the replacement parts to help you know when you must replace those components. Use the monitors in Table 7.8 to check replacement periods. When the monitor value is 100%, the component is at the end of its useful life and there is an increased risk of drive malfunction. Yaskawa recommends that you check the maintenance period regularly to make sure that you get the maximum performance life.

Table 7.8 Performance Life Monitors

Monitor No.	Parts	Description
U4-03	Cooling fan	Shows the total operation time of fans as 0 to 99999 hours. After this value is 99999, the drive automatically resets it to 0.
U4-04		Shows the total fan operation time as a percentage of the specified maintenance period.
U4-05	Electrolytic Capacitor	Shows the total capacitor usage time as a percentage of the specified maintenance period.
U4-06	Soft charge bypass relay	Shows the number of times the drive is energized as a percentage of the performance life of the inrush circuit.
U4-07	IGBT	Shows the percentage of the maintenance period for the IGBTs.

◆ Alarm Outputs for Maintenance Monitors

You can use *H2-xx [MFDO Function Selection]* to send a message that tells you when a specified component is near the end of its performance life estimate. Set the applicable value to *H2-xx* as shown in Table 7.9 for your component.

When the specified component is near the end of its performance life estimate, the MFDO terminals set for *H2-xx = 2F [Maintenance Notification]* will activate, and the keypad will show an alarm that identifies the component to replace.

Table 7.9 Maintenance Period Alarms

Display	Alarm Name	Cause	Possible Solutions	Digital Outputs (Setting Value in H2-xx)
LT-1	Cooling Fan Maintenance Time	The cooling fan is at 90% of its expected performance life.	Replace the cooling fan, then set *o4-03 = 0 [Fan Operation Time Setting = 0 h]* to reset the cooling fan operation time.	2F
LT-2	Capacitor Maintenance Time	The capacitors for the main circuit and control circuit are at 90% of expected performance life.	Replace the board or the drive. Contact Yaskawa or your nearest sales representative to replace the board.	
LT-3	SoftChargeBypassRelay MainteTime	The soft charge bypass relay is at 90% of its performance life estimate.	Replace the board or the drive. Contact Yaskawa or your nearest sales representative to replace the board.	
LT-4	IGBT Maintenance Time (50%)	The IGBT is at 50% of its expected performance life.	Check the load, carrier frequency, and output frequency.	
TrPC	IGBT Maintenance Time (90%)	The IGBT is at 90% of its expected performance life.	Replace the IGBT or the drive.	10

◆ Related Parameters

Replace the component, then set *o4-03, o4-05, o4-07, and o4-09 [Maintenance Setting] = 0* to reset the Maintenance Monitor. If these parameters are not reset after the corresponding parts have been replaced, the Maintenance Monitor function will continue to count down the performance life from the value that was reached with the old part. If the Maintenance Monitor is not reset, the drive will not have the correct value of the performance life for the new component.

Note:

The maintenance period changes for different operating environments.

Table 7.10 Maintenance Setting Parameters

No.	Name	Function
o4-03	Fan Operation Time Setting	Sets the value from which to start the cumulative drive cooling fan operation time in 10-hour units. **Note:** When *o4-03 = 30* has been set, the drive will count the operation time for the cooling fan from 300 hours and *U4-03 [Cooling Fan Ope Time]* will show *300 h*.
o4-05	Capacitor Maintenance Setting	Sets the value from which to start the count for the main circuit capacitor maintenance period as a percentage.
o4-07	Softcharge Relay Maintenance Set	Sets as a percentage the value from which to start the count for the soft charge bypass relay maintenance time.
o4-09	IGBT Maintenance Setting	Sets the value from which to start the count for the IGBT maintenance period as a percentage.

7.4 Replace Cooling Fans

To replace a cooling fan, contact Yaskawa or your nearest sales representative.

◆ Number of Cooling Fans

Table 7.11 Three-Phase 400 V Class

Model	Cooling Fans	Replacement Procedure	Ref.
4009	1	A	186
4015, 4018	1	B	188
4024 - 4045	2	A	186

◆ Replace the Cooling Fan (Procedure A)

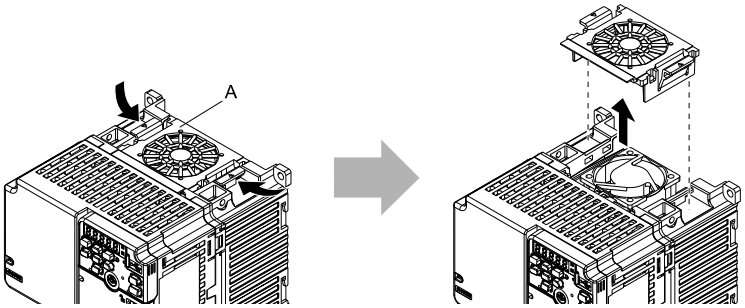
DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

CAUTION! *Burn Hazard. Do not touch a hot drive heatsink. De-energize the drive, wait for a minimum of 15 minutes, then make sure that the heatsink is cool before you replace the cooling fans. If you touch a hot drive heatsink, it can burn you.*

NOTICE: *Use the instructions in this manual to replace the cooling fans. When you do maintenance on the fans, replace all the fans to increase product life. If you install the fans incorrectly, it can cause damage to the drive.*

■ Remove a Fan

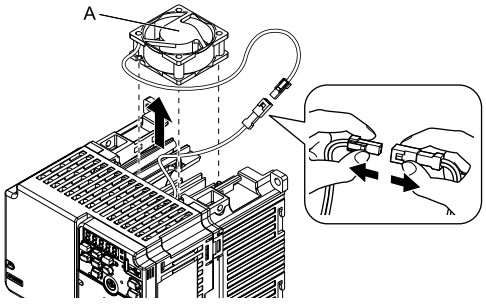
1. To remove the fan finger guard from the drive, push the hooks on the left and right sides of it and pull up.



A - Fan finger guard

Figure 7.1 Remove the Fan Finger Guard

2. Pull the cooling fan straight up from the drive. Disconnect the power supply connector and remove the fan from the drive.



A - Cooling fan

Figure 7.2 Remove the Cooling Fan

■ Install a Fan

Reverse the removal procedure to install a cooling fan.

1. Connect the power supply connector between the drive and cooling fan.

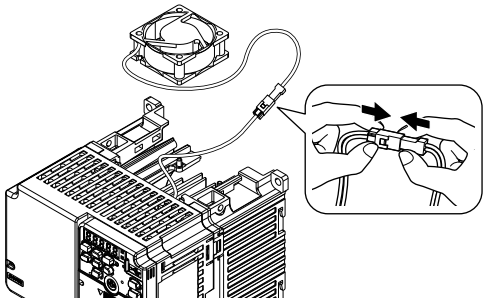
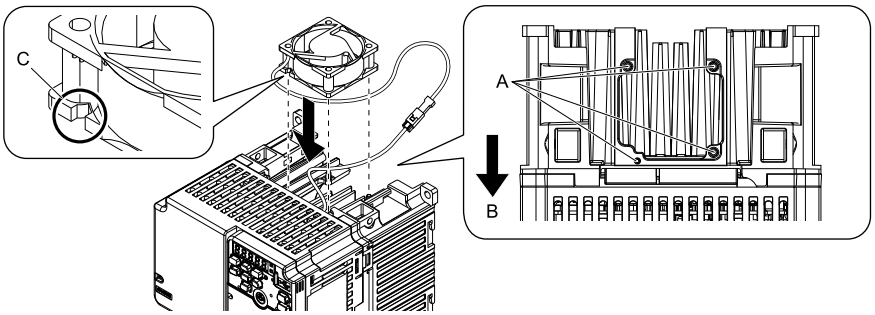


Figure 7.3 Connect the Power Supply Connector

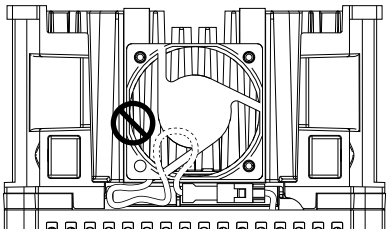
2. Align the notches on the fan with the pins on the drive and install the cooling fans in the drive.



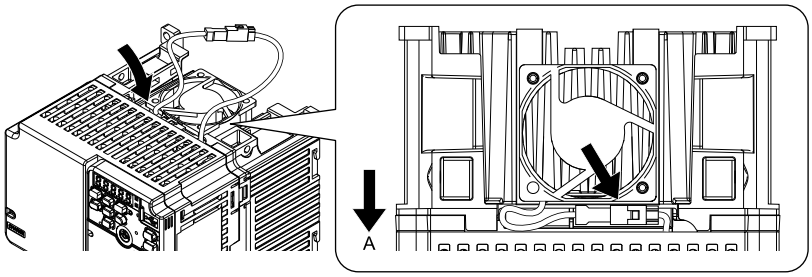
A - Alignment pins on drive
B - Front of drive
C - Notch on fan

Figure 7.4 Install the Cooling Fan

Note:
When you install the cooling fan, make sure that you do not pinch cables between the cooling fan and the drive.



3. Put the cable and connector in the recess of the drive.



A - Front of drive

Figure 7.5 Put the Cable and Connector in the Drive Recess

Note:
The connector installation position is different for different models.

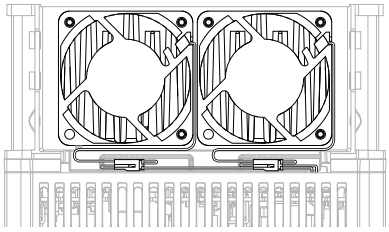


Figure 7.6 Put the Connector in the Recess

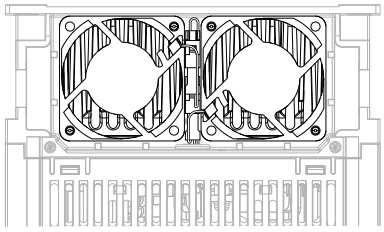


Figure 7.7 Put the Connector in Between the Fans

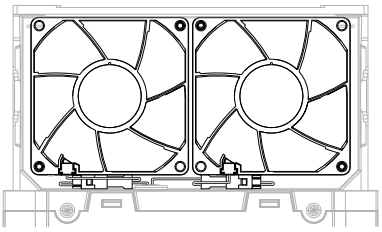


Figure 7.8 Put the Connector in Between the Drive and Fan

4. Insert the fan cover straight until the hook clicks into place.

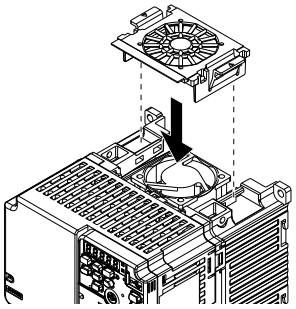


Figure 7.9 Reattach the Fan Finger Guard

5. Energize the drive and set *o4-03 = 0 [Fan Operation Time Setting = 0 h]* to reset the cooling fan operation time.

◆ Replace the Cooling Fan (Procedure B)

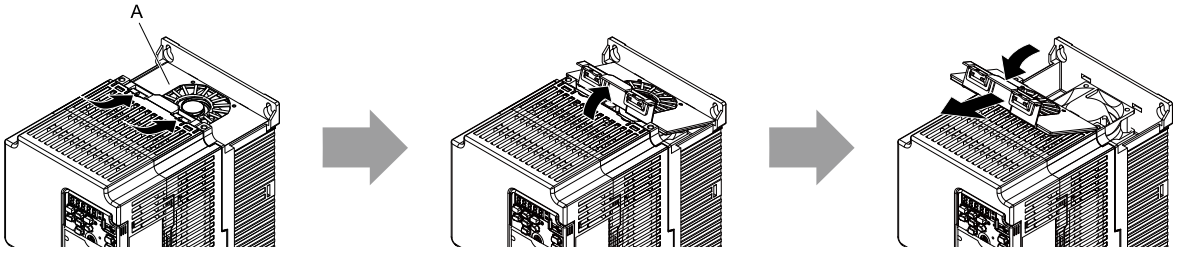
DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

CAUTION! *Burn Hazard. Do not touch a hot drive heatsink. De-energize the drive, wait for a minimum of 15 minutes, then make sure that the heatsink is cool before you replace the cooling fans. If you touch a hot drive heatsink, it can burn you.*

NOTICE: *Use the instructions in this manual to replace the cooling fans. When you do maintenance on the fans, replace all the fans to increase product life. If you install the fans incorrectly, it can cause damage to the drive.*

■ Remove a Fan

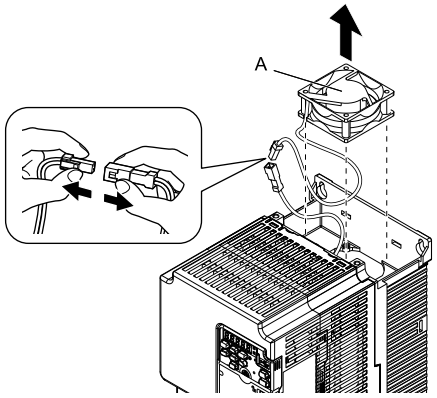
1. Push the tabs toward the back of the drive and pull up to remove the fan finger guard from the drive.



A - Fan finger guard

Figure 7.10 Remove the Fan Finger Guard

2. Pull the cooling fan straight up from the drive. Disconnect the power supply connector and remove the fan from the drive.



A - Cooling fan

Figure 7.11 Remove the Cooling Fan

■ Install the Cooling Fans

Reverse the removal procedure to install a cooling fan.

1. Connect the power supply connector between the drive and cooling fan.

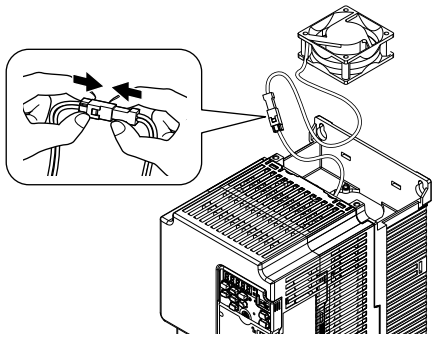
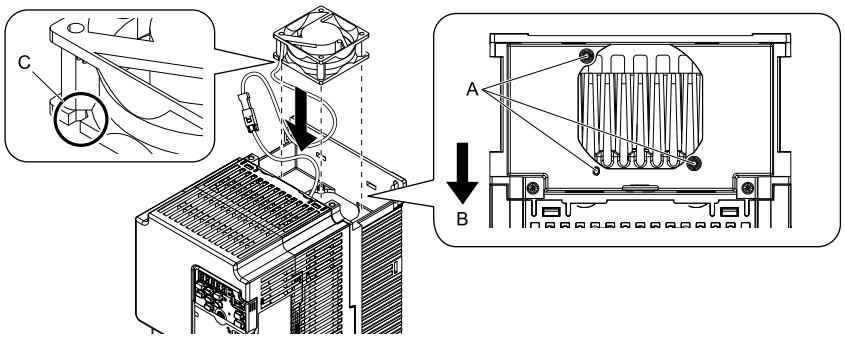


Figure 7.12 Connecting the power supply connector

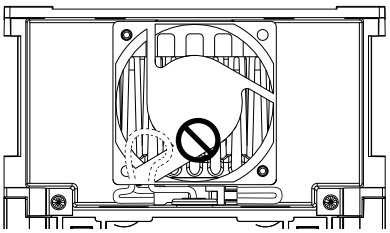
2. Install the cooling fans so that they align with the pins on the drive.



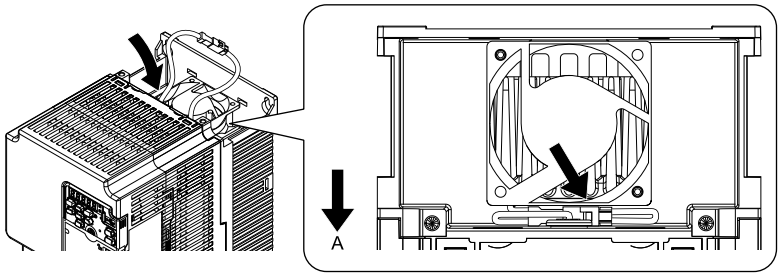
A - Alignment pins on drive
B - Front of drive
C - Notches

Figure 7.13 Installing the cooling fans

Note:
When you install the cooling fan, make sure that you do not pinch cables between the cooling fan and the drive.



3. Put the cable and connector in the recess of the drive.



A - Front of drive

Figure 7.14 Putting the cable and connector in the recess

4. Insert the tabs of the fan cover into the holes in the drive and press in the fan cover until the hook clicks into place.

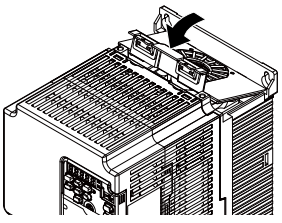


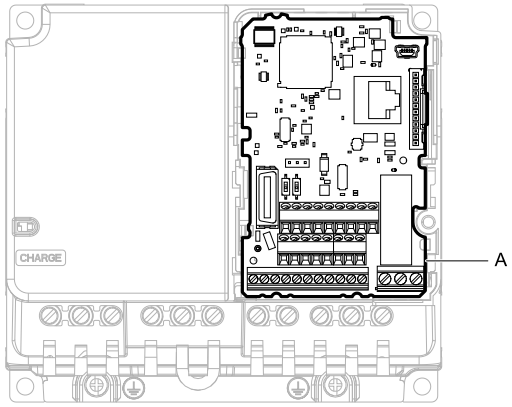
Figure 7.15 Reattach the Fan Finger Guard

5. Energize the drive and set *o4-03 = 0 [Fan Operation Time Setting = 0 h]* to reset the cooling fan operation time.

7.5 Replace the Drive

◆ About the Control Circuit Board

You can remove the control circuit board of the drive and install a new board. If there is a failure in the drive, you can use this feature to easily replace the control circuit board.



A - Control circuit board

Figure 7.16 Control Circuit Terminal Block

◆ Replace the Drive

DANGER! *Electrical Shock Hazard. Disconnect all power to the drive and wait for the time specified on the warning label before you remove covers. Check the drive for dangerous voltages before servicing or repair work. If you do work on the drive when it is energized and there is no cover over the electronic circuits, it will cause serious injury or death from electrical shock. The drive has internal capacitors that stay charged after you de-energize the drive.*

DANGER! *Electrical Shock Hazard. Do not examine, connect, or disconnect wiring on an energized drive. Before servicing, disconnect all power to the equipment and wait for the time specified on the warning label at a minimum. The internal capacitor stays charged after the drive is de-energized. The charge indicator LED extinguishes when the DC bus voltage decreases below 50 Vdc. When all indicators are OFF, measure for dangerous voltages to make sure that the drive is safe. If you do work on the drive when it is energized, it will cause serious injury or death from electrical shock.*

WARNING! *Electrical Shock Hazard. Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive. If personnel are not approved, it can cause serious injury or death.*

NOTICE: *Damage to Equipment. When you touch the drive and circuit boards, make sure that you observe correct electrostatic discharge (ESD) procedures. If you do not follow procedures, it can cause ESD damage to the drive circuitry.*

Before you wire the main circuit terminal block, also refer to *Notes on Wiring the Main Circuit Terminal Block on page 132*.

■ Remove the Control Circuit Board

Remove the front cover and keypad from the drive.

1. Push the tabs to the left that hold the control board to the drive.

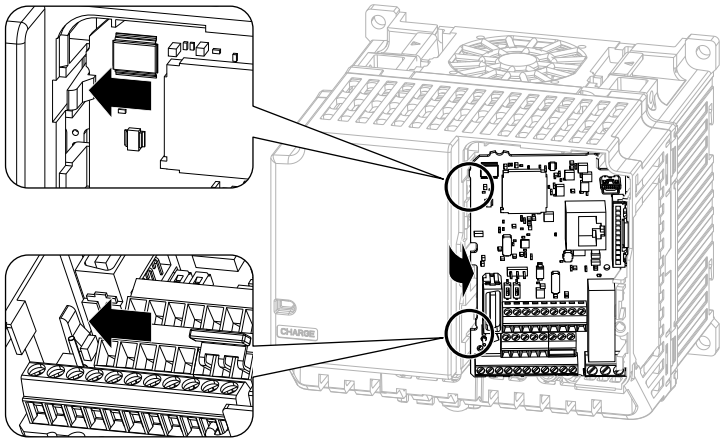


Figure 7.17 Unhook the Tabs

2. Pull the left side of the control circuit board out first.

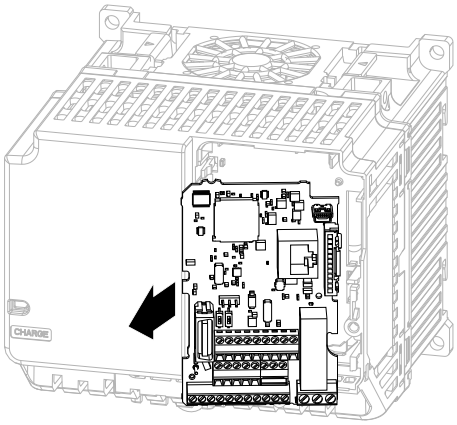


Figure 7.18 Remove the Control Circuit Board

■ Put the Control Circuit Board in a New Drive

Remove the keypad, front cover, and control circuit board of the new drive.
Wire the main circuit terminals of the new drive, then attach the wired control circuit board.

1. Wire the main circuit terminals.

Note:
To wire terminals +1 and +2, remove the jumper between terminals +1 and +2.

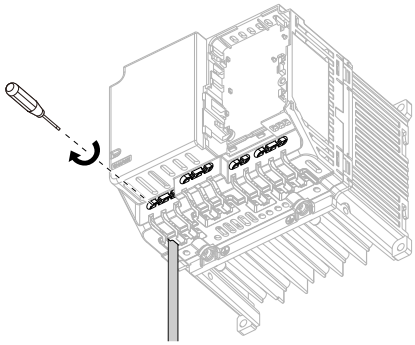


Figure 7.19 Wire the Main Circuit Terminals

2. Attach the wired control circuit board to the drive.
   Push the control circuit board until the hooks click into place on the drive.

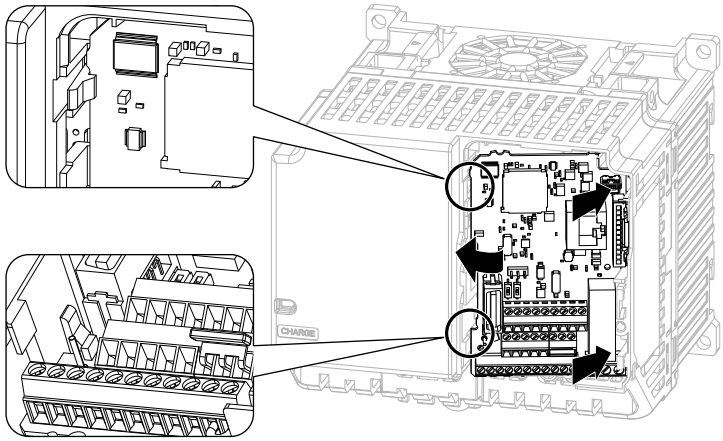


Figure 7.20 Attach the Control Circuit Board

3. Attach the keypad and front cover to the new drive.
4. Energize the drive and set these parameters:
   - *o2-04 [Drive Model (KVA) Selection]*: Set this parameter to the model number of the new drive.
   - *o4-01 to o4-13 [Maintenance Period]*: Reset the performance life monitors for the components.

7.6 Storage Guidelines

The chemicals in the electrolytic capacitors and other electronic parts of the drive change over time. When you store the drive for long periods of time, use the information in this section to help keep the performance life estimates.

◆ Storage Location

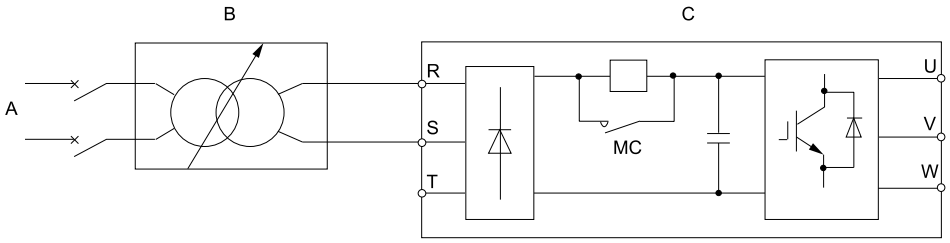
- Temperature and Humidity
  When you store the drive for approximately one month, for example during shipping, you can put the drive in a location where the temperature is -20 °C to +70 °C (-4 °F to +158 °F). Correctly package and store the drive during shipping to prevent vibration and impact damage.
  Do not put the drive in direct sunlight or where there will be condensation or ice. Put the drive in a location where the relative humidity is 95% or less.
- Dust and Oil Mist
  Do not keep the drive locations with dust or oil mist. For example, cement factories and cotton mills.
- Corrosive Gas
  Do not keep the drive in locations with corrosive gas. For example, chemical plants, refineries, and sewage plants.
- Salt Damage
  Do not keep the drive in salty locations. For example, locations near the ocean, and salt damage-designated locations.

Do not keep the drive in unsatisfactory locations. Keep all drives in storage rooms that are safe from unsatisfactory elements.

◆ Regular Application of Power

To prevent deterioration of the capacitors, Yaskawa recommends that you apply power to the drive a minimum of one time each year for a minimum of 30 minutes.

If you store the drive for longer than two years and do not apply power, Yaskawa recommends that you use a variable power source and gradually increase the power from 0 V to the rated drive voltage over a period of 2 to 3 minutes. Apply power for a minimum of 1 hour with no load to reform the main circuit electrolytic capacitor. When you operate the drive after you apply power, wire the drive correctly and check for drive faults, overcurrents, motor vibration, motor speed differences, and other defects during operation.



A - AC power supply
B - Variable power source
C - Drive

Figure 7.21 Power Distribution Method

8

Disposal

8.1 Section Safety 196
8.2 Disposal Instructions 197
8.3 WEEE Directive 198

8.1 Section Safety

⚠WARNING

Electrical Shock Hazard

De-energize the drive and wait 5 minutes minimum until the Charge LED turns off. Remove the front cover and terminal cover to do work on wiring, circuit boards, and other parts. Use terminals for their correct function only.

Incorrect wiring, incorrect ground connections, and incorrect repair of protective covers can cause death or serious injury.

Only let approved personnel install, wire, maintain, examine, replace parts, and repair the drive.

If personnel are not approved, it can cause serious injury or death.

Do not wear loose clothing or jewelry when you do work on the drive. Tighten loose clothing and remove all metal objects, for example watches or rings.

Loose clothing can catch on the drive and jewelry can conduct electricity and cause serious injury or death.

Crush Hazard

Wear eye protection when you do work on the drive.

If you do not use correct safety equipment, it can cause serious injury or death.

⚠CAUTION

Crush Hazard

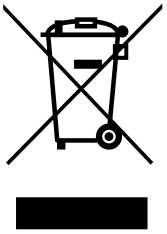
Tighten terminal cover screws and hold the case safely when you move the drive.

If the drive or covers fall, it can cause moderate injury.

8.2 Disposal Instructions

Correctly dispose of the product and packing material as specified by applicable regional, local, and municipal laws and regulations.

8.3 WEEE Directive



The wheelie bin symbol on this product, its manual, or its packaging identifies that you must recycle it at the end of its product life.

You must discard the product at an applicable collection point for electrical and electronic equipment (EEE). Do not discard the product with usual waste.

9

Specifications

9.1 Model Specifications (Three-Phase 400 V Class) 200
9.2 Drive Specifications 201
9.3 Drive Derating 203
9.4 Drive Exterior and Mounting Dimensions 204
9.5 Peripheral Devices and Options 206

9.1 Model Specifications (Three-Phase 400 V Class)

Table 9.1 Ratings (Three-Phase 400 V Class)

Model		Duty Rating	4009	4015	4023	4024	4031	4039	4045
Maximum Applicable Motor Output *1 (kW)		HD	4.0	5.5	7.5	11.0	15.0	18.5	22.0
Maximum Applicable Motor Output *1 (HP)		HD	5	10	10	15	20	25	30
Input	Rated Input Current (A)	HD	10.4	15	20	29	39	50.5	59.7
Output	Rated Output Capacity *2 (kVA)	HD	7	11.3	13.7	18.3	23.6	29.7	34.3
	Rated Output Current (A)	HD	9.2	14.8	18.0	24.0	31.0	39.0	45.0
	Overload Tolerance		• HD: 150% of the rated output current for 60 seconds or 165% for 30 seconds. The permitted frequency of overload is one time each 10 minutes. **Note:** Derating can be necessary for applications that start and stop frequently.						
	Carrier Frequency		HD: 8 kHz without derating the drive capacity. Derate the drive capacity to use values to 15 kHz maximum.						
	Maximum Output Voltage		Three-phase 380 V to 480 V **Note:** The maximum output voltage is proportional to the input voltage.						
	Maximum Output Frequency		• V/f, OLV: 200 Hz						
Measures for Harmonics	DC reactor		External options						
Braking Device	Braking Transistor		Standard internal characteristics						
EMC Filter	EMC Filter IEC61800-3		Factory option Models 4xxxE: Category C2 (Conducted emission)						
Power supply	Rated Voltage/Rated Frequency		Three-phase AC power supply 380 V to 480 V at 50/60 Hz						
	Permitted Voltage Fluctuation		-15% to +10%						
	Permitted Frequency Fluctuation		±5%						
	Input Power (kVA)	HD	9.5	14	18	27	36	47	55

*1 The maximum applicable motor output complies with 380 V motor ratings as specified in Annex G of IEC 60947-4-1. The rated output current of the drive output amps must be equal to or more than the motor rated current.

*2 The rated output capacity is calculated with a rated output voltage of 440 V.

9.2 Drive Specifications

Note:
- To get the OLV specifications, do Rotational Auto-Tuning.
- To get the longest product life, install the drive in an environment that meets the necessary specifications.

Table 9.2 Control Characteristics

Item	Specification
Control Methods	• V/f Control • Open Loop Vector
Frequency Control Range	• V/f, OLV: 0.01 Hz to 200 Hz
Frequency Accuracy (Temperature Fluctuation)	Digital inputs: ±0.01% of the maximum output frequency (-10 °C to +40 °C (14 °F to 104 °F)) Analog inputs: In ±0.1% of the maximum output frequency (25 °C ±10 °C (77 °F ±18 °F))
Frequency Setting Resolution	Digital inputs: 0.01 Hz Analog inputs: 1/2048 of the maximum output frequency (11-bit)
Output Frequency Resolution	0.001 Hz
Frequency Setting Signal	Main speed frequency reference: 0 Vdc to 10 Vdc (20 kΩ), 4 mA to 20 mA (250 Ω), 0 mA to 20 mA (250 Ω) Main speed reference: Pulse train input (maximum 32 kHz)
Starting Torque	• V/f: 150%/3 Hz • OLV: 150%/1 Hz **Note:** Correctly select the drive and motor capacity for this starting torque in these control methods: • OLV
Speed Control Range	• V/f: 1:40 • OLV: 1:100
Torque Limits	You can use parameter settings for different limits in four quadrants in these control methods: • OLV
Acceleration and Deceleration Times	0.0 s to 6000.0 s The drive can set four pairs of different acceleration and deceleration times.
Braking Torque	Approximately 20% without a resistor Approximately 125% with a dynamic braking option • Short-time average deceleration torque Motor output 0.1/0.2 kW: over 150% Motor output 0.4/0.75 kW: over 100% Motor output 1.5 kW: over 50% Motor output 2.2 kW and larger: over 20%, Overexcitation Braking/High Slip Braking allow for approximately 40% **WARNING!** *Set L3-04 = 0 [Stall Prevention during Decel = Disabled] when you operate the drive with:* • *a regenerative converter* • *regenerative unit* • *braking resistor* • *braking resistor unit.* *If you set the parameter incorrectly, the drive can decelerate for too long and cause serious injury or death.* **Note:** • Short-time average deceleration torque refers to the torque needed to decelerate the motor (uncoupled from the load) from the rated speed to zero. Motor characteristics can change the actual specifications. • Motor characteristics change the continuous regenerative torque and short-time average deceleration torque for motors 2.2 kW and larger.

Table 9.3 Protection Functions

Item	Specification
Motor Protection	Electronic thermal overload protection
Momentary Overcurrent Protection	Drive stops when the output current is more than 200% of the HD output current.
Overload Protection	Drive stops when the output current is more than these overload tolerances: • HD: 165% of the rated output current for 60 seconds. **Note:** If output frequency < 6 Hz, the drive can trigger the overload protection function when the output current is in the overload tolerance range.
Overvoltage Protection	400 V class: Stops when the DC bus voltage is more than approximately 820 V
Undervoltage Protection	Three-phase 400 V class: Stops when the DC bus voltage decreases to less than approximately 380 V

Item	Specification
Momentary Power Loss Ride-thru	Stops when power loss is longer than 15 ms and continues operation if power loss is shorter than 2 s (depending on parameter settings). **Note:** • Load size and motor speed can cause the stop time to be shorter. • Drive capacity will change the continuous operation time. A Momentary Power Loss Recovery Unit is necessary to continue operation through a 2 s power loss on models 4009 to 4018.
Heatsink Overheat Protection	Thermistor
Braking Resistor Overheat Protection	Overheat detection for braking resistor (optional ERF-type, 3% ED)
Stall Prevention	Stall prevention is available during acceleration, deceleration, and during run.
Ground Fault Protection	Electronic circuit protection **Note:** This protection detects ground faults during run. The drive will not provide protection when: • There is a low-resistance ground fault for the motor cable or terminal block • Energizing the drive when there is a ground fault.
DC Bus Charge LED	Charge LED illuminates when DC bus voltage is more than 50 V.

Table 9.4 Environment

Item	Specification
Area of Use	Indoors
Power Supply	Overvoltage Category III
Ambient Temperature Setting	IP20/UL Open Type: -10°C to +50 °C (14 °F to 122 °F) IP20/UL Type 1: -10 °C to +40 °C (14 °F to 104 °F) • When you install the drive in an enclosure, use a cooling fan or air conditioner to keep the internal air temperature in the permitted range. • Do not let the drive freeze.
Humidity	95% RH or less Do not let condensation form on the drive.
Storage Temperature	-20 °C to +70 °C (-4 °F to +158 °F) (short-term temperature during transportation)
Surrounding Area	Pollution degree 2 or less Install the drive in an area without: • Oil mist, corrosive or flammable gas, or dust • Metal powder, oil, water, or other unwanted materials • Radioactive materials or flammable materials, including wood • Harmful gas or fluids • Salt • Direct sunlight
Altitude	1000 m (3281 ft) Maximum **Note:** Derate the output current by 1% for each 100 m (328 ft) to install the drive in altitudes between 1000 m to 4000 m (3281 ft to 13123 ft). It is not necessary to derate the rated voltage in these conditions: • Installing the drive at 2000 m (6562 ft) or lower • Installing the drive between 2000 m to 4000 m (6562 ft to 13123 ft) and grounding the neutral point on the power supply. Contact Yaskawa or your nearest sales representative when not grounding the neutral point.
Vibration	• 10 Hz to 20 Hz: 1 G (9.8 m/s^2, 32.15 ft/s^2) • 20 Hz to 55 Hz: 0.6 G (5.9 m/s^2, 19.36 ft/s^2)
Installation Orientation	Install the drive vertically for sufficient airflow to cool the drive.

Table 9.5 Standard

Item	Specification
Harmonized Standard	• UL 61800-5-1 • EN 61800-3 • EN 61800-5-1 • Two Safe Disable inputs and one EDM output according to EN ISO 13849-1 (Cat.3, PL e), EN 61800-5-2 SIL3
Enclosure protection design	IP20/UL Open Type IP20/UL Type 1 **Note:** Install a UL Type 1 kit (optional) on an IP20/UL Open Type drive to change the drive to an IP20/UL Type 1 enclosure.

9.3 Drive Derating

You must derate the drive capacity to operate the drive above the rated temperature, altitude, and default carrier frequency.

◆ Carrier Frequency Settings and Rated Current Values

The tables in this section show how the drive rated output current changes when the *C6-02 [Carrier Frequency Selection]* value changes. The output current value changes linearly as the carrier frequency changes. You can use the values from the tables to calculate a frequency that is not shown.

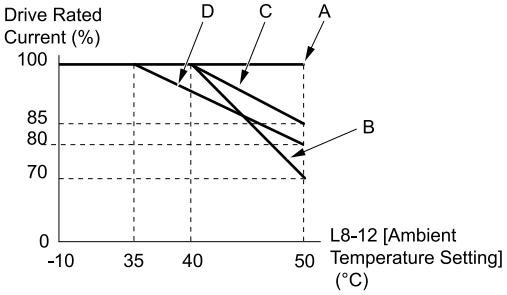
■ Three-Phase 400 V Class

Table 9.6 Carrier Frequency and Rated Current Derating (Three-Phase 400 V)

Model	Rated Current (A)					
	2 kHz	5 kHz	8 kHz	10 kHz	12.5 kHz	15 kHz
4009	9.2	9.2	9.2	8.1	6.8	5.5
4015	14.8	14.8	14.8	13.1	11.0	8.9
4018	18.0	18.0	18.0	13.1	11.0	11
4024	24.0	24.0	24.0	21.3	17.8	14
4031	31.0	31.0	31.0	27.5	23.0	19
4039	39.0	39.0	39.0	34.5	29.0	23
4045	45.0	45.0	45.0	39.9	33.4	27

◆ Derating Depending on Ambient Temperature

When you install drives in a place where ambient temperatures are higher than the rated conditions or install drives side-by-side in the enclosure panel, set *L8-12 [Ambient Temperature Setting]* and *L8-35 [Installation Method Selection]*. Derate the output current as specified in Figure 9.1.



A - L8-35 = 0 [IP20/UL Open Type]
B - L8-35 = 1 [Side-by-Side Mounting]
C - L8-35 = 2 [IP20/UL Type 1]
D - L8-35 = 3 [Finless Installation]

Figure 9.1 Derating Depending on Drive Installation Method

Refer to *L8-12: Ambient Temperature Setting on page 308* and *L8-35: Installation Method Selection on page 309* for details.

◆ Altitude Derating

Install the drive in a location that has an altitude of 1000 m (3281 ft) or lower.

Derate the output current by 1% for each 100 m (328 ft) to install the drive in altitudes between 1000 to 4000 m (3281 to 13123 ft).

It is not necessary to derate the rated voltage in these conditions:

- Installing the drive at 2000 m (6562 ft) or lower
- Installing the drive between 2000 to 4000 m (6562 to 13123 ft) and grounding the neutral point on the power supply.
  If you do not ground the drive with a neutral network, contact Yaskawa or your nearest sales representative.

9.4 Drive Exterior and Mounting Dimensions

◆ IP20/UL Open Type

■ 4009

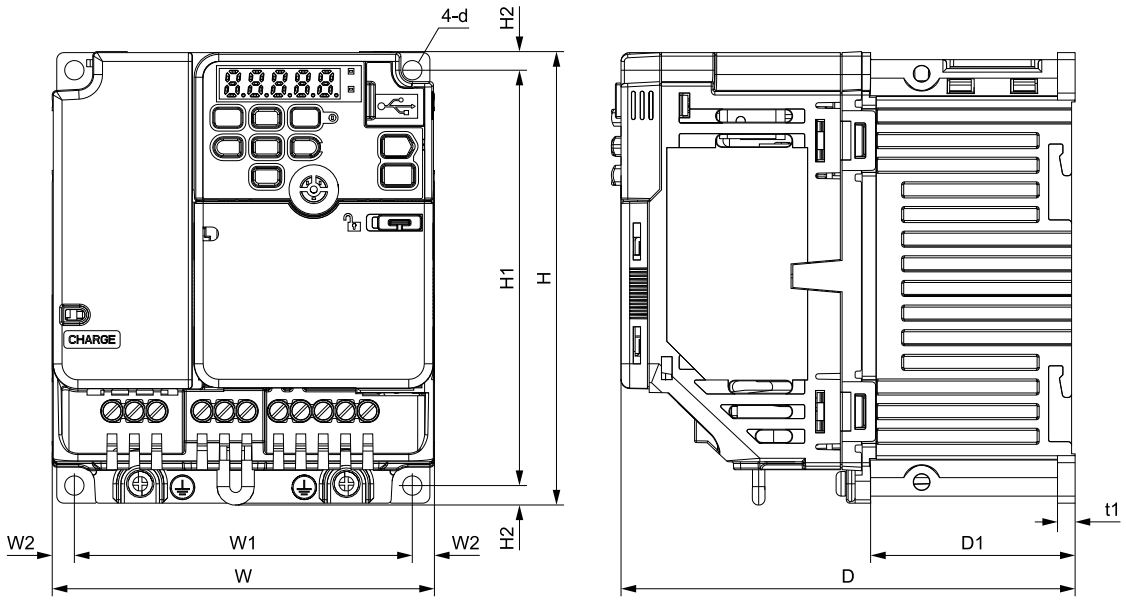


Figure 9.2 Exterior and Mounting Dimensions

Table 9.7 Three-Phase 400 V Class (IP20/UL Open Type, without Built-in EMC Filter)

Model	Dimensions mm (in)										Est. Weight kg (lb)
	W	H	D	W1	W2	H1	H2	D1	t1	d	
4009A	140 (5.51)	128 (5.04)	143 (5.63)	128 (5.04)	6 (0.24)	118 (4.65)	5 (0.20)	65 (2.56)	5 (0.20)	M5	2.0 (4.4)

Table 9.8 Three-Phase 400 V Class (IP20/UL Open Type, with Built-in EMC Filter)

Model	Dimensions mm (in)										Est. Weight kg (lb)
	W	H	D	W1	W2	H1	H2	D1	t1	d	
4009E	140 (5.51)	128 (5.04)	193 (7.60)	128.0 (5.04)	6 (0.24)	118 (4.65)	5 (0.20)	65 (2.56)	5 (0.20)	M5	2.6 (5.7)

■ 4015 to 4045

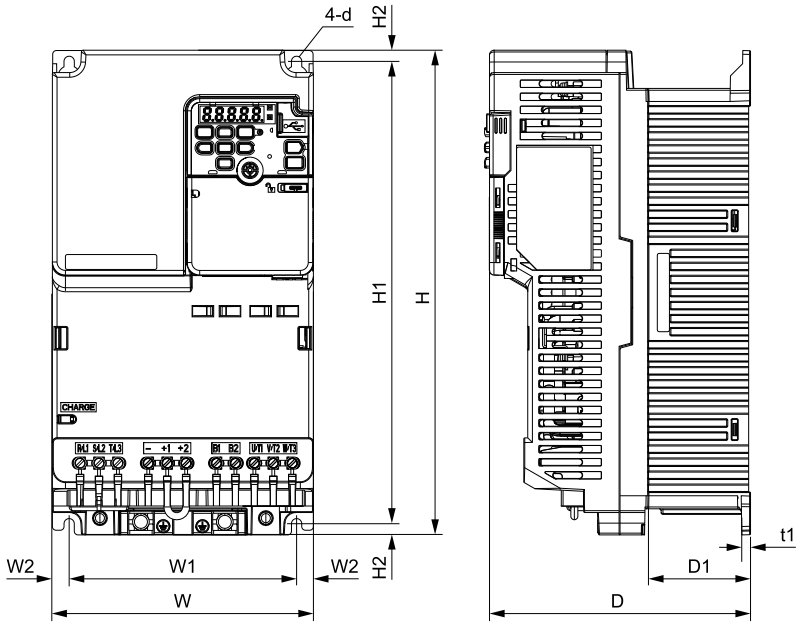


Figure 9.3 Exterior and Mounting Dimensions

Table 9.9 Three-Phase 400 V Class (IP20/UL Open Type, without Built-in EMC Filter)

Model	Dimensions mm (in)										Est. Weight kg (lb)
	W	H	D	W1	W2	H1	H2	D1	t1	d	
4015A	140 (5.51)	260 (10.24)	140 (5.51)	122 (4.80)	9 (0.35)	248 (9.76)	6 (0.24)	55 (2.17)	5 (0.20)	M5	3.0 (6.6)
4018A	140 (5.51)	260 (10.24)	140 (5.51)	122 (4.80)	9 (0.35)	248 (9.76)	6 (0.24)	55 (2.17)	5 (0.20)	M5	3.2 (7.1)
4024A	180 (7.09)	300 (11.81)	143 (5.63)	160 (6.30)	10 (0.39)	284 (11.18)	8 (0.31)	55 (2.17)	5 (0.20)	M5	4.6 (10.2)
4031A	180 (7.09)	300 (11.81)	143 (5.63)	160 (6.30)	10 (0.39)	284 (11.18)	8 (0.31)	55 (2.17)	5 (0.20)	M5	4.8 (10.6)
4039A	190 (7.48)	350 (13.78)	204 (8.03)	160 (6.30)	15 (0.59)	336 (13.23)	7 (0.28)	94 (3.70)	5 (0.20)	M6	6.5 (14.3)
4045A	190 (7.48)	350 (13.78)	204 (8.03)	160 (6.30)	15 (0.59)	336 (13.23)	7 (0.28)	94 (3.70)	5 (0.20)	M6	6.5 (14.3)

Table 9.10 Three-Phase 400 V Class (IP20/UL Open Type, with Built-in EMC Filter)

Model	Dimensions mm (in)										Est. Weight kg (lb)
	W	H	D	W1	W2	H1	H2	D1	t1	d	
4015E	140 (5.51)	260 (10.24)	196 (7.72)	122 (4.80)	9 (0.35)	248 (9.76)	6 (0.24)	55 (2.17)	5 (0.20)	M5	3.9 (8.6)
4018E	140 (5.51)	260 (10.24)	196 (7.72)	122 (4.80)	9 (0.35)	248 (9.76)	6 (0.24)	55 (2.17)	5 (0.20)	M5	3.9 (8.6)
4024E	180 (7.09)	300 (11.81)	196 (7.72)	160 (6.30)	10 (0.39)	284 (11.18)	8 (0.31)	55 (2.17)	5 (0.20)	M5	5.5 (12.1)
4031E	180 (7.09)	300 (11.81)	196 (7.72)	160 (6.30)	10 (0.39)	284 (11.18)	8 (0.31)	55 (2.17)	5 (0.20)	M5	5.5 (12.1)
4039E	190 (7.48)	350 (13.78)	251 (9.88)	160 (6.30)	15 (0.59)	336 (13.23)	7 (0.28)	94 (3.70)	5 (0.20)	M6	8.0 (17.6)
4045E	190 (7.48)	350 (13.78)	251 (9.88)	160 (6.30)	15 (0.59)	336 (13.23)	7 (0.28)	94 (3.70)	5 (0.20)	M6	8.5 (18.7)

9.5 Peripheral Devices and Options

These tables show the available peripheral devices and options for the drive. Contact Yaskawa or your nearest sales representative to make an order.

- Selection: Refer to the drive catalog for information about available products.
- Installation and wiring: Refer to the instruction manual for each option.

Table 9.11 Main Circuit Options

Name	Model	Intended Use
AC reactor	LR3 series and ALR3 series	To improve the drive input power factor. • To prevent damage to the drive when the power supply capacity is large. You must only use this option when the power supply capacity is more than 600 kVA. • To decrease harmonic current. • To improve the power supply total power factor.
Braking Resistor	RH Series	To decrease the regenerative energy of the motor and decrease the deceleration time (duty cycle of 3% ED). You must also use the installation attachment.

Table 9.12 Attachments

Name	Model	Intended Use
Heatsink External Mounting Kit	ZPSA-GA50Vx-x	Use this option to install the heatsink outside of the control panel.
UL Type 1 Kit	ZBAA-GA50Vx-x	To change an IP20/UL Open Type drive to an IP20/UL Type 1 drive.
DIN Rail Attachment	ZPZ-GA50Vx	To use a DIN rail to install the drive.

Table 9.13 Engineering Tools

Name	Model	Intended Use
DriveWizard	-	To use a PC to program drives and manage parameters.

10

Parameter List

10.1 How to Read the Parameter List 208
10.2 Parameter Groups 209
10.3 A: Initialization Parameters 210
10.4 b: Application 213
10.5 C: Tuning 214
10.6 d: Reference Settings 216
10.7 E: Motor Parameters 218
10.8 H: Terminal Functions 220
10.9 L: Protection Functions 226
10.10 n: Special Adjustment 231
10.11 o: Keypad-Related Settings 232
10.12 S: Elevator Parameters 235
10.13 T: Motor Tuning 237
10.14 U: Monitors 238
10.15 Parameters that Change from the Default Settings with A1-02 [Control Method Selection] 247
10.16 Defaults by Drive Model 248

10.1 How to Read the Parameter List

◆ Icons and Terms that Identify Parameters and Control Modes

Icon	Description
V/f	The parameter is available when operating the drive with V/f Control.
OLV	The parameter is available when operating the drive with Open Loop Vector Control.
Hex.	Hexadecimal numbers that represent MEMOBUS addresses to change parameters over network communication.
RUN	The parameter can be changed settings during run.
Expert	The parameter that is available in Expert Mode only. *1

*1 Set *A1-01 = 3 [Access Level Selection = Expert Level]* to display and set Expert Mode parameters on the keypad.

Note:
Gray icons identify parameters that are not available in the specified control method.

10.2 Parameter Groups

Represents the type of product parameters.

Parameter	Name	Ref.
A	**Initialization Parameters**	210
A1	Initialization	210
A2	User Parameters	210
B	**Application**	213
b1	Operation Mode Selection	213
b4	Timer Function	213
b6	Dwell Function	213
C	**Tuning**	214
C1	Accel & Decel Time	214
C2	Jerk Characteristics	214
C3	Slip Compensation	215
C4	Torque Compensation	215
C6	Carrier Frequency	215
D	**Reference Settings**	216
d1	Speed Reference	216
E	**Motor Parameters**	218
E1	V/f Pattern for Motor 1	218
E2	Motor Parameters	218
H	**Terminal Functions**	220
H1	Digital Inputs	220
H2	Digital Outputs	222
H4	Analog Outputs	225
L	**Protection Functions**	226
L1	Motor Protection	226
L2	Power Loss Ride Through	226
L3	Stall Prevention	226

Parameter	Name	Ref.
L4	Speed Detection	227
L5	Fault Restart	227
L6	Torque Detection	227
L7	Torque Limit	228
L8	Drive Protection	228
n	**Special Adjustment**	231
n2	Auto Freq Regulator (AFR)	231
n6	Online Tuning	231
o	**Keypad-Related Settings**	232
o1	Keypad Display	232
o2	Keypad Operation	232
o3	Copy Keypad Function	233
o4	Maintenance Monitors	233
S	**Elevator Parameters**	235
S1	Brake Sequence	235
S2	Slip Compensation for Elevators	235
S4	Rescue Operation	236
S6	Error Detection	236
T	**Motor Tuning**	237
T1	Induction Motor Auto-Tuning	237
U	**Monitors**	238
U1	Operation Status Monitors	238
U2	Fault Trace	239
U3	Fault History	241
U4	Maintenance Monitors	243
U6	Operation Status Monitors	245

10.3 A: Initialization Parameters

◆ A1: Initialization

No. (Hex.)	Name	Description	Default (Range)	Ref.
A1-00 (0100) RUN	Language Selection	V/f OLV Sets the language for the LCD keypad. **Note:** • When you use *A1-03 [Initialize Parameters]* to initialize the drive, the drive will not reset this parameter. 0 : English 1 : Japanese 2 : German 3 : French 4 : Italian 5 : Spanish 6 : Portuguese 7 : Chinese 8 : Czech 9 : Russian 10 : Turkish 11 : Polish 12 : Greek	0 (0 - 12)	250
A1-01 (0101) RUN	Access Level Selection	V/f OLV Sets user access to parameters. The access level controls which parameters the keypad will display, and which parameters the user can set. 0 : Operation Only 1 : User Parameters 2 : Advanced Level 3 : Expert Level	2 (0 - 3)	250
A1-02 (0102)	Control Method Selection	V/f OLV Sets the control method for the drive application and the motor. 0 : V/f Control 2 : Open Loop Vector	0 (0, 2)	251
A1-03 (0103)	Initialize Parameters	V/f OLV Sets parameters to default values. 0 : No Initialization 1110 : User Initialization 2220 : 2-Wire Initialization	0 (0 - 2220)	251
A1-04 (0104)	Password	V/f OLV Entry point for the password set in *A1-05 [Password Setting]*. The user can view the settings of parameters that are locked without entering the password. Enter the correct password in this parameter to change parameter settings.	0000 (0000 - 9999)	252
A1-05 (0105)	Password Setting	V/f OLV Set the password to lock parameters and prevent changes to parameter settings. Enter the correct password in *A1-04 [Password]* to unlock parameters and accept changes.	0000 (0000 - 9999)	252

◆ A2: User Parameters

No. (Hex.)	Name	Description	Default (Setting Range)	Ref.
A2-01 (0106)	User Parameter 1	V/f OLV Sets the parameter number to show on the first line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	A1-02 (Determined by A1-06)	253
A2-02 (0107)	User Parameter 2	V/f OLV Sets the parameter number to show on the second line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	b1-01 (Determined by A1-06)	253
A2-03 (0108)	User Parameter 3	V/f OLV Sets the parameter number to show on the third line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	b1-02 (Determined by A1-06)	253
A2-04 (0109)	User Parameter 4	V/f OLV Sets the parameter number to show on the fourth line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	b1-03 (Determined by A1-06)	253
A2-05 (010A)	User Parameter 5	V/f OLV Sets the parameter number to show on the fifth line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	C1-01 (Determined by A1-06)	253

No. (Hex.)	Name	Description	Default (Setting Range)	Ref.
A2-06 (010B)	User Parameter 6	V/f OLV Sets the parameter number to show on the sixth line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	C1-02 (Determined by A1-06)	253
A2-07 (010C)	User Parameter 7	V/f OLV Sets the parameter number to show on the seventh line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	C6-01 (Determined by A1-06)	253
A2-08 (010D)	User Parameter 8	V/f OLV Sets the parameter number to show on the eighth line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	C6-02 (Determined by A1-06)	253
A2-09 (010E)	User Parameter 9	V/f OLV Sets the parameter number to show on the ninth line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	d1-01 (Determined by A1-06)	253
A2-10 (010F)	User Parameter 10	V/f OLV Sets the parameter number to show on the 10th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	d1-02 (Determined by A1-06)	253
A2-11 (0110)	User Parameter 11	V/f OLV Sets the parameter number to show on the 11th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	d1-03 (Determined by A1-06)	253
A2-12 (0111)	User Parameter 12	V/f OLV Sets the parameter number to show on the 12th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	d1-04 (Determined by A1-06)	253
A2-13 (0112)	User Parameter 13	V/f OLV Sets the parameter number to show on the 13th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	d1-17 (Determined by A1-06)	253
A2-14 (0113)	User Parameter 14	V/f OLV Sets the parameter number to show on the 14th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E1-01 (Determined by A1-06)	253
A2-15 (0114)	User Parameter 15	V/f OLV Sets the parameter number to show on the 15th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E1-03 (Determined by A1-06)	253
A2-16 (0115)	User Parameter 16	V/f OLV Sets the parameter number to show on the 16th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E1-04 (Determined by A1-06)	253
A2-17 (0116)	User Parameter 17	V/f OLV Sets the parameter number to show on the 17th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E1-05 (Determined by A1-06)	253
A2-18 (0117)	User Parameter 18	V/f OLV Sets the parameter number to show on the 18th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E1-06 (Determined by A1-06)	253
A2-19 (0118)	User Parameter 19	V/f OLV Sets the parameter number to show on the 19th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E1-09 (Determined by A1-06)	253
A2-20 (0119)	User Parameter 20	V/f OLV Sets the parameter number to show on the 20th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E1-13 (Determined by A1-06)	253
A2-21 (011A)	User Parameter 21	V/f OLV Sets the parameter number to show on the 21st line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E2-01 (Determined by A1-06)	253
A2-22 (011B)	User Parameter 22	V/f OLV Sets the parameter number to show on the 22nd line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E2-04 (Determined by A1-06)	253
A2-23 (011C)	User Parameter 23	V/f OLV Sets the parameter number to show on the 23rd line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	E2-11 (Determined by A1-06)	253
A2-24 (011D)	User Parameter 24	V/f OLV Sets the parameter number to show on the 24th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	H4-02 (Determined by A1-06)	253
A2-25 (011E)	User Parameter 25	V/f OLV Sets the parameter number to show on the 25th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	L1-01 (Determined by A1-06)	253
A2-26 (011F)	User Parameter 26	V/f OLV Sets the parameter number to show on the 26th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	L3-04 (Determined by A1-06)	253
A2-27 (0120)	User Parameter 27	V/f OLV Sets the parameter number to show on the 27th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	- (Determined by A1-06)	253

No. (Hex.)	Name	Description	Default (Setting Range)	Ref.
A2-28 (0121)	User Parameter 28	V/f OLV Sets the parameter number to show on the 28th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	- (Determined by A1-06)	253
A2-29 (0122)	User Parameter 29	V/f OLV Sets the parameter number to show on the 29th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	- (Determined by A1-06)	253
A2-30 (0123)	User Parameter 30	V/f OLV Sets the parameter number to show on the 30th line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	- (Determined by A1-06)	253
A2-31 (0124)	User Parameter 31	V/f OLV Sets the parameter number to show on the 31st line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	- (Determined by A1-06)	253
A2-32 (0125)	User Parameter 32	V/f OLV Sets the parameter number to show on the 32nd line in Setup Mode. You can select a maximum of 32 parameters for the drive and set them to parameters *A2-01* to *A2-32*.	- (Determined by A1-06)	253
A2-33 (0126)	User Parameter Auto Selection	V/f OLV Sets the automatic save feature for changes to parameters *A2-17 to A2-32 [User Parameters 17 to 32]*. 0 : Disabled: Manual Entry Required 1 : Enabled: Auto Save Recent Parms	Determined by A1-06 (0, 1)	253

10.4 b: Application

◆ b1: Operation Mode Selection

No. (Hex.)	Name	Description	Default (Range)	Ref.
b1-02 (0181)	Run Command Selection 1	V/f OLV Sets the Up/Down command source in Remote mode. Wire the motor so the elevator goes up when an Up command is issued. 0 : Keypad 1 : Control Circuit Terminal	1 (0 - 1)	254
b1-03 (0182)	Stopping Method Selection	V/f OLV Sets the method to stop the motor after removing a Run command or entering a Stop command. 0 : Ramp to Stop 1 : Coast to Stop	0 (0 - 1)	254
b1-08 (0187)	Run Command Select in PRG Mode	V/f OLV Sets the conditions for the drive to accept an Up/Down command entered from an external source when using the keypad to set parameters. 0 : Disregard RUN while Programming 1 : Accept RUN while Programming 2 : Allow Programming Only at Stop	0 (0 - 2)	254
b1-14 (01C3)	Phase Order Selection	V/f OLV Sets the phase order for output terminals U/T1, V/T2, and W/T3. This parameter can align the Forward Run command from the drive and the forward direction of the motor without changing wiring. 0 : Standard 1 : Switch Phase Order	0 (0, 1)	255

◆ b4: Timer Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
b4-01 (01A3)	Timer Function ON-Delay Time	V/f OLV Sets the ON-delay time for the timer input.	0.0 s (0.0 - 3000.0 s)	256
b4-02 (01A4)	Timer Function OFF-Delay Time	V/f OLV Sets the OFF-delay time for the timer input.	0.0 s (0.0 - 3000.0 s)	256

◆ b6: Dwell Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
b6-01 (01B6)	Dwell Reference at Start	V/f OLV Sets the output frequency that the drive will hold momentarily when the motor starts.	0.0 (Determined by A1-02)	256
b6-02 (01B7)	Dwell Time at Start	V/f OLV Sets the length of time that the drive will hold the output frequency when the motor starts.	0.0 s (0.0 - 10.0 s)	257
b6-03 (01B8)	Dwell Reference at Stop	V/f OLV Sets the output frequency that the drive will hold momentarily when ramping to stop the motor.	0.0 (Determined by A1-02)	257
b6-04 (01B9)	Dwell Time at Stop	V/f OLV Sets the length of time for the drive to hold the output frequency when ramping to stop the motor.	0.0 s (0.0 - 10.0 s)	257

10.5 C: Tuning

◆ C1: Accel & Decel Time

No. (Hex.)	Name	Description	Default (Range)	Ref.
C1-01 (0200) RUN	Acceleration Ramp 1	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)	259
C1-02 (0201) RUN	Deceleration Ramp 1	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)	259
C1-03 (0202) RUN	Acceleration Ramp 2	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)	259
C1-04 (0203) RUN	Deceleration Ramp 2	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)	259
C1-05 (0204) RUN	Acceleration Ramp 3	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)	259
C1-06 (0205) RUN	Deceleration Ramp 3	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)	259
C1-07 (0206) RUN	Acceleration Ramp 4	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)	259
C1-08 (0207) RUN	Deceleration Ramp 4	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)	260
C1-09 (0208) RUN	Fast Stop Ramp	V/f OLV Sets the ramp that the drive will decelerate to zero for Emergency Stop. **Note:** • Decelerating too quickly can cause an *ov [Overvoltage]* fault that shuts off the drive while the motor to coasts to a stop. Set a Fast Stop time in *C1-09* that prevents motor coasting and makes sure that the motor stops quickly and safely. • When *L2-29 = 0 [Kinetic Energy Backup Method = Single Drive KEB Ride-Thru 1]* and you do KEB Auto-Tuning, the drive will automatically set *C1-09*. If you must not change the Fast Stop time, do not do KEB Tuning.	1.50 s (0.00 - 600.00 s)	260
C1-10 (0209)	Accel/Decel Time Setting Units	V/f OLV Sets the setting units for *C1-01 to C1-08 [Accel/Decel Times 1 to 4]*, *C1-09 [Fast Stop Time]*, *L2-06 [Kinetic Energy Backup Decel Time]*, and *L2-07 [Kinetic Energy Backup Accel Time]*. 0 : 0.01 s (0.00 to 600.00 s) 1 : 0.1 s (0.0 to 6000.0 s)	0 (0, 1)	260
C1-11 (020A)	Accel/Decel Switching Speed	V/f OLV Sets the speed at which the drive will automatically change acceleration and deceleration times.	0.0% (0.0 - 100.0%)	260
C1-15 (0260)	Inspect Dec Ramp	V/f OLV Sets the deceleration ramp used for inspection run.	0.00 s (0.00 - 2.00 s)	261

◆ C2: Jerk Characteristics

No. (Hex.)	Name	Description	Default (Range)	Ref.
C2-01 (020B)	Jerk at Accel Start	V/f OLV Sets the jerks acceleration time at start.	0.50 s (0.00 - 10.00 s)	261
C2-02 (020C)	Jerk at Accel End	V/f OLV Sets the jerk acceleration time at completion.	0.50 s (0.00 - 10.00 s)	261
C2-03 (020D)	Jerk at Decel Start	V/f OLV Sets the jerk deceleration time at start.	0.50 s (0.00 - 10.00 s)	261
C2-04 (020E)	Jerk at Decel End	V/f OLV Sets the jerk deceleration time at completion.	0.50 s (0.00 - 10.00 s)	261
C2-05 (020F)	Jerk below Leveling Speed	V/f OLV Sets the jerk deceleration time below leveling speed.	0.50 s (0.00 - 10.00 s)	262

◆ C3: Slip Compensation

No. (Hex.)	Name	Description	Default (Range)	Ref.
C3-01 (020F) RUN	Slip Compensation Gain	V/f OLV Sets the gain for the slip compensation function. Usually it is not necessary to change this setting. **Note:** Correctly set these parameters before changing the slip compensation gain: • *E2-01 [Motor Rated Current (FLA)]* • *E2-02 [Motor Rated Slip]* • *E2-03 [Motor No-Load Current]*	1.0 (0.0 - 2.5)	262
C3-02 (0210) RUN	Slip Compensation Delay Time	V/f OLV Sets the slip compensation delay time when speed is unstable or when the slip compensation response is too slow. Usually it is not necessary to change this setting.	2000 ms (0 - 10000 ms)	262
C3-03 (0211)	Slip Compensation Limit	V/f OLV Sets the upper limit for the slip compensation function as a percentage of the motor rated slip.	200% (0 - 250%)	262
C3-04 (0212)	Slip Compensation at Regen	V/f OLV Sets the slip compensation function during regenerative operation. 0 : Disabled 1 : Enabled Above 6Hz 2 : Enabled Above C3-15	0 (0 - 2)	263
C3-05 (0213)	Output Voltage Limit Selection	V/f OLV Sets the automatic reduction of motor magnetic flux when the output voltage is saturated. 0 : Disabled 1 : Enabled	0 (0, 1)	263

◆ C4: Torque Compensation

No. (Hex.)	Name	Description	Default (Range)	Ref.
C4-01 (0215) RUN	Torque Compensation Gain	V/f OLV Sets the gain for the torque compensation function.	1.00 (0.00 - 2.50)	264
C4-02 (0216) RUN	Torque Compensation Delay Time	V/f OLV Sets the torque compensation delay time. Usually it is not necessary to change this setting.	Determined by A1-02 (0 - 60000 ms)	264
C4-03 (0217)	Torque Compensation @ FWD Start	V/f OLV Set the amount of torque reference for forward start as a percentage of the motor rated torque.	0.0% (0.0 - 200.0%)	264
C4-04 (0218)	Torque Compensation @ REV Start	V/f OLV Sets the amount of torque reference for reverse start as a percentage of the motor rated torque.	0.0% (-200.0 - 0.0%)	264
C4-05 (0219)	Torque Compensation Time	V/f OLV Sets the starting torque constant to use with *C4-03 and C4-04 [Torque Compensation @ FWD/REV Start]*.	10 ms (0 - 200 ms)	264

◆ C6: Duty & Carrier Frequency

No. (Hex.)	Name	Description	Default (Range)	Ref.
C6-02 (0224)	Carrier Frequency Selection	V/f OLV Sets the carrier frequency for the transistors in the drive. 1 : 2.0 kHz 2 : 5.0 kHz 3 : 8.0 kHz 4 : 10.0 kHz 5 : 12.5 kHz 6 : 15.0 kHz	3 (1 - 6)	265

10.6 d: Reference Settings

◆ d1: Speed Reference

No. (Hex.)	Name	Description	Default (Range)	Ref.
d1-01 (0280) RUN	Speed Reference 1	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)	266
d1-02 (0281) RUN	Speed Reference 2	V/f OLV Sets the speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)	266
d1-03 (0282) RUN	Speed Reference 3	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 590.00 Hz)	266
d1-04 (0283) RUN	Speed Reference 4	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)	266
d1-05 (0284) RUN	Speed Reference 5	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)	266
d1-06 (0285) RUN	Speed Reference 6	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)	266
d1-07 (0286) RUN	Speed Reference 7	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)	267
d1-08 (0287) RUN	Speed Reference 8	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)	267
d1-18 (02C0)	Spd Ref Sel Mode	V/f OLV Sets the priority of the speed reference digital inputs. 0 : Use Multi-Speed references d1-01 to d1-08 1 : High speed reference has priority 2 : Leveling speed reference has priority	1 (0, 1)	267
d1-19 (02C1) RUN	Nominal Speed	V/f OLV Sets the speed reference when multi-function input “Nominal Speed” is on.	50.00 Hz (0.00 - 120.00 Hz)	267
d1-20 (02C2) RUN	Intermed Speed 1	V/f OLV Sets the speed reference when intermediate speed 1 is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)	267
d1-21 (02C3) RUN	Intermed Speed 2	V/f OLV Sets the speed reference when intermediate speed 2 is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)	267
d1-22 (02C4) RUN	Intermed Speed 3	V/f OLV Sets the speed reference when intermediate speed 3 is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)	268
d1-23 (02C5) RUN	Releveling Speed	V/f OLV Sets the speed reference when releveling speed is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)	268
d1-24 (02C6) RUN	Inspect Oper Spd	V/f OLV Sets the speed reference when inspection speed is selected by digital inputs.	30.00 Hz (0.00 - 120.00 Hz)	268
d1-25 (02C7) RUN	Rescue Oper Spd	V/f OLV Sets the speed reference when rescue speed is selected by digital inputs.	5.00 Hz (0.00 - 15.00 Hz)	268
d1-26 (02C8) RUN	Leveling Speed	V/f OLV Sets the speed reference when leveling speed is selected by digital inputs.	4.00 Hz (0.00 - 120.00 Hz)	268

No. (Hex.)	Name	Description	Default (Range)	Ref.
d1-28 (02CA)	Nom/Lev Spd Det	V/f OLV Sets the speed reference level to detect that leveling speed has been selected when multi speed input commands are used.	0.00 Hz (0.00 - 120.00 Hz)	268
d1-29 (02CB)	Inspect Spd Det	V/f OLV Sets the speed reference level to detect that inspection speed has been selected when multi speed input commands are used.	0.00 Hz (0.00 - 120.00 Hz)	268

10.7 E: Motor Parameters

◆ E1: V/f Pattern for Motor 1

No. (Hex.)	Name	Description	Default (Range)	Ref.
E1-01 (0300)	Input AC Supply Voltage	V/f OLV Sets the drive input voltage. **NOTICE:** *Set parameter E1-01 to align with the drive input voltage (not motor voltage). If this parameter is incorrect, the protective functions of the drive will not operate correctly and it can cause damage to the drive.*	400 V: 400 V (400 V Class: 310 - 510 V)	269
E1-04 (0303)	Maximum Output Frequency	V/f OLV Sets the maximum output frequency for the V/f pattern.	Determined by A1-02 (10.0 - 120.0 Hz)	269
E1-05 (0304)	Maximum Output Voltage	V/f OLV Sets the maximum output voltage for the V/f pattern.	400 V Class: 380.0 V (400 V Class: 0.0 - 510.0 V)	269
E1-06 (0305)	Base Frequency	V/f OLV Sets the base frequency for the V/f pattern.	Determined by A1-02 (0.0 - 120.0 Hz)	269
E1-07 (0306)	Middle Output Frequency	V/f OLV Sets a middle output frequency for the V/f pattern.	Determined by A1-02 (0.0 - 120.0 Hz)	269
E1-08 (0307)	Middle Output Frequency Voltage	V/f OLV Sets a middle output voltage for the V/f pattern.	Determined by A1-02 and o2-04 (400 V Class: 0.0 - 510.0 V)	270
E1-09 (0308)	Minimum Output Frequency	V/f OLV Sets the minimum output frequency for the V/f pattern.	Determined by A1-02 (0.0 - 120.0 Hz)	270
E1-10 (0309)	Minimum Output Voltage	V/f OLV Sets the minimum output voltage for the V/f pattern.	Determined by o2-04 (400 V Class: 0.0 - 510.0 V)	270
E1-11 (030A) Expert	Middle Output Frequency 2	V/f OLV Sets a middle output frequency for the V/f pattern.	0.0 Hz (0.0 - 120.0 Hz)	270
E1-12 (030B) Expert	Middle Output Frequency Voltage 2	V/f OLV Sets a middle point voltage for the V/f pattern.	0.0 V (400 V Class: 0.0 - 510.0 V)	270
E1-13 (030C) Expert	Base Voltage	V/f OLV Sets the base voltage for the V/f pattern.	0.0 V (400 V Class: 0.0 - 510.0 V)	270

◆ E2: Motor Parameters

No. (Hex.)	Name	Description	Default (Range)	Ref.
E2-01 (030E)	Motor Rated Current (FLA)	V/f OLV Sets the motor rated current in amps.	Determined by o2-04 (10% to 200% of the drive rated current)	271
E2-02 (030F)	Motor Rated Slip	V/f OLV Sets motor rated slip.	Determined by o2-04 (0.00 - 20.00 Hz)	271
E2-03 (0310)	Motor No-Load Current	V/f OLV Sets the no-load current for the motor in amps when operating at the rated frequency and the no-load voltage.	Determined by o2-04 (0 to E2-01)	271
E2-04 (0311)	Motor Pole Count	V/f OLV Sets the number of motor poles.	4 (2 - 48)	271
E2-05 (0312)	Motor Line-to-Line Resistance	V/f OLV Sets the line-to-line resistance for the motor stator windings.	Determined by o2-04 (0.000 - 65.000 Ω)	272
E2-06 (0313)	Motor Leakage Inductance	V/f OLV Sets the voltage drop from motor leakage inductance when the motor is operating at the rated frequency and rated current. This value is a percentage of Motor Rated Voltage.	Determined by o2-04 (0.0 - 60.0%)	272
E2-07 (0314)	Motor Saturation Coefficient 1	V/f OLV Sets the motor iron-core saturation coefficient at 50% of the magnetic flux.	0.50 (0.00 - 0.50)	272
E2-08 (0315)	Motor Saturation Coefficient 2	V/f OLV Sets the motor iron-core saturation coefficient at 75% of the magnetic flux.	0.75 (E2-07 - 0.75)	272

No. (Hex.)	Name	Description	Default (Range)	Ref.
E2-09 (0316) Expert	Motor Mechanical Loss	V/f OLV Sets the mechanical loss of the motor. It is set as a percentage of *E2-11 [Motor Rated Power]*. Usually it is not necessary to change this setting.	0.0% (0.0 - 10.0%)	272
E2-10 (0317)	Motor Iron Loss	V/f OLV Sets the motor iron loss.	Determined by o2-04 (0 - 65535 W)	273
E2-11 (0318)	Motor Rated Power	V/f OLV Sets the motor rated output in the units from *o1-58 [Motor Power Unit Selection]*.	Determined by o2-04 (0.00 - 650.00 kW)	273

10.8 H: Terminal Functions

◆ H1: Digital Inputs

No. (Hex.)	Name	Description	Default (Range)	Ref.
H1-03 (0400)	Terminal S3 Function Selection	V/f OLV Sets the function for MFDI terminal S3.	50 (3 - 79)	274
H1-04 (0401)	Terminal S4 Function Selection	V/f OLV Sets the function for MFDI terminal S4.	54 (3 - 79)	274
H1-05 (0402)	Terminal S5 Function Selection	V/f OLV Sets the function for MFDI terminal S5.	51 (3 - 79)	274
H1-06 (0403)	Terminal S6 Function Selection	V/f OLV Sets the function for MFDI terminal S6.	53 (3 - 79)	275
H1-07 (0404)	Terminal S7 Function Selection	V/f OLV Sets the function for MFDI terminal S7.	F (3 - 79)	275

■ H1-xx: MFDI Setting Values

Setting Value	Function	Description	Ref.
3	Multi-Step Speed Reference 1	V/f OLV Uses speed references *d1-01 to d1-16* to set a multi-step speed reference.	275
4	Multi-Step Speed Reference 2	V/f OLV Uses speed references *d1-01 to d1-16* to set a multi-step speed reference.	275
5	Multi-Step Speed Reference 3	V/f OLV Uses speed references *d1-01 to d1-16* to set a multi-step speed reference.	275
7	Accel/Decel Time Selection 1	V/f OLV Sets the drive to use *Acceleration/Deceleration Time 1 [C1-01, C1-02]* or *Acceleration/Deceleration Time 2 [C1-03, C1-04]*.	275
8	Baseblock Command (N. O.)	V/f OLV Sets the command that stops drive output and coasts the motor to stop when the input is ON. ON : Baseblock (drive output stop) OFF : Normal operation	275
9	Baseblock Command (N. C.)	V/f OLV Sets the command that stops drive output and coasts the motor to stop when the input terminal is OFF. ON : Normal operation OFF : Baseblock (drive output stop)	276
F	Not Used	V/f OLV Use this setting for unused terminals or to use terminals in through mode.	276
14	Fault Reset	V/f OLV Sets the command to reset the current fault when the Run command is inactive. **Note:** The drive ignores the fault reset command when the Run command is active. Remove the Run command before trying to reset a fault.	276
15	Fast Stop (N.O.)	V/f OLV Sets the command to ramp to stop in the deceleration time set in *C1-09 [Fast Stop Time]* when the input terminal is ON while the drive is operating.	276
17	Fast Stop (N.C.)	V/f OLV Sets the command to ramp to stop in the deceleration time set in *C1-09 [Fast Stop Time]* when the input terminal is ON while the drive is operating.	277
18	Timer Function	V/f OLV Sets the command to start the timer function. Use this setting with *Timer Output [H2-xx = 12]*.	277
1A	Accel/Decel Time Selection 2	V/f OLV Set this function and *H1-xx = 7 [Accel/Decel Time Selection 1]* together. Sets the drive to use *Acceleration/Deceleration Time 3 [C1-05, C1-06]* or *Acceleration/Deceleration Time 4 [C1-07, C1-08]*.	277
20	External Fault (NO-Always-Ramp)	V/f OLV When the terminal activates, the drive ramps to stop in the selected deceleration time. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive always detects external faults whether the drive is stopped or running.	277
21	External Fault (NC-Always-Ramp)	V/f OLV When the terminal deactivates, the drive ramps to stop in the selected deceleration time. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive always detects external faults whether the drive is stopped or running.	277

Setting Value	Function	Description	Ref.
22	External Fault (NO-@Run-Ramp)	V/f OLV When the terminal activates during run, the drive ramps to stop in the selected deceleration time. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive does not detect external faults while the drive is stopped.	277
23	External Fault (NC-@Run-Ramp)	V/f OLV When the terminal deactivates during run, the drive ramps to stop in the selected deceleration time. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive does not detect external faults while the drive is stopped.	277
24	External Fault (NO-Always-Coast)	V/f OLV When the terminal activates, the drive shuts off the output and the motor coasts to stop. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive always detects external faults whether the drive is stopped or running.	277
25	External Fault (NC-Always-Coast)	V/f OLV When the terminal deactivates, the drive shuts off the output and the motor coasts to stop. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive always detects external faults whether the drive is stopped or running.	277
26	External Fault (NO-@Run-Coast)	V/f OLV When the terminal activates during run, the drive shuts off the output and the motor coasts to stop. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive does not detect external faults while the drive is stopped.	277
27	External Fault (NC-@Run-Coast)	V/f OLV When the terminal deactivates during run, the drive shuts off the output and the motor coasts to stop. Fault relay output terminal MA-MC will turn ON, and MB-MC will turn OFF. The drive does not detect external faults while the drive is stopped.	277
28	External Fault (NO-Always-FStop)	V/f OLV When the terminal activates, the drive stops the motor in the deceleration time set to *C1-09 [Fast Stop Time]*. Fault relay output terminal MA-MC turns ON, and MB-MC turns OFF. Stopped drives and running drives will detect external faults.	277
29	External Fault (NC-Always-FStop)	V/f OLV When the terminal deactivates, the drive stops the motor in the deceleration time set to *C1-09 [Fast Stop Time]*. Fault relay output terminal MA-MC turns ON, and MB-MC turns OFF. Stopped drives and running drives will detect external faults.	277
2A	External Fault (NO-@Run-FStop)	V/f OLV When the terminal activates during run, the drive stops the motor in the deceleration time set to *C1-09 [Fast Stop Time]*. Fault relay output terminal MA-MC turns ON, and MB-MC turns OFF. Stopped drives will not detect external faults.	277
2B	External Fault (NC-@Run-FStop)	V/f OLV When the terminal deactivates during run, the drive stops the motor in the deceleration time set to *C1-09 [Fast Stop Time]*. Fault relay output terminal MA-MC turns ON, and MB-MC turns OFF. Stopped drives will not detect external faults.	277
2C	External Fault (NO-Always-Alarm)	V/f OLV When the terminal activates, the keypad shows *EFx [External Fault (Input Terminal Sx)]* and the output terminal set for *Alarm [H2-01 to H2-03 = 10]* activates. The drive continues operation. Stopped drives and running drives will detect external faults.	277
2D	External Fault (NC-Always-Alarm)	V/f OLV When the terminal deactivates, the keypad shows *EFx [External Fault (Input Terminal Sx)]* and the output terminal set for *Alarm [H2-01 to H2-03 = 10]* activates. The drive continues operation. Stopped drives and running drives will detect external faults.	277
2E	External Fault (NO-@Run-Alarm)	V/f OLV When the terminal activates during run, the keypad shows *EFx [External Fault (Input Terminal Sx)]* and the output terminal set for *Alarm [H2-01 to H2-03 = 10]* activates. The drive continues operation. Stopped drives will not detect external faults.	277
2F	External Fault (NC-@Run-Alarm)	V/f OLV When the terminal deactivates during run, the keypad shows *EFx [External Fault (Input Terminal Sx)]* and the output terminal set for *Alarm [H2-01 to H2-03 = 10]* activates. The drive continues operation. Stopped drives will not detect external faults.	277
40	Forward RUN (2-Wire)	V/f OLV Sets the Forward Run command for 2-wire sequence 1. Set this function and *H1-xx = 41 [Reverse Run Command (2-Wire Seq)]* together. ON : Forward Run OFF : Run Stop **Note:** • If you turn ON the Forward Run command terminal and the Reverse Run command terminal, it will cause an *EF [FWD/REV Run Command Input Error]* alarm and the motor will ramp to stop. • Initialize the drive with a 2-wire sequence to set the Forward Run command to terminal S1. • This function will not operate at the same time as *H1-xx = 42, 43 [Run Command/FWD/REV Command (2-Wire Seq 2)]*.	278

Setting Value	Function	Description	Ref.
41	Reverse RUN (2-Wire)	V/f OLV Sets the Forward Run command for 2-wire sequence 1. Set this function and *H1-xx = 40 [Forward Run Command (2-Wire Seq)]* together. ON : Reverse Run OFF : Run Stop **Note:** • If you turn ON the Forward Run command terminal and the Reverse Run command terminal, it will cause an *EF [FWD/REV Run Command Input Error]* alarm and the motor will ramp to stop. • Initialize the drive with a 2-wire sequence to set the Reverse Run command to terminal S2. • This function will not operate at the same time as *H1-xx = 42, 43 [Run Command/FWD/REV Command (2-Wire Seq 2)]*.	279
50	Nominal Speed	V/f OLV Sets the speed reference nominal speed.	279
51	Intermed Speed	V/f OLV Sets the speed reference intermediate speed 1. In combination with nominal and releveling speed you can set intermediate speed 2 and 3.	279
52	Releveling Speed	V/f OLV Sets the speed reference releveling speed.	279
53	Leveling Speed	V/f OLV Sets the speed reference leveling speed.	279
54	Inspection Speed	V/f OLV Sets the speed reference inspection speed.	279
55	Rescue Operation	V/f OLV Activates rescue operation.	279
56	Motor Contractor Feedback	V/f OLV Used for motor contactor supervision and fault detection.	280
79	Brake Feedback	V/f OLV Used for brake supervision and detection of incorrect operation.	280

◆ H2: Digital Outputs

No. (Hex.)	Name	Description	Default (Range)	Ref.
H2-01 (040B)	Term MA/MB-MC Function Selection	V/f OLV Sets the function set for MFDO terminal MA-MC or MB-MC. **Note:** Set this parameter to *F* when the terminal is not being used or to use the terminal in through mode.	E (0 - 158)	281
H2-02 (040C)	Term P1-C1 Function Selection	V/f OLV Sets the function for MFDO terminal P1-C1. **Note:** Set this parameter to *F* when the terminal is not being used or to use the terminal in through mode.	6 (0 - 158)	281
H2-03 (040D)	Term P2-C2 Function Selection	V/f OLV Sets the function for MFDO terminal P2-C2. **Note:** Set this parameter to *F* when the terminal is not being used or to use the terminal in through mode.	50 (0 - 158)	281

■ H2-xx: MFDO Setting Values

<table>
<tr><th>Setting Value</th><th>Function</th><th>Description</th><th>Ref.</th></tr>
<tr><td>0</td><td>During Run</td><td>V/f OLV
The terminal activates when you input a Run command and when the drive is outputting voltage.
ON : Drive is running
OFF : Drive is stopping</td><td>281</td></tr>
<tr><td>1</td><td>Zero Speed</td><td>V/f OLV
The terminal activates when the output frequency is less than E1-09 [Minimum Output Frequency] or b2-01 [DC Injection/Zero SpeedThreshold].
Note:
Parameter A1-02 [Control Method Selection] selects which parameter is the reference.
<table>
<tr><th>A1-02 Setting</th><th>Control Method Selection</th><th>Parameter Used as the Reference</th></tr>
<tr><td>0</td><td>V/f</td><td>E1-09</td></tr>
<tr><td>2</td><td>OLV</td><td>b2-01</td></tr>
</table>
ON : Output frequency < value of E1-09 or b2-01.
OFF : Output frequency ≥ E1-09 or b2-01.</td><td>281</td></tr>
<tr><td>2</td><td>Speed Agree 1</td><td>V/f OLV
The terminal activates when the output frequency is in the range of the frequency reference ± L4-02 [Speed Agree Detection Width].
ON : The output frequency is in the range of “frequency reference ± L4-02”.
OFF : The output frequency does not align with the frequency reference although the drive is running.</td><td>282</td></tr>
<tr><td>3</td><td>User-Set Speed Agree 1</td><td>V/f OLV
The terminal activates when the output frequency is in the range of L4-01 [Speed Agree Detection Level] ± L4-02 [Speed Agree Detection Width] and in the range of the frequency reference ± L4-02.
Note:
The detection function operates in the two motor rotation directions. The drive uses the L4-01 value as the forward/reverse detection level.
ON : The output frequency is in the range as defined by the result of “L4-01 ± L4-02” and the range of frequency reference ± L4-02.
OFF : The output frequency is not in the range of “L4-01 ± L4-02” nor the range of frequency reference ± L4-02.</td><td>282</td></tr>
<tr><td>4</td><td>Frequency Detection 1</td><td>V/f OLV
The terminal deactivates when the output frequency is higher than the value of L4-01 [Speed Agree Detection Level] + L4-02 [Speed Agree Detection Width]. After the terminal deactivates, the terminal stays deactivated until the output frequency = L4-01.
Note:
The detection function operates in the two motor rotation directions. The drive uses the L4-01 value as the forward/reverse detection level.
ON : The output frequency is less than L4-01 or is not more than L4-01 + L4-02.
OFF : The output frequency is more than L4-01 + L4-02.</td><td>283</td></tr>
<tr><td>5</td><td>Frequency Detection 2</td><td>V/f OLV
The terminal activates when the output frequency is higher than the value of L4-01 [Speed Agree Detection Level]. After the terminal activates, the terminal stays activated until the output frequency is at the value of L4-01 - L4-02.
Note:
The detection function operates in the two motor rotation directions. The drive uses the L4-01 value as the forward/reverse detection level.
ON : The output frequency is more than L4-01.
OFF : The output frequency is less than “L4-01 - L4-02,” or it is not more than L4-01.</td><td>283</td></tr>
<tr><td>6</td><td>Drive Ready</td><td>V/f OLV
The terminal activates when the drive is ready and running.</td><td>284</td></tr>
<tr><td>7</td><td>DC Bus Undervoltage</td><td>V/f OLV
The terminal activates when the DC bus voltage or control circuit power supply is at the voltage set in L2-05 [Undervoltage Detection Lvl (Uv1)] or less. The terminal also activates when there is a fault with the DC bus voltage.
ON : The DC bus voltage ≤ L2-05
OFF : The DC bus voltage > L2-05</td><td>284</td></tr>
<tr><td>8</td><td>During Baseblock (N.O.)</td><td>V/f OLV
The terminal activates during baseblock. When the drive is in baseblock, the drive output transistor stops switching and does not make DC bus voltage.
ON : During baseblock
OFF : The drive is not in baseblock.</td><td>284</td></tr>
<tr><td>B</td><td>Torque Detection 1 (N. O.)</td><td>V/f OLV
The terminal activates when the drive detects overtorque or undertorque.
ON : The output current/torque > L6-02 [Torque Detection Level 1], or the output current/torque < L6-02 for longer than the time set in L6-03 [Torque Detection Time 1].</td><td>284</td></tr>
<tr><td>E</td><td>Fault</td><td>V/f OLV
The terminal activates when the drive detects a fault.
Note:
The terminal will not activate for CPF00 and CPF01 [Control Circuit Error] faults.</td><td>285</td></tr>
</table>

Setting Value	Function	Description	Ref.
F	Not Used	V/f OLV Use this setting for unused terminals or to use terminals in through mode. Also use this setting as the PLC contact output via MEMOBUS/Modbus or the communication option. This signal does not function if you do not configure signals from the PLC.	285
10	Alarm	V/f OLV The terminal activates when the drive detects a minor fault.	285
11	Fault Reset Command Active	V/f OLV The terminal activates when the drive receives the Reset command from the control circuit terminal, serial communications, or the communication option.	285
12	Timer Output	V/f OLV Use this setting when the drive uses the timer function as an output terminal.	285
13	Speed Agree 2	V/f OLV The terminal activates when the output frequency is in the range of the frequency reference ± *L4-04 [Speed Agree Detection Width(+/-)]*. **Note:** The detection function operates in the two motor rotation directions. ON : The output frequency is in the range of "frequency reference ± *L4-04*". OFF : The output frequency is not in the range of "frequency reference ± *L4-04*".	285
14	User-Set Speed Agree 2	V/f OLV The terminal activates when the output frequency is in the range of *L4-03 [Speed Agree Detection Level(+/-)]* ± *L4-04 [Speed Agree Detection Width(+/-)]* and in the range of the frequency reference ± *L4-04*. **Note:** The detection level set in *L4-03* is a signed value. The drive will only detect in one direction. ON : The output frequency is in the range as defined by the result of "*L4-03* ± *L4-04*" and the range of frequency reference ± *L4-04*. OFF : The output frequency is not in the range of "*L4-03* ± *L4-04*" nor the range of frequency reference ± *L4-04*.	286
15	Frequency Detection 3	V/f OLV The terminal deactivates when the output frequency is more than "*L4-03 [Speed Agree Detection Level(+/-)]* + *L4-04 [Speed Agree Detection Width(+/-)]*". After the terminal deactivates, the terminal stays deactivated until the output frequency is at the value of *L4-03*. **Note:** The detection level set in *L4-03* is a signed value. The drive will only detect in one direction. ON : The output frequency is less than *L4-03* or is not more than *L4-03* + *L4-04*. OFF : The output frequency is higher than *L4-03* + *L4-04*.	286
16	Frequency Detection 4	V/f OLV The terminal activates when the output frequency is higher than the value of *L4-03 [Speed Agree Detection Level (+/-)]*. After the terminal activates, the terminal stays activated until the output frequency = *L4-03* - *L4-04*. **Note:** The detection level set in *L4-03* is a signed value. The drive will only detect in one direction. ON : The output frequency is more than *L4-03*. OFF : The output frequency is less than "*L4-03* - *L4-04*," or it is not more than *L4-03*.	287
18	Torque Detection 2 (N. O.)	V/f OLV The terminal activates when the drive detects overtorque or undertorque. ON : The output current/torque > *L6-05 [Torque Detection Level 2]*, or the output current/torque < *L6-05* for longer than the time set in *L6-06 [Torque Detection Time 2]*.	287
1A	During Reverse	V/f OLV The terminal activates when the motor operates in the reverse direction. ON : The motor is operating in the reverse direction. OFF : The motor is operating in the forward direction or the motor stopped.	287
1B	During Baseblock (N.C.)	V/f OLV The terminal deactivates during baseblock. When the drive is in baseblock, the drive output transistor stops switching and does not make DC bus voltage. ON : The drive is not in baseblock. OFF : During baseblock	288
1E	Executing Auto-Restart	V/f OLV The terminal activates when the Auto Restart function is trying to restart after a fault.	288
1F	Motor Overload Alarm (oL1)	V/f OLV The terminal activates when the electronic thermal protection value of the motor overload protective function is a minimum of 90% of the detection level.	288
20	Drive Overheat Pre-Alarm (oH)	V/f OLV The terminal activates when the drive heatsink temperature is at the level set with *L8-02 [Overheat Alarm Level]*.	288
21	Safety Monitor Output Status	V/f OLV The terminal activates (safety stop state) when the safety circuit and safety diagnosis circuit are operating correctly and when terminals H1-HC and H2-HC are OFF (Open). ON : Safety stop state OFF : Safety circuit fault or RUN/READY	288

Setting Value	Function	Description	Ref.
2F	Maintenance Notification	V/f OLV The terminal activates when drive components are at their estimated maintenance period. Tells you about the maintenance period for these items: • IGBT • Cooling Fan • Capacitor • Soft charge bypass relay	288
30	During Torque Limit	V/f OLV The terminal activates when the torque reference is the torque limit set with *L7 parameters* or *H3-02 or H3-10 [MFAI Function Selection]*.	289
37	During Frequency Output	V/f OLV The terminal activates when the drive outputs frequency. ON : The drive outputs frequency. OFF : The drive does not output frequency.	289
47	PF Detection	V/f OLV Indicates input phase loss. 0 : No error 1 : Input Phase Loss error detected	289
50	Brake Control	V/f OLV This setting can be used in the brake sequence for the elevator application. Closing the output terminal should cause the brake to release, and opening the terminal should apply the brake. Close : Release brake. Open : Apply brake	289
51	Output Contactor Control	V/f OLV Assigning this command to an output terminal can send a signal to the controller to close the output contactor. The output contactor should open when the terminal is released. Closed : Close output contactor	290
54	Light Load Direction	V/f OLV Indicates the light load direction detected during emergency operation with light load search. When the terminal is closed the light load direction is up, when it is open the light load direction is down. Closed : Light load direction is up Open : Light load direction is down	290
58	Safe Disable Status	V/f OLV This terminal closes if the Safe Disable inputs H1-HC are closed and opens when terminals H1-HC are open. Closed : Safe Disable terminals H1-HC and H2-HC are open, drive is in a baseblock state Open : Safe Disable terminals H1-HC and H2-HC are closed (normal operation)	290
100 - 158	Inverse output of 0 to 58	V/f OLV Causes inverse output of the function for the selected MFDO. Uses the last two digits of 1xx to select which function to inversely output.	290

◆ H4: Analog Outputs

No. (Hex.)	Name	Description	Default (Range)	Ref.
H4-01 (041D)	Terminal AM Analog Output Select	V/f OLV Sets the monitoring number to be output from the MFAO terminal AM. Set the *x-xx* part of the *Ux-xx [Monitor]*. For example, set *H4-01 = 102* to monitor *U1-02 [Output Frequency]*.	102 (000 - 999)	291
H4-02 (041E) RUN	Terminal AM Analog Output Gain	V/f OLV Sets the gain of the monitor signal that is sent from MFAO terminal AM. Sets the analog signal output level from the terminal AM at 10 V or 20 mA as 100% when an output for monitoring items is 100%.	100.0% (-999.9 - +999.9%)	291
H4-03 (041F) RUN	Terminal AM Analog Output Bias	V/f OLV Sets the bias of the monitor signal that is sent from MFAO terminal AM. When an output for monitoring items is 0%, this parameter sets the analog signal output level from the AM terminal at 10 V or 20 mA as 0%.	0.0% (-999.9 - +999.9%)	292

10.9 L: Protection Functions

◆ L1: Motor Protection

No. (Hex.)	Name	Description	Default (Range)	Ref.
L1-01 (0480)	Motor Overload (oL1) Protection	V/f OLV Sets the motor overload protection with electronic thermal protectors. 0 : No 1 : Variable Torque 2 : Constant Torque 10:1 Speed Range 3 : Constant Torque 100:1 SpeedRange	1 (0 - 3)	293
L1-02 (0481)	Motor Overload Protection Time	V/f OLV Sets the operation time for the electronic thermal protector of the drive to prevent damage to the motor. Usually it is not necessary to change this setting.	1.0 min (0.1 - 5.0 min)	295
L1-13 (046D)	Motor Overload Memory Selection	V/f OLV Sets the function that keeps the current electronic thermal protector value when the drive stops receiving power. 0 : Disabled 1 : Enabled	1 (0, 1)	295

◆ L2: Power Loss Ride Through

No. (Hex.)	Name	Description	Default (Range)	Ref.
L2-05 (0489)	Undervoltage Detection Lvl (Uv1)	V/f OLV Sets the voltage at which a *Uv1 [DC Bus Undervoltage]* fault is triggered. Usually it is not necessary to change this setting. **NOTICE:** *Damage to Equipment. When you set this parameter to a value lower than the default, you must install an AC reactor on the input side of the power supply. If you do not install an AC reactor, it will cause damage to the drive circuitry.*	Determined by E1-01 (400 V Class: 300 - 420 V)	295

◆ L3: Stall Prevention

No. (Hex.)	Name	Description	Default (Range)	Ref.
L3-01 (048F)	Stall Prevention during Accel	V/f OLV Sets the method of the Stall Prevention During Acceleration. 0 : Disabled 1 : Enabled	1 (0 - 1)	296
L3-02 (0490)	Stall Prevent Level during Accel	V/f OLV Sets the output current level at which the Stall Prevention function operates during acceleration where the drive rated output current is 100%.	Determined by L8-38 (0 - 165%)	297
L3-03 (0491)	Stall Prevent Limit during Accel	V/f OLV Sets the lower limit for the stall prevention level used in the constant output range as a percentage of the drive rated output current.	50% (0 - 100%)	297
L3-05 (0493)	Stall Prevention during RUN	V/f OLV Sets the function to enable and disable Stall Prevention During Run. **Note:** An output frequency less than 6 Hz disables Stall Prevention during Run. The setting values of *L3-05* and *L3-06 [Stall Prevent Level during Run]* do not have an effect. 0 : No 1 : Deceleration Time 1 (C1-02)	1 (0 - 1)	297
L3-06 (0494)	Stall Prevent Level during Run	V/f OLV Sets the output current level at which the Stall Prevention function is enabled during run when the drive rated output current is 100%. **Note:** This parameter is applicable when *L3-05 = 1 [Stall Prevention during RUN = Deceleration Time 1 (C1-02)]*.	Determined by L8-38 (30 - 165%)	298

◆ L4: Speed Detection

No. (Hex.)	Name	Description	Default (Range)	Ref.
L4-01 (0499)	Speed Agree Detection Level	V/f OLV Sets the level to detect speed agree or motor speed. Sets the level to detect speed agree or motor speed when *H2-01 to H2-03 = 2, 3, 4, 5 [MFDO Function Selection = Speed Agree 1, User-set Speed Agree 1, Frequency Detection 1, Frequency Detection 2]*.	0.0 Hz (0.0 - 120.0 Hz)	298
L4-02 (049A)	Speed Agree Detection Width	V/f OLV Sets the width to detect speed agree or motor speed. Sets the width to detect speed agree or motor speed when *H2-01 to H2-03 = 2, 3, 4, 5 [MFDO Function Selection = Speed Agree 1, User-set Speed Agree 1, Frequency Detection 1, Frequency Detection 2]*.	2.0 Hz (0.0 - 20.0 Hz)	298
L4-03 (049B)	Speed Agree Detection Level(+/-)	V/f OLV Sets the level to detect speed agree or motor speed. Sets the level to detect speed agree or motor speed when *H2-01 to H2-03 = 13, 14, 15, 16 [MFDO Function Selection = Speed Agree 2, User-set Speed Agree 2, Frequency Detection 3, Frequency Detection 4]*.	0.0 Hz (-120.0 - 120.0 Hz)	299
L4-04 (049C)	Speed Agree Detection Width(+/-)	V/f OLV Sets the width to detect speed agree or motor speed. Sets the width to detect speed agree or motor speed when *H2-01 to H2-03 = 13, 14, 15, 16 [MFDO Function Selection = Speed Agree 2, User-set Speed Agree 2, Frequency Detection 3, Frequency Detection 4]*.	2.0 Hz (0.0 - 20.0 Hz)	299

◆ L5: Fault Restart

No. (Hex.)	Name	Description	Default (Range)	Ref.
L5-01 (049E)	Number of Auto-Restart Attempts	V/f OLV Sets the number of times that the drive will try to restart.	2 (0 - 10 times)	299
L5-02 (049F)	Fault Contact at Restart Select	V/f OLV Sets the function that sends signals to the MFDO terminal set for *Fault [H2-xx = E]* while the drive is automatically restarting. 0 : Active Only when Not Restarting 1 : Always Active	1 (0, 1)	300
L5-04 (046C)	Interval Method Restart Time	V/f OLV Sets the time interval between each Auto Restart attempt. This function is enabled when *L5-05 = 1 [Auto Restart Operation Selection = Use L5-04 Time]*.	10.0 s (0.5 - 600.0 s)	300
L5-06 (046E)	Undervoltage Fault Restart Selection	V/f OLV Determines whether a limit should be placed on the number of reset attempts after a Uv1 fault. 0 : Restrict auto-reset attempts to L5-01 after Uv1 1 : No Limit on auto-reset attempts to L5-01	0 (0, 1)	300

◆ L6: Torque Detection

No. (Hex.)	Name	Description	Default (Range)	Ref.
L6-01 (04A1)	Torque Detection Selection 1	V/f OLV Sets the speed range that detects overtorque and undertorque and the operation of drives (operation status) after detection. 0 : Disabled 1 : oL @ Speed Agree - Alarm only 2 : oL @ RUN - Alarm only 3 : oL @ Speed Agree - Fault 4 : oL @ RUN - Fault 5 : UL @ Speed Agree - Alarm only 6 : UL @ RUN - Alarm only 7 : UL @ Speed Agree - Fault 8 : UL @ RUN - Fault	0 (0 - 8)	301
L6-02 (04A2)	Torque Detection Level 1	V/f OLV Sets the detection level for Overtorque/Undertorque Detection 1. In V/f control, drive rated output current = 100% value. In vector control, motor rated torque = 100% value.	150% (0 - 300%)	302
L6-03 (04A3)	Torque Detection Time 1	V/f OLV Sets the detection time for Overtorque/Undertorque Detection 1.	0.1 s (0.0 - 10.0 s)	302

No. (Hex.)	Name	Description	Default (Range)	Ref.
L6-04 (04A4)	Torque Detection Selection 2	V/f OLV Sets the speed range that detects overtorque and undertorque and the operation of drives (operation status) after detection. 0 : Disabled 1 : oL @ Speed Agree - Alarm only 2 : oL @ RUN - Alarm only 3 : oL @ Speed Agree - Fault 4 : oL @ RUN - Fault 5 : UL @ Speed Agree - Alarm only 6 : UL @ RUN - Alarm only 7 : UL @ Speed Agree - Fault 8 : UL @ RUN - Fault	0 (0 - 8)	303
L6-05 (04A5)	Torque Detection Level 2	V/f OLV Sets the detection level for Overtorque/Undertorque Detection 2. In V/f control, drive rated output current = 100% value. In vector control, motor rated torque = 100% value.	150% (0 - 300%)	303
L6-06 (04A6)	Torque Detection Time 2	V/f OLV Sets the detection time for Overtorque/Undertorque Detection 2.	0.1 s (0.0 - 10.0 s)	303

◆ L7: Torque Limit

No. (Hex.)	Name	Description	Default (Range)	Ref.
L7-01 (04A7) RUN	Forward Torque Limit	V/f OLV Sets the torque limit value for forward motoring as a percentage, where motor rated torque is the 100% value.	200% (0 - 300%)	304
L7-02 (04A8) RUN	Reverse Torque Limit	V/f OLV Sets the torque limit value for reversed motoring as a percentage, where motor rated torque is the 100% value.	200% (0 - 300%)	305
L7-03 (04A9) RUN	Forward Regenerative Trq Limit	V/f OLV Sets the torque limit value for forward regenerative conditions as a percentage of the motor rated torque.	200% (0 - 300%)	305
L7-04 (04AA) RUN	Reverse Regenerative Trq Limit	V/f OLV Sets the torque limit value for reversed regenerative conditions as a percentage of the motor rated torque.	200% (0 - 300%)	305
L7-06 (04AC)	Torque Limit Integral Time	V/f OLV Sets the integral time constant for the torque limit function.	200 ms (5 - 10000 ms)	305
L7-07 (04C9)	Torque Limit during Accel/Decel	V/f OLV Sets the torque limit function during acceleration and deceleration. 0 : Proportional only 1 : Proportional & Integral control	0 (0, 1)	305

◆ L8: Drive Protection

No. (Hex.)	Name	Description	Default (Range)	Ref.
L8-02 (04AE)	Overheat Alarm Level	V/f OLV Sets the *oH* detection level in temperature.	Determined by o2-04 (50 - 130 °C)	306
L8-03 (04AF)	Overheat Pre-Alarm Selection	V/f OLV Sets the operation of drives when an *oH* alarm is detected. 0 : Ramp to Stop 1 : Coast to Stop 2 : Fast Stop (Use C1-09) 3 : Alarm Only 4 : Operate at Reduced Speed (L8-19)	3 (0 - 4)	306
L8-05 (04B1)	Input Phase Loss Protection Sel	V/f OLV Sets the function to enable and disable input phase loss detection. 0 : Disable 1 : Enabled 2 : Standard + dv/dt	Depends on o2-04 (0 - 2)	307
L8-06 (04B2)	Input Phase Loss Detection Level	V/f OLV When ripple is observed in the DC bus, expansion of the input bias is calculated and becomes the input phase if the difference between the max and minimum values of the ripple are greater than L8-06.	Determined by o2-04 (0.0 - 50.0%)	307

No. (Hex.)	Name	Description	Default (Range)	Ref.
L8-07 (04B3)	Output Phase Loss Protection Sel	V/f OLV Sets the function to enable and disable output phase loss detection. The drive starts output phase loss detection when the output current decreases to less than 5% of the drive rated current. **Note:** The drive can incorrectly start output phase loss detection in these conditions: • The motor rated current is very small compared to the drive rating. • The drive is operating a PM motor with a small load. 0 : Disabled 1 : Fault when one phase is lost 2 : Fault when two phases are lost	1 (0 - 2)	307
L8-09 (04B5)	Output Ground Fault Detection	V/f OLV Sets the function to enable and disable ground fault protection. 0 : Disabled 1 : Enabled	1 (0, 1)	308
L8-10 (04B6)	Heatsink Fan Operation Selection	V/f OLV Sets operation of the heatsink cooling fan. 0 : During Run, w/ L8-11 Off-Delay 1 : Always On	0 (0 - 1)	308
L8-11 (04B7)	Heatsink Fan Off-Delay Time	V/f OLV Sets the length of time that the drive will wait before it stops the cooling fan after it cancels the Run command when *L8-10 = 0 [Heatsink Cooling Fan Ope Select = Dur Run (OffDly)]*.	60 s (0 - 300 s)	308
L8-12 (04B8)	Ambient Temperature Setting	V/f OLV Sets the ambient temperature of the drive installation area. **Note:** The setting range changes when the *L8-35 [Installation Method Selection]* value changes: • *0 [IP20/UL Open Type]*: -10 °C to +60 °C • *1 [Side-by-Side Mounting]*: -10 °C to +50 °C • *2 [IP20/UL Type 1]*: -10 °C to +50 °C • *3 [Finless Installation]*: -10 °C to +50 °C	40 °C (-10 °C - +60 °C)	308
L8-15 (04BB)	Drive oL2 @ Low Speed Protection	V/f OLV Sets the function to decrease drive overload at low speeds to prevent damage to the main circuit transistor during low speed operation (at 6 Hz or slower) to prevent *oL2 [Drive Overloaded]*. **Note:** Contact Yaskawa or your nearest sales representative before disabling this function at low speeds. If you frequently operate drives with high output current in low speed ranges, it can cause heat stress and decrease the life span of drive IGBTs. 0 : Disabled (No Additional Derate) 1 : Enabled (Reduced oL2 Level)	1 (0, 1)	308
L8-18 (04BE)	Software Current Limit Selection	V/f OLV Set the software current limit selection function to prevent damage to the main circuit transistor caused by too much current. 0 : Disabled 1 : Enabled	0 (0, 1)	309
L8-19 (04BF)	Freq Reduction @ oH Pre-Alarm	V/f OLV Sets the ratio at which the drive derates the frequency reference during an *oH* alarm.	0.8 (0.1 - 0.9)	309
L8-35 (04EC)	Installation Method Selection	V/f OLV Sets the type of drive installation. 0 : IP20/UL Open Type 1 : Side-by-Side Mounting 2 : IP20/UL Type 1 3 : Finless Installation	Determined by o2-04 (0 - 3)	309
L8-38 (04EF)	Carrier Frequency Reduction Selection	V/f OLV Selects the operation of the carrier frequency reduction function. 0 : Disabled 1 : Enabled for Output Frequencies below 6 Hz 2 : Enabled for Entire Frequency Range	Determined by o2-04 (0 - 2)	310
L8-40 (04F1)	Carrier Freq Reduction Off-Delay	V/f OLV Sets the length of time until the automatically reduced carrier frequency returns to the condition before the reduction.	0.50 s (0.00 - 2.00 s)	310
L8-41 (04F2)	High Current Alarm Selection	V/f OLV Sets the function to cause an *HCA [Current Alarm]* when the output current is more than 150% of the drive rated current. 0 : No 1 : Yes	0 (0, 1)	310

No. (Hex.)	Name	Description	Default (Range)	Ref.
L8-42 (04F3)	Input Phase Loss Detection Time 2	V/f OLV Sets the amount of time that an Input Phase Loss condition has to be present before a fault is triggered.	2 s	310
L8-43 (04F4)	High Current Alarm Selection	V/f OLV Sets the amount of time that an Output Phase Loss condition has to be present before a fault is triggered.	0.5 s (0.0 - 2.0 s)	311
L8-88 (04F2)	Safe Disable Operation Mode	V/f OLV Determines the operation performed by the drive when the Safe Disable input is activated. 0 : Mode 0 1 : Mode 1	0 (0, 1)	310

10.10 n: Special Adjustment

◆ n2: Speed Feedback Detection Control (AFR) Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
n2-01 (0584)	Speed Feedback Detection Control (AFR) Gain	V/f OLV Sets the gain of the AFR function as a magnification value. Usually it is not necessary to change this setting.	1.50 (0.00 - 10.00)	312
n2-02 (0585)	Speed Feedback Detection Control (AFR) Time Constant 1	V/f OLV Sets the time constant that sets the rate of change for the AFR function. Usually it is not necessary to change this setting.	50 ms (0 - 2000 ms)	312
n2-03 (0586)	Speed Feedback Detection Control (AFR) Time Constant 2	V/f OLV Sets the time constant that sets the speed difference of the AFR function. Use this parameter for speed searches or regeneration. Usually it is not necessary to change this setting.	750 ms (0 - 2000 ms)	312

◆ n6: Online Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
n6-01 (0570)	Online Tuning Selection	V/f OLV Sets the type of motor data that Online Tuning uses for OLV control. 0 : Disabled 1 : Line-to-Line Resistance Tuning 2 : Voltage Correction Tuning	2 (0 - 2)	313

10.11 o: Keypad-Related Settings

◆ o1: Keypad Display

No. (Hex.)	Name	Description	Default (Range)	Ref.
o1-01 (0500) RUN	User Monitor Selection	V/f OLV Sets the *U monitor* for the Drive Mode. This parameter is only available with an LED keypad.	106 (104 - 855)	315
o1-02 (0501) RUN	Monitor Selection at Power-up	V/f OLV Sets the monitor item that the keypad screen shows after you energize the drive. Refer to "U: Monitors" for information about the monitor items that the keypad screen can show. This parameter is only available with an LED keypad. 1 : Frequency Reference [U1-01] 2 : Direction 3 : Output Frequency [U1-02] 4 : Output Current [U1-03] 5 : User Monitor (*o1-01*)	1 (1 - 5)	315
o1-03 (0502)	Keypad Display Unit Selection	V/f OLV Sets the display units for the frequency reference and output frequency. 0 : 0.01 Hz units 1 : 0.01% units 2 : Revolutions Per Minute (RPM) 3 : User Units	0 (0 - 3)	315
o1-10 (0520)	User Units Maximum Value	V/f OLV Sets the value that the drive shows as the maximum output frequency.	Determined by o1-03 (1 - 60000)	316
o1-11 (0521)	User Units Decimal Position	V/f OLV Sets the number of decimal places for frequency reference and monitor values. 0 : No Decimal Places (XXXXX) 1 : One Decimal Places (XXXX.X) 2 : Two Decimal Places (XXX.XX) 3 : Three Decimal Places (XX.XXX)	Determined by o1-03 (0 - 3)	316

◆ o2: Keypad Operation

No. (Hex.)	Name	Description	Default (Range)	Ref.
o2-01 (0505)	LO/RE Key Function Selection	V/f OLV Sets the function that lets you use LO/RE to switch between LOCAL and REMOTE Modes. 0 : Disabled 1 : Enabled	0 (0, 1)	316
o2-02 (0506)	STOP Key Function Selection	V/f OLV Sets the function to use STOP on the keypad to stop the drive when the Run command source for the drive is REMOTE (external) and not assigned to the keypad. 0 : Disabled 1 : Enabled	0 (0, 1)	317
o2-03 (0507)	User Parameter Default Value	V/f OLV Sets the function to keep the settings of changed parameters as user parameter defaults to use during initialization. 0 : No change 1 : Set defaults 2 : Clear all	0 (0 - 2)	317
o2-04 (0508)	Drive Model (KVA) Selection	V/f OLV Sets the Drive Model code. Set this parameter after replacing the control board.	Determined by the drive (-)	317
o2-05 (0509)	Home Mode Freq Ref Entry Mode	V/f OLV Sets the function that makes it necessary to push ENTER to use the keypad to change the frequency reference value while in Drive Mode. 0 : ENTER Key Required 1 : Immediate / MOP-style	0 (0, 1)	318

No. (Hex.)	Name	Description	Default (Range)	Ref.
o2-06 (050A)	Keypad Disconnect Detection	V/f OLV Sets the function that stops the drive if you disconnect the keypad connection cable from the drive or if you damage the cable while the keypad is the Run command source. 0 : Disabled 1 : Enabled	0 (0, 1)	318
o2-07 (0527)	Keypad RUN Direction @ Power-up	V/f OLV Sets the direction of motor rotation when the drive is energized and the keypad is the Run command source. 0 : Forward 1 : Reverse	0 (0, 1)	318

◆ o3: Copy Keypad Function

No. (Hex.)	Name	Description	Default (Range)	Ref.
o3-01 (0515)	Copy Keypad Function Selection	V/f OLV Sets the function that saves and copies drive parameters to a different drive with the keypad. 0 : Copy Select 1 : Backup (drive → keypad) 2 : Restore (keypad → drive) 3 : Verify (check for mismatch)	0 (0 - 3)	319
o3-02 (0516)	Copy Allowed Selection	V/f OLV Sets the copy function when *o3-01 = 1 [Copy Keypad Function Selection = Backup (drive → keypad)]*. 0 : Disabled 1 : Enabled	0 (0, 1)	319

◆ o4: Maintenance Monitor Settings

No. (Hex.)	Name	Description	Default (Range)	Ref.
o4-01 (050B)	Elapsed Operating Time Setting	V/f OLV Sets the initial value of the cumulative drive operation time in 10-hour units.	0 h (0 - 9999 h)	319
o4-02 (050C)	Elapsed Operating Time Selection	V/f OLV Sets the condition that counts the cumulative operation time. 0 : U4-01 Shows Total Power-up Time 1 : U4-01 Shows Total RUN Time	1 (0, 1)	319
o4-03 (050E)	Fan Operation Time Setting	V/f OLV Sets the value from which to start the cumulative drive cooling fan operation time in 10-hour units.	0 h (0 - 9999 h)	320
o4-05 (051D)	Capacitor Maintenance Setting	V/f OLV Sets the *U4-05 [CapacitorMaintenance]* monitor value.	0% (0 - 150%)	320
o4-07 (0523)	Softcharge Relay Maintenance Set	V/f OLV Sets the *U4-06 [PreChargeRelayMainte]* monitor value.	0% (0 - 150%)	320
o4-09 (0525)	IGBT Maintenance Setting	V/f OLV Sets the *U4-07 [IGBT Maintenance]* monitor value.	0% (0 - 150%)	320
o4-11 (0510)	Fault Trace/History Init (U2/U3)	V/f OLV Resets the records of Monitors *U2-xx [Fault Trace]* and *U3-xx [Fault History]*. 0 : Disabled 1 : Enabled	0 (0, 1)	320
o4-12 (0512)	kWh Monitor Initialization	V/f OLV Resets the monitor values for *U4-10 [kWh, Lower 4 Digits]* and *U4-11 [kWh, Upper 5 Digits]*. 0 : No Reset 1 : Reset	0 (0, 1)	321
o4-13 (0528)	RUN Command Counter @ Initialize	V/f OLV Resets the monitor values for *U4-02 [Num of Run Commands]*, *U4-24 [Number of Runs (Low)]*, and *U4-25 [Number of Runs (High)]*. 0 : No Reset 1 : Reset	0 (0, 1)	321

No. (Hex.)	Name	Description	Default (Range)	Ref.
o4-22 (154F) RUN	Time Format	V/f OLV Sets the time display format. This parameter is only available when using an LCD keypad. 0 : 24 Hour Clock 1 : 12 Hour Clock 2 : 12 Hour JP Clock	0 (0 - 2)	321
o4-23 (1550) RUN	Date Format	V/f OLV Sets the date display format. This parameter is only available when using an LCD keypad. 0 : YYYY/MM/DD 1 : DD/MM/YYYY 2 : MM/DD/YYYY	0 (0 - 2)	321
o4-24 (310F) RUN	bAT Detection Selection	V/f OLV Sets the operation when the drive detects *bAT [Keypad Battery Low Voltage]* and *TiM [Keypad Time Not Set]*. This parameter is only available when you use an LCD keypad. 0 : Disable 1 : Enable (Alarm Detected) 2 : Enable (Fault Detected)	0 (0 - 2)	321

10.12 S: Elevator Parameters

◆ S1: Brake Sequence

No. (Hex.)	Name	Description	Default (Range)	Ref.
S1-01 (0680)	Zero Speed Level at Stop	V/f OLV Determines the speed to begin applying DC Injection when the drive is ramping to stop (b1-03 = 0). Set as a percentage of the maximum output frequency (E1-04).	Determined by A1-02 (0.000 - 9.999%)	323
S1-02 (0681)	DC Injection Current at Start	V/f OLV Determines the amount of current to use for DC Injection at start. Set as a percentage of the drive rated current.	50% (0 - 75%)	323
S1-03 (0682)	DC Injection Current at Stop	V/f OLV Determines the amount of current to use for DC Injection at stop. Set as a percentage of the drive rated current. When using OLV Control, the DC injection current is determined by multiplying S1-03 by S3-25 or S3-26.	50% (0 - 75%)	323
S1-04 (0683)	DC Injection Time at Start	V/f OLV Determines how long the drive should perform DC Injection at start. During this time, the drive allows motor flux to develop, which is essential for applying torque quickly once the brake is released. A setting of 0.00 disables S1-04.	0.40 s (0.00 - 10.00 s)	323
S1-05 (0684)	DC Injection Time at Stop	V/f OLV Determines how long the drive should perform DC Injection at stop. A setting of 0.00 disables S1-05.	0.60 s (0.00 - 10.00 s)	323
S1-06 (0685)	Brake Release Delay Time	V/f OLV Determines the time that must pass after an Up/Down command is entered before the output terminal set for "Brake control" (H2-xx = 50) is triggered. Adjusting this delay time can help when there is not enough time to develop the appropriate amount of motor flux. Be sure to also increase the time S1-04 when setting S1-06 to relatively long delay time.	0.20 s (0.00 - 10.00 s)	324
S1-07 (0686)	Brake Apply Delay Time	V/f OLV Determines the time that must pass after zero speed is reached before the output terminal set for "Brake control" (H2-xx = 50) is released.	0.10 s (0.00 - 10.00 s)	324
S1-10 (0687)	Run Command Delay Time	V/f OLV Determines the time that must pass after zero speed is reached before the output terminal set for "Brake control" (H2-xx = 50) is released.	0.10 s (0.00 - 10.00 s)	324
S1-11 (0688)	Output Reactor Open Delay Time	V/f OLV Determines the time that must pass for an output terminal set for “Output contactor control” (H2-xx = 51) to be released after the drive has stopped and drive output has been shut off.	0.10 s (0.00 - 10.00 s)	324

◆ S2: Slip Compensation for Elevators

No. (Hex.)	Name	Description	Default (Range)	Ref.
S2-01 (068F)	Motor Rated Speed	V/f OLV Sets the rated speed of the motor.	1380 rpm (300 - 1800 rpm)	324
S2-02 (0690)	Slip Compensation Gain in Motoring Mode	V/f OLV Slip compensation for leveling speed can be set separately for motoring and regenerative states to help improve the accuracy of leveling.	0.7 (0.0 - 5.0 rpm)	324
S2-03 (0691)	Slip Compensation Gain in Regenerative Mode	V/f OLV Slip compensation for leveling speed can be set separately for motoring and regenerative states to help improve the accuracy of leveling.	1.0 (0.0 - 5.0)	324
S2-05 (0693)	Slip Compensation Torque Detection Delay Time	V/f OLV Sets a delay time before detecting torque for slip compensation.	1000 ms (0 - 10000 ms)	324
S2-06 (0694)	Slip Compensation Torque Detection Filter Time Constant	V/f OLV Sets the filter time constant applied to the torque signal used for the slip compensation value calculation.	500 ms (0 - 2000 ms)	325

◆ S4: Rescue Operation

No. (Hex.)	Name	Description	Default (Range)	Ref.
S4-01 (06A6)	Light Load Direction Search Selection	V/f OLV Enables and disables the Light Load Direction Search. 0 : Disabled 1 : Advanced	0 (0, 1)	325
S4-20 (06B1)	Light Load Search Override	V/f OLV Defines the source that determines the light load search direction. 0 : Controller Direction 1 : Inverter Direction	0 (0 - 1)	325

◆ S6: Error Detection

No. (Hex.)	Name	Description	Default (Range)	Ref.
S6-01 (06B3)	Motor Contactor Response Error (SE1) Detection/Reset Selection	V/f OLV Determines when the drive should detect a motor contactor response error (SE1). SE1 is triggered if there is no response from the motor contactor within the time set in S6-10 after the contactor control output has been set. 0 : Detect during stop, SE1 must be manually reset 1 : Detect during stop, SE1 can be automatically reset 2 : No SE1 detection	0 (0 - 2)	325
S6-02 (06B4)	Starting Current Error (SE2) Detection Delay Time	V/f OLV Sets a delay time for starting current error (SE2). SE2 is detected when the drive output current is below 25% after the Up/Down command has been entered and the brake release time and the time set to S6-02 have both passed. The brake control command will not be issued (brake stays applied).	200 ms (0.00 - [S1-04 - S1-06] ms)	325
S6-04 (06B6)	Output Current Error (SE3) Detection Delay Time	V/f OLV Sets a delay time for detecting an output current fault (SE3). SE3 is detected when the drive output current drops below 25% after the brake has released.	200 ms (0 - 5000 ms)	325
S6-05 (06B7)	Brake Response Error (SE4) Detection Time	V/f OLV Sets a delay time for detecting a brake response error (SE4). SE4 is detected when an output terminal set for “Brake release” (H2-xx = 50) and an input terminal set for “Brake feedback” (H1-xx = 79) do match for the time set to S6- 05.	500 ms (0 - 10000 ms)	326
S6-15 (06BB)	Speed Reference Loss Detection	V/f OLV Enables speed reference loss detection when D1-18 = 1 and H1-xx ≠ 53. 0 : Disabled 1 : Enabled	0 (0, 1)	326
S6-16 (06BC)	Restart after Baseblock Selection	V/f OLV Allows the drive to restart the motor after returning to normal operation from Baseblock state (H1-xx = 8/9) or from Safe Torque-Off state (Safe Disable inputs H1 and H2 enabled) while the Up/Down command is still active.	0 (0, 1)	326

10.13 T: Motor Tuning

◆ T1: Induction Motor Auto-Tuning

No. (Hex.)	Name	Description	Default (Range)	Ref.
T1-01 (0701)	Tuning Mode Selection	V/f OLV Sets the type of Auto-Tuning. 0 : Rotational Auto-Tuning 1 : Stationary Auto-Tuning 1 2 : Stationary Line-Line Resistance	Determined by A1-02 (Determined by A1-02)	327
T1-02 (0702)	Motor Rated Power	V/f OLV Uses the units set in *o1-58 [Motor Power Unit Selection]* to set the motor rated output power.	Determined by o2-04 (0.00 - 650.00 kW)	327
T1-03 (0703)	Motor Rated Voltage	V/f OLV Sets the rated voltage (V) of the motor. Enter the base speed voltage for constant output motors.	400 V Class: 400.0 V (400 V Class: 0.0 - 511.0 V)	327
T1-04 (0704)	Motor Rated Current	V/f OLV Sets the rated current (A) of the motor.	Determined by o2-04 (10% to 200% of the drive rated current)	327
T1-05 (0705)	Motor Base Frequency	V/f OLV Sets the base frequency (Hz) of the motor.	50.0 Hz (0.0 - 120.0 Hz)	328
T1-06 (0706)	Number of Motor Poles	V/f OLV Sets the number of motor poles.	4 (2 to 48)	328
T1-07 (0707)	Motor Base Speed	V/f OLV Sets the motor base speed for Auto-Tuning (min^{-1} (r/min)).	1450 min^{-1} (r/min) (0 - 24000 min^{-1} (r/min))	328
T1-08 (0708)	Motor no-load voltage	V/f OLV	40% of motor nominal current	328

10.14 U: Monitors

◆ U1: Operation Status Monitors

No. (Hex.)	Name	Description	MFAO Signal Level
U1-01 (0040)	Frequency Reference	V/f OLV Shows the frequency reference value. Parameter *o1-03 [Keypad Display Unit Selection]* sets the display units. Unit: 0.01 Hz	10 V = Maximum frequency (0 V to +10 V)
U1-02 (0041)	Output frequency	V/f OLV Shows the output frequency. Parameter *o1-03 [Keypad Display Unit Selection]* sets the display units. Unit: 0.01 Hz	10 V = Maximum frequency (0 V to +10 V)
U1-03 (0042)	Output Current	V/f OLV Shows the actual output current. The keypad shows the value of *U1-03* in amperes (A). When looking at the monitor through MEMOBUS/Modbus communications, the current is "8192 = drive rated current (A)". Calculate the current from the monitor value that is in at MEMOBUS/Modbus communications using "Numerals being displayed / 8192 × drive rated current (A). Unit: When the drive model changes, the display units for this parameter also change. • 0.01 A units: B001 - B018, 2001 - 2042, 4001 - 4023 • 0.1A units: 2056 - 2082, 4031 - 4060	10 V = Drive rated current
U1-04 (0043)	Control method selection	V/f OLV Shows the drive control method. 0 : V/f Control 2 : Open Loop Vector 5 : PM Open Loop Vector 6 : PM Advanced Open Loop Vector 8 : EZ Vector Control	No signal output available
U1-05 (0044)	Motor Speed	V/f OLV Shows the detected motor speed. Parameter *o1-03 [Keypad Display Unit Selection]* sets the display units. Unit: 0.01 Hz	10 V = Maximum frequency (0 V to +10 V)
U1-06 (0045)	Output Voltage Ref	V/f OLV Shows the output voltage reference. Unit: 0.1 V	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms
U1-07 (0046)	DC Bus Voltage	V/f OLV Shows the DC bus voltage. Unit: 1 V	200 V class: 10 V = 400 V 400 V class: 10 V = 800 V
U1-08 (0047)	Output Power	V/f OLV Shows the internally-calculated output power. When you change *A1-02 [Control Method Selection]*, it will also change the signal level of the analog output. • A1-02 = 0: Drive capacity (kW) • A1-02 = 2: *Motor Rated Power [E2-11]* (kW) • A1-02 = 5, 6: *PM Motor Rated Power [E5-02]* (kW) • A1-02 = 8: *Motor Rated Power [E9-07]* (kW) Unit: When the drive model changes, the display units for this parameter also change. • 0.01 A units: B001 to B018, 2001 to 2042, 4001 to 4023 • 0.1 A units: 2056 - 2082, 4031 - 4060	10 V: Drive capacity (motor rated power) kW (0 V to +10 V)
U1-09 (0048)	Torque Reference	V/f OLV Shows the internal torque reference value. Unit: 0.1%	10 V = Motor rated torque (0 V to +10 V)
U1-10 (0049)	Input Terminal Status	V/f OLV Shows the status of the MFDO terminal where 1 = ON and 0 = OFF. For example, *U1-10* shows 00000101 when terminals S1 and S3 are ON. bit0 : Terminal S1 (MFDI 1) bit1 : Terminal S2 (MFDI 2) bit2 : Terminal S3 (MFDI 3) bit3 : Terminal S4 (MFDI 4) bit4 : Terminal S5 (MFDI 5) bit5 : Terminal S6 (MFDI 6) bit6 : Terminal S7 (MFDI 7) bit7 : Not used (normal value of [0]).	No signal output available

No. (Hex.)	Name	Description	MFAO Signal Level
U1-11 (004A)	Output Terminal Status	V/f OLV Shows the status of the MFDO terminal where 1 = ON and 0 = OFF. For example, *U1-11* shows when terminals MA and P2 are ON. **Note:** When *H2-xx = 100 to 1A7 [U1-11 Inverse U1-11 Output of Function]*, *U1-11* does not show the status in inverse. bit0 : Terminal MA/MB-MC bit1 : Terminal P1-C1 bit2 : Terminal P2-C2 bit3 : Not used (normal value of [0]). bit4 : Not used (normal value of [0]). bit5 : Not used (normal value of [0]). bit6 : Not used (normal value of [0]). bit7 : Not used (normal value of [0]).	No signal output available
U1-12 (004B)	Drive Status	V/f OLV Shows drive status where 1 = ON and 0 = OFF. For example, *U1-12* shows during run with the Reverse Run command. bit0 : During Run bit1 : During zero-speed bit2 : During reverse bit3 : During fault reset signal input bit4 : During speed agreement bit5 : Drive Ready bit6 : During minor fault detection bit7 : During fault detection	No signal output available
U1-16 (0053)	SFS Output Frequency	V/f OLV Shows the output frequency after soft start. Shows the frequency with acceleration and deceleration times and S-curves. Parameter *o1-03 [Keypad Display Unit Selection]* sets the display units. Unit: 0.01 Hz	10 V = Maximum frequency (0 V to +10 V)
U1-18 (0061)	oPE Fault Parameter	V/f OLV Shows the parameter number that caused the *oPE02 [Parameter Range Setting Error]* or *oPE08 [Parameter Selection Error]*.	No signal output available
U1-19 (0066)	MEMOBUS/Modbus Error Code	V/f OLV Shows the contents of the MEMOBUS/Modbus communication error where 1 =”error” and 0 = “no error”. For example, *U1-19* shows when the drive detects a CRC error. bit0 : CRC Error bit1 : Data Length Error bit2 : Not used (normal value of [0]). bit3 : Parity Error bit4 : Overrun Error bit5 : Framing Error bit6 : Timed Out bit7 : Not used (normal value of [0]).	No signal output available
U1-25 (004D)	Software number	V/f OLV Shows the ID.	No signal output available
U1-26 (005B)	SoftwareNumber ROM	V/f OLV Shows the ROM ID.	No signal output available

◆ U2: Fault Trace

No. (Hex.)	Name	Description	MFAO Signal Level
U2-01 (0080)	Current Fault	V/f OLV Shows the fault that the drive has when viewing the monitor.	No signal output available
U2-02 (0081)	Previous Fault	V/f OLV Shows the fault that occurred most recently.	No signal output available

No. (Hex.)	Name	Description	MFAO Signal Level
U2-03 (0082)	Freq Reference@Fault	V/f OLV Shows the frequency reference at the fault that occurred most recently. Use *U1-01 [Frequency Reference]* to monitor the frequency reference value. Unit: 0.01 Hz	No signal output available
U2-04 (0083)	Output Freq @ Fault	V/f OLV Shows the output frequency at the fault that occurred most recently. Use *U1-02 [Output Frequency]* to monitor the actual output frequency. Unit: 0.01 Hz	No signal output available
U2-05 (0084)	Output Current@Fault	V/f OLV Shows the output current at the fault that occurred most recently. Use *U1-03 [Output Current]* to monitor the actual output current. The keypad shows the value of *U1-03* in amperes (A). When looking at the monitor through MEMOBUS/Modbus communications, the current is "8192 = drive rated current (A)". Calculate the current from the monitor value that is in at MEMOBUS/ Modbus communications using "Numerals being displayed / 8192 × drive rated current (A). Unit: When the drive model changes, the display units for this parameter also change. • 0.01 A units: B001 - B018, 2001 - 2042, 4001 - 4023 • 0.1A units: 2056 - 2082, 4031 - 4060	No signal output available
U2-06 (0085)	Motor Speed @ Fault	V/f OLV Shows the motor speed at the fault that occurred most recently. Use *U1-05 [Motor Speed]* to monitor the motor speed. Unit: 0.01 Hz	No signal output available
U2-07 (0086)	Output Voltage@Fault	V/f OLV Shows the output voltage reference at the fault that occurred most recently. Use *U1-06 [Output Voltage Ref]* to monitor the output voltage reference. Unit: 0.1 V	No signal output available
U2-08 (0087)	DC Bus Voltage@Fault	V/f OLV Shows the DC bus voltage at the fault that occurred most recently. Use *U1-07 [DC Bus Voltage]* to monitor the DC bus voltage. Unit: 1 V	No signal output available
U2-09 (0088)	Output Power @ Fault	V/f OLV Shows the output power at the fault that occurred most recently. Use *U1-08 [Output Power]* to monitor the output power. Unit: 0.1 kW	No signal output available
U2-10 (0089)	Torque Ref @ Fault	V/f OLV Shows the torque reference at the fault that occurred most recently as a percentage of the motor rated torque. Use *U1-09 [Torque Reference]* to monitor the torque reference. Unit: 0.1%	No signal output available
U2-11 (008A)	Input Terminal Status @ Fault	V/f OLV Shows the status of the MFDI terminals at the most recent fault where 1 = ON and 0 = OFF. For example, *U2-11* shows when terminals S1 and S3 are ON. Use *U1-10 [Input Terminal Status]* to monitor the actual MFDI terminal status. bit0 : Terminal S1 bit1 : Terminal S2 bit2 : Terminal S3 bit3 : Terminal S4 bit4 : Terminal S5 bit5 : Terminal S6 bit6 : Terminal S7 bit7 : Not used (normal value of [1]).	No signal output available

No. (Hex.)	Name	Description	MFAO Signal Level
U2-12 (008B)	Output Terminal Status @ Fault	V/f OLV Shows the status of the MFDO terminals at the most recent fault where 1 = ON and 0 = OFF. For example, *U2-12* shows when terminals MA and P2 are ON. Use *U1-11 [Output Terminal Status]* to monitor the actual MFDO terminal status. bit0 : Terminal MA/MB-MC bit1 : Terminal P1-C1 bit2 : Terminal P2-C2 bit3 : Not used (normal value of [0]). bit4 : Not used (normal value of [0]). bit5 : Not used (normal value of [0]). bit6 : Not used (normal value of [0]). bit7 : Not used (normal value of [0]).	No signal output available
U2-13 (008C)	Operation Status @ Fault	V/f OLV Shows the status of the MFDO terminals at the most recent fault where 1 = ON and 0 = OFF. For example, *U2-13* shows during run. Use *U1-12 [Drive Status]* to monitor the actual MFDO terminal status. bit0 : During Run bit1 : During zero-speed bit2 : During reverse bit3 : During fault reset signal input bit4 : During speed agreement bit5 : Drive Ready bit6 : During minor fault detection bit7 : During fault detection	No signal output available
U2-14 (008D)	Elapsed Time @ Fault	V/f OLV Shows the cumulative operation time of the drive at the fault that occurred most recently. Use *U4-01 [Cumulative Ope Time]* to monitor the cumulative operation time. Unit: 1 h	No signal output available
U2-15 (07E0)	SFS Output @ Fault	V/f OLV Shows the output frequency after soft start at the fault that occurred most recently. Use *U1-16 [SFS Output Frequency]* to monitor the output frequency after soft start. Unit: 0.01 Hz	No signal output available
U2-16 (07E1)	q-Axis Current@Fault	V/f OLV Shows the q-Axis current of the motor at the fault that occurred most recently. Use *U6-01 [Iq Secondary Current]* to monitor the q-Axis current of the motor. Unit: 0.1 %	No signal output available
U2-17 (07E2)	d-Axis Current@Fault	V/f OLV Shows the d-Axis current of the motor at the fault that occurred most recently. Use *U6-02 [Id ExcitationCurrent]* to monitor the d-Axis current of the motor. Unit: 0.1 %	No signal output available

◆ U3: Fault History

No. (Hex.)	Name	Description	MFAO Signal Level
U3-01 (0090)	1st MostRecent Fault	V/f OLV Shows the fault history of the most recent fault. **Note:** The drive saves this fault history to two types of registers at the same time for the MEMOBUS/Modbus communications.	No signal output available
U3-02 (0091)	2nd MostRecent Fault	V/f OLV Shows the fault history of the second most recent fault. **Note:** The drive saves this fault history to two types of registers at the same time for the MEMOBUS/Modbus communications.	No signal output available
U3-03 (0092)	3rd MostRecent Fault	V/f OLV Shows the fault history of the third most recent fault. **Note:** The drive saves this fault history to two types of registers at the same time for the MEMOBUS/Modbus communications.	No signal output available

No. (Hex.)	Name	Description	MFAO Signal Level
U3-04 (0093)	4th MostRecent Fault	V/f OLV Shows the fault history of the fourth most recent fault. **Note:** The drive saves this fault history to two types of registers at the same time for the MEMOBUS/Modbus communications.	No signal output available
U3-05 (0804)	5th MostRecent Fault	V/f OLV Shows the fault history of the fifth most recent fault.	No signal output available
U3-06 (0805)	6th MostRecent Fault	V/f OLV Shows the fault history of the sixth most recent fault.	No signal output available
U3-07 (0806)	7th MostRecent Fault	V/f OLV Shows the fault history of the seventh most recent fault.	No signal output available
U3-08 (0807)	8th MostRecent Fault	V/f OLV Shows the fault history of the eighth most recent fault.	No signal output available
U3-09 (0808)	9th MostRecent Fault	V/f OLV Shows the fault history of the ninth most recent fault.	No signal output available
U3-10 (0809)	10th MostRecentFault	V/f OLV Shows the fault history of the tenth most recent fault.	No signal output available
U3-11 (0094)	ElapsedTime@1stFault	V/f OLV Shows the cumulative operation time when the most recent fault occurred. **Note:** The drive saves this cumulative operation time to two types of registers at the same time for the MEMOBUS/Modbus communications. Unit: 1 h	No signal output available
U3-12 (0095)	ElapsedTime@2ndFault	V/f OLV Shows the cumulative operation time when the second most recent fault occurred. **Note:** The drive saves this cumulative operation time to two types of registers at the same time for the MEMOBUS/Modbus communications. Unit: 1 h	No signal output available
U3-13 (0096)	ElapsedTime@3rdFault	V/f OLV Shows the cumulative operation time when the third most recent fault occurred. **Note:** The drive saves this cumulative operation time to two types of registers at the same time for the MEMOBUS/Modbus communications. Unit: 1 h	No signal output available
U3-14 (0097)	ElapsedTime@4thFault	V/f OLV Shows the cumulative operation time when the fourth most recent fault occurred. **Note:** The drive saves this cumulative operation time to two types of registers at the same time for the MEMOBUS/Modbus communications. Unit: 1 h	No signal output available
U3-15 (080E)	ElapsedTime@5thFault	V/f OLV Shows the cumulative operation time when the fifth most recent fault occurred. Unit: 1 h	No signal output available
U3-16 (080F)	ElapsedTime@6thFault	V/f OLV Shows the cumulative operation time when the sixth most recent fault occurred. Unit: 1 h	No signal output available
U3-17 (0810)	ElapsedTime@7thFault	V/f OLV Shows the cumulative operation time when the seventh most recent fault occurred. Unit: 1 h	No signal output available
U3-18 (0811)	ElapsedTime@8thFault	V/f OLV Shows the cumulative operation time when the eighth most recent fault occurred. Unit: 1 h	No signal output available
U3-19 (0812)	ElapsedTime@9thFault	V/f OLV Shows the cumulative operation time when the ninth most recent fault occurred. Unit: 1 h	No signal output available
U3-20 (0813)	ElapsedTime@10 Fault	V/f OLV Shows the cumulative operation time when the tenth most recent fault occurred. Unit: 1 h	No signal output available

◆ U4: Maintenance Monitors

No. (Hex.)	Name	Description	MFAO Signal Level
U4-01 (004C)	Cumulative Ope Time	V/f OLV Shows the cumulative operation time of the drive. Use parameter *o4-01 [Elapsed Operating Time Setting]* to reset this monitor. Use parameter *o4-02 [Elapsed Operating Time Selection]* to select the cumulative operation times from: • The time from when the drive is energized until it is de-energized. • The time at which the Run command is turned ON. The maximum value that the monitor will show is *99999*. After this value is more than *99999*, the drive automatically resets it and starts to count from *0* again. Unit: 1 h **Note:** The MEMOBUS/Modbus communication data is shown in 10 h units. Use register 0099H for data in 1 h units.	10 V: 99999 h
U4-02 (0075)	Num of Run Commands	V/f OLV Shows how many times that the drive has received a Run command. Use parameter *o4-13 [RUN Command Counter @ Initialize]* to reset this monitor. The maximum value that the monitor will show is *65535*. After this value is more than *65535*, the drive automatically resets it and starts to count from *0* again. Unit: 1	10 V: 65535 times
U4-03 (0067)	Cooling Fan Ope Time	V/f OLV Shows the cumulative operation time of the cooling fans. Use parameter *o4-03 [Fan Operation Time Setting]* to reset this monitor. The maximum value that the monitor will show is *99999*. After this value is more than *99999*, the drive automatically resets it and starts to count from *0* again. Unit: 1 h **Note:** The MEMOBUS/Modbus communication data is shown in 10 h units. Use register 009BH for data in 1 h units.	10 V: 99999 h
U4-04 (007E)	Cool Fan Maintenance	V/f OLV Shows the cumulative operation time of the cooling fans as a percentage of the replacement life of the cooling fans. Use parameter *o4-03 [Fan Operation Time Setting]* to reset this monitor. Unit: 1% **Note:** Replace the cooling fans when this monitor is 90%.	10 V: 100%
U4-05 (007C)	CapacitorMaintenance	V/f OLV Shows the operation time of the electrolytic capacitors for the main circuit and control circuit as a percentage of the replacement life of the electrolytic capacitors. Use parameter *o4-05 [Capacitor Maintenance Setting]* to reset this monitor. Unit: 1% **Note:** Replace the electrolytic capacitor when this monitor is 90%.	10 V: 100%
U4-06 (07D6)	PreChargeRelayMainte	V/f OLV Shows the operation time of the soft charge bypass relay as a percentage of the replacement life of the soft charge bypass relay. Use parameter *o4-07 [Softcharge Relay Maintenance Set]* to reset this monitor. Unit: 1% **Note:** Replace the drive when this monitor is 90%.	10 V: 100%
U4-07 (07D7)	IGBT Maintenance	V/f OLV Shows the operation time of the IGBTs as a percentage of the replacement life of the IGBTs. Set parameter *o4-09 [IGBT Maintenance Setting]* to reset this monitor. Unit: 1% **Note:** Replace the drive when this monitor is 90%.	10 V: 100%
U4-08 (0068)	Heatsink Temperature	V/f OLV Shows the heatsink temperature of the drive. Unit: 1 °C	10 V: 100 °C
U4-09 (005E)	LED Check	V/f OLV Turns on all of the keypad LEDs to make sure that the LEDs operate correctly. 1. With *U4-09* displayed, press ↵. All LEDs on the keypad will turn on. **Note:** When Safety input 2 CH is open (STo), READY will flash.	No signal output available

No. (Hex.)	Name	Description	MFAO Signal Level
U4-10 (005C)	kWh, Lower 4 Digits	V/f OLV Shows the lower 4 digits of the watt hour value for the drive. Unit: 1 kWh **Note:** The watt hour is displayed in 9 digits. Monitor *U4-11 [kWh, Upper 5 Digits]* shows the upper 5 digits and *U4-10* shows the lower 4 digits. Example for 12345678.9 kWh: *U4-10*: *678.9 kWh* *U4-11*: *12345 MWh*	No signal output available
U4-11 (005D)	kWh, Upper 5 Digits	V/f OLV Shows the upper 5 digits of the watt hour value for the drive. Unit: 1 MWh **Note:** Monitor *U4-11* shows the upper 5 digits and *U4-10 [kWh, Lower 4 Digits]* shows the lower 4 digits. Example for 12345678.9 kWh: *U4-10*: 678.9 kWh *U4-11*: 12345 MWh	No signal output available
U4-13 (07CF)	Peak Hold Current	V/f OLV Shows the hold value of the peak value (rms) for the drive output current. Use *U4-14 [PeakHold Output Freq]* to show the drive output frequency at the time that the drive holds the output current. The drive will hold the peak hold current at the next start up and restart of the power supply. The drive keeps the value that was under hold during baseblock (during stop). The keypad shows the value of *U4-13* in amperes (A). When looking at the monitor through MEMOBUS/Modbus communications, the current is “8192 = drive rated current (A)”. Calculate the current from the monitor value that is in at MEMOBUS/Modbus communications using “Numerals being displayed / 8192 × drive rated current (A). Unit: When the drive model changes, the display units for this parameter also change. • 0.01 A units: B001 - B018, 2001 - 2042, 4001 - 4023 • 0.1A units: 2056 - 2082, 4031 - 4060	No signal output available
U4-14 (07D0)	PeakHold Output Freq	V/f OLV Shows the output frequency at which the peak value (rms) of the drive output current is held. The peak hold current can be monitored by *U4-13 [Peak Hold Current]*. The peak hold output frequency will be cleared at the next startup and restart of the power supply. The drive keeps the value that was under hold during baseblock (during stop). Unit: 0.01 Hz	No signal output available
U4-15 (07E4)	CASE SW Number	V/f OLV Shows the CASE Software number.	No signal output available
U4-16 (07D8)	Motor oL1 Level	V/f OLV Shows the integrated value of *oL1 [Motor Overload]* as a percentage of *oL1* detection level. Unit: 0.1%	10 V: 100%
U4-17 (07D9)	Drive Overload Calculations (OL2)	V/f OLV Shows the level of the drive overload detection (oL2). A value of 100% equals the oL2 detection level. Unit: 0.1%	10 V: 100%

No. (Hex.)	Name	Description	MFAO Signal Level
U4-18 (07DA)	Reference Source	V/f OLV Shows the selected frequency reference source. The keypad shows the frequency reference source as “XY-nn” as specified by these rules: X: *External Reference 1/2 Selection [H1-xx = 2]* selection status • 1: *b1-01 [Frequency Reference Selection 1]* • 2: *b1-15 [Frequency Reference Selection 2]* Y-nn: Frequency reference source • 0-01: Keypad (*d1-01 [Reference 1]*) • 1-01: MFAI terminal A1 • 1-02: MFAI terminal A2 • 2-02 to 2-08: Multi-step speed reference (*d1-02 to d1-08 [Reference 2 to 8, Jog Reference]*) • 3-01: MEMOBUS/Modbus communications • 4-01: Communication option card	No signal output available
U4-21 (07DD)	Run Command Source	V/f OLV Shows the selected Run command source. The keypad shows the Run command source as “XY-nn” as specified by these rules: X: *External Reference 1/2 Selection [H1-xx = 2]* selection status • 1: *b1-02 [Run Command Selection 1]* Y: Run command source • 0: Keypad • 1: Control circuit terminal • 3: MEMOBUS/Modbus communications • 4: Communication option card nn: Run command limit status data • 00: No limit status. • 01: The Run command was left ON when the drive stopped in the Programming Mode. • 02: The Run command was left ON when switching from LOCAL Mode to REMOTE Mode. • 03: The Run command is in standby after the drive was energized until the soft charge bypass contactor turns ON. **Note:** The drive will detect *Uv1 [DC Bus Undervoltage]* or *Uv [Undervoltage]* if the soft charge bypass contactor does not turn ON after 10 s. • 04: Restart after run stop is prohibited. • 05: Fast stop has been executed using the MFDI terminal. Or, the motor has ramped to stop by pressing the STOP key on the keypad. • 07: During baseblock while coast to stop with timer. • 08: Frequency reference is below *E1-09 [Minimum Output Frequency]* during baseblock. • 09: Waiting for the Enter command from PLC.	No signal output The keypad shows the Run command source as "XY-nn" as specified by these rules: available

◆ U6: Operation Status Monitors

No. (Hex.)	Name	Description	MFAO Signal Level
U6-01 (0051)	Iq Secondary Current	V/f OLV Shows the value calculated for the motor secondary current (q-Axis) as a percentage of the motor rated secondary current. Unit: 0.1%	10 V: Motor secondary rated current (0 V to +10 V)
U6-02 (0052)	Id ExcitationCurrent	V/f OLV Shows the value calculated for the motor excitation current (d-Axis) as a percentage of the motor rated secondary current. Unit: 0.1%	10 V: Motor secondary rated current (0 V to +10 V)
U6-03 (0054)	ASR Input	V/f OLV Shows the ASR input value as a percentage of the maximum frequency. Unit: 0.01%	10 V: Maximum frequency (0 V to +10 V)
U6-04 (0055)	ASR Output	V/f OLV Shows the ASR output value as a percentage of the motor rated secondary current. Unit: 0.01%	10 V: Motor secondary rated current (0 V to +10 V)
U6-05 (0059)	OutputVoltageRef: Vq	V/f OLV Shows the drive internal voltage reference for motor secondary current control (q-Axis). Unit: 0.1 V	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (0 V to +10 V)
U6-06 (005A)	OutputVoltageRef: Vd	V/f OLV Shows the drive internal voltage reference for motor excitation current control (d-Axis). Unit: 0.1 V	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (0 V to +10 V)

No. (Hex.)	Name	Description	MFAO Signal Level
U6-07 (005F) Expert	q-Axis ACR Output	V/f OLV Shows the output value for current control related to motor secondary current (q axis). Unit: 0.1%	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (0 V to +10 V)
U6-08 (0060) Expert	d-Axis ACR Output	V/f OLV Shows the output value for current control related to motor excitation current (d axis). Unit: 0.1%	200 V class: 10 V = 200 Vrms 400 V class: 10 V = 400 Vrms (0 V to +10 V)

10.15 Parameters that Change from the Default Settings with A1-02 [Control Method Selection]

The values for the parameters in these tables depend on the values for parameter *A1-02*. When you change the setting for *A1-02*, the default settings will change.

Table 10.1 A1-02 = 0, 2 [V/f, OLV]

No.	Name	Range	Unit	Control Method (A1-02 Setting)	
				V/f (0)	OLV (2)
C3-01	Slip Compensation Gain	0.0 - 2.5	0.1	0.0	1.0
C3-02	Slip Compensation Delay Time	0 - 10000	1 ms	2000	200
C3-05	Output Voltage Limit Operation Selection	0, 1	-	-	0
C4-02	Torque Compensation Delay Time	0 - 10000	1 ms	200	20
C5-01	Speed Control Loop Proportional Gain 1	0.00 - 300.00	0.01	-	0.20
C5-02	Speed Control Loop Integral Time 1	0 000 - 10.000	0.001 s	-	0.200
C5-03	Speed Control Loop Proportional Gain 2	0.00 - 300.00	0.01	-	0.02
C5-04	ASR Integral Time 2	0.000 - 10.000	0.001 s	-	0.05
C5-05	ASR Limit	0.0 - 20.0	0.1%	-	5.0
C5-06	ASR Primary Delay Time Constant	0.000 - 0.500	0.001 s	-	0.004
C5-07	Speed Control Loop Gain Switching Speed	0-0 - 100.0	0.1%	-	0.0
C5-13	Speed Control Loop Proportional Gain 3	0.00 - 300.00	0.01	-	0.20
C5-14	Speed Control Loop Integral Time 3	0.000 - 10.000	0.001 s	-	0.200
C5-40	ASR Start Speed	0.00 - 100.00	0.01%	-	2.50
E1-08	Mid Point A Voltage	0.0 - 510.0	0.1 V	15.0 *1	25.0
E1-09	Minimum Output Frequency	0.0 - 400.0	0.1 Hz	1.5 *1	0.3
E1-10	Minimum Output Voltage	0.0 - 510.0	0.1 V	9.0 *1	5.0
S1-01	Zero Speed Level at Stop	0.0000 to 9.9999	0.001%	1.000	2.400

*1 The default setting changes when the drive model and *E1-03 [V/f Pattern Selection]* change.

10.16 Defaults by Drive Model

The values for the parameters in these tables depend on the values for parameter *o2-04*. Changing the settings for *o2-04* will change the default settings.

◆ Three-Phase 400 V Class

Parameter	Name	Unit	Default						
-	Drive Model	-	4009	4015	4018	4024	4031	4039	4060
o2-04	Drive Model Selection	Hex.	72	73	74	75	76	77	78
E2-11	Motor Rated Power	kW	4.0	5.5	7.5	11	15	18.5	22
C6-02	Carrier Frequency Selection	-	3	3	3	3	3	3	3
E2-01	Motor Rated Current (FLA)	A	7	9.8	13.3	19.9	26.5	32.9	38.6
E2-02	Motor Rated Slip	Hz	2.7	1.5	1.3	1.7	1.6	1.67	1.7
E2-03	Motor No-Load Current	A	2.3	2.6	4	5.6	7.6	7.8	9.2
E2-05	Motor Line-to-Line Resistance	Ω	3.333	1.595	1.152	0.922	0.55	0.403	0.316
E2-06	Motor Leakage Inductance	%	19.3	18.2	15.5	19.6	17.2	20.1	23.5
E2-10	Motor Iron Loss	W	130	193	263	385	440	508	586
L2-05	Undervoltage Detection Lvl (Uv1)	-	380	380	380	380	380	380	380
L8-02	Overheat Alarm Level	°C	110	120	120	120	120	125	115
L8-09	Output Ground Fault Detection	-	0	0	0	1	1	1	1
L8-38	Carrier Frequency Reduction	-	1	2	2	2	2	2	2
T1-02	Motor Rated Power	kW	4.0	5.5	7.5	11	15	18.5	22
T1-04	Motor Rated Current	A	7	9.8	13.3	19.9	26.5	32.9	38.6

11

Parameter Details

11.1 A: Initialization Parameters 250
11.2 b: Application 254
11.3 C: Tuning 258
11.4 d: References 266
11.5 E: Motor Parameters 269
11.6 H: Terminal Function Selection 274
11.7 L: Protection Functions 293
11.8 n: Special Adjustment 312
11.9 o: Keypad-Related Settings 314
11.10 S: Elevator Parameters 323
11.11 T: Auto-Tuning 327

11.1 A: Initialization Parameters

A parameters [Initialization Parameters] set the operating environment and operating conditions for the drive.

◆ A1: Initialization

A1 parameters set the operating environment and operating conditions for the drive. For example, these parameters set the keypad language, the control method, and the parameter access level for the drive.

■ A1-00: Language Selection

No. (Hex.)	Name	Description	Default (Range)
A1-00 (0100) RUN	Language Selection	V/f OLV Sets the language for the LCD keypad.	0 (0 - 12)

Note:
- This parameter is only available when you use an LCD keypad or a Bluetooth LCD Keypad.
- When you use *A1-03 [Initialize Parameters]* to initialize the drive, the drive will not reset this parameter.

0 : English
1 : Japanese
2 : German
3 : French
4 : Italian
5 : Spanish
6 : Portuguese
7 : Chinese
8 : Czech
9 : Russian
10 : Turkish
11 : Polish
12 : Greek

■ A1-01: Access Level Selection

No. (Hex.)	Name	Description	Default (Range)
A1-01 (0101) RUN	Access Level Selection	V/f OLV Sets user access to parameters. The access level controls which parameters the keypad will display, and which parameters the user can set.	2 (0 - 3)

0 : Operation Only
Access to *A1-00, A1-01, A1-04 [Password]*, and the *U Monitors*.
1 : User Parameters
Access to *A1-00, A1-01, A1-04*, and *A2-01 to A2-32 [User Parameters 1 to 32]*.
2 : Advanced Level
Access to all parameters, but not Expert Mode parameters.
3 : Expert Level
Access to all parameters including Expert Mode parameters.
Table 11.1 shows which keypad screens are available for each *A1-01* settings.

Table 11.1 Access Level and Available Keypad Screens

Mode	Keypad Screen	*A1-01 [Access Level Selection]* Setting			
		0	1	2	3
Drive Mode	Monitors	Yes	Yes	Yes	Yes
Programming Mode	Parameters	Yes	Yes	Yes	Yes
	User Custom Parameters	No	Yes	Yes	Yes
	Parameter Backup/Restore	No	No	Yes	Yes
	Modified Parameters/Fault Log	No	No	Yes	Yes
	Auto-Tuning	No	No	Yes	Yes
	Initial Setup Screen	No	No	Yes	Yes
	Diagnostic Tools	No	No	Yes	Yes

Note:
- When you use *A1-04* and *A1-05 [Password Setting]* to set a password, you cannot change the values set in *A1-01 to A1-03, A1-06, A1-07*,or *A2-01 to A2-32*.
- When *H1-xx = 1B [MFDI Function Select = Program Lockout]*, you must activate the terminal to change parameter settings.
- When you use MEMOBUS/Modbus communications, you must send the Enter command from the controller to the drive and complete the serial communication write process before you can use the keypad to change parameter settings.

■ A1-02: Control Method Selection

No. (Hex.)	Name	Description	Default (Range)
A1-02 (0102)	Control Method Selection	V/f OLV Sets the control method for the drive application and the motor.	0 (0, 2)

Note:
When you change the control methods, the parameter values specified by *A1-02* are changed to their default values.

Sets the control method for the drive application and the motor.

0 : V/f Control

Use this control method in these applications and conditions:

- For general variable-speed control applications in which a high level of responsiveness or high-precision speed control is not necessary.
- Applications in which more than one motor are connected to one drive
- When there is not sufficient data to set the motor parameters
- When it is not possible to do Auto-Tuning. The speed control range is 1:40.

2 : Open Loop Vector

Use this control method for general variable-speed control applications in which high-precision speed control is necessary. In this control method, a feedback signal from the motor is not necessary to have high torque response and high torque when operating at low speeds. The speed control range is 1:120.

■ A1-03: Initialize Parameters

No. (Hex.)	Name	Description	Default (Range)
A1-03 (0103)	Initialize Parameters	V/f OLV Sets parameters to default values.	0 (0 - 2220)

Note:
- After you initialize the drive, the drive automatically sets *A1-03 = 0*.
- User Parameters can save the parameter values for your application and use these values as default values for drive initialization.

0 : No Initialization

1110 : User Initialization

Sets parameters to the values set by the user as user settings. Set *o2-03 = 1 [User Parameter Default Value = Set defaults]* to save the user settings.

You can save the parameter settings that were adjusted for the test run as user-set default values to the drive. When you make changes to the parameter values after you save the settings as User Parameter Settings, the drive will set the parameters to the User Parameter Setting value when you initialize with *A1-03 = 1110*.

Follow this procedure to save User Parameter setting values, and to do a User Initialization.

1. Set parameters correctly for the application.
2. Set *o2-03 = 1 [User Parameter Default Value = Set defaults]*.
   This saves parameter settings for a User Initialization.
   The drive will then automatically set *o2-03 = 0*.
3. Set *A1-03 = 1110* to reset to the saved parameter settings.
   When you initialize the drive, the drive sets the parameter values to the User Parameter setting values.

2220 : 2-Wire Initialization

Sets MFDI terminal S1 to Forward Run and terminal S2 to Reverse Run, and resets all parameters to default settings.

■ A1-04: Password

No. (Hex.)	Name	Description	Default (Range)
A1-04 (0104)	Password	V/f OLV Entry point for the password set in *A1-05 [Password Setting]*. The user can view the settings of parameters that are locked without entering the password. Enter the correct password in this parameter to change parameter settings.	0000 (0000 - 9999)

If the password entered in *A1-04* does not agree with the password setting in *A1-05*, you cannot change these parameters:

- *A1-01 [Access Level Selection]*
- *A1-02 [Control Method Selection]*
- *A1-03 [Initialize Parameters]*
- *A2-01 to A2-32 [User Parameter 1 to 32]*

To lock parameter settings after making changes without changing the password, enter the incorrect password in *A1-04* and push ⏎.

Enter the Password to Unlock Parameters

Use this procedure to unlock parameter settings.

Set the password in *A1-05 [Password Setting]*, and show the Parameter Setting Mode screen on the keypad.

This procedure verifies the password, and makes sure that the parameter settings are unlocked.

1. Push ^ or v to select “A: Initialization Parameters”, then push ⏎.
2. Push ^ or v to select *[A1-04]*, then push ⏎.
   You can now change parameter settings.
3. Push > or < to move the digit and enter the password.
4. Push ⏎ to confirm the password.
   The drive unlocks the parameters and automatically shows the Parameters Screen.
5. Push ^ or v to show *[A1-02]*, then push ⏎.
   The keypad shows the setting value for *[A1-02]*.
6. Push ^ or v to make sure that you can change the setting value.

Push F1 (Back) until the keypad shows the Parameter Setup Mode screen.

■ A1-05: Password Setting

No. (Hex.)	Name	Description	Default (Range)
A1-05 (0105)	Password Setting	V/f OLV Set the password to lock parameters and prevent changes to parameter settings. Enter the correct password in *A1-04 [Password]* to unlock parameters and accept changes.	0000 (0000 - 9999)

This parameter can lock these parameter settings:

- *A1-01 [Access Level Selection]*
- *A1-02 [Control Method Selection]*
- *A1-03 [Initialize Parameters]*
- *A2-01 to A2-32 [User Parameter 1 to 32]*

Note:

- Usually, the keypad will not show *A1-05*. To show and set *A1-05*, show *A1-04 [Password]* and then push STOP and ∨ on the keypad at the same time.
- After you set *A1-05*, the keypad will not show it again until you enter the correct password in *A1-04*. Make sure that you remember the *A1-05* setting value. If you do not know the *A1-05* setting value, contact the manufacturer or your nearest sales representative.
- When *A1-03 = 2220 [2-Wire Initialization]*, the drive is initialized to *A1-05 = 0000*. Be sure to set the password again when a password is necessary for the application.
- Change the setting value in *A1-05* to change the password. The new setting value becomes the new password.
- When you use the password to unlock and change a parameter, enter a value other than the password in *A1-04* to lock the parameter again with the same password.
- If *A1-04 ≠ A1-05*, MEMOBUS Communication cannot read or write *A1-05*.

◆ A2: User Parameters

You can register frequently used parameters and recently changed parameters here to access them quickly. Use Setup Mode to show the saved parameters.

■ A2-01 to A2-32: User Parameters 1 to 32

No. (Hex.)	Name	Description	Default (Range)
A2-01 to A2-32 (0106 - 0125)	User Parameters 1 to 32	V/f OLV You can select a maximum of 32 parameters for the drive and save the values to parameters *A2-01 to A2-32*. Use Setup Mode to show the saved parameters. You can immediately access these saved parameters.	Parameters in General-Purpose Setup Mode (Determined by A1-06)

Note:

- When the *A1-06 [Application Preset]* value changes, the settings for *A2-01 to A2-32* change.
- You must set *A1-01 = 1 [Access Level Selection = User Parameters]* to access parameters *A2-01 to A2-32*.
- When *A1-07 = 1 or 2 [DriveWorksEZ Function Selection = DWEZ Enabled or Enabled/Disabled wDigital Input]*, the drive saves *qx-xx [DriveWorksEZ Parameters]* to *A2-01 to A2-32*.

The drive saves these parameters to *A2-01 to A2-32*.

- The drive saves a maximum of 32 parameters.

  Note:

  Set *A1-01 = 2 [Advanced Level]* or *A1-01 = 3 [Expert Level]* to save the necessary parameters.

- The drive automatically saves changed parameters to *A2-17 to A2-32*.

  Note:

  Set *A2-33 = 1 [User Parameter Auto Selection = Enabled]*.

■ A2-33: User Parameter Auto Selection

No. (Hex.)	Name	Description	Default (Range)
A2-33 (0126)	User Parameter Auto Selection	V/f OLV Sets the automatic save feature for changes to parameters *A2-17 to A2-32 [User Parameters 17 to 32]*.	Determined by A1-06 (0, 1)

0 : Disabled: Manual Entry Required

Set User Parameters manually.

1 : Enabled: Auto Save Recent Parms

The drive automatically registers changed parameter *A2-17 to A2-32*. The drive automatically saves the most recently changed parameter to *A2-17*, and saves a maximum of 16 parameters. After the drive registers 16 parameters, when you save a new parameter, the drive will remove a parameter from the User Parameter list to make space for the new parameter. The drive removes parameters with First In, First Out.

Use Setup Mode to show the saved parameters.

Note:

In General-Purpose Setup Mode, the drive saves parameters starting with *A2-27* because the drive saves parameters *A2-26* and lower by default.

11.2 b: Application

b parameters set these functions:

- Frequency reference source/Run command source
- Stopping method settings
- DC Injection Braking
- Speed Search
- Timer Function
- PID control
- Dwell function
- Energy-Saving Control

◆ b1: Operation Mode Selection

b1 parameters set the operation mode for the drive.

■ b1-02: Up/Down Command Selection

No. (Hex.)	Name	Description	Default (Range)
b1-02 (0181)	Run Command Selection 1	V/f OLV Sets the Up/Down command source in Remote mode. Wire the motor so the elevator goes up when an Up command is issued.	1 (0 - 1)

0 : Keypad

Use the keypad to enter the Up/Down command. Use this setting only when performing a test tun.

Note:

The LORE on the keypad is on while keypad is the Up/Down command source.

1 : Control Circuit Terminal

Use the control circuit terminals to enter Up/Down commands.

■ b1-03: Stopping Method Selection

No. (Hex.)	Name	Description	Default (Range)
b1-03 (0182)	Stopping Method Selection	V/f OLV Sets the method to stop the motor after removing a Run command or entering a Stop command.	0 (0 - 1)

Select the applicable stopping method for the application from these options:

0 : Ramp to Stop

When you enter the Stop command or turn OFF the Run command, the drive ramps the motor to stop.

The drive ramps the motor to stop as specified by the deceleration time. The default setting for the deceleration time is *C1-02 [Deceleration Time 1]*. The actual deceleration time changes as the load conditions change (for example, mechanical loss and inertia).

1 : Coast to Stop

When you enter the Stop command or turn OFF the Run command, the drive turns OFF the output and coasts the motor to stop.

Load conditions will have an effect on the deceleration rate as the motor coasts to stop (for example, mechanical loss and inertia).

■ b1-08: Up/Down Command Select in PRG Mode

No. (Hex.)	Name	Description	Default (Range)
b1-08 (0187)	Run Command Select in PRG Mode	V/f OLV Sets the conditions for the drive to accept an Up/Down command entered from an external source when using the keypad to set parameters.	0 (0 - 2)

As a safety precaution, when the drive is in Programming Mode, it will not respond to a Up/Down command.

This parameter helps prevent accidents that can occur if the motor starts to rotate because the drive received a Up/Down command from an external source while the user is programming the drive. You can also set the drive to not show the Programming Mode when a Up/Down command is active.

Note:

Refer to this table for Drive Mode and Programming Mode functions.

Mode	Keypad Screen	Function
Drive Mode	Monitors	Sets monitor display.
Programming Mode	Parameters	Changes parameter settings.
	User Custom Parameters	Shows the User Parameters.
	Parameter Backup/Restore	Saves parameters to the keypad as backup.
	Modified Parameters/Fault Log	Shows modified parameters and fault history.
	Auto-Tuning	Auto-Tunes the drive.
	Initial Setup	Changes initial settings.
	Diagnostic Tools	Sets data logs and backlight.

0 : Disregard RUN while Programming

The drive does not accept the Up/Down command when setting the parameters in the Programming Mode.

1 : Accept RUN while Programming

The drive accepts a Up/Down command entered from an external source when setting the parameters in Programming Mode.

2 : Allow Programming Only at Stop

The drive does not allow the user to enter the Programming Mode while the drive is operating. The keypad does not display the Programming Mode while the drive is operating.

■ b1-14: Phase Order Selection

No. (Hex.)	Name	Description	Default (Range)
b1-14 (01C3)	Phase Order Selection	V/f OLV Sets the phase order for output terminals U/T1, V/T2, and W/T3. This parameter can align the Forward Run command from the drive and the forward direction of the motor without changing wiring.	0 (0, 1)

0 : Standard

1 : Switch Phase Order

◆ b4: Timer Function

The drive uses timers to delay activating and deactivating MFDO terminals.

Timers prevent sensors and switches from making chattering noise.

There are two types of timers:

- Timers that set a delay for timer inputs and timer outputs.
  These timers delay activating and deactivating of the MFDIs and MFDOs.
  To enable this function, set *H1-xx = 18 [MFDI Function Selection = Timer Function], and set H2-01 to H2-03 = 12 [MFDO Function Selection = Timer Output]*.
- Timers that set a delay to activate and deactivate MFDO terminals.
  These timers delay activating and deactivating MFDO terminals.
  To enable this function, set delay times in parameters *b4-03 to b4-08*.

■ Timer Function Operation

- Timers that Set a Delay for Timer Inputs and Timer Outputs
  Triggers timer output if the timer input is active for longer than the time set in *b4-01 [Timer Function ON-Delay Time]*. Triggers timer output late for the time set in *b4-02 [Timer Function OFF-Delay Time]*. Figure 11.1 shows an example of how the timer function works.

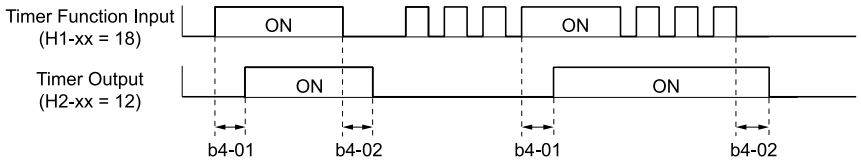


Figure 11.1 Example of Timer Function Operation

■ b4-01: Timer Function ON-Delay Time

No. (Hex.)	Name	Description	Default (Range)
b4-01 (01A3)	Timer Function ON-Delay Time	V/f OLV Sets the ON-delay time for the timer input.	0.0 s (0.0 - 3000.0 s)

■ b4-02: Timer Function OFF-Delay Time

No. (Hex.)	Name	Description	Default (Range)
b4-02 (01A4)	Timer Function OFF-Delay Time	V/f OLV Sets the OFF-delay time for the timer input.	0.0 s (0.0 - 3000.0 s)

◆ b6: Dwell Function

The Dwell function momentarily holds the output frequency at start and stop.

This prevents motor speed loss when you start and stop heavy loads. The Dwell function is also enabled when backlash on the machine side causes sudden movement at the start of acceleration and deceleration.

At the start of acceleration, the drive uses the output frequency and acceleration time set for the Dwell function to automatically operate at low speed to minimize the effects of backlash. Then, the drive can accelerate again. The Dwell function operates the same for deceleration.

For conveyor applications, the Dwell function also lets the drive interlock the output frequency and a delay time for the holding brake on the load side.

The Dwell function momentarily stops during acceleration to prevent a PM motor from stepping out. Figure 11.2 shows how the Dwell function works.

Note:

When you use the Dwell function at stop, set *b1-03 = 0 [Stopping Method Selection = Ramp to Stop]*.

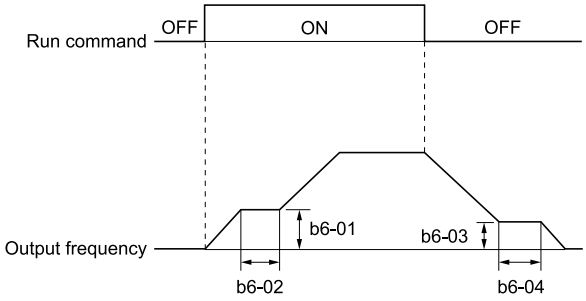


Figure 11.2 Time Chart for the Dwell Function at Start/Stop

■ b6-01: Dwell Reference at Start

No. (Hex.)	Name	Description	Default (Range)
b6-01 (01B6)	Dwell Reference at Start	V/f OLV Sets the output frequency that the drive will hold momentarily when the motor starts.	0.0 (Determined by A1-02)

When the drive accelerates to the output frequency set in *b6-01*, it holds that frequency for the time set in *b6-02 [Dwell Time at Start]*, and starts to accelerate again.

■ b6-02: Dwell Time at Start

No. (Hex.)	Name	Description	Default (Range)
b6-02 (01B7)	Dwell Time at Start	V/f OLV Sets the length of time that the drive will hold the output frequency when the motor starts.	0.0 s (0.0 - 10.0 s)

■ b6-03: Dwell Reference at Stop

No. (Hex.)	Name	Description	Default (Range)
b6-03 (01B8)	Dwell Reference at Stop	V/f OLV Sets the output frequency that the drive will hold momentarily when ramping to stop the motor.	0.0 (Determined by A1-02)

When the drive decelerates to the output frequency set in *b6-03*, it holds that frequency for the time set in *b6-04 [Dwell Time at Stop]* and starts to decelerate again.

■ b6-04: Dwell Time at Stop

No. (Hex.)	Name	Description	Default (Range)
b6-04 (01B9)	Dwell Time at Stop	V/f OLV Sets the length of time for the drive to hold the output frequency when ramping to stop the motor.	0.0 s (0.0 - 10.0 s)

11.3 C: Tuning

C parameters adjust drive operation, including:

- Acceleration Time
- Deceleration Time
- Slip Compensation
- Torque Compensation
- Carrier Frequency

◆ C1: Accel & Decel Speed

You can set four different acceleration and deceleration time pairs in the drive. When you activate and deactivate H1-xx = 7, 1A [MFDI Function Select = Accel/Decel Ramp Selection 1, Accel/Decel Time Selection 2], you can switch acceleration and deceleration times during run.

Acceleration time parameters always set the time to accelerate from 0 Hz to *E1-04 [Maximum Output Frequency]*. Deceleration time parameters always set the time to decelerate from *E1-04* to 0 Hz.

C1-01 [Acceleration Ramp 1] and *C1-02 [Deceleration Ramp 1]* are the default active accel/decel settings.

■ Use MFDIs to Switch Acceleration Ramps

Table 11.2 shows the different acceleration and deceleration times.

Table 11.2 Accel/Decel Ramps and Active Parameters

H1-xx = 7 [Accel/Decel Ramp Selection 1]	H1-xx = 1A [Accel/Decel Ramp Selection 2]	Active Parameter	
		Acceleration Time	Deceleration Time
OFF	OFF	C1-01 [Acceleration Ramp 1]	C1-02 [Deceleration Ramp 1]
ON	OFF	C1-03 [Acceleration Ramp 2]	C1-04 [Deceleration Ramp 2]
OFF	ON	C1-05 [Acceleration Ramp 3]	C1-06 [Deceleration Ramp 3]
ON	ON	C1-07 [Acceleration Ramp 4]	C1-08 [Deceleration Ramp 4]

Figure 11.3 shows an operation example to change acceleration and deceleration times. It is necessary to set *b1-03 = 0 [Stopping Method Selection = Ramp to Stop]* for this example.

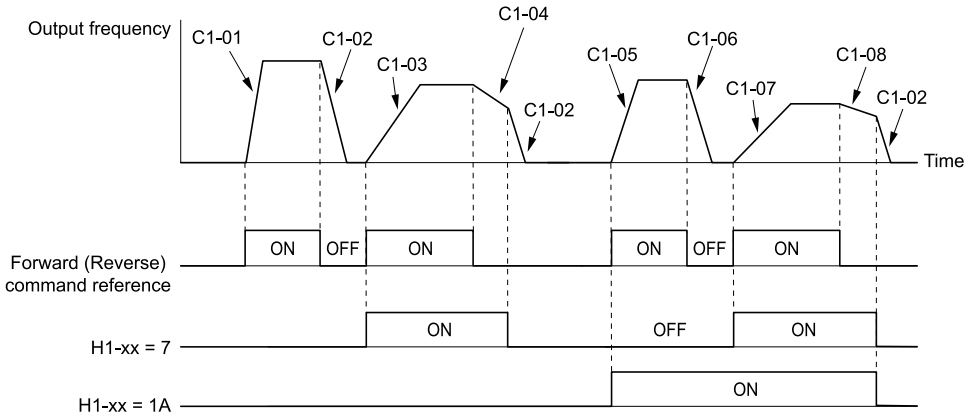


Figure 11.3 Timing Diagram of Acceleration and Deceleration Times

■ Use Output Speed Level to Switch Acceleration and Deceleration Times

The drive can use output Speed to automatically switch between different acceleration and deceleration times. When the output Speed = *C1-11 [Accel/Decel Time Switching Speed]*, the drive automatically switches the acceleration and deceleration times. Set *C1-11 = 0.0%* to disable this function.

Note:

- Acceleration and deceleration times set to MFDIs are more important than the automatic switch using the speed level set in *C1-11*. For example, if you set the switchover speed to *C1-11*, the drive will not automatically switch acceleration and deceleration times when the MFDI terminal set for *Accel/Decel Speed Selection 1 [H1-xx = 7]* is activated.

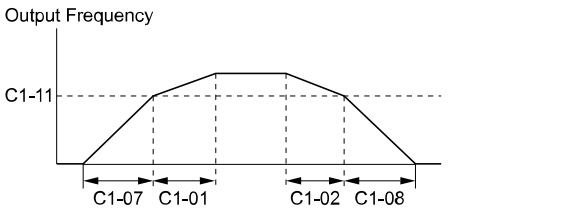


Figure 11.4 Accel/Decel Time Switching Speed

■ C1-01: Acceleration Ramp 1

No. (Hex.)	Name	Description	Default (Range)
C1-01 (0200) RUN	Acceleration Ramp 1	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)

■ C1-02: Deceleration Ramp 1

No. (Hex.)	Name	Description	Default (Range)
C1-02 (0201) RUN	Deceleration Ramp 1	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)

■ C1-03: Acceleration Ramp 2

No. (Hex.)	Name	Description	Default (Range)
C1-03 (0202) RUN	Acceleration Ramp 2	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)

■ C1-04: Deceleration Ramp 2

No. (Hex.)	Name	Description	Default (Range)
C1-04 (0203) RUN	Deceleration Ramp 2	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)

■ C1-05: Acceleration Ramp 3

No. (Hex.)	Name	Description	Default (Range)
C1-05 (0204) RUN	Acceleration Ramp 3	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)

■ C1-06: Deceleration Ramp 3

No. (Hex.)	Name	Description	Default (Range)
C1-06 (0205) RUN	Deceleration Ramp 3	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)

■ C1-07: Acceleration Ramp 4

No. (Hex.)	Name	Description	Default (Range)
C1-07 (0206) RUN	Acceleration Ramp 4	V/f OLV Sets the ramp to accelerate from zero to maximum output speed.	1.50 s (0.00 - 600.00 s)

■ C1-08: Deceleration Ramp 4

No. (Hex.)	Name	Description	Default (Range)
C1-08 (0207) RUN	Deceleration Ramp 4	V/f OLV Sets the ramp to decelerate from maximum output speed to zero.	1.50 s (0.00 - 600.00 s)

■ C1-09: Fast Stop Ramp

No. (Hex.)	Name	Description	Default (Range)
C1-09 (0208) RUN	Fast Stop Ramp	V/f OLV Sets the ramp that the drive will decelerate to zero for Emergency Stop.	1.50 s (0.00 - 600.00 s)

The Emergency Stop function will be triggered in the following circumstances.

- The Emergency Stop operation will be triggered by the input of the Emergency Stop command via the multi-function digital input terminal.
- The Emergency Stop operation is will be triggered when by the input of the Fast Stop command is input via the multi-function digital input terminal.

Set *H1-xx = 15, 17 [MFDI Function Select = Fast Stop (N.O.), Fast Stop (N.C.)]*.

When the Emergency Stop command is input, the Emergency Stop operation will be triggered at the deceleration time set to *C1-09*. The drive cannot be restarted after initiating a Emergency Stop operation until deceleration is complete. Complete deceleration and cycle the Run command to clear the Emergency Stop input.

The terminal set for *H2-xx = 4C [MFDO Function Select = During Fast Stop]* will be ON during Emergency Stop.

Note:

Decelerating too quickly can cause an *ov [Overvoltage]* fault that shuts off the drive while the motor to coasts to a stop. Set a Emergency Stop time in *C1-09* that prevents motor coasting and makes sure that the motor stops quickly and safely.

■ C1-10: Accel/Decel Time Setting Units

No. (Hex.)	Name	Description	Default (Range)
C1-10 (0209)	Accel/Decel Time Setting Units	V/f OLV Sets the setting units for *C1-01 to C1-08 [Accel/Decel Times 1 to 4]*, *C1-09 [Fast Stop Time]*, *L2-06 [Kinetic Energy Backup Decel Time]*, and *L2-07 [Kinetic Energy Backup Accel Time]*.	0 (0, 1)

0 : 0.01 s (0.00 to 600.00 s)

Sets acceleration and deceleration times in 0.01 s units. The setting range is 0.0 to 6000.0 s.

If one of these parameters is set to 1000.0 s or longer, you cannot set *C1-10 = 0*:

- *C1-01 to C1-09*
- *L2-06*
- *L2-07*

When one of those parameters is set to a value between 600.1 s and 1000.0 s, you can set *C1-10 = 0*, but the time will change to 600.00 s.

1 : 0.1 s (0.0 to 6000.0 s)

Sets acceleration and deceleration times in 0.1 s units. The setting range is 0.0 to 6000.0 s.

■ C1-11: Accel/Decel Switching Speed

No. (Hex.)	Name	Description	Default (Range)
C1-11 (020A)	Accel/Decel Switching Speed	V/f OLV Sets the speed at which the drive will automatically change acceleration and deceleration times.	0.0% (0.0 - 100.0%)

When the output speed is at the *C1-11* value, the drive automatically switches the acceleration and deceleration times. Set this parameter to *0.0%* to disable this function.

■ C1-15: Inspect Dec Ramp

No. (Hex.)	Name	Description	Default (Range)
C1-15 (0260)	Inspect Dec Ramp	V/f OLV Sets the deceleration ramp used for inspection run.	0.00 s (0.00 - 2.00 s)

◆ C2: Jerk Characteristics

Use Jerk characteristics to smooth acceleration and deceleration and to minimize abrupt shock to the load.
Set Jerk characteristic time during acceleration/deceleration at start and acceleration/deceleration at stop.
The following figure explains how Jerk are applied.

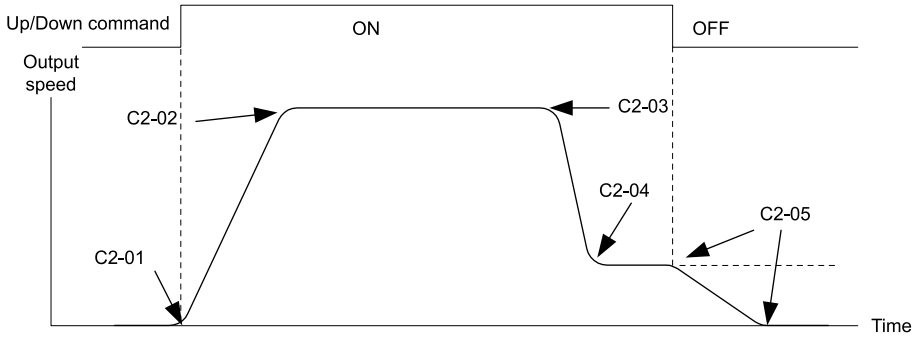


Figure 11.5 Jerk Timing Diagram - Forward/Reverse Operation

When o1-03 is set to between 0 and 4, the jerk settings are expressed in seconds. Then the actual accel/decel time including jerk settings can be calculated as follows:

- Actual accel ramp = accel ramp setting + (C2-01 + C2-02) / 2
- Actual decel ramp = decel ramp setting + (C2-03 + C2-04) / 2

■ C2-01: Jerk at Accel Start

No. (Hex.)	Name	Description	Default (Range)
C2-01 (020B)	Jerk at Accel Start	V/f OLV Sets the jerks acceleration time at start.	0.50 s (0.00 - 10.00 s)

■ C2-02: Jerk at Accel End

No. (Hex.)	Name	Description	Default (Range)
C2-02 (020C)	Jerk at Accel End	V/f OLV Sets the jerk acceleration time at completion.	0.50 s (0.00 - 10.00 s)

■ C2-03: Jerk at Decel Start

No. (Hex.)	Name	Description	Default (Range)
C2-03 (020D)	Jerk at Decel Start	V/f OLV Sets the jerk deceleration time at start.	0.50 s (0.00 - 10.00 s)

■ C2-04: Jerk at Decel End

No. (Hex.)	Name	Description	Default (Range)
C2-04 (020E)	Jerk at Decel End	V/f OLV Sets the jerk deceleration time at completion.	0.50 s (0.00 - 10.00 s)

■ C2-05: Jerk below Leveling Speed

No. (Hex.)	Name	Description	Default (Range)
C2-05 (020F)	Jerk below Leveling Speed	V/f OLV Sets the jerk deceleration time below leveling speed.	0.50 s (0.00 - 10.00 s)

◆ C3: Slip Compensation

The Slip Compensation function improves the speed accuracy of an induction motor. As loads on induction motors increase, motor slip increases and motor speed decreases. By adjusting the output frequency in accordance with the motor load, it compensates the slip and makes the motor speed equal to the frequency reference.

■ C3-01: Slip Compensation Gain

No. (Hex.)	Name	Description	Default (Range)
C3-01 (020F) RUN	Slip Compensation Gain	V/f OLV Sets the gain for the slip compensation function. Usually it is not necessary to change this setting.	1.0 (0.0 - 2.5)

Note:

Correctly set these parameters before changing the slip compensation gain:

- *E2-01 [Motor Rated Current (FLA)]*
- *E2-02 [Motor Rated Slip]*
- *E2-03 [Motor No-Load Current]*

Adjust this parameter as follows if necessary:

- If the motor speed is slower than the speed reference, increase this parameter in 0.1-unit increments.
- If the motor speed is slower than the speed reference value, decrease this parameter value in 0.1-unit increments.

■ C3-02: Slip Compensation Delay Time

No. (Hex.)	Name	Description	Default (Range)
C3-02 (0210) RUN	Slip Compensation Delay Time	V/f OLV Sets the slip compensation delay time when speed is unstable or when the slip compensation response is too slow. Usually it is not necessary to change this setting.	2000 ms (0 - 10000 ms)

Use these settings to adjust this parameter as necessary:

- When the speed is not stable, increase the setting.
- When the slip compensation response is too slow, decrease the setting.

■ C3-03: Slip Compensation Limit

No. (Hex.)	Name	Description	Default (Range)
C3-03 (0211)	Slip Compensation Limit	V/f OLV Sets the upper limit for the slip compensation function as a percentage of the motor rated slip.	200% (0 - 250%)

If you increase the value of *C3-01 [Slip Compensation Gain]* and the motor speed is slow, use this parameter. The drive uses this parameter when the slip is at the upper limit of slip compensation. Make sure that you measure the motor speed when you increase this parameter value. Set this parameter to make the frequency reference and the slip compensation limit less than the permitted range of the machine.

The slip compensation limit is constant in the constant torque range (frequency reference ≤ *E1-06 [Base Frequency]*). In the constant output range where the frequency reference > *E1-06*, the slip compensation limit increases with the *C3-03* value and the output frequency as shown in Figure 11.6.

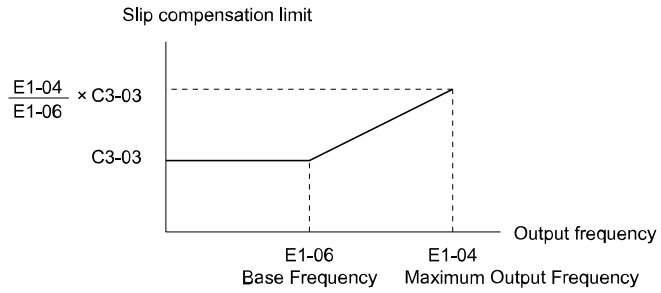


Figure 11.6 Slip Compensation Limit

■ C3-04: Slip Compensation at Regen

No. (Hex.)	Name	Description	Default (Range)
C3-04 (0212)	Slip Compensation at Regen	V/f OLV Sets the slip compensation function during regenerative operation.	0 (0 - 1)

If you apply a regenerative load when slip compensation function during regeneration is active, the quantity of regeneration can increase immediately. In this condition, it is necessary to use a dynamic braking option (braking resistor or braking resistor unit).

0 : Disabled

The drive does not provide slip compensation during regeneration.

The load and operation status (regenerative operation) can cause the motor speed to be higher or lower than the frequency reference.

1 : Enabled Above 6Hz

Slip compensation function is enabled during regeneration. Slip compensation is disabled at output frequencies of 6 Hz or less.

■ C3-05: Output Voltage Limit Selection

No. (Hex.)	Name	Description	Default (Range)
C3-05 (0213)	Output Voltage Limit Selection	V/f OLV Sets the automatic reduction of motor magnetic flux when the output voltage is saturated.	0 (0, 1)

The drive will decrease flux and increase current to compensate torque when voltage is saturated. Make sure that the drive has sufficient output current capacity before you enable this parameter. When this parameter = *1 [Enabled]*, the output current will increase by 10% at a maximum (at rated load) before it is enabled.

Enable this parameter to increase speed precision when you move heavy loads at high speeds in these conditions:

- Power supply voltage is low
- Motor rated voltage is high

Do not enable this parameter in these conditions:

- Operating a motor in the middle speed range or low speed range
- Power supply voltage is a minimum of 10% more than the motor rated voltage

When you enable this parameter, if the power supply voltage is much less than the motor rated voltage, torque control will not be accurate.

0 : Disabled

1 : Enabled

◆ C4: Torque Compensation

Torque compensation is a function that increases voltage to increase output torque as compensation for insufficient torque production at start-up or low-speed operation.

Voltage drops due to motor winding resistance cause torque generating voltage to decrease, which causes insufficient torque. If the main circuit cable connecting the drive and motor is long, this can also cause insufficient torque due to voltage drops.

Note:
Set the motor parameters and V/f pattern properly before setting *C4 parameters*.

■ C4-01: Torque Compensation Gain

No. (Hex.)	Name	Description	Default (Range)
C4-01 (0215) RUN	Torque Compensation Gain	V/f OLV Sets the gain for the torque compensation function.	1.00 (0.00 - 2.50)

For these control methods and states, adjust the setting value.

A1-02 [Control Method Selection]	Status	Adjustment
0 [V/f Control] 2 [Open Loop Vector]	Torque is not sufficient during low-speed operation of 10 Hz or less.	Increase the setting in 0.05-unit increments.
	There is vibration in the motor or the motor hunts when operating the drive with a light load.	Decrease the setting in 0.05-unit decrements.
	The cable length between the drive and motor is too long.	Increase the setting in 0.05-unit increments.

Note:
- Adjust *C4-01* to make sure that output current is not more than the drive rated current while the drive operates at low speed.
- When *A1-02 = 2 [Open Loop Vector]*, do not change this parameter under normal conditions. Torque accuracy will decrease.

■ C4-02: Torque Compensation Delay Time

No. (Hex.)	Name	Description	Default (Range)
C4-02 (0216) RUN	Torque Compensation Delay Time	V/f OLV Sets the torque compensation delay time. Usually it is not necessary to change this setting.	Determined by A1-02 (0 - 60000 ms)

Set this parameter in these conditions:
- If there is vibration in the motor, increase the setting.
- If the motor speed or motor torque response is too slow, decrease the setting.

■ C4-03: Torque Compensation @ FWD Start

No. (Hex.)	Name	Description	Default (Range)
C4-03 (0217)	Torque Compensation @ FWD Start	V/f OLV Set the amount of torque reference for forward start as a percentage of the motor rated torque.	0.0% (0.0 - 200.0%)

The torque compensation function is performed using the time constant set in *C4-05 [Torque Compensation Time]*.

This is available only when you start the motor with the forward Up/Down command. Set this parameter to 0.0 to disable this function.

■ C4-04: Torque Compensation @ REV Start

No. (Hex.)	Name	Description	Default (Range)
C4-04 (0218)	Torque Compensation @ REV Start	V/f OLV Sets the amount of torque reference for reverse start as a percentage of the motor rated torque.	0.0% (-200.0 - 0.0%)

The torque compensation function is performed using the time constant set in *C4-05 [Torque Compensation Time]*.

This is available only when you start the motor with the reverse Up/Down command. Set this parameter to 0.0 to disable this function.

■ C4-05: Torque Compensation Time

No. (Hex.)	Name	Description	Default (Range)
C4-05 (0219)	Torque Compensation Time	V/f OLV Sets the starting torque constant to use with *C4-03 and C4-04 [Torque Compensation @ FWD/REV Start]*.	10 ms (0 - 200 ms)

◆ C6: Carrier Frequency

C6 parameters are used to set the selection of carrier frequency.

■ C6-02: Carrier Frequency

No. (Hex.)	Name	Description	Default (Range)
C6-02 (0224)	Carrier Frequency Selection	V/f OLV Sets the carrier frequency for the transistors in the drive.	3 (1 - 6)

Changes to the switching frequency will decrease audible noise and decrease leakage current.

Note:
Increasing the carrier frequency to more than the default setting will automatically decrease the drive current rating.

1 : 2.0 kHz
2 : 5.0 kHz
3 : 8.0 kHz
4 : 10.0 kHz
5 : 12.5 kHz
6 : 15.0 kHz

11.4 d: References

d parameters [References] set the frequency reference input method and dead band range. They also set torque control, field weakening, and field forcing functions.

WARNING! *Sudden Movement Hazard. Use fast stop circuits to safely and quickly stop the drive. After you wire the fast stop circuits, you must check their operation. Test the operation of the fast stop function before you use the drive. If you do not test the fast stop circuit before you operate the drive, it can cause serious injury or death.*

WARNING! *Sudden Movement Hazard. When you use the drive in a lifting application, you must also install external safety circuitry. The drive does not have protection against accidental load drops in lifting applications. Install electrical and/or mechanical safety circuit mechanisms that are isolated from the drive circuitry. If you do not use external safety circuitry, the drive could drop the load and cause serious injury or death.*

◆ d1: Speed Reference

■ d1-01: Speed Reference 1

No. (Hex.)	Name	Description	Default (Range)
d1-01 (0280) RUN	Speed Reference 1	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-02: Speed Reference 2

No. (Hex.)	Name	Description	Default (Range)
d1-02 (0281) RUN	Speed Reference 2	V/f OLV Sets the speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-03: Speed Reference 3

No. (Hex.)	Name	Description	Default (Range)
d1-03 (0282) RUN	Speed Reference 3	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 590.00 Hz)

■ d1-04: Speed Reference 4

No. (Hex.)	Name	Description	Default (Range)
d1-04 (0283) RUN	Speed Reference 4	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-05: Speed Reference 5

No. (Hex.)	Name	Description	Default (Range)
d1-05 (0284) RUN	Speed Reference 5	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-06: Speed Reference 6

No. (Hex.)	Name	Description	Default (Range)
d1-06 (0285) RUN	Speed Reference 6	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-07: Speed Reference 7

No. (Hex.)	Name	Description	Default (Range)
d1-07 (0286) RUN	Speed Reference 7	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-08: Speed Reference 8

No. (Hex.)	Name	Description	Default (Range)
d1-08 (0287) RUN	Speed Reference 8	V/f OLV Sets the Speed reference in the units from *o1-03 [Frequency Display Unit Selection]*.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-18: Spd Ref Sel Mode

No. (Hex.)	Name	Description	Default (Range)
d1-18 (02C0)	Spd Ref Sel Mode	V/f OLV Sets the priority of the speed reference digital inputs.	1 (0, 1)

0 : Use Multi-Speed references d1-01 to d1-08

Up to eight separate preset speed references can be programmed to the drive using parameters d1-01 through d1-08 and can be selected using binary coded digital inputs. When *d1-18 = 0*, parameters d1-19 through d1-23 are not displayed.

1 : High speed reference has priority

Six different speeds (d1-19 to d1-23, d1-26) can be programmed to the drive and can be selected using dedicated digital inputs. Each of the speed references set to d1-19 through d1-23 takes priority over the leveling speed set to d1-26. When *d1-18 = 1*, parameter d1-01 to d1-08 are not displayed.

2 : Leveling speed reference has priority

Six different speeds (d1-19 to d1-23, d1-26) can be programmed to the drive and can be selected using dedicated digital inputs. The leveling speed reference in d1-26, however, takes priority over all other speed references when enabled via one of the multi-function input terminals *[H1-xx = 53]*. When *d1-18 = 2*, parameters d1-01 to d1-08 are not displayed.

■ d1-19: Nominal Speed

No. (Hex.)	Name	Description	Default (Range)
d1-19 (02C1) RUN	Nominal Speed	V/f OLV Sets the speed reference when multi-function input “Nominal Speed” is on.	50.00 Hz (0.00 - 120.00 Hz)

■ d1-20: Intermed Speed 1

No. (Hex.)	Name	Description	Default (Range)
d1-20 (02C2) RUN	Intermed Speed 1	V/f OLV Sets the speed reference when intermediate speed 1 is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-21: Intermed Speed 2

No. (Hex.)	Name	Description	Default (Range)
d1-21 (02C3) RUN	Intermed Speed 2	V/f OLV Sets the speed reference when intermediate speed 2 is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-22: Intermed Speed 3

No. (Hex.)	Name	Description	Default (Range)
d1-22 (02C4) RUN	Intermed Speed 3	V/f OLV Sets the speed reference when intermediate speed 3 is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-23: Releveling Speed

No. (Hex.)	Name	Description	Default (Range)
d1-23 (02C5) RUN	Releveling Speed	V/f OLV Sets the speed reference when releveling speed is selected by digital inputs.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-24: Inspect Oper Spd

No. (Hex.)	Name	Description	Default (Range)
d1-24 (02C6) RUN	Inspect Oper Spd	V/f OLV Sets the speed reference when inspection speed is selected by digital inputs.	30.00 Hz (0.00 - 120.00 Hz)

■ d1-25: Rescue Oper Spd

No. (Hex.)	Name	Description	Default (Range)
d1-25 (02C7) RUN	Rescue Oper Spd	V/f OLV Sets the speed reference when rescue speed is selected by digital inputs.	5.00 Hz (0.00 - 15.00 Hz)

■ d1-26: Leveling Speed

No. (Hex.)	Name	Description	Default (Range)
d1-26 (02C8) RUN	Leveling Speed	V/f OLV Sets the speed reference when leveling speed is selected by digital inputs.	4.00 Hz (0.00 - 120.00 Hz)

■ d1-28: Nom/Lev Spd Det

No. (Hex.)	Name	Description	Default (Range)
d1-28 (02CA)	Nom/Lev Spd Det	V/f OLV Sets the speed reference level to detect that leveling speed has been selected when multi speed input commands are used.	0.00 Hz (0.00 - 120.00 Hz)

■ d1-29: Inspect Spd Det

No. (Hex.)	Name	Description	Default (Range)
d1-29 (02CB)	Inspect Spd Det	V/f OLV Sets the speed reference level to detect that inspection speed has been selected when multi speed input commands are used.	0.00 Hz (0.00 - 120.00 Hz)

11.5 E: Motor Parameters

E parameters cover drive input voltage, V/f pattern, and motor parameters.

◆ E1: V/f Pattern for Motor 1

E1 parameters set the drive input voltage and motor V/f characteristics. To switch drive operation from one motor to another motor, set the V/f characteristics for motor 1.

■ E1-01: Input AC Supply Voltage

No. (Hex.)	Name	Description	Default (Range)
E1-01 (0300)	Input AC Supply Voltage	V/f OLV Sets the drive input voltage.	400 V: 400 V (400 V Class: 310 - 510 V)

NOTICE: *Set parameter E1-01 to align with the drive input voltage (not motor voltage). If this parameter is incorrect, the protective functions of the drive will not operate correctly and it can cause damage to the drive.*

Values Related to the Drive Input Voltage

The value set in *E1-01* is the base value that the drive uses for the motor protective functions in Table 11.3. With a 400 V class drive, the detection level changes for some motor protective functions.

Table 11.3 Values Related to the Drive Input Voltage

Voltage	E1-01 Setting	Approximate Values		
		ov Detection Level	BTR Operation Level (rr Detection Level) *1	L2-05 [Undervoltage Detection Lvl (Uv1)]
400 V class	Setting value ≥ 400 V	820 V	788 V	380 V
	Setting value < 400 V	820 V	788 V	350 V

*1 This is the protection function enabled in drives with built-in braking transistors. These values show the level that will trigger the built-in braking transistor. Refer to "YASKAWA AC Drive 72060001 Series Option Braking Unit and Braking Resistor Unit Installation Manual (TOBPC72060001)" for more information.

■ E1-04: Maximum Output Frequency

No. (Hex.)	Name	Description	Default (Range)
E1-04 (0303)	Maximum Output Frequency	V/f OLV Sets the maximum output frequency for the V/f pattern.	Determined by A1-02 (10.0 - 120.0 Hz)

■ E1-05: Maximum Output Voltage

No. (Hex.)	Name	Description	Default (Range)
E1-05 (0304)	Maximum Output Voltage	V/f OLV Sets the maximum output voltage for the V/f pattern.	400 V Class: 380.0 V (400 V Class: 0.0 - 510.0 V)

■ E1-06: Base Frequency

No. (Hex.)	Name	Description	Default (Range)
E1-06 (0305)	Base Frequency	V/f OLV Sets the base frequency for the V/f pattern.	Determined by A1-02 (0.0 - 120.0 Hz)

■ E1-07: Middle Output Frequency

No. (Hex.)	Name	Description	Default (Range)
E1-07 (0306)	Middle Output Frequency	V/f OLV Sets a middle output frequency for the V/f pattern.	Determined by A1-02 (0.0 - 120.0 Hz)

■ E1-08: Middle Output Frequency Voltage

No. (Hex.)	Name	Description	Default (Range)
E1-08 (0307)	Middle Output Frequency Voltage	V/f OLV Sets a middle output voltage for the V/f pattern.	Determined by A1-02 and o2-04 (400 V Class: 0.0 - 510.0 V)

Note:
Default setting is determined by *A1-02 [Control Method Selection]*, and *o2-04 [Drive Model Selection]*.

■ E1-09: Minimum Output Frequency

No. (Hex.)	Name	Description	Default (Range)
E1-09 (0308)	Minimum Output Frequency	V/f OLV Sets the minimum output frequency for the V/f pattern.	Determined by A1-02 (0.0 - 120.0 Hz)

■ E1-10: Minimum Output Voltage

No. (Hex.)	Name	Description	Default (Range)
E1-10 (0309)	Minimum Output Voltage	V/f OLV Sets the minimum output voltage for the V/f pattern.	Determined by o2-04 (400 V Class: 0.0 - 510.0 V)

■ E1-11: Middle Output Frequency 2

No. (Hex.)	Name	Description	Default (Range)
E1-11 (030A) Expert	Middle Output Frequency 2	V/f OLV Sets a middle output frequency for the V/f pattern.	0.0 Hz (0.0 - 120.0 Hz)

Note:
Set this parameter to *0.0 Hz* to disable the function.

■ E1-12: Middle Output Frequency Voltage 2

No. (Hex.)	Name	Description	Default (Range)
E1-12 (030B) Expert	Middle Output Frequency Voltage 2	V/f OLV Sets a middle point voltage for the V/f pattern.	0.0 V (400 V Class: 0.0 - 510.0 V)

Note:
Set this parameter to *0.0 V* to disable the function.

■ E1-13: Base Voltage

No. (Hex.)	Name	Description	Default (Range)
E1-13 (030C) Expert	Base Voltage	V/f OLV Sets the base voltage for the V/f pattern.	0.0 V (400 V Class: 0.0 - 510.0 V)

Note:
- After Auto-Tuning, the value of *E1-13 = E1-05 [Maximum Output Voltage]*.
- When *E1-13 = 0.0*, use the value of *E1-05* to control the voltage.

◆ E2: Motor Parameters

E2 parameters [Motor Parameters] are used to set induction motor data. To switch drive operation from one motor to another motor, configure the first motor (motor 1).

If Auto-Tuning cannot be performed, set the *E2 parameters* manually. Performing Auto-Tuning automatically sets the *E2 parameters* to the optimal values.

Note:

If *A1-02 [Control Method Selection]* is set to the following control modes, the keypad does not display *E2-xx*.

- 5 [PM Open Loop Vector]
- 6 [PM Advanced Open Loop Vector]
- 8 [EZ Vector Control]

■ E2-01: Motor Rated Current (FLA)

No. (Hex.)	Name	Description	Default (Range)
E2-01 (030E)	Motor Rated Current (FLA)	V/f OLV Sets the motor rated current in amps.	Determined by o2-04 (10% to 200% of the drive rated current)

Note:

- If *E2-01 < E2-03 [Motor No-Load Current]*, the drive will detect *oPE02 [Parameter Range Setting Error]*.
- When the drive model changes, the display units for this parameter also change.
  - 0.01 A: 4009
  - 0.1 A: 4015 to 4045

The value set for *E2-01* becomes the reference value for motor protection and the torque limit. Enter the motor rated current written on the motor nameplate. Auto-Tuning the drive will automatically set *E2-01* to the value input for *T1-04 [Motor Rated Current]*.

■ E2-02: Motor Rated Slip

No. (Hex.)	Name	Description	Default (Range)
E2-02 (030F)	Motor Rated Slip	V/f OLV Sets motor rated slip.	Determined by o2-04 (0.00 - 20.00 Hz)

This parameter value becomes the base slip compensation value. The drive automatically sets this parameter during Auto-Tuning. When you cannot do Auto-Tuning, calculate the motor rated slip with the information on the motor nameplate and this formula:

E2-02 = f - (n × p) / 120

- f: Motor rated frequency (Hz)
- n: Rated motor speed (min^{-1} (r/min))
- p: Number of motor poles

■ E2-03: Motor No-Load Current

No. (Hex.)	Name	Description	Default (Range)
E2-03 (0310)	Motor No-Load Current	V/f OLV Sets the no-load current for the motor in amps when operating at the rated frequency and the no-load voltage.	Determined by o2-04 (0 to E2-01)

Note:

When the drive model changes, the display units for this parameter also change.

- 0.01 A: 4009
- 0.1 A: 4015 - 4045

The drive automatically sets this parameter during Auto-Tuning. When you cannot do Auto-Tuning, you can also use the motor no-load current on the motor test report to enter this value manually. Get the test report from the motor manufacturer.

Note:

The default setting of the no-load current is for a 4-pole motor recommended by Yaskawa.

■ E2-04: Motor Pole Count

No. (Hex.)	Name	Description	Default (Range)
E2-04 (0311)	Motor Pole Count	V/f OLV Sets the number of motor poles.	4 (2 - 48)

Auto-Tuning automatically sets this parameter to the value of [Number of Motor Poles].

■ E2-05: Motor Line-to-Line Resistance

No. (Hex.)	Name	Description	Default (Range)
E2-05 (0312)	Motor Line-to-Line Resistance	V/f OLV Sets the line-to-line resistance for the motor stator windings.	Determined by o2-04 (0.000 - 65.000 Ω)

Note:

This value is the motor line-to-line resistance. Do not set this parameter with the per-phase resistance.

Auto-Tuning automatically sets this parameter. If you cannot do Auto-Tuning, use the test report from the motor manufacturer to configure the settings. You can calculate the motor line-to-line resistance with one of these formulas:

- E-type insulation: [the resistance value (Ω) shown on the test report at 75 °C] × 0.92
- B-type insulation: [the resistance value (Ω) shown on the test report at 75 °C] × 0.92
- F-type insulation: [the resistance value (Ω) shown on the test report at 115 °C] × 0.87

■ E2-06: Motor Leakage Inductance

No. (Hex.)	Name	Description	Default (Range)
E2-06 (0313)	Motor Leakage Inductance	V/f OLV Sets the voltage drop from motor leakage inductance when the motor is operating at the rated frequency and rated current. This value is a percentage of Motor Rated Voltage.	Determined by o2-04 (0.0 - 60.0%)

The drive automatically sets this parameter during Auto-Tuning.

Note:

The motor nameplate does not usually show the quantity of voltage drop. If you do not know the value of the motor leakage inductance, contact the motor manufacturer to receive a copy of the motor test report.

■ E2-07: Motor Saturation Coefficient 1

No. (Hex.)	Name	Description	Default (Range)
E2-07 (0314)	Motor Saturation Coefficient 1	V/f OLV Sets the motor iron-core saturation coefficient at 50% of the magnetic flux.	0.50 (0.00 - 0.50)

The drive uses this coefficient when it operates with constant output. The drive uses this coefficient when it operates the motor in the constant output range.

■ E2-08: Motor Saturation Coefficient 2

No. (Hex.)	Name	Description	Default (Range)
E2-08 (0315)	Motor Saturation Coefficient 2	V/f OLV Sets the motor iron-core saturation coefficient at 75% of the magnetic flux.	0.75 (E2-07 - 0.75)

The drive uses this coefficient when it operates with constant output. The drive uses this coefficient when it operates the motor in the constant output range.

■ E2-09: Motor Mechanical Loss

No. (Hex.)	Name	Description	Default (Range)
E2-09 (0316) Expert	Motor Mechanical Loss	V/f OLV Sets the mechanical loss of the motor. It is set as a percentage of *E2-11 [Motor Rated Power]*. Usually it is not necessary to change this setting.	0.0% (0.0 - 10.0%)

Adjust this parameter in these conditions. The drive adds the configured mechanical loss to the torque reference value as a torque compensation value:

- There is a large quantity of torque loss from motor bearing friction.
- There is a large quantity of torque loss in fans and pumps.

■ E2-10: Motor Iron Loss

No. (Hex.)	Name	Description	Default (Range)
E2-10 (0317)	Motor Iron Loss	V/f OLV Sets the motor iron loss.	Determined by o2-04 (0 - 65535 W)

■ E2-11: Motor Rated Power

No. (Hex.)	Name	Description	Default (Range)
E2-11 (0318)	Motor Rated Power	V/f OLV Sets the motor rated output in the units from *o1-58 [Motor Power Unit Selection]*.	Determined by o2-04 (0.00 - 650.00 kW)

The drive automatically sets this parameter to the value input for “Motor Rated Power” during Auto-Tuning.

11.6 H: Terminal Function Selection

H parameters set functions for external input and output terminals.

◆ H1: Digital Inputs

H1 Parameters set the MFDI terminal functions.

■ H1-03 to H1-07 Terminal S3 to S7 Function Selection

The drive has 7 MFDI terminals. Refer to Table 11.4 for drive default settings and functions.

Table 11.4 MFDI Default Settings and Functions

No.	Name	Default	Function
H1-03	Term S3 Function Selection	50	Nominal Speed
H1-04	Term S4 Function Selection	54	Inspection Operation
H1-05	Term S5 Function Selection	51	Intermediate Speed
H1-06	Term S6 Function Selection	53	Leveling Speed
H1-07	Term S7 Function Selection	F	Not Used (Through Mode)

Refer to the Table 11.5 and use *H1-xx [MFDI Function Select]* to set the function.

Table 11.5 MFDI Setting Values

Setting Value	Function	Reference
3	Multi-Step Speed Reference 1	275
4	Multi-Step Speed Reference 2	275
5	Multi-Step Speed Reference 3	275
7	Accel/Decel Time Selection 1	275
8	Baseblock Command (N.O.)	275
9	Baseblock Command (N.C.)	276
F	Through Mode	276
14	Fault Reset	276
15	Fast Stop (N.O.)	276
17	Fast Stop (N.C.)	277
18	Timer Function	277
1A	Accel/Decel Time Selection 2	277
20 to 2F	External fault	277
40	Forward RUN (2-Wire)	278
41	Reverse RUN (2-Wire)	279
50	Nominal Speed	279
51	Intermediate Speed	279
52	Releveling Speed	279
53	Leveling Speed	279
54	Inspection Operation	279
55	Rescue Operation	279
56	Motor Connector Feedback	280
79	Brake Feedback	280
103 - 179	Inverse Inputs of 1 to 79	-

■ H1-03: Terminal S3 Function Selection

No. (Hex.)	Name	Description	Default (Range)
H1-03 (0400)	Terminal S3 Function Selection	V/f OLV Sets the function for MFDI terminal S3.	50 (3 - 79)

■ H1-04: Terminal S4 Function Selection

No. (Hex.)	Name	Description	Default (Range)
H1-04 (0401)	Terminal S4 Function Selection	V/f OLV Sets the function for MFDI terminal S4.	54 (3 - 79)

■ H1-05: Terminal S5 Function Selection

No. (Hex.)	Name	Description	Default (Range)
H1-05 (0402)	Terminal S5 Function Selection	V/f OLV Sets the function for MFDI terminal S5.	51 (3 - 79)

■ H1-06: Terminal S6 Function Selection

No. (Hex.)	Name	Description	Default (Range)
H1-06 (0403)	Terminal S6 Function Selection	V/f OLV Sets the function for MFDI terminal S6.	53 (3 - 79)

■ H1-07: Terminal S7 Function Selection

No. (Hex.)	Name	Description	Default (Range)
H1-07 (0404)	Terminal S7 Function Selection	V/f OLV Sets the function for MFDI terminal S7.	F (3 - 79)

◆ MFDI Setting Values

Selects a function for MFDI.

■ 3: Multi-Step Speed Reference 1

Setting Value	Function	Description
3	Multi-Step Speed Reference 1	V/f OLV Uses speed references *d1-01 to d1-08* to set a multi-step speed reference.

■ 4: Multi-Step Speed Reference 2

Setting Value	Function	Description
4	Multi-Step Speed Reference 2	V/f OLV Uses speed references *d1-01 to d1-08* to set a multi-step speed reference.

■ 5: Multi-Step Speed Reference 3

Setting Value	Function	Description
5	Multi-Step Speed Reference 3	V/f OLV Uses speed references *d1-01 to d1-08* to set a multi-step speed reference.

■ 7: Accel/Decel Time Selection 1

Setting Value	Function	Description
7	Accel/Decel Time Selection 1	V/f OLV Sets the drive to use *Acceleration/Deceleration Time 1 [C1-01, C1-02]* or *Acceleration/Deceleration Time 2 [C1-03, C1-04]*.

Note:

Refer to *C1: Accel & Decel Speed on page 258* for more information.

■ 8: Baseblock Command (N.O.)

Setting Value	Function	Description
8	Baseblock Command (N.O.)	V/f OLV Sets the command that stops drive output and coasts the motor to stop when the input is ON.

The keypad flashes *bb [Baseblock]*. If you cancel the baseblock command when the Run command is active, the drive will restart the motor and use the speed search function.

WARNING! *Sudden Movement Hazard. When you use a mechanical holding brake with the drive in a lifting application, you must close the brake if an input terminal triggers the Baseblock command to stop drive output. If you enter the baseblock command, the motor will suddenly coast and the load will slip, which can cause serious injury or death.*

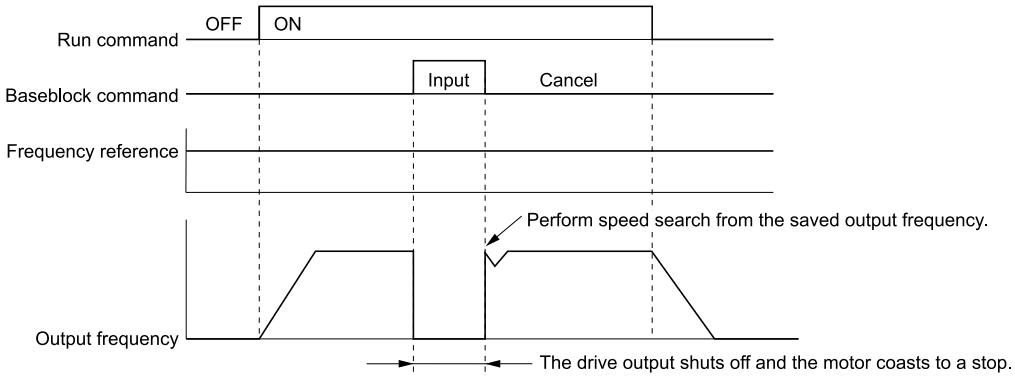


Figure 11.7 Baseblock Command Time Chart

ON : Baseblock (drive output stop)

OFF : Normal operation

■ 9: Baseblock Command (N.C.)

Setting Value	Function	Description
9	Baseblock Command (N. C.)	V/f OLV Sets the command that stops drive output and coasts the motor to stop when the input terminal is OFF.

The keypad flashes *bb [Baseblock]*. If you cancel the baseblock command when the Run command is active, the drive will restart the motor and use the speed search function.

ON : Normal operation

OFF : Baseblock (drive output stop)

WARNING! *Sudden Movement Hazard. When you use a mechanical holding brake with the drive in a lifting application, you must close the brake if an input terminal triggers the Baseblock command to stop drive output. If you enter the baseblock command, the motor will suddenly coast and the load will slip, which can cause serious injury or death.*

■ F: Not Used

Setting Value	Function	Description
F	Not Used	V/f OLV Use this setting for unused terminals or to use terminals in through mode.

Through Mode uses the signal input to the terminal as a digital input for the upper sequence through a communication option or MEMOBUS/Modbus communications. This input signal does not have an effect on drive operation.

■ 14: Fault Reset

Setting Value	Function	Description
14	Fault Reset	V/f OLV Sets the command to reset the current fault when the Run command is inactive.

If the drive detects a fault, the drive will activate the fault relay output, turn off the output, and the motor will coast to stop.

If the drive detects a fault for which you can set the stopping method, apply the appropriate Stopping Method. Then push (RESET) on the keypad to turn the Run command OFF, or activate the fault reset terminal to reset the fault.

Note:

The drive ignores the fault reset command when the Run command is active. Remove the Run command before trying to reset a fault.

■ 15: Fast Stop (N.O.)

Setting Value	Function	Description
15	Fast Stop (N.O.)	V/f OLV Sets the command to ramp to stop in the deceleration time set in *C1-09 [Fast Stop Time]* when the input terminal is ON while the drive is operating.

If you cancel the fast stop input, the drive will not restart the motor until you meet these conditions:

- Fully stop the motor

- Cancel the Run command
- Cancel the fast stop command

Note:
- To use the N.C. switch to input the fast stop command, set 17 (Fast Stop (N.C.)).
- Refer to *C1-09: Fast Stop Ramp on page 260* for more information.
- Set *C1-09 [Fast Stop Time]* to a correct deceleration time. If the deceleration time is too short, it can cause an overvoltage fault and failure to stop the motor from coasting.

■ 17: Fast Stop (N.C.)

Setting Value	Function	Description
17	Fast Stop (N.C.)	V/f OLV Sets the command to ramp to stop in the deceleration time set in *C1-09 [Fast Stop Time]* when the input terminal is ON while the drive is operating.

If you cancel the fast stop input, the drive will not restart the motor until you meet these conditions:

- Fully stop the motor
- Cancel the Run command
- Cancel the fast stop command

Note:
- To use the N.O. switch to input the fast stop command, set 15 (Fast Stop (N.O.)).
- Refer to *C1-09: Fast Stop Ramp on page 260* for more information.
- Set *C1-09 [Fast Stop Time]* to a correct deceleration time. If the deceleration time is too short, it can cause an overvoltage fault and failure to stop the motor from coasting.

Figure 11.8 shows an example of how fast stop operates.

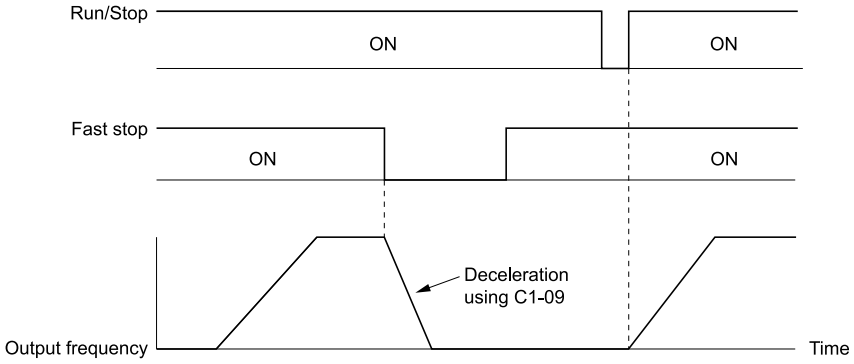


Figure 11.8 Fast Stop Time Chart

■ 18: Timer Function Input

Setting Value	Function	Description
18	Timer Function Input	V/f OLV Triggers the timer set up by parameters *b4-01* and *b4-02*. Mist be set in conjunction with the timer function output *[H2-xx = 2]*.

■ 1A: Accel/Decel Time Selection 2

Setting Value	Function	Description
1A	Accel/Decel Time Selection 2	V/f OLV Set this function and *H1-xx = 7 [Accel/Decel Time Selection 1]* together. Sets the drive to use *Acceleration/Deceleration Time 3 [C1-05, C1-06]* or *Acceleration/Deceleration Time 4 [C1-07, C1-08]*.

Note:
Refer to *C1: Accel & Decel Speed on page 258* for more information.

■ 20 to 2F: External Fault

Setting Value	Function	Description
20 to 2F	External fault	V/f OLV Sets a command to stop the drive when a failure or fault occurs on an external device.

If an external fault is input to the drive, the keypad will show *EFx [External Fault (Terminal Sx)]*, where x is the number of the terminal (terminal Sx) to which the external fault signal is assigned. For example, when an external fault signal is input to terminal S3, the keypad will show EF3.

Use these conditions to select the value to set in *H1-xx*:

- Signal input method from peripheral devices
- External fault detection method
- Motor stopping method (operation after external fault detection)

Table 11.6 shows the relation between the conditions and the value set to *H1-xx*.

Table 11.6 Stopping Methods for External Fault

Setting	Signal Input Method from Peripheral Devices *1		External Fault Detection Method *2		Stopping Method			
	N.O.	N.C.	Always Detected	Detected during RUN Only	Ramp to Stop (Fault)	Coast to Stop (Fault)	Fast Stop (Fault)	Continuous Operation (Alarm Only)
20	x	-	x	-	x	-	-	-
21	-	x	x	-	x	-	-	-
22	x	-	-	x	x	-	-	-
23	-	x	-	x	x	-	-	-
24	x	-	x	-	-	x	-	-
25	-	x	x	-	-	x	-	-
26	x	-	-	x	-	x	-	-
27	-	x	-	x	-	x	-	-
28	x	-	x	-	-	-	x	-
29	-	x	x	-	-	-	x	-
2A	x	-	-	x	-	-	x	-
2B	-	x	-	x	-	-	x	-
2C	x	-	x	-	-	-	-	x
2D	-	x	x	-	-	-	-	x
2E	x	-	-	x	-	-	-	x
2F	-	x	-	x	-	-	-	x

*1 Set the terminal to N.O. (detects external fault when switched ON) or N.C. (detects external fault when switched OFF).
*2 Set the drive to always detect each fault or to detect only during run.

■ 40: Forward RUN (2-Wire)

Setting Value	Function	Description
40	Forward RUN (2-Wire)	V/f OLV Sets the Forward Run command for 2-wire sequence 1. Set this function and *H1-xx = 41 [Reverse Run Command (2-Wire Seq)]* together.

ON : Forward Run
OFF : Run Stop

Note:
- Turning ON the Forward Run command terminal and the Reverse Run command terminal will cause alarm *EF [FWD/REV Run Command Input Error]* and the motor will ramp to stop.
- Initialize the drive with a 2-wire sequence to set the Forward Run command to terminal S1.
- This function will not operate at the same time as *H1-xx = 42, 43 [Run Command/FWD/REV Command (2-Wire Seq 2)]*.

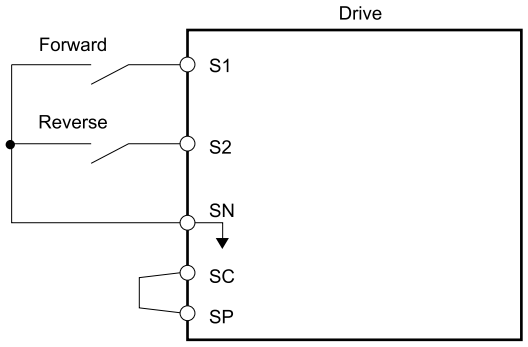


Figure 11.9 2-Wire Sequence Wiring Example

■ 41: Reverse RUN (2-Wire)

Setting Value	Function	Description
41	Reverse RUN (2-Wire)	V/f OLV Sets the Forward Run command for 2-wire sequence 1. Set this function and *H1-xx = 40 [Forward Run Command (2-Wire Seq)]* together.

ON : Reverse Run

OFF : Run Stop

Note:
- Turning ON the Forward Run command terminal and the Reverse Run command terminal will cause alarm EF [FWD/REV Run Command Input Error] and the motor will ramp to stop.
- Initialize the drive with a 2-wire sequence to set the Reverse Run command to terminal S2.
- This function will not operate at the same time as *H1-xx = 42, 43 [Run Command/FWD/REV Command (2-Wire Seq 2)]*.

■ 50: Nominal Speed

Setting Value	Function	Description
50	Nominal Speed	V/f OLV Sets the speed reference nominal speed.

■ 51: Intermed Speed

Setting Value	Function	Description
51	Intermed Speed	V/f OLV Sets the speed reference intermediate speed 1. In combination with nominal and releveling speed you can set intermediate speed 2 and 3.

■ 52: Releveling Speed

Setting Value	Function	Description
52	Releveling Speed	V/f OLV Sets the speed reference releveling speed.

■ 53: Leveling Speed

Setting Value	Function	Description
53	Leveling Speed	V/f OLV Sets the speed reference leveling speed.

■ 54: Inspection Speed

Setting Value	Function	Description
54	Inspection Speed	V/f OLV Sets the speed reference inspection speed.

■ 55: Rescue Operation

Setting Value	Function	Description
55	Rescue Operation	V/f OLV Activates rescue operation.

■ 56: Motor Contractor Feedback

Setting Value	Function	Description
56	Motor Contractor Feedback	V/f OLV Used for motor contactor supervision and fault detection.

■ 79: Brake Feedback

Setting Value	Function	Description
79	Brake Feedback	V/f OLV Used for brake supervision and detection of incorrect operation.

◆ H2: Digital Outputs

H2 parameters set the MFDO terminal functions.

■ H2-01 to H2-03 Terminal MA/MB-MC, P1-C1, P2-C2 Function Selection

The drive has three MFDO terminals. Table 11.7 shows the default function settings for the terminals.

Table 11.7 MFDO Terminals Default Function Settings

No.	Name	Default	Function
H2-01	Term MA/MB-MC Function Selection (Contact)	E	Fault
H2-02	Term P1-C1 Function Selection	6	Drive Ready
H2-03	Term P2-C2 Function Selection	50	Brake Control

Refer to Table 11.8 to set *H2-xx [MFDO Function Select]*.

Table 11.8 MFDO Setting Values

Setting Value	Function	Reference
0	During Run	281
1	Zero Speed	281
2	Speed Agree 1	282
3	User-Set Speed Agree 1	282
4	Frequency Detection 1	283
5	Frequency Detection 2	283
6	Drive Ready	284
7	DC Bus Undervoltage	284
8	During Baseblock (N.O.)	284
B	Torque Detection 1 (N.O.)	284
E	Fault	285
F *1	Not Used	285
10	Alarm	285
11	Fault Reset Command Active	285
12	Timer Output	285
13	Speed Agree 2	285
14	User-Set Speed Agree 2	286
15	Frequency Detection 3	286
16	Frequency Detection 4	287

Setting Value	Function	Reference
18	Torque Detection 2 (N.O.)	287
1A	During Reverse	287
1B	During Baseblock (N.C.)	288
1E	Executing Auto-Restart	288
1F	Motor Overload Alarm (oL1)	288
20	Drive Overheat Pre-Alarm (oH)	288
21	Safety Monitor Output Status	288
2F	Maintenance Notification	288
30	During Torque Limit	289
37	During Frequency Output	289
47	PF Detection	289
50	Brake Control	289
51	Output Contactor Control	290
54	Light Load Direction	290
58	Safe Disable Status	290
100 to 158	Inverse output of 0 to 58 Sets an inverse output of the function for the MFDO. Put a 1 at the front of the function setting to set inverse output. For example, set *H2-xx = 10E* for the inverse output of *E [Fault]*.	290

*1 Inverse output is not available.

■ H2-01: Term MA/MB-MC Function Selection

No. (Hex.)	Name	Description	Default (Range)
H2-01 (040B)	Term MA/MB-MC Function Selection	V/f OLV Sets the function set for MFDO terminal MA-MC or MB-MC.	E (0 - 158)

Note:
Set this parameter to *F* when the terminal is not being used or to use the terminal in through mode.

■ H2-02: Term P1-C1 Function Selection

No. (Hex.)	Name	Description	Default (Range)
H2-02 (040C)	Term P1-C1 Function Selection	V/f OLV Sets the function for MFDO terminal P1-C1.	6 (0 - 158)

Note:
Set this parameter to *F* when the terminal is not being used or to use the terminal in through mode.

■ H2-03: Term P2-C2 Function Selection

No. (Hex.)	Name	Description	Default (Range)
H2-03 (040D)	Term P2-C2 Function Selection	V/f OLV Sets the function for MFDO terminal P2-C2.	50 (0 - 158)

Note:
Set this parameter to *F* when the terminal is not being used or to use the terminal in through mode.

◆ MFDO Setting Values

Selects the function configured to MFDO.

■ 0: During Run

Setting Value	Function	Description
0	During Run	V/f OLV The terminal activates when you input a Run command and when the drive is outputting voltage.

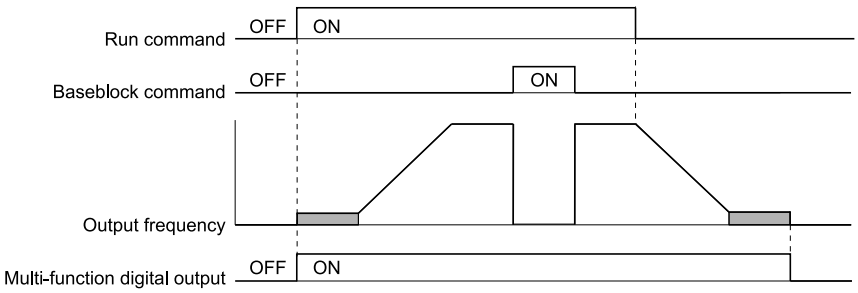


Figure 11.10 Drive Running Time Chart

ON : Drive is running

The drive is receiving a Run command or outputting voltage.

OFF : Drive is stopping

■ 1: Zero Speed

Setting Value	Function	Description
1	Zero Speed	V/f OLV The terminal activates when the output frequency is less than *E1-09 [Minimum Output Frequency]* or *b2-01 [DC Injection/Zero SpeedThreshold]*.

Note:
Parameter *A1-02 [Control Method Selection]* selects which parameter is the reference.

A1-02 Setting	Control Method Selection	Parameter Used as the Reference
0	V/f	*E1-09*
2	OLV	*b2-01*

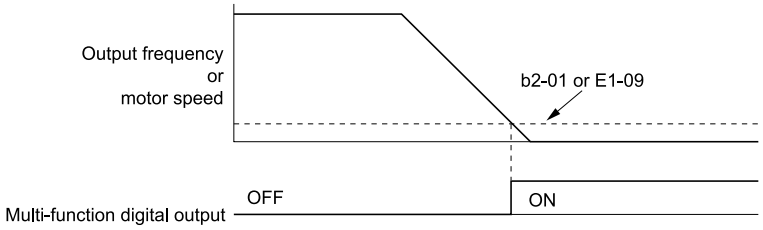


Figure 11.11 Zero Speed Time Chart

ON : Output frequency < value of *E1-09* or *b2-01*.
OFF : Output frequency ≥ value of *E1-09* or *b2-01*.

■ 2: Speed Agree 1

Setting Value	Function	Description
2	Speed Agree 1	V/f OLV The terminal activates when the output frequency is in the range of the frequency reference ± *L4-02 [Speed Agree Detection Width]*.

Note:
The detection function operates in the two motor rotation directions.

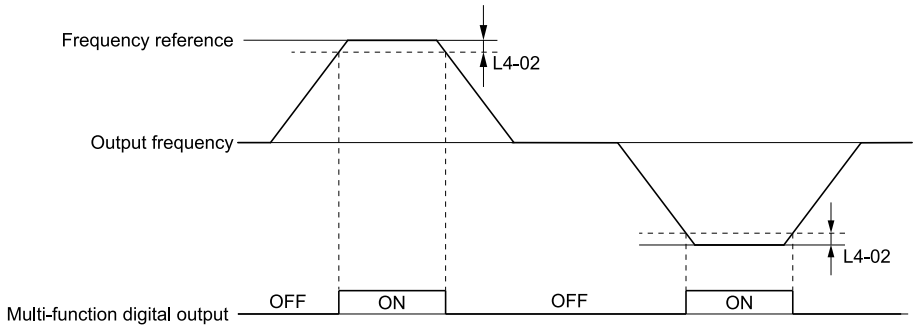


Figure 11.12 Speed Agree 1 Time Chart

ON : The output frequency is in the range of "frequency reference ± *L4-02*".
OFF : The output frequency does not align with the frequency reference although the drive is running.

■ 3: User-Set Speed Agree 1

Setting Value	Function	Description
3	User-Set Speed Agree 1	V/f OLV The terminal activates when the output frequency is in the range of *L4-01 [Speed Agree Detection Level]* ± *L4-02 [Speed Agree Detection Width]* and in the range of the frequency reference ± *L4-02*.

Note:
The detection function operates in the two motor rotation directions. The drive uses the *L4-01* value as the forward/reverse detection level.

ON : The output frequency is in the range of "*L4-01* ± *L4-02*" and the range of frequency reference ± *L4-02*.
OFF : The output frequency is not in the range of "*L4-01* ± *L4-02*" nor the range of frequency reference ± *L4-02*.

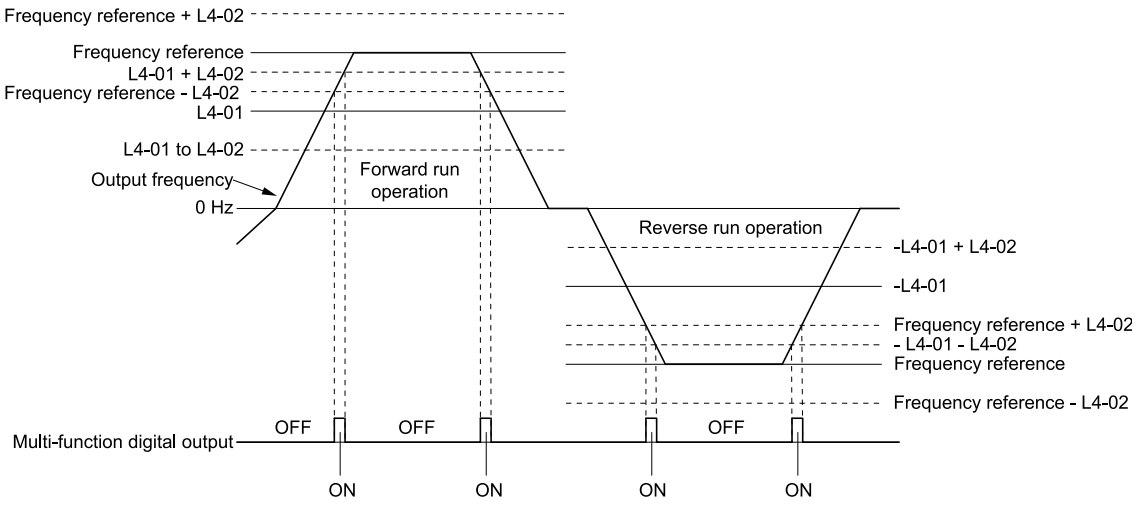


Figure 11.13 User-Defined Speed Agree 1 Time Chart

■ 4: Frequency Detection 1

Setting Value	Function	Description
4	Frequency Detection 1	V/f OLV The terminal deactivates when the output frequency is higher than the value of *L4-01 [Speed Agree Detection Level] + L4-02 [Speed Agree Detection Width]*. After the terminal deactivates, the terminal stays deactivated until the output frequency = *L4-01*.

Note:

The detection function operates in the two motor rotation directions. The drive uses the *L4-01* value as the forward/reverse detection level.

ON : The output frequency is less than the value of *L4-01* or does not exceed the value of *L4-01* + *L4-02*.

OFF : The output frequency is higher than the value of *L4-01* + *L4-02*.

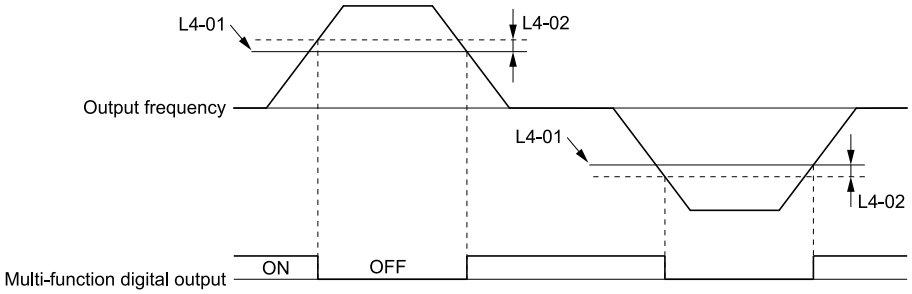


Figure 11.14 Frequency Detection 1 Time Chart

Note:

Figure 11.14 shows the result of the configuration when *L4-07 = 1 [Speed Agree Detection Selection = Detection Always Enabled]*. The default setting of *L4-07* is *0 [No detection during baseblock]*. When the speed agreement detection selection is "No Detection during Baseblock", the terminal is deactivated when the drive output stops.

■ 5: Frequency Detection 2

Setting Value	Function	Description
5	Frequency Detection 2	V/f OLV The terminal activates when the output frequency is higher than the value of *L4-01 [Speed Agree Detection Level]*. After the terminal activates, the terminal stays activated until the output frequency is at the value of *L4-01 - L4-02*.

Note:

The detection function operates in the two motor rotation directions. The drive uses the *L4-01* value as the forward/reverse detection level.

ON : The output frequency is higher than the value of *L4-01*.

OFF : The output frequency is less than the value of "*L4-01 - L4-02*," or it is not more than the value of *L4-01*.

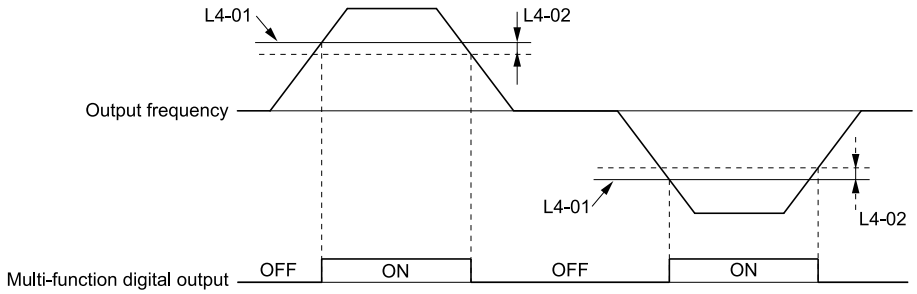


Figure 11.15 Frequency Detection 2 Time Chart

■ 6: Drive Ready

Setting Value	Function	Description
6	Drive Ready	V/f OLV The terminal activates when the drive is ready and running.

The terminal deactivates in these conditions:

- When the power supply is OFF
- During a fault
- When there is problem with the control power supply
- When there is a parameter setting error and the drive cannot operate although there is a Run command
- When you enter a Run command and it immediately triggers an overvoltage or undervoltage fault because the drive has an overvoltage or undervoltage fault during stop
- When the drive is in Programming Mode and will not accept a Run command
- When the Safe Disable function is active

■ 7: DC Bus Undervoltage

Setting Value	Function	Description
7	DC Bus Undervoltage	V/f OLV The terminal activates when the DC bus voltage or control circuit power supply is at the voltage set in *L2-05 [Undervoltage Detection Lvl (Uv1)]* or less. The terminal also activates when there is a fault with the DC bus voltage.

ON : The DC bus voltage ≤ *L2-05*

OFF : The DC bus voltage > *L2-05*

■ 8: During Baseblock (N.O.)

Setting Value	Function	Description
8	During Baseblock (N.O.)	V/f OLV The terminal activates during baseblock. When the drive is in baseblock, the drive output transistor stops switching and does not make DC bus voltage.

ON : During baseblock

OFF : The drive is not in baseblock.

■ B: Torque Detection 1 (N.O.)

Setting Value	Function	Description
B	Torque Detection 1 (N.O.)	V/f OLV The terminal activates when the drive detects overtorque or undertorque.

ON : The output current/torque > *L6-02 [Torque Detection Level 1]*, or the output current/torque < *L6-02* for longer than the time set in *L6-03 [Torque Detection Time 1]*.

Note:
- When *L6-01* ≥ *5*, the drive will detect when the output current/torque is less than *L6-02* for longer than *L6-03*.
- Refer to *L6: Torque Detection on page 300* for more information.

■ E: Fault

Setting Value	Function	Description
E	Fault	V/f OLV The terminal activates when the drive detects a fault.

Note:
The terminal will not activate for *CPF00* and *CPF01 [Control Circuit Error]* faults.

■ F: Not Used

Setting Value	Function	Description
F	Not Used	V/f OLV Use this setting for unused terminals or to use terminals in through mode. Also use this setting as the PLC contact output via MEMOBUS/Modbus or the communication option. This signal does not function if you do not configure signals from the PLC.

■ 10: Alarm

Setting Value	Function	Description
10	Alarm	V/f OLV The terminal activates when the drive detects a minor fault.

■ 11: Fault Reset Command Active

Setting Value	Function	Description
11	Fault Reset Command Active	V/f OLV The terminal activates when the drive receives the Reset command from the control circuit terminal, serial communications, or the communication option.

■ 12: Timer Output

Setting Value	Function	Description
12	Timer Output	V/f OLV Use this setting when the drive uses the timer function as an output terminal.

Note:
Refer to *b4: Timer Function on page 255* for more information.

■ 13: Speed Agree 2

Setting Value	Function	Description
13	Speed Agree 2	V/f OLV The terminal activates when the output frequency is in the range of the frequency reference ± *L4-04 [Speed Agree Detection Width(+/-)]*.

Note:
The detection function operates in the two motor rotation directions.

ON : The output frequency is in the range of "frequency reference ± *L4-04*".
OFF : The output frequency is not in the range of "frequency reference ± *L4-04*".

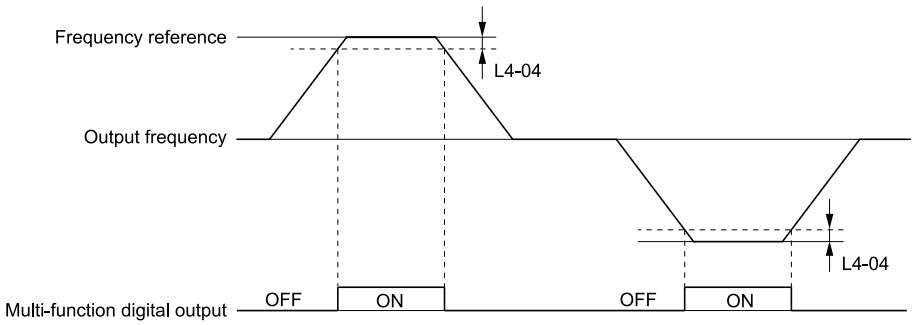


Figure 11.16 Speed Agree 2 Time Chart

■ 14: User-Set Speed Agree 2

Setting Value	Function	Description
14	User-Set Speed Agree 2	V/f OLV The terminal activates when the output frequency is in the range of *L4-03 [Speed Agree Detection Level(+/-)]* ± *L4-04 [Speed Agree Detection Width(+/-)]* and in the range of the frequency reference ± *L4-04*.

Note:

The detection level set with *L4-03* is a signed value. The drive will only detect in one direction.

ON : The output frequency is in the range of "*L4-03* ± *L4-04*" and the range of frequency reference ± *L4-04*.

OFF : The output frequency is not in the range of "*L4-03* ± *L4-04*" nor the range of frequency reference ± *L4-04*.

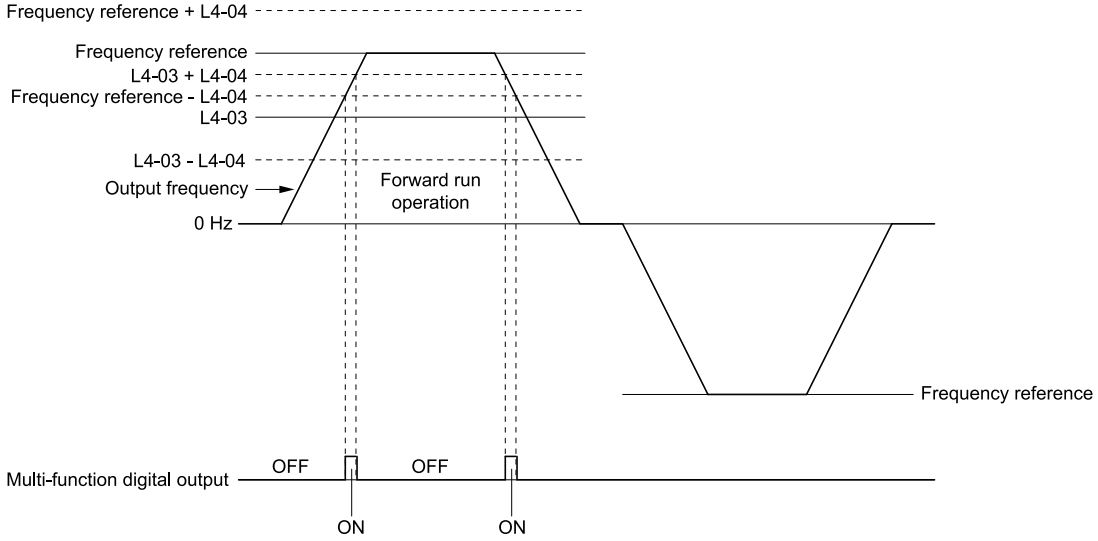


Figure 11.17 Example of User-set Speed Agree 2 (L4-03 Is Positive)

■ 15: Frequency Detection 3

Setting Value	Function	Description
15	Frequency Detection 3	V/f OLV The terminal deactivates when the output frequency is more than "*L4-03 [Speed Agree Detection Level(+/-)]* + *L4-04 [Speed Agree Detection Width(+/-)]*". After the terminal deactivates, the terminal stays deactivated until the output frequency is at the value of *L4-03*.

Note:

The detection level set with *L4-03* is a signed value. The drive will only detect in one direction.

ON : The output frequency is less than the value of *L4-03* or it is not more than the value of *L4-03* + *L4-04*.

OFF : The output frequency is higher than the value of *L4-03* + *L4-04*.

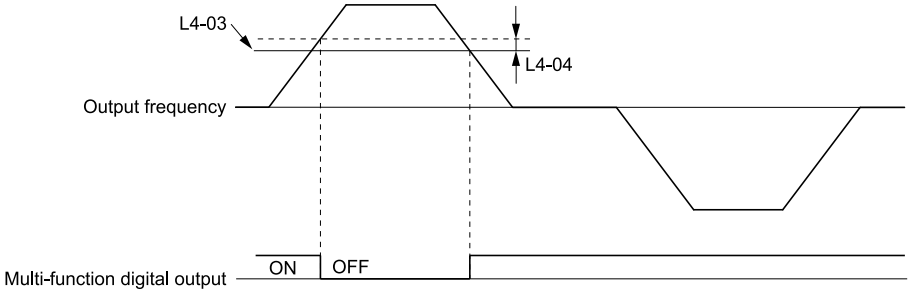


Figure 11.18 Example of Frequency Detection 3 (value of L4-03 Is Positive)

Note:

Figure 11.18 shows the result of the configuration when *L4-07 = 1 [Speed Agree Detection Selection = Detection Always Enabled]*. The default setting of *L4-07* is *0 [No detection during baseblock]*. When the speed agreement detection selection is "No Detection during Baseblock", the terminal is deactivated when the drive output stops.

■ 16: Frequency Detection 4

Setting Value	Function	Description
16	Frequency Detection 4	V/f OLV The terminal activates when the output frequency is higher than the value of *L4-03 [Speed Agree Detection Level(+/-)]*. After the terminal activates, the terminal stays activated until the output frequency = *L4-03 - L4-04*.

Note:

The detection level set with *L4-03* is a signed value. The drive will only detect in one direction.

ON : The output frequency is higher than the value of *L4-03*.

OFF : The output frequency is less than the value of "*L4-03 - L4-04*," or it is not more than the value of *L4-03*.

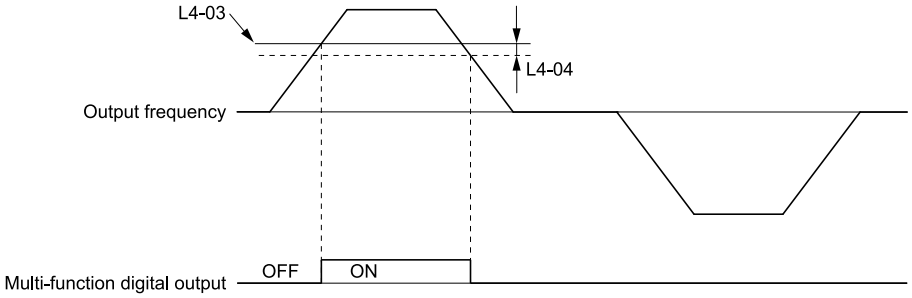


Figure 11.19 Example of Frequency Detection 4 (value of L4-03 Is Positive)

■ 18: Torque Detection 2 (N.O.)

Setting Value	Function	Description
18	Torque Detection 2 (N.O.)	V/f OLV The terminal activates when the drive detects overtorque or undertorque.

Use the *L6 [Torque Detection]* parameters to set torque detection.

ON : The output current/torque is more than *L6-05 [Torque Detection Level 2]*, or the level is less than *L6-05* for longer than*L6-06 [Torque Detection Time 2]*.

Note:

- When *L6-04* ≥ *5*, the drive will detect when the output current/torque is less than *L6-05* for longer than *L6-06*.
- Refer to *L6: Torque Detection on page 300* for more information.

■ 1A: During Reverse

Setting Value	Function	Description
1A	During Reverse	V/f OLV The terminal activates when the motor operates in the reverse direction.

ON : The motor is operating in the reverse direction.

OFF : The motor is operating in the forward direction or the motor stopped.

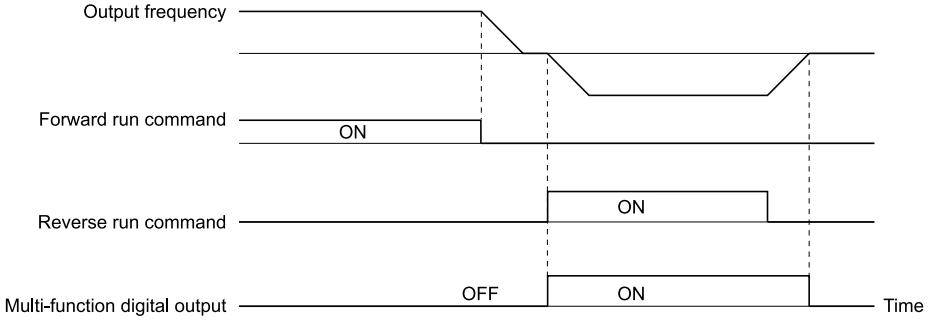


Figure 11.20 Reverse Operation Output Time Chart

■ 1B: During Baseblock (N.C.)

Setting Value	Function	Description
1B	During Baseblock (N.C.)	V/f OLV The terminal deactivates during baseblock. When the drive is in baseblock, the drive output transistor stops switching and does not make DC bus voltage.

ON : The drive is not in baseblock.

OFF : During baseblock

■ 1E: Executing Auto-Restart

Setting Value	Function	Description
1E	Executing Auto-Restart	V/f OLV The terminal activates when the Auto Restart function is trying to restart after a fault.

The terminal deactivates when the Auto Restart function automatically resets a fault. The terminal deactivates when the Auto Restart function detects the fault again because there were too many restart attempts as specified by *L5-01 [Number of Auto Restart Attempts]*.

Note:

Refer to *L5: Fault Restart on page 299* for more information.

■ 1F: Motor Overload Alarm (oL1)

Setting Value	Function	Description
1F	Motor Overload Alarm (oL1)	V/f OLV The terminal activates when the electronic thermal protection value of the motor overload protective function is a minimum of 90% of the detection level.

Note:

Refer to *L1-01: Motor Overload (oL1) Protection on page 293* for more information.

■ 20: Drive Overheat Pre-Alarm (oH)

Setting Value	Function	Description
20	Drive Overheat Pre-Alarm (oH)	V/f OLV The terminal activates when the drive heatsink temperature is at the level set with *L8-02 [Overheat Alarm Level]*.

Note:

Refer to *L8-02: Overheat Alarm Level on page 306* for more information.

■ 21: Safe Torque OFF

Setting Value	Function	Description
21	Safety Monitor Output Status	V/f OLV The terminal activates (safety stop state) when the safety circuit and safety diagnosis circuit are operating correctly and when terminals H1-HC and H2-HC are OFF (Open).

Note:

EDM = External Device Monitor

ON : Safety stop state

Terminals H1-HC and H2-HC are OFF (Open) (safety stop state).

OFF : Safety circuit fault or RUN/READY

Terminal H1-HC or terminal H2-HC is OFF (Open) (safety circuit fault), or the two terminals are ON or have short circuited (RUN/READY).

■ 2F: Maintenance Notification

Setting Value	Function	Description
2F	Maintenance Notification	V/f OLV The terminal activates when drive components are at their estimated maintenance period.

Tells you about the maintenance period for these items:

- IGBT
- Cooling Fan

- Capacitor
- Soft charge bypass relay

Note:

Refer to *Maintenance on page 184* for more information.

■ 30: During Torque Limit

Setting Value	Function	Description
30	During Torque Limit	V/f OLV The terminal activates when the torque reference is the torque limit set with *L7 parameters* or *H3-02 or H3-10 [MFAI Function Selection]*.

Note:

Refer to *L7: Torque Limit on page 304* for more information.

■ 37: During Frequency Output

Setting Value	Function	Description
37	During Frequency Output	V/f OLV The terminal activates when the drive outputs frequency.

ON : The drive outputs frequency.

OFF : The drive does not output frequency.

Note:

The terminal deactivates in these conditions:
- During Stop
- During baseblock
- During DC Injection Braking (initial excitation)
- During Short Circuit Braking

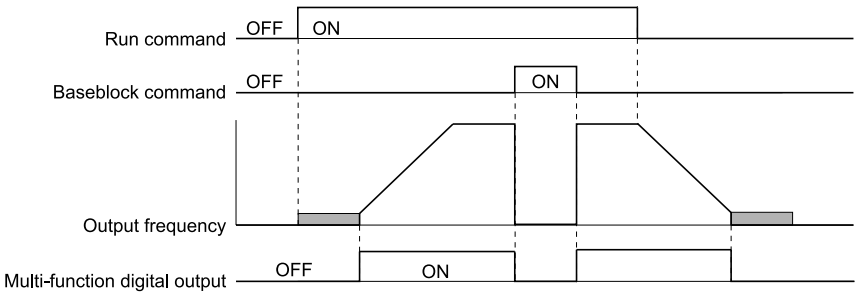


Figure 11.21 Active Frequency Output Time Chart

■ 47: PF Detection

Setting Value	Function	Description
47	PF Detection	V/f OLV Indicates input phase loss.

0 : No error

1 : Input Phase Loss error detected

■ 50: Brake Control

Setting Value	Function	Description
50	Brake Control	V/f OLV This setting can be used in the brake sequence for the elevator application. Closing the output terminal should cause the brake to release, and opening the terminal should apply the brake.

Close : Release brake.

Open : Apply brake

■ 51: Output Contactor Control

Setting Value	Function	Description
51	Output Contactor Control	V/f OLV Assigning this command to an output terminal can send a signal to the controller to close the output contactor. The output contactor should open when the terminal is released.

Closed : Close output contactor

■ 54: Light Load Direction

Setting Value	Function	Description
54	Light Load Direction	V/f OLV Indicates the light load direction detected during emergency operation with light load search. When the terminal is closed the light load direction is up, when it is open the light load direction is down.

Closed : Light load direction is up
Open : Light load direction is down

■ 58: Safe Disable Status

Setting Value	Function	Description
58	Safe Disable Status	V/f OLV This terminal closes if the Safe Disable inputs H1-HC are closed and opens when terminals H1-HC are open.

Closed : Safe Disable terminals H1-HC and H2-HC are open, drive is in a baseblock state
Open : Safe Disable terminals H1-HC and H2-HC are closed (normal operation)

■ 100 to 158: Inverse output of 0 to 58

Setting Value	Function	Description
100 - 158	Inverse output of 0 to 58	V/f OLV Causes inverse output of the function for the selected MFDO. Uses the last two digits of 1xx to select which function to inversely output.

For example, set *H2-xx = 10E* for the inverse output of *E [Fault]*.

◆ H4: Analog Outputs

H4 parameters set the drive analog monitors. These parameters select monitor parameters, adjust gain and bias, and select output signal levels.

■ Calibrate Meters Connected to MFAO Terminal AM

Use these parameters to calibrate meters connected to terminal AM:

- H4-02 [Terminal AM Analog Output Gain]
- H4-03 [Terminal AM Analog Output Bias]

Set these parameters where the output voltage of 10 V are 100% of the signal level.

No.	Name	Range	Default
H4-02	Terminal AM Analog Output Gain	-999.9 - +999.9%	100.0%
H4-03	Terminal AM Analog Output Bias	-999.9 - +999.9%	0.0%

Figure 11.22 and Figure 11.23 show the gain and bias.

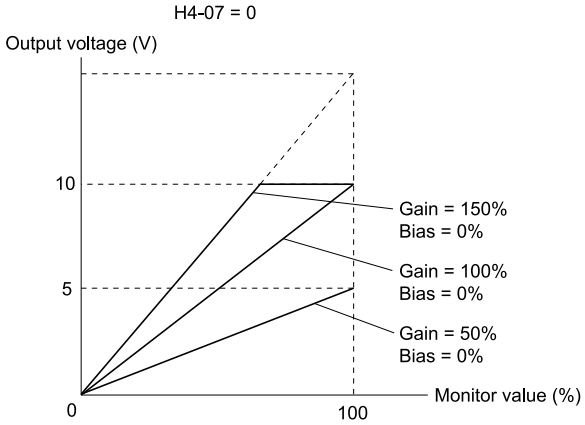


Figure 11.22 Analog Output Gain/Bias Configuration Example 1

For example, when the parameter value set to analog output is 0, and a 3 V signal is to be output to terminal AM, set *H4-03 [AM Analog Output Bias] = 30%*.

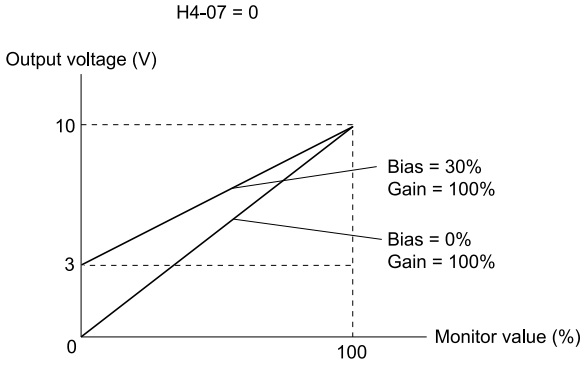


Figure 11.23 Analog Output Gain/Bias Configuration Example 2

Calibrate Terminal AM

Stop the drive to calibrate meters. Use this procedure to calibrate:

1. Show *H4-02 [Terminal AM Analog Output Gain]* on the keypad.
   Terminal AM outputs the analog signal when the monitor item that you set in *H4-01 [Terminal AM Analog Output Select]* is 100%.
2. Adjust *H4-02* while referencing the meter scale connected to terminal AM.
3. Show *H4-03 [Terminal AM Analog Output Bias]* on the keypad.
   The analog signal at the time when the monitor item selected with *H4-01* is 0% is output from terminal AM.
4. Adjust *H4-03* while referencing the meter scale connected to terminal AM.

■ H4-01: Terminal AM Analog Output Select

No. (Hex.)	Name	Description	Default (Range)
H4-01 (041D)	Terminal AM Analog Output Select	V/f OLV Sets the monitoring number to be output from the MFAO terminal AM.	102 (000 - 999)

Set the *x-xx* part of the *Ux-xx [Monitor]*. For example, set *H4-01 = 102* to monitor *U1-02 [Output Frequency]*.

Note:
- You cannot use all of the monitors in all of the control methods.
- When you use the terminal in through mode, set this parameter to *000* or *031*. You can set the terminal AM output level from the PLC through MEMOBUS/Modbus communications or the communication option.

■ H4-02: Terminal AM Analog Output Gain

No. (Hex.)	Name	Description	Default (Range)
H4-02 (041E) RUN	Terminal AM Analog Output Gain	V/f OLV Sets the gain of the monitor signal that is sent from MFAO terminal AM.	100.0% (-999.9 - +999.9%)

The analog signal output from the AM terminal is a maximum of 10 V (or 20 mA). Use *H4-07 [Terminal AM Signal Level Select]* to set the signal level.

■ H4-03: Terminal AM Analog Output Bias

No. (Hex.)	Name	Description	Default (Range)
H4-03 (041F) RUN	Terminal AM Analog Output Bias	V/f OLV Sets the bias of the monitor signal that is sent from MFAO terminal AM.	0.0% (-999.9 - +999.9%)

The analog signal output from the AM terminal is a maximum of 10 V (or 20 mA). Use *H4-07 [Terminal AM Signal Level Select]* to set the signal level.

11.7 L: Protection Functions

L parameters set the following functions.

- Motor Overload Protection
- Operation During Momentary Power Loss
- Stall Prevention
- Speed Detection
- Auto Restart
- Detection of Overtorque/Undertorque
- Torque Limit
- Hardware Protection

◆ L1: Motor Protection

L1 parameters set the motor overload protection function.

■ L1-01: Motor Overload (oL1) Protection

No. (Hex.)	Name	Description	Default (Range)
L1-01 (0480)	Motor Overload (oL1) Protection	V/f OLV Sets the motor overload protection with electronic thermal protectors.	1 (0 - 3)

This parameter enables and disables the motor overload protection with electronic thermal protectors.

The cooling capability of the motor changes when the speed control range of the motor changes. Use an electronic thermal protector that aligns with the permitted load characteristics of the motor to select motor protection.

The electronic thermal protector of the drive uses these items to calculate motor overload tolerance and supply overload protection for the motor:

- Output Current
- Output Frequency
- Motor thermal characteristics
- Time characteristics

If the drive detects motor overload, the drive will trigger an *oL1 [Motor Overload]* and stop the drive output.

Set *H2-01 = 1F [Term MA/MB-MC Function Selection = Motor Overload Alarm (oL1)]* to set a motor overload alarm. If the motor overload level is more than 90% of the *oL1* detection level, the output terminal activates and triggers an overload alarm.

0 : Disabled

Disable motor protection when motor overload protection is not necessary or when the drive is operating more than one motor.

Refer to Figure 11.24 for an example of the circuit configuration to connect more than one motor to one drive.

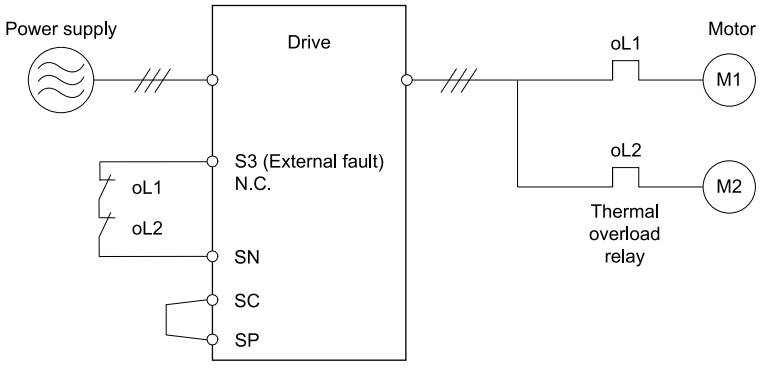


Figure 11.24 Protection Circuit Configuration to Connect More than One Motor to One Drive

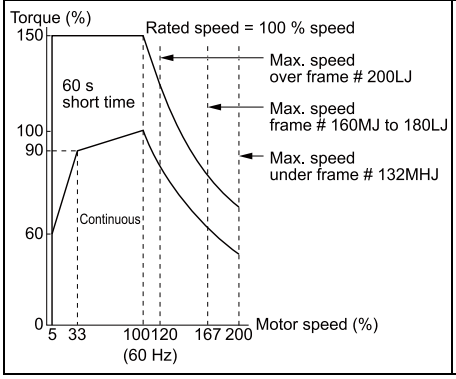
NOTICE: *When you connect more than one motor to one drive or when the motor amp rating is higher than the drive amp rating, set L1-01 =0 [Motor Overload (oL1) Protection = Disabled] and install thermal overload relays for each motor. The electronic thermal protection of the drive will not function and it can cause damage to the motor.*

1 : Variable Torque

Use this setting for general-purpose motors with a 60 Hz base frequency.

The overload tolerance decreases as motor speed decreases because the cooling fan speed decreases and the ability of the motor to cool decreases in the low speed range.

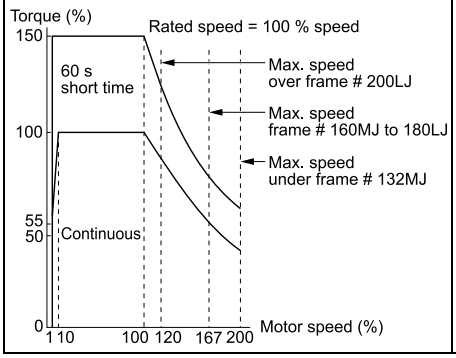
The overload tolerance characteristics of the motor change the trigger point for the electronic thermal protector. This provides motor overheat protection from low speed to high speed across the full speed range.

Load Tolerance	Cooling Capability	Overload Characteristics (at 100% motor load)
	This motor is designed to operate with commercial line power. Operate at a 60 Hz base frequency to maximize the motor cooling ability.	If the motor operates at frequencies less than 60 Hz, the drive will detect *oL1*. The drive triggers a fault relay output and the motor coasts to stop.

2 : Constant Torque 10:1 Speed Range

Use this setting for drive-dedicated motors with a speed range for constant torque of 1:10.

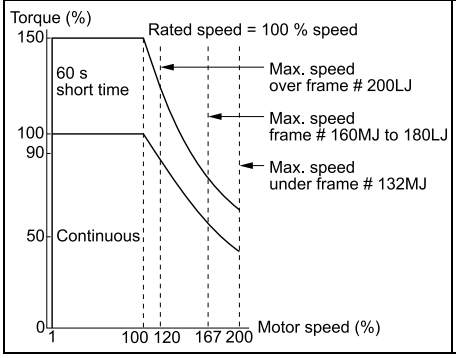
The speed control for this motor is 10% to 100% when at 100% load. Operating slower than 10% speed at 100% load will cause motor overload.

Load Tolerance	Cooling Capability	Overload Characteristics (at 100% motor load)
	This motor is designed to withstand increased temperatures during continuous operation in the low speed range (10% base frequency).	The motor operates continuously at 10% to 100% base frequency. Operating slower than 10% speed at 100% load will cause motor overload.

3 : Constant Torque 100:1 SpeedRange

Use this setting for vector motors with a speed range for constant torque of 1:100.

The speed control for this motor is 1% to 100% when at 100% load. Operating slower than 1% speed at 100% load will cause motor overload.

Load Tolerance	Cooling Capability	Overload Characteristics (at 100% motor load)
	This motor is designed to withstand increased temperatures during continuous operation in the low speed range (1% base frequency).	The motor operates continuously at 1% to 100% base frequency. Operating slower than 1% speed at 100% load will cause motor overload.

■ L1-02: Motor Overload Protection Time

No. (Hex.)	Name	Description	Default (Range)
L1-02 (0481)	Motor Overload Protection Time	V/f OLV Sets the operation time for the electronic thermal protector of the drive to prevent damage to the motor. Usually it is not necessary to change this setting.	1.0 min (0.1 - 5.0 min)

Set the overload tolerance time to the length of time that the motor can operate at 150% load from continuous operation at 100% load.

When the motor operates at 150% load continuously for 1 minute after continuous operation at 100% load (hot start), the default setting triggers the electronic thermal protector.

Figure 11.25 shows an example of the electronic thermal protector operation time. Motor overload protection operates in the range between a cold start and a hot start.

This example shows a general-purpose motor operating at the base frequency with *L1-02* set to 1.0 min.

- Cold start
  Shows the motor protection operation time characteristics when the overload occurs immediately after starting operation from a complete stop.
- Hot start
  Shows the motor protection operation time characteristics when overload occurs from continuous operation below the motor rated current.

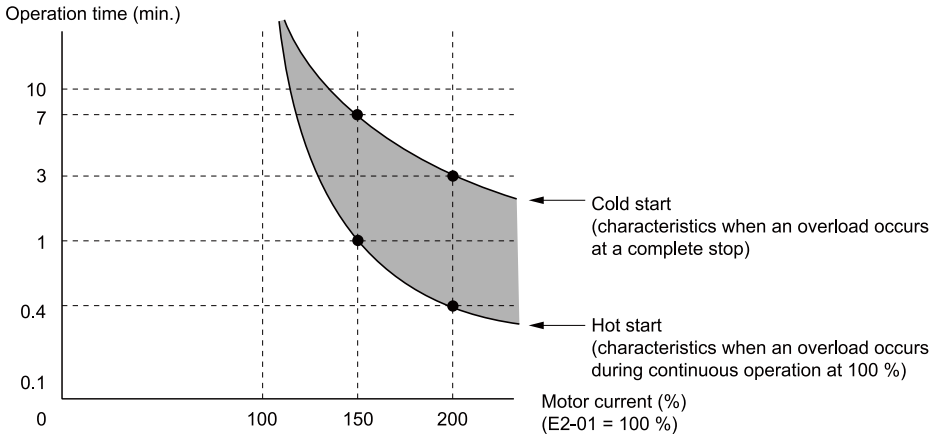


Figure 11.25 Protection Operation Time for a General-purpose Motor at Rated Output Frequency

■ L1-13: Motor Overload Memory Selection

No. (Hex.)	Name	Description	Default (Range)
L1-13 (046D)	Motor Overload Memory Selection	V/f OLV Sets the function that keeps the current electronic thermal protector value when the drive stops receiving power.	1 (0, 1)

0 : Disabled

1 : Enabled

Sets if the drive will calculate the motor again when the drive is energized again.

◆ L2: Undervoltage Detection

■ L2-05: Undervoltage Detection Lvl (Uv1)

No. (Hex.)	Name	Description	Default (Range)
L2-05 (0489)	Undervoltage Detection Lvl (Uv1)	V/f OLV Sets the voltage at which a *Uv1 [DC Bus Undervoltage]* fault is triggered. Usually it is not necessary to change this setting.	Determined by E1-01 (400 V Class: 300 - 420 V)

NOTICE: *Damage to Equipment. When you set this parameter to a value lower than the default, you must install an AC reactor on the input side of the power supply. If you do not install an AC reactor, it will cause damage to the drive circuitry.*

◆ L3: Stall Prevention

L3 parameters set the Stall Prevention function and overvoltage suppression function.

■ Stall Prevention

If the load is too heavy or the acceleration and deceleration times are too short, the motor can slip too much because it cannot work at the same rate as the frequency reference. If the motor stalls during acceleration, current increases as the slip increases to cause an *oC [Overcurrent]*, *oL2 [Drive Overload]*, or *oL1 [Motor Overload]* and the drive will stop. If the motor stalls during deceleration, too much regenerative power will flow back into the DC bus capacitors, and cause the drive to fault out from *ov [Overvoltage]* and the drive will stop.

The stall prevention function will let the motor get to the set speed without stalling and it is not necessary for you to change the acceleration or deceleration time settings. You can set a separate stall prevention functions for acceleration, operating at constant speeds, and deceleration.

■ L3-01: Stall Prevention during Accel

No. (Hex.)	Name	Description	Default (Range)
L3-01 (048F)	Stall Prevention during Accel	V/f OLV Sets the method of the Stall Prevention During Acceleration.	1 (0 - 1)

Stall prevention during acceleration prevents the stalling and stopping of motors when the drive detects *oC [Overcurrent]*, *oL2 [Drive Overloaded]*, or *oL1 [Motor Overload]* when you apply a large load during acceleration or when you set sudden acceleration times related to load inertia.

0 : Disable

The Stall Prevention function does not operate during acceleration, and acceleration occurs for the set acceleration time. If the acceleration time is too short, the motor does not fully accelerate during the set time, which causes the drive to detect *oL1* or *oL2* and the motor to stop.

1 : Enabled

Enables the Stall Prevention During Acceleration function. Operation is different for different control methods.

- **V/f Control, Open Loop Vector Control, or EZ Open Loop Vector Control**
  When the output current is more than the value set in *L3-02 [Stall Prevent Level during Accel]*, the drive stops acceleration. The drive stops deceleration when the output current is less than *L3-02* - 15%. The Stall Prevention function level automatically decreases for constant output ranges.

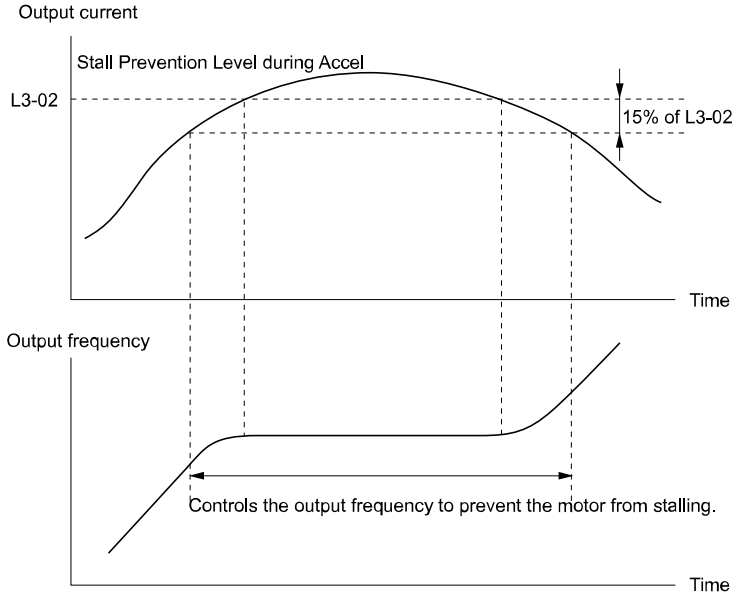


Figure 11.26 Stall Prevention During Acceleration when Using Induction Motors

- **Open Loop Vector Control for PM**
  When the output current is more than *L3-02*, the drive stops acceleration. When the time set in *L3-27 [Stall Prevention Detection Time]* is expired and the output current = *L3-02*, the drive will start deceleration as specified by the value set in *L3-22 [PM Stall Prevention Decel Time]*. The drive starts acceleration again when the output current is less than *L3-02* - 15%. When the time set in *L3-27* is expired, the drive starts acceleration again.

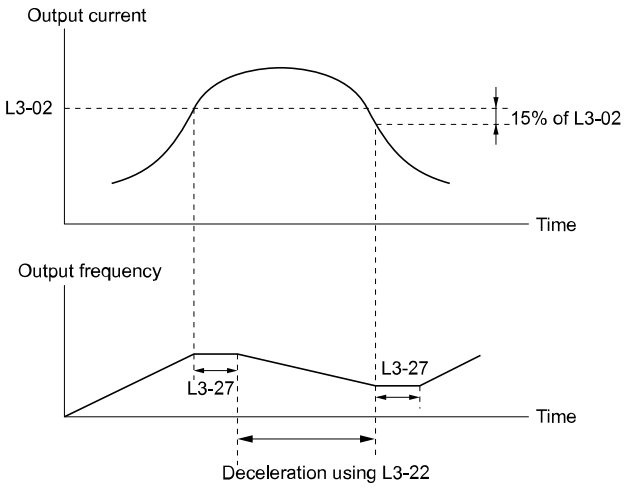


Figure 11.27 Stall Prevention During Acceleration Function in OLV/PM

■ L3-02: Stall Prevent Level during Accel

No. (Hex.)	Name	Description	Default (Range)
L3-02 (0490)	Stall Prevent Level during Accel	V/f OLV Sets the output current level at which the Stall Prevention function operates during acceleration where the drive rated output current is 100%.	Determined by L8-38 (0 - 165%)

Note:
- If you use a motor that is small compared to the drive and the motor stalls, decrease the setting value.
- When you operate the motor in the constant power range, set *L3-03 [Stall Prevent Limit during Accel]*.

■ L3-03: Stall Prevent Limit during Accel

No. (Hex.)	Name	Description	Default (Range)
L3-03 (0491)	Stall Prevent Limit during Accel	V/f OLV Sets the lower limit for the stall prevention level used in the constant output range as a percentage of the drive rated output current.	50% (0 - 100%)

The stall prevention level set in *L3-02 [Stall Prevent Level during Accel]* is automatically reduced when the motor is running within the constant output range. Parameter *L3-03* is the limit value used to prevent the stall prevention level during constant output ranges to fall below the minimum required level.

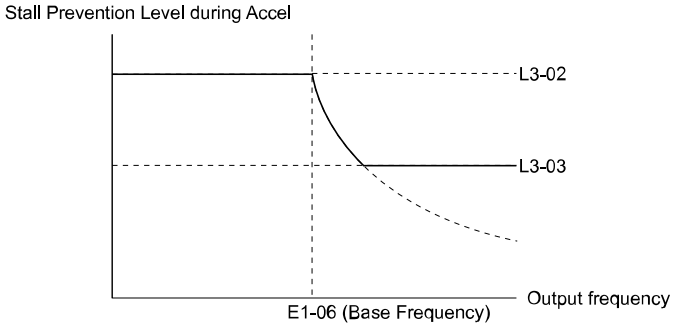


Figure 11.28 Stall Prevent Level during Accel/Limit

■ L3-05: Stall Prevention during RUN

No. (Hex.)	Name	Description	Default (Range)
L3-05 (0493)	Stall Prevention during RUN	V/f OLV Sets the function to enable and disable Stall Prevention During Run.	1 (0 - 1)

Stall Prevention function during run prevents the motor from stalling by automatically reducing the speed when an *oL1 [Motor Overload]* occurs while the motor is running at constant speed.

Note:

An output frequency less than 6 Hz disables Stall Prevention during Run. The setting values of *L3-05* and *L3-06 [Stall Prevent Level during Run]* do not have an effect.

0 : No

The drive runs at the set frequency reference. A heavy load can cause the motor to stall and trip the drive with *oC [Overcurrent]* or *oL1*.

1 : Deceleration Time 1 (C1-02)

The drive will decelerate for the time set in *C1-02 [Deceleration Time 1]* when the current is more than the Stall Prevention level set in *L3-06*. When the current level is less than the "*L3-06* setting value - 2%" for 100 ms, the drive accelerates again for the acceleration time applicable at that time until it is at the set frequency.

■ L3-06: Stall Prevent Level during Run

No. (Hex.)	Name	Description	Default (Range)
L3-06 (0494)	Stall Prevent Level during Run	V/f OLV Sets the output current level at which the Stall Prevention function is enabled during run when the drive rated output current is 100%.	Determined by L8-38 (30 - 165%)

Note:

This parameter is applicable when *L3-05 = 1 [Stall Prevention during RUN = Deceleration Time 1 (C1-02)]*.

Use an Analog Input to Change the Stall Prevent Level during Run

When *H3-xx = 8 [MFAI Function Select = Stall Prevent Level During Run]*, you can use the input gain and bias settings for terminals A1 and A2 to change the stall prevention level during run.

If you set the input level for terminals A1 and A2 *[H3-xx = 8]* and *L3-06*, the drive will use the smaller value for Stall Prevent Level during Run.

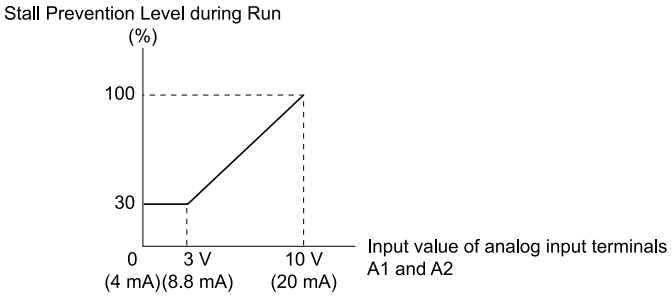


Figure 11.29 Stall Prevention Level during Run with Analog Input

◆ L4: Speed Detection

L4 parameters set the output of signals to the MFDO terminals, for example frequency agree and speed detection.

■ L4-01: Speed Agree Detection Level

No. (Hex.)	Name	Description	Default (Range)
L4-01 (0499)	Speed Agree Detection Level	V/f OLV Sets the level to detect speed agree or motor speed.	0.0 Hz (0.0 - 120.0 Hz)

Sets the level to detect speed agree or motor speed when *H2-01 to H2-03 = 2, 3, 4, 5 [MFDO Function Selection = Speed Agree 1, User-set Speed Agree 1, Frequency Detection 1, Frequency Detection 2]*.

■ L4-02: Speed Agree Detection Width

No. (Hex.)	Name	Description	Default (Range)
L4-02 (049A)	Speed Agree Detection Width	V/f OLV Sets the width to detect speed agree or motor speed.	2.0 Hz (0.0 - 20.0 Hz)

Sets the width to detect speed agree or motor speed when *H2-01 to H2-03 = 2, 3, 4, 5 [MFDO Function Selection = Speed Agree 1, User-set Speed Agree 1, Frequency Detection 1, Frequency Detection 2]*.

■ L4-03: Speed Agree Detection Level(+/-)

No. (Hex.)	Name	Description	Default (Range)
L4-03 (049B)	Speed Agree Detection Level(+/-)	V/f OLV Sets the level to detect speed agree or motor speed.	0.0 Hz (-120.0 - 120.0 Hz)

Sets the level to detect speed agree or motor speed when *H2-01 to H2-03 = 13, 14, 15, 16 [MFDO Function Selection = Speed Agree 2, User-set Speed Agree 2, Frequency Detection 3, Frequency Detection 4]*.

■ L4-04: Speed Agree Detection Width(+/-)

No. (Hex.)	Name	Description	Default (Range)
L4-04 (049C)	Speed Agree Detection Width(+/-)	V/f OLV Sets the width to detect speed agree or motor speed.	2.0 Hz (0.0 - 20.0 Hz)

Sets the width to detect speed agree or motor speed when *H2-01 to H2-03 = 13, 14, 15, 16 [MFDO Function Selection = Speed Agree 2, User-set Speed Agree 2, Frequency Detection 3, Frequency Detection 4]*.

◆ L5: Fault Restart

The Auto Restart function tries to keep machines operating when the drive detects a transient fault.

The drive can do a self-diagnostic check and continue the operation after a fault. If the cause of the fault goes away, the drive does speed search and restarts. It will not stop and the drive will not record a fault history. Use *L5-02 [Fault Contact at Restart Select]* to select the operation of fault relay signals during Auto Restart operation.

Sets if the drive will do Auto Restart and the number of times the drive will try to do Auto Restart in a set time. If the number of Auto Restart tries is more than the set value during the set time, drive output shuts off and operation stops. If this happens, remove the cause of the fault and manually restart the drive.

WARNING! *Sudden Movement Hazard. Do not use the fault restart function in hoist or lifting applications. Failure to obey can cause death or serious injury.*

The drive can do Auto Restart when these faults occur:

Table 11.9 List of Faults during which Auto Restart is Available

Fault	Name	Parameters to Disable Auto Restart
GF	Ground Fault	L5-08
LF	Output Phase Loss	-
oC	Overcurrent	-
oH1	Heatsink Overheat	L5-08
oL1	Motor Overload	L5-07
oL2	Drive Overload	L5-07
oL3	Overtorque Detection 1	L5-07
oL4	Overtorque Detection 2	L5-07

Fault	Name	Parameters to Disable Auto Restart
ov	Overvoltage	L5-08
rr	Dynamic Braking Transistor Fault	-
SE1	Sequence Error 1	-
SE2	Sequence Error 2	-
SE3	Sequence Error	-
UL3	Undertorque Detection 1	-
UL4	Undertorque Detection 2	-

■ L5-01: Number of Auto-Restart Attempts

No. (Hex.)	Name	Description	Default (Range)
L5-01 (049E)	Number of Auto-Restart Attempts	V/f OLV Sets the number of times that the drive will try to restart.	2 (0 - 10 times)

The drive resets the number of Auto Restart attempts to 0 in these conditions:

- The drive operates correctly for 10 minutes after a fault restart.
- When you manually clear a fault after the drive triggers protective functions.
- When you re-energize the drive.

■ L5-02: Fault Contact at Restart Select

No. (Hex.)	Name	Description	Default (Range)
L5-02 (049F)	Fault Contact at Restart Select	V/f OLV Sets the function that sends signals to the MFDO terminal set for *Fault [H2-xx = E]* while the drive is automatically restarting.	1 (0, 1)

0 : Active Only when Not Restarting

1 : Always Active

■ L5-04: Interval Method Restart Time

No. (Hex.)	Name	Description	Default (Range)
L5-04 (046C)	Interval Method Restart Time	V/f OLV Sets the time interval between each Auto Restart attempt. This function is enabled when *L5-05 = 1 [Auto Restart Operation Selection = Use L5-04 Time]*.	10.0 s (0.5 - 600.0 s)

■ L5-06: Undervoltage Fault Restart Selection

No. (Hex.)	Name	Description	Default (Range)
L5-06 (046E)	Undervoltage Fault Restart Selection	V/f OLV Determines whether a limit should be placed on the number of reset attempts after a Uv1 fault.	0 (0, 1)

0 : Restrict auto-reset attempts to L5-01 after Uv1

1 : No Limit on auto-reset attempts to L5-01

◆ L6: Torque Detection

The overtorque/undertorque detection function prevents damage to machinery and loads.

Overtorque is when there is too much load on the machine. If the motor current or output torque is at the overtorque detection level for the overtorque detection time, the drive will output an alarm and turn off the output.

Undertorque is when a load suddenly decreases. When the motor current or output torque is at the undertorque detection level for the undertorque detection time, the drive will output an alarm and turn off the output.

You can use the undertorque detection function to detect these conditions, for example:

- Machine belt cuts
- Unusual operation of the electromagnetic contactor on the drive output side
- Clogged output side air filters in fans and blowers
- Damage to blade tips and broken string

Note:

If there is *oC [Overcurrent]* or *oL1 [Motor Overload]*, the drive can stop during overtorque conditions. Use torque detection to identify overload conditions before the drive detects *oC* or *oL1* and stops. Use this function to detect issues that occur in the application.

■ Parameter Settings

You can individually set the two overtorque/undertorque detection functions with the drive. Use the information in Table 11.10 to set the parameters.

Table 11.10 Overtorque/Undertorque Detection Parameters

Configuration Parameter	Overtorque/Undertorque Detection 1	Overtorque/Undertorque Detection 2
MFDO Function Select • Terminal MA-MC • Terminal P1-C1 • Terminal P2-C2	H2-01, H2-02, and H2-03 = B N.O.: Activated when detected	H2-01, H2-02, and H2-03 = 18 N.O.: Activated when detected
	H2-01, H2-02, and H2-03 = 17 N.C.: Disactivated when detected	H2-01, H2-02, and H2-03 = 19 N.C.: Disactivated when detected
Detection conditions and selection of operation after detection	L6-01	L6-04
Detection Level	L6-02	L6-05
	Analog Input Terminal *1 H3-xx = 7	-
Detection Time	L6-03	L6-06

*1 You can also use an analog input terminal to supply the torque detection level. To enable this function, set *H3-xx = 7 [MFAI Function Selection = Overtorque/Undertorque DetectLvl]*. If both *L6-02* and *H3-xx = 7* are set, the analog input has priority and the setting of *L6-02* is disabled.

You cannot use Overtorque/Undertorque Detection 2 to set the detection level for the analog input terminals.

Note:

In V/f Control, the drive uses the current level (100% of the drive rated output current) to detect overtorque/undertorque. In vector control, the drive uses the motor torque (100% of the motor rated torque)to detect overtorque/undertorque. When you enable the mechanical weakening detection function, the overtorque/undertorque detection level for all control modes is the current level (100% of the drive rated output current).

■ Time Chart for Detection of Overtorque/Undertorque

Overtorque Detection Time Chart

When you use Overtorque/Undertorque Detection 1, the drive detects overtorque if the motor current or motor torque is at the detection level set in *L6-02 [Torque Detection Level 1]* for the time set in *L6-03 [Torque Detection Time 1]*. Parameter *L6-01 [Torque Detection Selection 1]* sets the operation after detection.

When you use Overtorque/Undertorque Detection 2, set *L6-04 [Torque Detection Selection 2]*, *L6-05 [Torque Detection Level 2]*, and *L6-06 [Torque Detection Time 2]*.

Set the terminal that outputs the alarm in *H2-01 to H2-03 [MFDO Function Select]*.

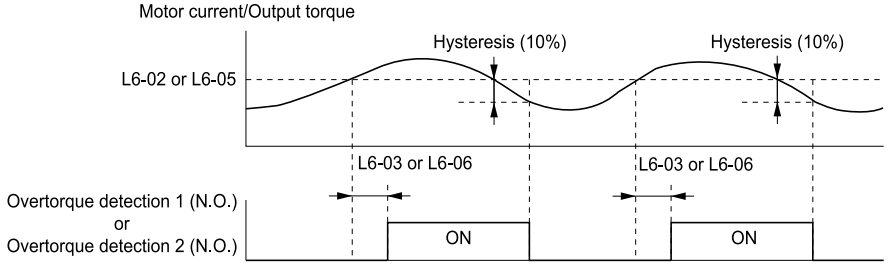


Figure 11.30 Overtorque Detection Time Chart

Note:

The drive applies a hysteresis of approximately 10% of the drive rated output current or the motor rated torque to the overtorque/undertorque detection function.

Undertorque Detection Time Chart

When you use Overtorque/Undertorque Detection 1, the drive detects undertorque if the motor current or motor torque is less than or equal to the detection level set in *L6-02* for the time set in *L6-03*. Parameter *L6-01* sets the operation after detection.

When you use Overtorque/Undertorque Detection 2, set the operation in *L6-04, L6-05, and L6-06*.

Set the terminal that outputs an alarm in *H2-01 to H2-03*.

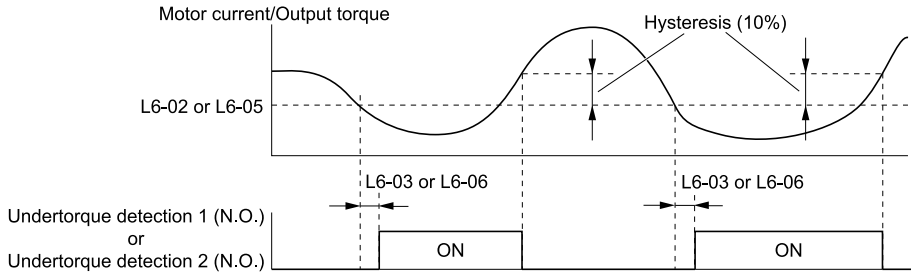


Figure 11.31 Undertorque Detection Time Chart

Note:

The drive applies a hysteresis of approximately 10% of the drive rated output current or the motor rated torque to the overtorque/undertorque detection function.

■ L6-01: Torque Detection Selection 1

No. (Hex.)	Name	Description	Default (Range)
L6-01 (04A1)	Torque Detection Selection 1	V/f OLV Sets the speed range that detects overtorque and undertorque and the operation of drives (operation status) after detection.	0 (0 - 8)

The drive detects overtorque if the motor current or output torque is more than the level set in *L6-02 [Torque Detection Level 1]* for the length of time set in *L6-03 [Torque Detection Time 1]*. The drive detects undertorque if the motor current or output torque is less than the level set in *L6-02* for the length the time set in *L6-03*.

0 : Disabled

The drive will not detect overtorque or undertorque.

1 : oL @ Speed Agree - Alarm only

The drive detects overtorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs an *oL3 [Overtorque Detection 1]* and operation continues.

2 : oL @ RUN - Alarm only

When the Run command is enabled, the drive constantly detects overtorque. The drive outputs an *oL3* and operation continues.

3 : oL @ Speed Agree - Fault

The drive detects overtorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs an *oL3 [Overtorque Detection 1]* and operation stops.

4 : oL @ RUN - Fault

When the Run command is enabled, the drive constantly detects overtorque. The drive outputs an *oL3* and operation stops.

5 : UL @ Speed Agree - Alarm only

The drive detects undertorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs a *UL3 [Undertorque Detection 1]* and operation continues.

6 : UL @ RUN - Alarm only

When the Run command is enabled, the drive constantly detects undertorque. The drive outputs a *UL3* and operation continues.

7 : UL @ Speed Agree - Fault

The drive detects undertorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs a *UL3* and operation stops.

8 : UL @ RUN - Fault

When the Run command is enabled, the drive constantly detects undertorque. The drive outputs a *UL3* and operation stops

■ L6-02: Torque Detection Level 1

No. (Hex.)	Name	Description	Default (Range)
L6-02 (04A2)	Torque Detection Level 1	V/f OLV Sets the detection level for Overtorque/Undertorque Detection 1. In V/f control, drive rated output current = 100% value. In vector control, motor rated torque = 100% value.	150% (0 - 300%)

Note:

- Set the torque detection level as a percentage of the drive rated output current in all control methods to set the mechanical weakening detection level.
- You can also use an analog input terminal to supply the torque detection level. To enable this function, set *H3-xx = 7 [MFAI Function Select = Overtorque/Undertorque DetectLvl]*. If you set *L6-02* and *H3-x = 7*, the analog input is most important and the drive disables *L6-02*.

■ L6-03: Torque Detection Time 1

No. (Hex.)	Name	Description	Default (Range)
L6-03 (04A3)	Torque Detection Time 1	V/f OLV Sets the detection time for Overtorque/Undertorque Detection 1.	0.1 s (0.0 - 10.0 s)

■ L6-04: Torque Detection Selection 2

No. (Hex.)	Name	Description	Default (Range)
L6-04 (04A4)	Torque Detection Selection 2	V/f OLV Sets the speed range that detects overtorque and undertorque and the operation of drives (operation status) after detection.	0 (0 - 8)

The drive detects overtorque if the motor current or output torque is more than the level set in *L6-05 [Torque Detection Level 2]* for the length of time set in *L6-06 [Torque Detection Time 2]*. The drive detects undertorque if the motor current or output torque is less than the level set in *L6-05* for the length the time set in *L6-06*.

0 : Disabled

The drive will not detect overtorque or undertorque.

1 : oL @ Speed Agree - Alarm only

The drive detects overtorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs an *oL4 [Overtorque Detection 2]* and operation continues.

2 : oL @ RUN - Alarm only

When the Run command is enabled, the drive constantly detects overtorque. The drive outputs an *oL4* and operation continues.

3 : oL @ Speed Agree - Fault

The drive detects overtorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs an *oL4 [Overtorque Detection 2]* and operation stops.

4 : oL @ RUN - Fault

When the Run command is enabled, the drive constantly detects overtorque. The drive outputs an *oL4* and operation stops.

5 : UL @ Speed Agree - Alarm only

The drive detects undertorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs a *UL4 [Undertorque Detection 2]* and operation continues.

6 : UL @ RUN - Alarm only

When the Run command is enabled, the drive constantly detects undertorque. The drive outputs a *UL4* and operation continues.

7 : UL @ Speed Agree - Fault

The drive detects undertorque when the output frequency aligns with the frequency reference. Detection does not occur during acceleration/deceleration. The drive outputs a *UL4* and operation stops.

8 : UL @ RUN - Fault

When the Run command is enabled, the drive constantly detects undertorque. The drive outputs a *UL4* and operation stops

■ L6-05: Torque Detection Level 2

No. (Hex.)	Name	Description	Default (Range)
L6-05 (04A5)	Torque Detection Level 2	V/f OLV Sets the detection level for Overtorque/Undertorque Detection 2. In V/f control, drive rated output current = 100% value. In vector control, motor rated torque = 100% value.	150% (0 - 300%)

Note:

Overtorque/Undertorque Detection 2 cannot set the detection level for the analog input terminal.

■ L6-06: Torque Detection Time 2

No. (Hex.)	Name	Description	Default (Range)
L6-06 (04A6)	Torque Detection Time 2	V/f OLV Sets the detection time for Overtorque/Undertorque Detection 2.	0.1 s (0.0 - 10.0 s)

◆ L7: Torque Limit

The torque limit function limits the internal torque reference for the drive to and keeps the torque from the motor constant. This function limits the torque applied to loads and regenerative torque to a value less than a set quantity. This function also prevents damage to machinery and increases the reliability of continuous operation. You can set torque limits individually for the four quadrants, which include torque direction (motoring/regeneration) and direction of motor rotation (forward/reverse). When the torque reference value is at the set torque limit, the MFDO terminal set for *During Torque Limit [H2-xx = 30]* activates.

Note:
- The drive output current limits maximum output torque. The drive limits torque to 150% of the rated output current for Heavy Duty Rating (HD) and to 110% of the rated output current for Normal Duty Rating (ND). The actual output torque is not more than the limits of the drive rated output current when you set the torque limit to a high value.
- When you use torque limits for lifting applications, do not lower the torque limit value too much. When the torque limit function is triggered, falls and rollbacks can occur because of sudden acceleration stops and stalls of the motor.

■ Configuring Settings

Use one of these methods to set torque limits:

- Individually set the four torque limit quadrants using *L7-01 to L7-04 [Torque Limit]*.
- Use MFAI to individually set the four torque limit quadrants. Set *H3-02, H3-10 = 10, 11, 12 [MFAI Function Select = Forward Torque Limit, Reverse Torque Limit, Regenerative Torque Limit]*.
- Use MFAI to set all four torque limit quadrants together. Set *H3-02, H3-10 = 15 [General Torque Limit]*.
- Use a communication option to set all four torque limit quadrants together.

Figure 11.32 shows the configuration method for each quadrant.

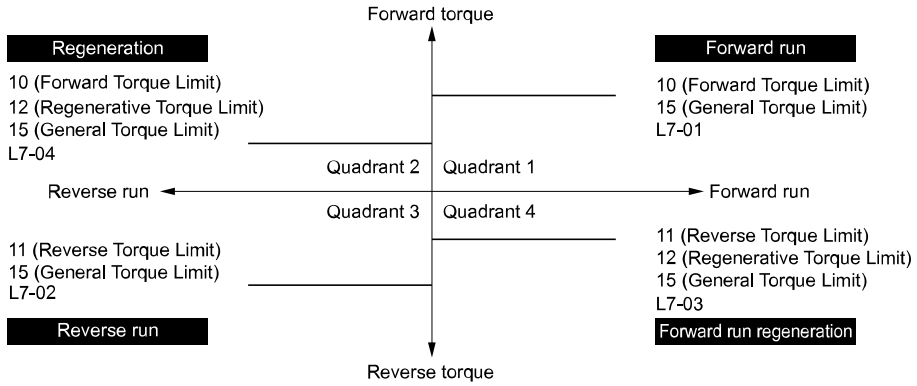


Figure 11.32 Torque Limits and Analog Input Setting Parameters

Note:
- When *L7-01 to L7-04* and analog inputs or communication option torque limits set torque limits for the same quadrant, the drive enables the lowest value.
  In this example of parameter settings, the torque limit for quadrant 1 is 130% and the torque limit for quadrants 2, 3, and 4 is 150%.
  Settings: *L7-01 = 130%*, *L7-02 to L7-04 = 200%*, and MFAI torque limit = 150%
- The drive output current limits maximum output torque. The torque limit is 150% of the rated output current for HD and to 120% of the rated output current for ND. The actual output torque is not more than the limits of the drive rated output current when you set the torque limit to a high value.

■ L7-01: Forward Torque Limit

No. (Hex.)	Name	Description	Default (Range)
L7-01 (04A7) RUN	Forward Torque Limit	V/f OLV Sets the torque limit value for forward motoring as a percentage, where motor rated torque is the 100% value.	200% (0 - 300%)

Note:
- The lower torque limit is enabled when you set the torque limit by the following method.
  –Set *H3-02 or H3-10 = 10, 15 [MFAI Function Select = Forward Torque Limit, General Torque Limit]*.
  –Use a communication option to set the torque limits
- You must think about drive capacity when a large quantity of torque is necessary. If you set the value too high, the drive can detect *oC [Overcurrent]*.
- If you set the value too low with large loads, the motor can stall.

■ L7-02: Reverse Torque Limit

No. (Hex.)	Name	Description	Default (Range)
L7-02 (04A8) RUN	Reverse Torque Limit	V/f OLV Sets the torque limit value for reversed motoring as a percentage, where motor rated torque is the 100% value.	200% (0 - 300%)

Note:
- The lower torque limit is enabled when you set the torque limit by the following method.
  – Set *H3-02 or H3-10 = 10, 15 [MFAI Function Select = Forward, Reverse/Regenerative Torque Limit]*.
  – Use a communication option to set the torque limits
- You must think about drive capacity when a large quantity of torque is necessary. If you set the value too high, the drive can detect *oC [Overcurrent]*.
- If you set the value too low with large loads, the motor can stall.

■ L7-03: Forward Regenerative Trq Limit

No. (Hex.)	Name	Description	Default (Range)
L7-03 (04A9) RUN	Forward Regenerative Trq Limit	V/f OLV Sets the torque limit value for forward regenerative conditions as a percentage of the motor rated torque.	200% (0 - 300%)

Note:
- The lower torque limit is enabled when you set the torque limit by the following method.
  – Set *H3-02 or H3-10 = 10, 15 [MFAI Function Select = Forward, Reverse/Regenerative Torque Limit]*.
  – Use a communication option to set the torque limits
- You must think about drive capacity when a large quantity of torque is necessary. If you set the value too high, the drive can detect *oC [Overcurrent]*.
- If you set the value too low with large loads, the motor can stall.

■ L7-04: Reverse Regenerative Trq Limit

No. (Hex.)	Name	Description	Default (Range)
L7-04 (04AA) RUN	Reverse Regenerative Trq Limit	V/f OLV Sets the torque limit value for reversed regenerative conditions as a percentage of the motor rated torque.	200% (0 - 300%)

Note:
- Use this method to set the torque limit and enable the lower torque limit:
  – Set *H3-02 or H3-10 = 10, 15 [MFAI Function Select = Forward, Reverse/Regenerative Torque Limit]*.
  – Use a communication option to set the torque limits
- You must think about drive capacity when a large quantity of torque is necessary. If you set the value too high, the drive can detect *oC [Overcurrent]*.
- If you set the value too low with large loads, the motor can stall.

■ L7-06: Torque Limit Integral Time

No. (Hex.)	Name	Description	Default (Range)
L7-06 (04AC)	Torque Limit Integral Time	V/f OLV Sets the integral time constant for the torque limit function.	200 ms (5 - 10000 ms)

Decrease the setting value to increase torque limit responsiveness when you use torque limits and *L7-07 = 1 [Torque Limit during Accel/Decel = Proportional & Integral control]*.

If there is hunting when torque limits are active, increase the setting value.

■ L7-07: Torque Limit during Accel/Decel

No. (Hex.)	Name	Description	Default (Range)
L7-07 (04C9)	Torque Limit during Accel/Decel	V/f OLV Sets the torque limit function during acceleration and deceleration.	0 (0, 1)

0 : Proportional only

The torque limit function works with proportional control during acceleration and deceleration, and switches to integral control at constant speed. Use this setting when quickly reaching the target speed is more important than the torque limit during speed changes.

1 : Proportional & Integral control

The torque limit function always uses integral control. Use this setting when a very accurate torque limit is necessary during speed changes, for example in winding machine applications.

If you make the torque limit the most important, it can:

- Increase the acceleration and deceleration times.
- Not let the motor speed reach the frequency reference value during run at constant speed.

◆ L8: Drive Protection

L8 parameters set protective functions that prevent faults such as overheating, phase loss, and ground faults.

■ L8-02: Overheat Alarm Level

No. (Hex.)	Name	Description	Default (Range)
L8-02 (04AE)	Overheat Alarm Level	V/f OLV Sets the *oH* detection level in temperature.	Determined by o2-04 (50 - 130 °C)

If the heatsink temperature is more than the temperature set in this parameter, the drive detects an overheat pre-alarm. To enable this function, set one of *H2-01 to H2-03 [MFDO Function Select]* to *20 [Drive Overheat Pre-Alarm (oH)]*.

If the temperature increases to the overheat fault level, the drive will trigger an *oH1 [Heatsink Overheat]* fault and stop operation.

■ L8-03: Overheat Pre-Alarm Selection

No. (Hex.)	Name	Description	Default (Range)
L8-03 (04AF)	Overheat Pre-Alarm Selection	V/f OLV Sets the operation of drives when an *oH* alarm is detected.	3 (0 - 4)

0 : Ramp to Stop

The drive ramps the motor to stop in the deceleration time. The output terminal set for *Fault [H2-01 to H2-03 = E]* activates.

1 : Coast to Stop

The output turns off and the motor coasts to stop. The output terminal set for *Fault [H2-01 to H2-03 = E]* activates.

2 : Fast Stop (Use C1-09)

The drive stops the motor in the deceleration time set in *C1-09 [Fast Stop Time]*. The output terminal set for *Fault [H2-01 to H2-03 = E]* activates.

3 : Alarm Only

The keypad shows *oH* and the drive continues operation. The output terminal set for *Alarm [H2-01 to H2-03 = 10]* activates.

4 : Operate at Reduced Speed (L8-19)

The drive decelerates to the level set in *L8-19 [Freq Reduction @ oH Pre-Alarm]* and continues operation. *oH* flashes on the keypad.

oH flashes on the keypad. When the alarm is output, the drive decelerates each 10 seconds. If the drive decelerates 10 times and the alarm continues to be output, the output terminal set for *oH Pre-Alarm Reduction Limit [H2-01 to H2-03 = 4D]* activates. When the alarm is not output during deceleration, the drive accelerates until it is at the frequency reference that was applicable before the alarm was turned off. Figure 11.33 shows the output of the alarm and the drive operation at a decreased output frequency.

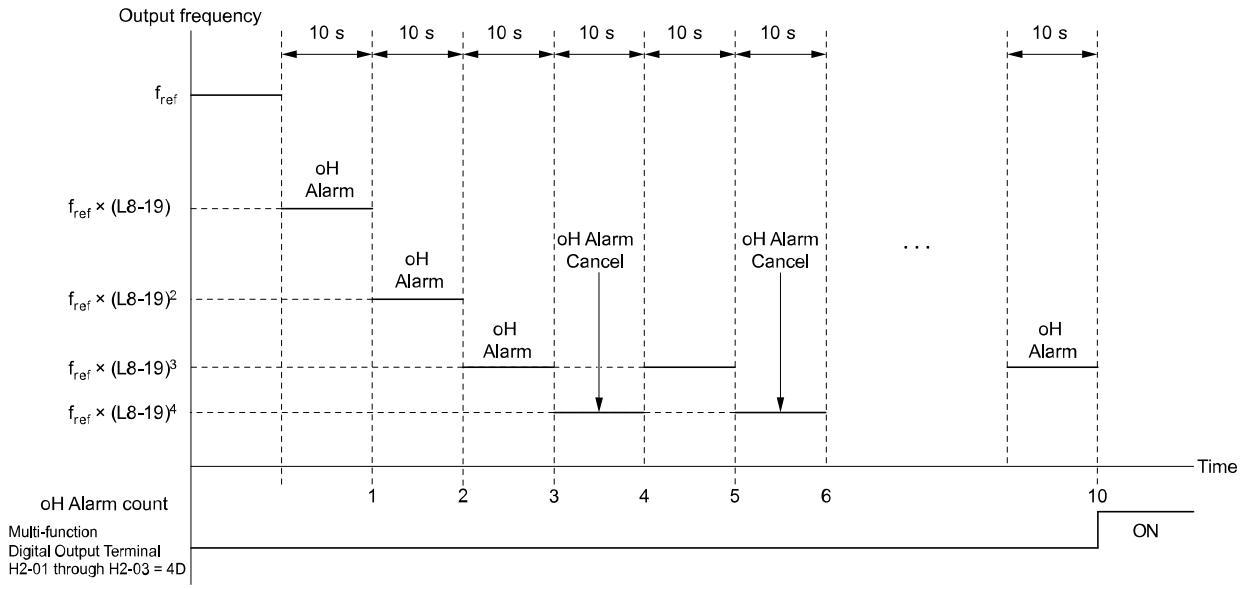


Figure 11.33 Drive Operation at a Decreased Output Frequency when the Overheat Alarm is Output

■ L8-05: Input Phase Loss Protection Sel

No. (Hex.)	Name	Description	Default (Range)
L8-05 (04B1)	Input Phase Loss Protection Sel	V/f OLV Sets the function to enable and disable input phase loss detection.	Depends on o2-04 (0 - 2)

0 : Disable

1 : Enabled

2 : Standard + dv/dt

■ L8-06: Input Phase Loss Detection Level

No. (Hex.)	Name	Description	Default (Range)
L8-06 (04B2)	Input Phase Loss Detection Level	V/f OLV When ripple is observed in the DC bus, expansion of the input bias is calculated and becomes the input phase if the difference between the max and minimum values of the ripple are greater than L8-06.	Determined by o2-04 (0.0 - 50.0%)

100% detection level = voltage (200V or 400 V) × √2

■ L8-07: Output Phase Loss Protection Sel

No. (Hex.)	Name	Description	Default (Range)
L8-07 (04B3)	Output Phase Loss Protection Sel	V/f OLV Sets the function to enable and disable output phase loss detection. The drive starts output phase loss detection when the output current decreases to less than 5% of the drive rated current.	1 (0 - 2)

Note:

The drive can incorrectly start output phase loss detection in these conditions:

- The motor rated current is very small compared to the drive rating.
- The drive is operating a PM motor with a small load.

0 : Disabled

1 : Fault when one phase is lost

If the drive loses one output phase, it will trigger *LF [Output Phase Loss]*.

The output turns off and the motor coasts to stop.

2 : Fault when two phases are lost

If the drive loses more than one output phase, it will trigger *LF [Output Phase Loss]*.

The output turns off and the motor coasts to stop.

■ L8-09: Output Ground Fault Detection

No. (Hex.)	Name	Description	Default (Range)
L8-09 (04B5)	Output Ground Fault Detection	V/f OLV Sets the function to enable and disable ground fault protection.	1 (0, 1)

0 : Disable

The drive will not detect ground faults.

1 : Enabled

If there is high leakage current or a ground short circuit in one or two output phases, the drive will detect *GF [Ground Fault]*.

Note:

If the ground path impedance is low, the drive can detect *oC [Overcurrent]*, *SC [Out Short Circuit or IGBT Fault]*, or *ov [DC Bus Overvoltage]* instead of *GF*.

■ L8-10: Heatsink Fan Operation Selection

No. (Hex.)	Name	Description	Default (Range)
L8-10 (04B6)	Heatsink Fan Operation Selection	V/f OLV Sets operation of the heatsink cooling fan.	0 (0 - 1)

0 : During Run, w/ L8-11 Off-Delay

The drive turns on the fan when a Run command is active.

1 : Always On

The fan turns on when you supply power to the drive. When you release the Run command and the delay time set in *L8-11 [HeatsinkCoolingFan Off DelayTime]* is expired, the fan stops. his setting extends the fan lifetime.

■ L8-11: Heatsink Fan Off-Delay Time

No. (Hex.)	Name	Description	Default (Range)
L8-11 (04B7)	Heatsink Fan Off-Delay Time	V/f OLV Sets the length of time that the drive will wait before it stops the cooling fan after it cancels the Run command when *L8-10 = 0 [Heatsink Cooling Fan Ope Select = Dur Run (OffDly)]*.	60 s (0 - 300 s)

■ L8-12: Ambient Temperature Setting

No. (Hex.)	Name	Description	Default (Range)
L8-12 (04B8)	Ambient Temperature Setting	V/f OLV Sets the ambient temperature of the drive installation area.	40 °C (-10 °C - +60 °C)

Note:

The setting range changes when the *L8-35 [Installation Method Selection]* value changes:

- *0 [IP20/UL Open Type]*: -10 °C to +60 °C
- *1 [Side-by-Side Mounting]*: -10 °C to +50 °C
- *2 [IP20/UL Type 1]*: -10 °C to +50 °C
- *3 [Finless Installation]*: -10 °C to +50 °C

The drive automatically adjusts the drive rated current to the best value as specified by the set temperature. Set the ambient temperature of the area where you install the drive to a value that is more than the drive rating.

Refer to *Derating Depending on Ambient Temperature on page 203* for information about derating as specified by ambient temperature.

■ L8-15: Drive oL2 @ Low Speed Protection

No. (Hex.)	Name	Description	Default (Range)
L8-15 (04BB)	Drive oL2 @ Low Speed Protection	V/f OLV Sets the function to decrease drive overload at low speeds to prevent damage to the main circuit transistor during low speed operation (at 6 Hz or slower) to prevent *oL2 [Drive Overloaded]*.	1 (0, 1)

Note:

Contact Yaskawa or your nearest sales representative for consultation before disabling this function at low speeds. Frequent operation of drives under conditions of high output current in low speed ranges may shorten the service life of the drive IGBT due to heat stress.

0 : Disabled (No Additional Derate)

The drive does not decrease the overload protection level.

1 : Enabled (Reduced oL2 Level)

When the drive detects *oL2* during low speed operation, it automatically decreases the overload detection level.

At zero speed, the drive derates the overload by 50%.

■ L8-18: Software Current Limit Selection

No. (Hex.)	Name	Description	Default (Range)
L8-18 (04BE)	Software Current Limit Selection	V/f OLV Set the software current limit selection function to prevent damage to the main circuit transistor caused by too much current.	0 (0, 1)

0 : Disabled

When the output current is at the software current limit value, the drive does not restrict the output voltage.

Note:

The drive may detect an *oC [Overcurrent]* when loads are particularly heavy or the acceleration time is particularly short.

1 : Enabled

When the output current is at the software current limit value, the drive decreases output voltage to decrease output current.

When the output current decreases to the software current limit level, the drive starts usual operation.

■ L8-19: Freq Reduction @ oH Pre-Alarm

No. (Hex.)	Name	Description	Default (Range)
L8-19 (04BF)	Freq Reduction @ oH Pre-Alarm	V/f OLV Sets the ratio at which the drive derates the frequency reference during an *oH* alarm.	0.8 (0.1 - 0.9)

When these two conditions are correct, this function is enabled:

- *L8-03 = 4 [Overheat Pre-Alarm Ope Selection = Run@L8-19 Rate]*
- *oH* alarm is output

■ L8-35: Installation Method Selection

No. (Hex.)	Name	Description	Default (Range)
L8-35 (04EC)	Installation Method Selection	V/f OLV Sets the type of drive installation.	Determined by o2-04 (0 - 3)

Note:

- Parameter *A1-03 [Initialize Parameters]* does not initialize this parameter.
- This parameter is set to the correct value when the drive is shipped. Change the value only in these conditions:
  - Side-by-Side installation
  - When you install a UL Type 1 kit on an IP20/UL Open Type drive to convert the drive to an IP20/UL Type 1 drive.
  - Mounting a standard drive with the heatsink outside the enclosure panel.

The overload protection detection level for the drive is automatically adjusted to the optimal value in accordance with the setting value. Refer to *Derating Depending on Ambient Temperature on page 203* for information about derating as specified by ambient temperature.

0 : IP20/UL Open Type

Use this setting to install IP20/UL Open Type drives.

Make sure that there is 30 mm (1.18 in) minimum of space between drives or between the drive and side of the enclosure panel.

1 : Side-by-Side Mounting

Use this setting to install more than one drive side-by-side.

2 : IP20/UL Type 1

Use this setting to install IP20/UL Type 1 or IP55 drives.

3 : Finless Installation

Use this setting to install Finless Type drives.

■ L8-38: Carrier Frequency Reduction Selection

No. (Hex.)	Name	Description	Default (Range)
L8-38 (04EF)	Carrier Frequency Reduction Selection	V/f OLV Selects the operation of the carrier frequency reduction function.	Determined by o2-04 (0 - 2)

The drive can reduce the carrier frequency when the output current exceeds a certain level. This temporarily increases the overload capability (oL2 detection) and the drive can run through transient load peaks without tripping.

0 : Disabled

The carrier frequency is not automatically reduced.

1 : Enabled for Output Frequencies below 6 Hz

The carrier frequency is reduced at speeds below 6 Hz when output current exceeds 100% of drive rated current. The drive returns to its normal carrier frequency when output current falls below 88% of drive rated current or when the output frequency exceeds 7 Hz.

2 : Enabled for Entire Frequency Range

The carrier frequency is reduced at the following speeds:

- At or below 6 Hz when output current exceeds 100% of drive rated current.
- Above 6 Hz when output current exceeds 112% of drive rated current.

The drive uses the delay time set in parameter L8-40 and a hysteresis of 12% when switching the carrier frequency back to its normal value.

■ L8-40: Carrier Freq Reduction Off-Delay

No. (Hex.)	Name	Description	Default (Range)
L8-40 (04F1)	Carrier Freq Reduction Off-Delay	V/f OLV Sets the length of time until the automatically reduced carrier frequency returns to the condition before the reduction.	0.50 s (0.00 - 2.00 s)

Set *L8-40 ≠ 0.00* to enable the carrier frequency reduction function during start-up. When operation starts, the drive automatically decreases the carrier frequency. When the time set in *L8-40* is expired, the carrier frequency returns to the value set in *C6-02 [Carrier Frequency Selection]*.

■ L8-41: High Current Alarm Selection

No. (Hex.)	Name	Description	Default (Range)
L8-41 (04F2)	High Current Alarm Selection	V/f OLV Sets the function to cause an *HCA [Current Alarm]* when the output current is more than 150% of the drive rated current.	0 (0, 1)

0 : No

The drive will not detect *HCA*.

1 : Yes

If the output current is more than 150% of the drive rated current, the drive will detect *HCA*.

The MFDO terminal set for an *Alarm [H2-01 to H2-03 = 10]* activates.

■ L8-42: Input Phase Loss Detection Time 2

No. (Hex.)	Name	Description	Default (Range)
L8-42 (04F3)	Input Phase Loss Detection Time 2	V/f OLV Sets the amount of time that an Input Phase Loss condition has to be present before a fault is triggered.	2 s

■ L8-43: Output Phase Loss Detection Time

No. (Hex.)	Name	Description	Default (Range)
L8-43 (04F4)	High Current Alarm Selection	V/f OLV Sets the amount of time that an Output Phase Loss condition has to be present before a fault is triggered.	0.5 s (0.0 - 2.0 s)

■ L8-62: Input Phase Loss Detection

No. (Hex.)	Name	Description	Default (Range)
L8-62 (04D8)	Input Phase Loss Detection	V/f OLV Sets stopping method when an input phase loss fault (PF) occurs.	0 (0 - 3)

0 : Ramp to stop

Decelerate to stop using the active deceleration time.

1 : Coast to stop

2 : Stop (Use C1-09)

Decelerate to stop using the deceleration time in C1-09.

3 : Alarm Only

Drive continues operation.

■ L8-65: PF Min Current

No. (Hex.)	Name	Description	Default (Range)
L8-65 (04DB)	PF Detection Minimum Current Level	V/f OLV Percentage of rated current. Used for standard method and dv/dt method. The parameter is only active during run, and is not evaluated for dv/dt detection method at start.	10% (0 - 100%)

■ L8-70: PF dv/dt Level

No. (Hex.)	Name	Description	Default (Range)
L8-70 (04E0)	PF dv/dt Level	V/f OLV	2100 V/sec (0 - 10,000 V/sec)

■ L8-79: dv/dt Tune Factor

No. (Hex.)	Name	Description	Default (Range)
L8-79 (04E9)	dv/dt Tune Factor	V/f OLV	130% (99 - 150%)

■ L8-88: Safe Disable Operation Mode

No. (Hex.)	Name	Description	Default (Range)
L8-88 (04F2)	Safe Disable Operation Mode	V/f OLV Determines the operation performed by the drive when the Safe Disable input is activated.	0 (0, 1)

When the Safe Disabled Input is triggered, the operator displays and alarm, and the corresponding output terminal will react as follows:

L8-88 Setting	Safe Disable Operation Selection	Alarm Display during Safety Disable	Alarm Output *[H2-xx = 10]*	Drive Ready *[H2-xx = 6]*
0 [Mode 0]	Hbb	ALM flashes	ON	OFF
1 [Mode 1]	Hbb	ALM flashes	OFF	ON

0 : Mode 0

1 : Mode 1

L7 compatible

11.8 n: Special Adjustment

n parameters set these functions:

- Function to prevent hunting
- High-slip braking
- Motor line-to-line resistance online tuning
- Fine-tune the parameters that adjust motor control

◆ n2: Speed Feedback Detection Control (AFR) Tuning

The speed feedback detection reduction function (or AFR: Automatic Frequency Regulator) helps the speed become stable when you suddenly apply or remove a load.

Note:

Before you change *n2-xx parameters*, do one of these procedures:
- Set the motor parameters and V/f pattern correctly.
- Do Rotational Auto-Tuning.

■ n2-01: Speed Feedback Detection Control (AFR) Gain

No. (Hex.)	Name	Description	Default (Range)
n2-01 (0584)	Speed Feedback Detection Control (AFR) Gain	V/f OLV Sets the gain of the AFR function as a magnification value. Usually it is not necessary to change this setting.	1.50 (0.00 - 10.00)

Adjust this parameter in these conditions:

- If hunting or oscillation occurs with light loads, increase the setting value in 0.05-unit increments and examine the response.
- When torque is not sufficient with heavy loads or to make the torque or speed response better, decrease the setting value in 0.05-unit increments and examine the response.

■ n2-02: Speed Feedback Detection Control (AFR) Time Constant 1

No. (Hex.)	Name	Description	Default (Range)
n2-02 (0585)	Speed Feedback Detection Control (AFR) Time Constant 1	V/f OLV Sets the time constant that sets the rate of change for the AFR function. Usually it is not necessary to change this setting.	50 ms (0 - 2000 ms)

Adjust this parameter in these conditions:

- If there is hunting or oscillation with a light load, increase the setting value in 50 ms increments and examine the response. If the load inertia is large, increase the setting value in 50 ms increments and examine the response.
- If torque is not sufficient with a heavy load or if you must increase torque or speed responsiveness, decrease the setting value in 10 ms increments and examine the response.

Note:
- Set *n2-02* ≤ *n2-03*. If *n2-02* > *n2-03*, the drive will detect *oPE08 [Parameter Selection Error]*.
- When you change the value in *n2-02*, also change the value in *C4-02 [Torque Compensation Delay Time]* by the same ratio.

■ n2-03: Speed Feedback Detection Control (AFR) Time Constant 2

No. (Hex.)	Name	Description	Default (Range)
n2-03 (0586)	Speed Feedback Detection Control (AFR) Time Constant 2	V/f OLV Sets the time constant that sets the speed difference of the AFR function. Use this parameter for speed searches or regeneration. Usually it is not necessary to change this setting.	750 ms (0 - 2000 ms)

Adjust this parameter in these conditions:

- If the drive detects *ov [Overvoltage]* when acceleration stops under high-inertia loads, increase the setting value in 50 ms increments.
  If the drive detects *ov* when the load changes suddenly, increase the setting value in 50 ms increments.
- To increase the responsiveness of torque and speed, decrease the setting value in 10 ms increments and examine the response.

Note:
• Set *n2-02* ≤ *n2-03*. If *n2-02* > *n2-03*, the drive will detect *oPE08 [Parameter Selection Error]*.

◆ n6: Online Tuning

n6 parameters are used to set the online tuning function for motor line-to-line resistance.

The Online Tuning for motor line-to-line resistance is used to prevent degradation of speed control accuracy due to motor temperature fluctuation and motor stalls due to insufficient torque.

■ n6-01: Online Tuning Selection

No. (Hex.)	Name	Description	Default (Range)
n6-01 (0570)	Online Tuning Selection	V/f OLV Sets the type of motor data that Online Tuning uses for OLV control.	2 (0 - 2)

0 : Disabled

1 : Line-to-Line Resistance Tuning

The drive adjusts the motor line-to-line resistance during run. This procedure is applicable for speed values 6 Hz and less. It also adjusts the motor resistance value to increase the overload capacity in the low speed range.

2 : Voltage Correction Tuning

The drive adjusts the output voltage during run to increase overload tolerance and minimize the effects of high temperatures on speed precision.

Note:
Setting 2 is enabled only when *b8-01 = 0 [Energy Saving Control Selection = Disabled]*.

11.9 o: Keypad-Related Settings

o parameters set keypad functions.

Note:
You cannot use the optional LED keypad to set these parameters.

Table 11.11 Parameters You Cannot Set with the LED Keypad

No.	Name
o1-05	LCD Contrast Adjustment
o1-24 to o1-35	Custom Monitor 1 to 12
o1-36	LCD Backlight Brightness
o1-37	LCD Backlight ON/OFF Selection
o1-38	LCD Backlight Off-Delay
o1-39	Show Initial Setup Screen
o1-40	Home Screen Display Selection
o1-41 to o1-46	1st to 3rd Monitor Area Selections/Settings
o1-47 to o1-51	Trend Plot 1 or 2 Scale Settings
o1-55 to o1-56	Analog Gauge Area Selection/Setting
o2-27	bCE Detection Selection

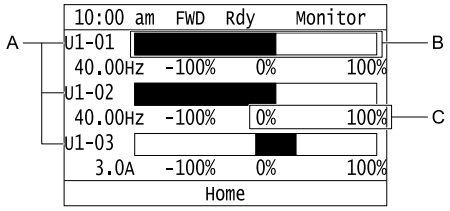
No.	Name
o3-04	Select Backup/Restore Location
o3-05	Select Items to Backup/Restore
o3-06	Auto Parameter Backup Selection
o3-07	Auto Parameter Backup Interval
o4-22	Time Format
o4-23	Date Format
o4-24	bAT Detection Selection
o5-01	Log Start/Stop Selection
o5-02	Log Sampling Interval
o5-03 to o5-12	Log Monitor Data 1 to 10

◆ o1: Keypad Display Selection

o1 parameters select the parameters shown on the initial keypad screen and to configure the parameter setting units and display units. These parameters also adjust the backlight and contrast of the LCD display.

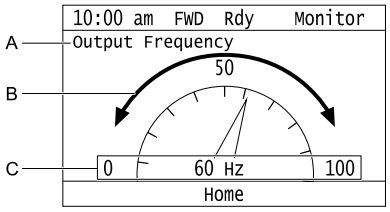
■ Home Screen Display Format

o1-40 [Home Screen Display Selection] changes the display of the monitor shown on the Home screen. You can show numerical values or one of these three displays on the Home screen monitor:



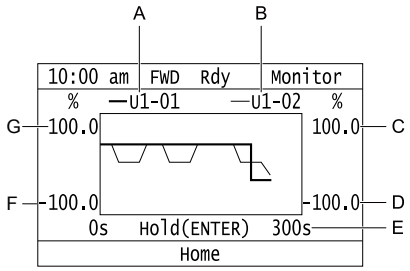
A - Set *Ux-xx [Monitors]* to *o1-24*, *o1-25*, and *o1-26*.
B - Set display regions to *o1-41*, *o1-43*, and *o1-45*.
C - Set display ranges to *o1-42*, *o1-44*, and *o1-46*.

Figure 11.34 Bar Graph Display



A - Set *Ux-xx [Monitors]* to *o1-24*.
B - Set display regions to *o1-56*.
C - Set display ranges to *o1-55*.

Figure 11.35 Analog Gauge Display



A - Set *Ux-xx [Monitors]* (Monitor 1) to *o1-24*.
B - Set *Ux-xx [Monitors]* (Monitor 2) to *o1-25*.
C - Set the maximum value of Monitor 2 to *o1-50*.
D - Set the minimum value of Monitor 2 to *o1-49*.
E - Set the time scale to *o1-51*.
F - Set the minimum value of Monitor 1 to *o1-47*.
G - Set the maximum value of Monitor 1 to *o1-48*.

Figure 11.36 Trend Plot Display

■ o1-01: User Monitor Selection

No. (Hex.)	Name	Description	Default (Range)
o1-01 (0500) RUN	User Monitor Selection	V/f OLV Sets the *U monitor* for the Drive Mode. This parameter is only available with an LED keypad.	106 (104 - 855)

When the drive is in Drive Mode, push ^ on the keypad to cycle through this data: frequency reference → rotational direction → output frequency → output current → *o1-01* selection.

Set the *x-xx* part of *Ux-xx* that is shown in the fifth position in Drive Mode. For example, to show *U1-05 [Motor Speed]*, set *o1-01 = 105*.

Note:

The monitors that you can select are different for different control methods.

■ o1-02: Monitor Selection at Power-up

No. (Hex.)	Name	Description	Default (Range)
o1-02 (0501) RUN	Monitor Selection at Power-up	V/f OLV Sets the monitor item that the keypad screen shows after you energize the drive. Refer to "U: Monitors" for information about the monitor items that the keypad screen can show. This parameter is only available with an LED keypad.	1 (1 - 5)

1 : Frequency Reference (U1-01)
2 : Direction
3 : Output Frequency (U1-02)
4 : Output Current (U1-03)
5 : User Monitor (o1-01)

Shows the monitor item selected in *o1-01 [User Monitor Selection]*.

■ o1-03: Keypad Display Unit Selection

No. (Hex.)	Name	Description	Default (Range)
o1-03 (0502)	Keypad Display Unit Selection	V/f OLV Sets the display units for the frequency reference and output frequency.	0 (0 - 3)

Note:

- Select the setting unit of these parameters:
  - d1-01 to d1-08 [Reference 1 to 8]
  - U1-01 [Frequency Reference]
  - U1-02 [Output Frequency]
  - U1-05 [Motor Speed]
  - U1-16 [SFS Output Frequency]
  - U4-14 [PeakHold Output Freq]
- For monitor 2, the setting value is always *0 [Hz Unit]*.

0 : 0.01 Hz units

1 : 0.01% units

The maximum output frequency is 100%.

2 : Revolutions Per Minute (RPM)

The drive uses the maximum output frequency and number of motor poles calculate this value automatically.

Note:

When you set o1-03 = 2 [r/min], make sure that you set the number of motor poles in *E2-04 [Motor Pole Count]*.

3 : User Units

Uses *o1-10* and *o1-11* to set the unit of measure. The value of parameter *o1-10* is the value when you remove the decimal point from the maximum output frequency. Parameter *o1-11* is to the number of digits after the decimal point in the maximum output frequency.

To display a maximum output frequency of 100.00, set parameters to these values:

- o1-10 = 10000
- o1-11 = 2 [User Units Decimal Position = 2 Dec (XXX.XX)]

■ o1-10: User Units Maximum Value

No. (Hex.)	Name	Description	Default (Range)
o1-10 (0520)	User Units Maximum Value	V/f OLV Sets the value that the drive shows as the maximum output frequency.	Determined by o1-03 (1 - 60000)

To display a maximum output frequency of 100.00, set parameters to these values:

- *o1-10 = 10000*
- *o1-11 = 2 [User Units Decimal Position = 2 Dec (XXX.XX)]*

Note:

Set *o1-03 = 3 [Frequency Display Unit Selection = User Units (o1-10 & o1-11)]* before you set *o1-10* and *o1-11*.

■ o1-11: User Units Decimal Position

No. (Hex.)	Name	Description	Default (Range)
o1-11 (0521)	User Units Decimal Position	V/f OLV Sets the number of decimal places for frequency reference and monitor values.	Determined by o1-03 (0 - 3)

0 : No Decimal Places (XXXXX)

1 : One Decimal Places (XXXX.X)

2 : Two Decimal Places (XXX.XX)

3 : Three Decimal Places (XX.XXX)

Note:

Set *o1-03 = 3 [Frequency Display Unit Selection = User Units (o1-10 & o1-11)]* before you set *o1-10 [User Units Maximum Value]* and *o1-11*.

◆ o2: Keypad Operation

■ o2-01: LO/RE Key Function Selection

No. (Hex.)	Name	Description	Default (Range)
o2-01 (0505)	LO/RE Key Function Selection	V/f OLV Sets the function that lets you use LO/RE to switch between LOCAL and REMOTE Modes.	0 (0, 1)

0 : Disabled

You cannot use LO/RE to switch between LOCAL and REMOTE Modes.

1 : Enabled

You can use LO/RE to switch between LOCAL and REMOTE Modes when the drive is stopped. When LOCAL Mode is selected, LO/RE on the keypad will come on.

WARNING! *Sudden Movement Hazard. If you change the control source when b1-07 = 1 [LOCAL/REMOTE Run Selection = Accept Existing RUN Command], the drive can start suddenly. Before you change the control source, remove all personnel from the area around the drive, motor, and load. Sudden starts can cause serious injury or death.*

WARNING! *Sudden Movement Hazard. Fully examine all mechanical and electrical connections before you change o2-01 [LO/RE Key Function Selection] or b1-07 [LOCAL/REMOTE Run Selection]. Sudden starts can cause serious injury or death. If b1-07 = 1 [Accept Existing RUN Command] and there is an active Run command when you switch from LOCAL to REMOTE Mode, the drive can start suddenly.*

Table 11.12 Function Settings with o2-01 and b1-07

LO/RE Function Selection	LOCAL/REMOTE Run Selection	Switching from LOCAL Mode to REMOTE Mode	Switching from REMOTE Mode to LOCAL Mode
o2-01 = 0 [Disabled]	b1-07 = 0 [Disregard Existing RUN Command]	The drive will not switch modes.	The drive will not switch modes.
	b1-07 = 1 [Accept Existing RUN Command]		
o2-01 = 1 [Enabled]	b1-07 = 0 [Disregard Existing RUN Command]	The drive will not start operating although the Run command is active. When you set Run command to active again, the drive will start to run.	The drive cannot operate because the Run command is not enabled.
	b1-07 = 1 [Accept Existing RUN Command]	When the Run command is active, the drive will start to run immediately when the mode switches from LOCAL to REMOTE.	The drive cannot operate because the Run command is not enabled.

■ o2-02: STOP Key Function Selection

No. (Hex.)	Name	Description	Default (Setting Range)
o2-02 (0506)	STOP Key Function Selection	V/f OLV Sets the function to use STOP on the keypad to stop the drive when the Run command source for the drive is REMOTE (external) and not assigned to the keypad.	0 (0, 1)

0 : Disabled

1 : Enabled

STOP stays enabled when the Run command source has not been assigned to the keypad.

To start the drive again after you push STOP to stop operation, turn the external Run command OFF and ON again.

■ o2-03: User Parameter Default Value

No. (Hex.)	Name	Description	Default (Range)
o2-03 (0507)	User Parameter Default Value	V/f OLV Sets the function to keep the settings of changed parameters as user parameter defaults to use during initialization.	0 (0 - 2)

When you set *o2-03 = 1 [Set defaults]*, the drive saves changed parameter settings as user parameter setting values in a part of the memory that is isolated from drive parameters.

When you set *A1-03 = 1110 [Initialize Parameters = User Initialization]* to initialize the drive, the drive resets the internal parameter setting values to those user parameter setting values.

0 : No change

1 : Set defaults

Saves changed parameter setting values as user default settings.

Set *o2-03 = 1* then push ↵ to save the user parameter setting values. After the drive saves the setting value, *o2-03* automatically resets to *0*.

2 : Clear all

Deletes all of the saved user parameter setting values.

Set *o2-03 = 2* then push ↵ to clear the user parameter setting values. The drive will automatically reset *o2-03* to *0*. If you delete the user parameter setting values, you cannot set *A1-03 = 1110* to initialize parameters.

■ o2-04: Drive Model (KVA) Selection

No. (Hex.)	Name	Description	Default (Setting Range)
o2-04 (0508)	Drive Model (KVA) Selection	V/f OLV Sets the Drive Model code. Set this parameter after replacing the control board.	Determined by the drive (-)

NOTICE: *Set o2-04 [Drive Model Selection] correctly. If you set this parameter incorrectly, it will decrease drive performance, cause the protection function to operate incorrectly, and cause damage to the drive.*

Note:

When the setting value of *o2-04* changes, related parameter setting values also change.

Table 11.13 lists the relation between *o2-04* setting values and drive models.

Table 11.13 Drive Models and *o2-04* Settings

Drive Model	o2-04 Setting	Drive Model	o2-04 Setting
4009	6E	4031	9D
4015	6F	4039	9E
4018	9A	4045	9F
4024	9C		

■ o2-05: Home Mode Freq Ref Entry Mode

No. (Hex.)	Name	Description	Default (Setting Range)
o2-05 (0509)	Home Mode Freq Ref Entry Mode	V/f OLV Sets the function that makes it necessary to push ENTER to use the keypad to change the frequency reference value while in Drive Mode.	0 (0, 1)

0 : ENTER Key Required

You must push ENTER to use the keypad to change the frequency reference value.

1 : Immediate / MOP-style

The frequency reference changes when you enter it with the keypad. This then changes the output frequency. It is not necessary to push ENTER. The drive keeps the frequency reference for 5 seconds after you use ▲ and ▼ on the keypad to change the frequency reference value.

■ o2-06: Keypad Disconnect Detection

No. (Hex.)	Name	Description	Default (Range)
o2-06 (050A)	Keypad Disconnect Detection	V/f OLV Sets the function that stops the drive if you disconnect the keypad connection cable from the drive or if you damage the cable while the keypad is the Run command source.	0 (0, 1)

This parameter continues to operate if the keypad installed to the drive becomes disconnected.

This parameter is enabled in these conditions:

- When *b1-02 = 0 [Run Command Selection 1 = Keypad]* or *b1-16 = 0 [Run Command Selection 2 = Keypad]*
- In LOCAL Mode

0 : Disabled

The drive continues operation when it detects a keypad disconnection.

1 : Enabled

The drive stops operation, detects *oPr [Keypad Connection Fault]*, and the motor coasts to stop when the drive detects a keypad disconnection.

■ o2-07: Keypad RUN Direction @ Power-up

No. (Hex.)	Name	Description	Default (Range)
o2-07 (0527)	Keypad RUN Direction @ Power-up	V/f OLV Sets the direction of motor rotation when the drive is energized and the keypad is the Run command source.	0 (0, 1)

This parameter is enabled in these conditions:

- When *b1-02 = 0 [Run Command Selection 1 = Keypad]* or *b1-16 = 0 [Run Command Selection 2 = Keypad]*
- In LOCAL Mode

0 : Forward

1 : Reverse

◆ o3: Copy Function

o3 parameters set the operation of the parameter backup function.

■ o3-01: Copy Keypad Function Selection

No. (Hex.)	Name	Description	Default (Range)
o3-01 (0515)	Copy Keypad Function Selection	V/f OLV Sets the function that saves and copies drive parameters to a different drive with the keypad.	0 (0 - 3)

0 : Copy Select

1 : Backup (drive → keypad)

The parameter setting values are read from the drive and saved in the keypad.

2 : Restore (keypad → drive)

Copies the parameter setting values saved in the keypad to a different drive.

3 : Verify (check for mismatch)

Makes sure that the parameter setting values in the drive agree with the parameters saved in the keypad.

■ o3-02: Copy Allowed Selection

No. (Hex.)	Name	Description	Default (Range)
o3-02 (0516)	Copy Allowed Selection	V/f OLV Sets the copy function when *o3-01 = 1 [Copy Keypad Function Selection = Backup (drive → keypad)]*.	0 (0, 1)

Note:

When you select [Parameter Backup] on the keypad menu screen to do the backup function, the drive automatically sets *o3-02 = 1*.

0 : Disabled

1 : Enabled

◆ o4: Maintenance Mon Settings

o4 parameters set the expected service life to help you know when to replace parts. The drive will show an alarm to tell you when the replacement part interval is near.

■ o4-01: Elapsed Operating Time Setting

No. (Hex.)	Name	Description	Default (Range)
o4-01 (050B)	Elapsed Operating Time Setting	V/f OLV Sets the initial value of the cumulative drive operation time in 10-hour units.	0 h (0 - 9999 h)

When you select *o4-01* on the keypad, it will show the current value of *U4-01* in units of 10 hours (h). When you change the setting of *o4-01* through the monitor, the *U4-01* count starts again as specified by the setting of *o4-01*.

Note:

Set this parameter in 10-hour (h) units. When *o4-01 = 30*, *U4-01 [Cumulative Ope Time] = 300 h*.

■ o4-02: Elapsed Operating Time Selection

No. (Hex.)	Name	Description	Default (Range)
o4-02 (050C)	Elapsed Operating Time Selection	V/f OLV Sets the condition that counts the cumulative operation time.	1 (0, 1)

0 : U4-01 Shows Total Power-up Time

Counts the time from when the drive is energized to when it is de-energized.

1 : U4-01 Shows Total RUN Time

Counts the time that the drive outputs voltage.

■ o4-03: Fan Operation Time Setting

No. (Hex.)	Name	Description	Default (Range)
o4-03 (050E)	Fan Operation Time Setting	V/f OLV Sets the value from which to start the cumulative drive cooling fan operation time in 10-hour units.	0 h (0 - 9999 h)

Use monitor *U4-03 [Cooling Fan Ope Time]* to view the total operation time of the cooling fan. When you replace a cooling fan, set *o4-03 = 0* to reset *U4-03*. Select *o4-03* on the keypad to show the current value of *U4-03* in 10-hour (h) units. If you use the monitor to change *o4-03*, the recount of *U4-03* starts with the *o4-03* setting.

Note:

The drive sets *o4-03* in 10-hour (h) units. When *o4-03 = 30*, *U4-03 [Cooling Fan Ope Time]* will show "300 h".

■ o4-05: Capacitor Maintenance Setting

No. (Hex.)	Name	Description	Default (Range)
o4-05 (051D)	Capacitor Maintenance Setting	V/f OLV Sets the *U4-05 [CapacitorMaintenance]* monitor value.	0% (0 - 150%)

When you replace a drive, set *o4-05 = 0* to reset *U4-05*. When the *o4-05* setting changes, the count of *U4-05* starts again as specified by the setting of *o4-05*. After you complete the configuration, *o4-05* automatically resets to *0*.

Note:

The maintenance period changes for different operating environments.

■ o4-07: Softcharge Relay Maintenance Set

No. (Hex.)	Name	Description	Default (Range)
o4-07 (0523)	Softcharge Relay Maintenance Set	V/f OLV Sets the *U4-06 [PreChargeRelayMainte]* monitor value.	0% (0 - 150%)

When you replace a drive, set *o4-07 = 0* to reset *U4-06*. When the *o4-07* setting changes, the count of *U4-06* starts again as specified by the setting of *o4-07*. After you complete the configuration, *o4-07* automatically resets to *0*.

Note:

The maintenance period changes for different operating environments.

■ o4-09: IGBT Maintenance Setting

No. (Hex.)	Name	Description	Default (Range)
o4-09 (0525)	IGBT Maintenance Setting	V/f OLV Sets the *U4-07 [IGBT Maintenance]* monitor value.	0% (0 - 150%)

When you replace a drive, set *o4-09 = 0* to reset *U4-07*. When the *o4-09* setting changes, the count of *U4-07* starts again as specified by the setting of *o4-09*. After you complete the configuration, *o4-09* automatically resets to *0*.

Note:

The maintenance period changes for different operating environments.

■ o4-11: Fault Trace/History Init (U2/U3)

No. (Hex.)	Name	Description	Default (Range)
o4-11 (0510)	Fault Trace/History Init (U2/U3)	V/f OLV Resets the records of Monitors *U2-xx [Fault Trace]* and *U3-xx [Fault History]*.	0 (0, 1)

Note:

When you initialize the drive with *A1-03 [Initialize Parameters]*, the drive will not reset the records for *U2-xx* and *U3-xx*.

0 : Disabled

Keeps the records of Monitors *U2-xx* and *U3-xx*.

1 : Enabled

Resets the records for Monitors *U2-xx* and *U3-xx*. After the reset, the drive automatically resets *o4-11* to *0*.

■ o4-12: kWh Monitor Initialization

No. (Hex.)	Name	Description	Default (Range)
o4-12 (0512)	kWh Monitor Initialization	V/f OLV Resets the monitor values for *U4-10 [kWh, Lower 4 Digits]* and *U4-11 [kWh, Upper 5 Digits]*.	0 (0, 1)

Note:

When you initialize the drive with *A1-03 [Initialize Parameters]*, the drive will not reset *U4-10* and *U4-11*.

0 : No Reset

Keeps the monitor values for *U4-10* and *U4-11*.

1 : Reset

Resets the values of *U4-10* and *U4-11*. After the reset, the drive automatically resets *o4-12* to *0*.

■ o4-13: RUN Command Counter @ Initialize

No. (Hex.)	Name	Description	Default (Range)
o4-13 (0528)	RUN Command Counter @ Initialize	V/f OLV Resets the monitor values for *U4-02 [Num of Run Commands]*, *U4-24 [Number of Runs (Low)]*, and *U4-25 [Number of Runs (High)]*.	0 (0, 1)

0 : No Reset

Keeps the monitor values for *U4-02*, *U4-24, and U4-25*.

1 : Reset

Resets the values of *U4-02*, *U4-24*, and *U4-25*. After the reset, the drive automatically resets *o4-13* to *0*.

■ o4-22: Time Format

No. (Hex.)	Name	Description	Default (Range)
o4-22 (154F) RUN	Time Format	V/f OLV Sets the time display format. This parameter is only available when using an LCD keypad.	0 (0 - 2)

Sets the display of the time shown in the upper-left of the LCD keypad screen.

0 : 24 Hour Clock

1 : 12 Hour Clock

2 : 12 Hour JP Clock

■ o4-23: Date Format

No. (Hex.)	Name	Description	Default (Range)
o4-23 (1550) RUN	Date Format	V/f OLV Sets the date display format. This parameter is only available when using an LCD keypad.	0 (0 - 2)

Sets the date format that the drive uses for the fault history and other records.

0 : YYYY/MM/DD

1 : DD/MM/YYYY

2 : MM/DD/YYYY

■ o4-24: bAT Detection Selection

No. (Hex.)	Name	Description	Default (Range)
o4-24 (310F) RUN	bAT Detection Selection	V/f OLV Sets the operation when the drive detects *bAT [Keypad Battery Low Voltage]* and *TiM [Keypad Time Not Set]*. This parameter is only available when you use an LCD keypad.	0 (0 - 2)

0 : Disable

The drive will not detect *bAT* or *TiM*.

1 : Enable (Alarm Detected)

The keypad shows *bAT* or *TiM* and the drive continues operation. The output terminal set to *Alarm [H2-01 to H2-03 = 10]* activates.

2 : Enable (Fault Detected)

The output turns off and the motor coasts to stop. The output terminal set for *Fault [H2-01 to H2-03 = E]* activates.

11.10 S: Elevator Parameters

S parameters describe various functions and faults needed to operate an elevator application:

- Braking Sequence
- Slip Compensation
- Optimal Adjustments at Start and Stop
- Rescue Operation
- Elevator-related Faults

◆ S1: Brake Sequence

The drive supports braking sequences using an analog input terminal to control torque compensation at start (H3-xx = 14), and braking sequences that do not require an analog input to set the torque compensation level.

■ S1-01: Zero Speed Level at Stop

No. (Hex.)	Name	Description	Default (Range)
S1-01 (0680)	Zero Speed Level at Stop	V/f OLV Determines the speed to begin applying DC Injection when the drive is ramping to stop (b1-03 = 0). Set as a percentage of the maximum output frequency (E1-04).	Determined by A1-02 (0.000 - 9.999%)

Parameter S1-01 sets the starting speed for DC Injection Braking at stop. Once the output speed falls below the setting of S1-01, the amount of DC Injection Braking current set in S1-03 is injected into the motor for the time set in parameter S1-05.

■ S1-02: DC Injection Current at Start

No. (Hex.)	Name	Description	Default (Range)
S1-02 (0681)	DC Injection Current at Start	V/f OLV Determines the amount of current to use for DC Injection at start. Set as a percentage of the drive rated current.	50% (0 - 75%)

■ S1-03: DC Injection Current at Stop

No. (Hex.)	Name	Description	Default (Range)
S1-03 (0682)	DC Injection Current at Stop	V/f OLV Determines the amount of current to use for DC Injection at stop. Set as a percentage of the drive rated current. When using OLV Control, the DC injection current is determined by multiplying S1-03 by S3-25 or S3-26.	50% (0 - 75%)

■ S1-04: DC Injection Time at Start

No. (Hex.)	Name	Description	Default (Range)
S1-04 (0683)	DC Injection Time at Start	V/f OLV Determines how long the drive should perform DC Injection at start. During this time, the drive allows motor flux to develop, which is essential for applying torque quickly once the brake is released. A setting of 0.00 disables S1-04.	0.40 s (0.00 - 10.00 s)

■ S1-05: DC Injection Time at Stop

No. (Hex.)	Name	Description	Default (Range)
S1-05 (0684)	DC Injection Time at Stop	V/f OLV Determines how long the drive should perform DC Injection at stop. A setting of 0.00 disables S1-05.	0.60 s (0.00 - 10.00 s)

■ S1-06: Brake Release Delay Time

No. (Hex.)	Name	Description	Default (Range)
S1-06 (0685)	Brake Release Delay Time	V/f OLV Determines the time that must pass after an Up/Down command is entered before the output terminal set for "Brake control" (H2-xx = 50) is triggered. Adjusting this delay time can help when there is not enough time to develop the appropriate amount of motor flux. Be sure to also increase the time S1-04 when setting S1-06 to relatively long delay time.	0.20 s (0.00 - 10.00 s)

■ S1-07: Brake Apply Delay Time

No. (Hex.)	Name	Description	Default (Range)
S1-07 (0686)	Brake Apply Delay Time	V/f OLV Determines the time that must pass after zero speed is reached before the output terminal set for "Brake control" (H2-xx = 50) is released.	0.10 s (0.00 - 10.00 s)

■ S1-10: Run Command Delay Time

No. (Hex.)	Name	Description	Default (Range)
S1-10 (0687)	Run Command Delay Time	V/f OLV Determines the time that must pass after zero speed is reached before the output terminal set for "Brake control" (H2-xx = 50) is released.	0.10 s (0.00 - 10.00 s)

■ S1-11: Output Reactor Open Delay Time

No. (Hex.)	Name	Description	Default (Range)
S1-11 (0688)	Output Reactor Open Delay Time	V/f OLV Determines the time that must pass for an output terminal set for “Output contactor control” (H2-xx = 51) to be released after the drive has stopped and drive output has been shut off.	0.10 s (0.00 - 10.00 s)

◆ S2: Slip Compensation for Elevators

The slip compensation function automatically adjusts the speed reference for leveling operation depending on the load measured at constant speed. S2 parameters tune the slip compensation function to improve the landing accuracy. Slip Compensation requires that the drive be set for V/f Control or Open Loop Vector Control.

■ S2-01: Motor Rated Speed

No. (Hex.)	Name	Description	Default (Range)
S2-01 (068F)	Motor Rated Speed	V/f OLV Sets the rated speed of the motor.	1380 rpm (300 - 1800 rpm)

■ S2-02: Slip Compensation Gain in Motoring Mode

No. (Hex.)	Name	Description	Default (Range)
S2-02 (0690)	Slip Compensation Gain in Motoring Mode	V/f OLV Slip compensation for leveling speed can be set separately for motoring and regenerative states to help improve the accuracy of leveling.	0.7 (0.0 - 5.0 rpm)

■ S2-03: Slip Compensation Gain in Regenerative Mode

No. (Hex.)	Name	Description	Default (Range)
S2-03 (0691)	Slip Compensation Gain in Regenerative Mode	V/f OLV Slip compensation for leveling speed can be set separately for motoring and regenerative states to help improve the accuracy of leveling.	1.0 (0.0 - 5.0)

■ S2-05: Slip Compensation Torque Detection Delay Time

No. (Hex.)	Name	Description	Default (Range)
S2-05 (0693)	Slip Compensation Torque Detection Delay Time	V/f OLV Sets a delay time before detecting torque for slip compensation.	1000 ms (0 - 10000 ms)

■ S2-06: Slip Compensation Torque Detection Filter Time Constant

No. (Hex.)	Name	Description	Default (Range)
S2-06 (0694)	Slip Compensation Torque Detection Filter Time Constant	V/f OLV Sets the filter time constant applied to the torque signal used for the slip compensation value calculation.	500 ms (0 - 2000 ms)

◆ S4: Rescue Operation

Rescue Operation switches to a backup battery or some other UPS during a power outage.

■ S4-01: Light Load Direction Search Selection

No. (Hex.)	Name	Description	Default (Range)
S4-01 (06A6)	Light Load Direction Search Selection	V/f OLV Enables and disables the Light Load Direction Search.	0 (0, 1)

0 : Disabled
1 : Advanced

■ S4-20: Light Load Search Override

No. (Hex.)	Name	Description	Default (Range)
S4-20 (06B1)	Light Load Search Override	V/f OLV Defines the source that determines the light load search direction.	0 (0 - 1)

0 : Controller Direction
1 : Inverter Direction

◆ S6: Faults for Elevator Applications

■ S6-01: Motor Contactor Response Error (SE1) Detection/Reset Selection

No. (Hex.)	Name	Description	Default (Range)
S6-01 (06B3)	Motor Contactor Response Error (SE1) Detection/Reset Selection	V/f OLV Determines when the drive should detect a motor contactor response error (SE1). SE1 is triggered if there is no response from the motor contactor within the time set in S6-10 after the contactor control output has been set.	0 (0 - 2)

0 : Detect during stop, SE1 must be manually reset
1 : Detect during stop, SE1 can be automatically reset
2 : No SE1 detection

■ S6-02: Starting Current Error (SE2) Detection Delay Time

No. (Hex.)	Name	Description	Default (Range)
S6-02 (06B4)	Starting Current Error (SE2) Detection Delay Time	V/f OLV Sets a delay time for starting current error (SE2). SE2 is detected when the drive output current is below 25% after the Up/Down command has been entered and the brake release time and the time set to S6-02 have both passed. The brake control command will not be issued (brake stays applied).	200 ms (0.00 - [S1-04 - S1-06] ms)

■ S6-04: Output Current Error (SE3) Detection Delay Time

No. (Hex.)	Name	Description	Default (Range)
S6-04 (06B6)	Output Current Error (SE3) Detection Delay Time	V/f OLV Sets a delay time for detecting an output current fault (SE3). SE3 is detected when the drive output current drops below 25% after the brake has released.	200 ms (0 - 5000 ms)

■ S6-05: Brake Response Error (SE4) Detection Time

No. (Hex.)	Name	Description	Default (Range)
S6-05 (06B7)	Brake Response Error (SE4) Detection Time	V/f OLV Sets a delay time for detecting a brake response error (SE4). SE4 is detected when an output terminal set for “Brake release” (H2-xx = 50) and an input terminal set for “Brake feedback” (H1-xx = 79) do match for the time set to S6- 05.	500 ms (0 - 10000 ms)

■ S6-15: Speed Reference Loss Detection

No. (Hex.)	Name	Description	Default (Range)
S6-15 (06BB)	Speed Reference Loss Detection	V/f OLV Enables speed reference loss detection when D1-18 = 1 and H1-xx ≠ 53.	0 (0, 1)

0 : Disabled

1 : Enabled

■ S6-16: Restart after Baseblock Selection

No. (Hex.)	Name	Description	Default (Range)
S6-16 (06BC)	Restart after Baseblock Selection	V/f OLV Allows the drive to restart the motor after returning to normal operation from Baseblock state (H1-xx = 8/9) or from Safe Torque-Off state (Safe Disable inputs H1 and H2 enabled) while the Up/Down command is still active.	0 (0, 1)

11.11 T: Auto-Tuning

Numbers identifying the *T parameters* are displayed when an LED keypad is used. The names of the parameters are displayed on the LCD screen of the LCD keypad. Set the following.

- Induction Motor Auto-Tuning
- PM Motor Auto-Tuning
- ASR and Inertia Tuning

◆ T1: Induction Motor Auto-Tuning

T1 parameters set the Auto-Tuning input data for induction motor tuning.

Note:
- The base frequency of drive dedicated motors and special motors for use with vector control may be lower than the base frequency of general-purpose motors, which is 50 Hz or 60 Hz. In such cases, this lower frequency is used as the value for *E1-06 [Base Frequency]* and *E1-04 [Maximum Output Frequency]* after Auto-Tuning completes. If the maximum output frequency is too low and causes problems, change the setting of *E1-04* after Auto-Tuning completes.
- The following induction motor parameters are set automatically.
  – *E1-xx [V/f Pattern for Motor 1]*
  – *E2-xx [Motor Parameters]*

■ T1-01: Auto-Tuning Mode Selection

No. (Hex.)	Name	Description	Default (Range)
T1-01 (0701)	Tuning Mode Selection	V/f OLV Sets the type of Auto-Tuning.	Determined by A1-02 (Determined by A1-02)

0 : Rotational Auto-Tuning

Default setting if *A1-02 = 2 [Open Loop Vector]*.

1 : Stationary Auto-Tuning 1

2 : Stationary Line-Line Resistance

Default setting if *A1-02 = 0 [V/f Control]*.

■ T1-02: Motor Rated Power

No. (Hex.)	Name	Description	Default (Range)
T1-02 (0702)	Motor Rated Power	V/f OLV Uses the units set in *o1-58 [Motor Power Unit Selection]* to set the motor rated output power.	Determined by o2-04 (0.00 - 650.00 kW)

■ T1-03: Motor Rated Voltage

No. (Hex.)	Name	Description	Default (Range)
T1-03 (0703)	Motor Rated Voltage	V/f OLV Sets the rated voltage (V) of the motor. Enter the base speed voltage for constant output motors.	400 V Class: 400.0 V (400 V Class: 0.0 - 511.0 V)

If you do Auto-Tuning on a drive-dedicated motor or a specialized motor for vector control, the voltage or frequency can be lower than that of a general-purpose motor. Always compare the data from the nameplate or test report with the Auto-Tuning results and check for differences. Enter the voltage necessary to operate the motor in no-load conditions at rated speed for better control precision around rated speed. If the motor test report or the motor nameplate is not available, enter approximately 90% of the motor rated voltage.

If the drive input power supply voltage is low, enter approximately 90% of the input voltage. When the input power supply voltage is low, the current will increase. Make sure that the main power supply capacity is correct and use a molded-case circuit breaker for the drive.

■ T1-04: Motor Rated Current

No. (Hex.)	Name	Description	Default (Range)
T1-04 (0704)	Motor Rated Current	V/f OLV Sets the rated current (A) of the motor.	Determined by o2-04 (10% to 200% of the drive rated current)

Set the motor rated current between 50% and 100% of the drive rated current for the best performance. Enter the current at the motor base speed.

■ T1-05: Motor Base Frequency

No. (Hex.)	Name	Description	Default (Range)
T1-05 (0705)	Motor Base Frequency	V/f OLV Sets the base frequency (Hz) of the motor.	50.0 Hz (0.0 - 120.0 Hz)

When Auto-Tuning is carried out, the value of *T1-05* is set to *E1-04 [Maximum Output Frequency]*. If *T1-05 < 40 Hz*, *E1-04* = 40 Hz. If you operate the drive at a speed that is higher than the base frequency, or if you operate in the field weakening range, set *E1-04* to the maximum output frequency after you complete Auto-Tuning.

■ T1-06: Number of Motor Poles

No. (Hex.)	Name	Description	Default (Range)
T1-06 (0706)	Number of Motor Poles	V/f OLV Sets the number of motor poles.	4 (2 to 48)

■ T1-07: Motor Base Speed

No. (Hex.)	Name	Description	Default (Range)
T1-07 (0707)	Motor Base Speed	V/f OLV Sets the motor base speed for Auto-Tuning (min^{-1} (r/min)).	1450 min^{-1} (r/min) (0 - 24000 min^{-1} (r/min))

■ T1-08: Motor no-load voltage

No. (Hex.)	Name	Description	Default (Range)
T1-08 (0708)	Motor no-load voltage	V/f OLV	40% of motor nominal current

Index

Symbols

Numerics

24 V power supply
  Power supply input terminals 54

A

AC reactor
  Wiring 69
Acceleration and deceleration times
  Switching by external input 258
  Switching by Motor 2 Selection commands 258
  Switching by output frequency 258
Acceleration time
  Parameter 258
  Unit of measurement setting 260
Adjustment of control functionality 109
AFR
  Parameter 312
Alarm 142, 158
Altitude
  Derating 203
  Environment 25
Ambient humidity 25
Ambient Temperature Setting 25
  Derating 203
  Parameter 308
Analog Output
  Gain/Bias Adjustment 290
  Monitor Parameter Selection 290
  Signal Level Selection 290
  Terminal AM Monitor Selection 291
Auto-Tuning 91, 107
  Induction Motor Parameters 327
  Input Data 93
  Interruption 92
  Precautions 91, 107
  Rotational Auto-Tuning Precautions 107
  Stationary Auto-Tuning Precautions 107
  Types 91
Auto-Tuning Error 142, 168

B

Backup
  Parameters (drive to keypad) 104
Backup function 319
Base frequency
  Parameter 269
Base voltage
  Parameter 270
Basic operation
  Get started 88–89
bAT
  Fault 146, 150
  Minor Fault 158
Battery 100
bb 158
bCE
  Fault 146
  Minor Fault 158
boL
  Fault 146
  Minor Fault 158
Brake Sequence 97
Braking Resistor
  Wiring 64
bUS
  Fault 146
  Minor Fault 158
bUSy 158

C

CALL 159
Capacitor Maintenance Setting 320
Carrier frequency 265
  Derating 203
  Parameter 265
CE
  Fault 146
  Minor Fault 159
CF 147
Circulation Fan
  Replacement 186
CoF 147
Control circuit terminal block
  Replacement 191
Control circuit terminals
  Configuration of terminal block 55
  I/O terminals function selection switches 58
  Terminal functions 53
  Wire gauge 56
  Wiring 52
  Wiring procedure for terminal block 57
Control method
  Selection 251
Cooling Fan
  Activation Conditions Setting 308
  Estimated Lifespan 184
  Off Delay Timer 308
  Replacement 186
Copy Function Error 142, 171
CPEr 171
CPF00 147–148
CPF01 147–148
CPF02 148
CPF03 148
CPF06 148–149
CPF08 148
CPF11 148
CPF11 to CPF14 148
CPF14 148
CPF16 148
CPF16 to CPF24 148
CPF24 149
CPF25 149
CPF38 148–149
CPyE 171
Crimp ferrule 56
CrST 159
CSEr 171
Cumulative Operation TimeSetting 319

D

DC reactor
  Wiring 69
dCE1 149
dCE2 149
Deceleration time
  Parameter 258
  Unit of measurement setting 260
Delete
  Delete Backed-up Parameters 106
Derating

Altitude....203
Ambient Temperature Setting....203, 308
Carrier frequency....203
Enclosure Type....309–310
External Cooling Fin....309–310
Side-by-side....309–310
dFPS....171
Diagnosing and Resetting Faults....172
DIP switch....58
Disposal
Drive....197–198
Packing material....197–198
Drive
Control Circuit Terminal Block Replacement....191
Disposal....197–198
Exterior Dimension Diagram (IP20)....204
Exterior Dimensions Diagram (IP20)....205
Initialization....251
Initialize Parameters....251
Inspection....182
Long-Term Storage....193
Rating (400 V)....200
Drive mode....84
Drive Mode Unit Monitor Select....315
Drive Model Selection....317
Drive watt loss....30
dv/dt Tune Factor....311
dv7....149
Dwell function
Parameter....256

E

EF....159
EF0
Minor Fault....159
EF1
Fault....149
Minor Fault....159
EF2
Fault....149
Minor Fault....160
EF3
Fault....150
Minor Fault....160
EF4
Fault....150
Minor Fault....160
EF5
Fault....150
Minor Fault....160
EF6
Fault....150
Minor Fault....160
EF7
Fault....150
Minor Fault....160
Elapsed Operating Time Selection....319
Electrolytic Capacitor
Estimated Lifespan....184
Enclosure Type
Derating....309–310
End1....168
End2....168
End3....168
End4....168
End5....168
End6....168
End7....168
EP24v....161
Er-01....168
Er-02....169
Er-03....169
Er-04....169
Er-05....169
Er-08....169
Er-09....170
Er-11....170
Er-12....170
Er-13....170
Er-15....170
ERF type braking resistor
Wiring....64
Err....150
Error Code List....143
Exterior Dimension Diagram (IP20)
Drive....204
Exterior Dimensions Diagram (IP20)
Drive....205
External 24 V power supply
Power supply input terminals....54
External Cooling Fin
Derating....309–310

F

Fan Operation Time Setting....320
Fast Stop Time
Parameter....260
Fault....142, 146
Fault Code List....143
Fault Reset....172
Fault Restart
Parameter....299
Fine tuning....109
Freq Ref Setting Method Select....318
Frequency Agreement
Parameter....298
Fuse rating....73

G

Getting set up....88–89
GF....151
Protection Functions....308
Ground
Drive....45
Ground Fault Detection
Protection Functions....308

H

HCA....161
Alarm Settings....310–311
How to read the model number....20
Humidity
Environment....25

I

iFEr....171
IGBT Maintenance Setting....320
Induction Motor
Motor parameters....270
Input Phase Detection
Protection Functions....307
Input Phase Loss Detection....311
Input Phase Loss Detection Time....307, 310
Input voltage
Parameter....269
Inspection
Drive....182
Inspection Operation....97

Installation
Front cover 31
Keypad 33
Installation environment 25
J
Jerk characteristics
Parameter 261
Jumper switch 58
K
Keypad
Installation 33
Language selection 250
LED status 79
Method of operation 79
Remove 33
Keypad Display 314
Keypad Display Selection 315
Keypad Operation 316
Keypad-related settings 314
kWh Monitor Initialization 321
L
L24v 161
Language selection 250
LF 151
Protective function 307
LF2 151
LKEB type braking resistor unit
Wiring 64
LO/RE Key Function Selection 316
LOCAL/REMOTE indicator 79
LoG 161
LT-1 161
LT-2 161
LT-3 161
LT-4 162
M
Main circuit terminals
Configuration of terminal block 42
Line voltage drop 45
Wire gauge 45
Wiring 41
Wiring procedure for terminal block 48
Maintenance Period 319
Maximum Output Frequency
Parameter 269
Maximum Output Voltage
Parameter 269
MCCB 66
MEMOBUS/Modbus Communications
Serial communication terminals 55
Setting for termination resistor 61
Middle Output Frequency
Parameter 269
Middle Output Frequency 2
Parameter 270
Middle Output Frequency Voltage
Parameter 270
Middle Output Frequency Voltage 2
Parameter 270
Minimum output frequency
Parameter 270
Minimum Output Voltage
Parameter 270
Minor Fault 142, 158
Minor Fault Code List 143
Molded-case circuit breaker 66
Monitors
Display procedure 103
Motor
Wiring 41
Wiring distance 45
Motor Overload
Electric Thermal Protection Operation Time 295
Protection Functions 293
Motor parameters 270
Motor Parameters 269
Motor parameters (induction motors) 270
Leakage Inductance 272
Line-to-Line Resistance 272
Motor Iron Loss 273
Motor Iron-Core Saturation Coefficient 1 272
Motor Iron-Core Saturation Coefficient 2 272
Motor rated power (kW) 273
No-load Current 271
Number of motor poles 271
Rated current 271
Rated Slip 271
MotorDirect@PowUpWhenUsingKeypad 318
N
Nameplate 20
ndAT 171
Noise filter
Wiring 71
NumOfRunCommands Counter Initial 321
O
oC 151
Off-Delay Timer 255
oH
Alarm Settings 306
Fault 152
Minor Fault 162
oH1 152
oH2 162
Alarm Settings 306
oH3
Minor Fault 162
oL1 152
oL2 153
Protective function 308
oL3
Fault 154
Minor Fault 162
oL4
Fault 154
Minor Fault 163
oL5
Fault 154
On-Delay Timer 255
Ope Select @Keypad is Disconnect 318
oPE01 166
oPE02 166
oPE03 166
oPE08 167
oPE10 167
oPE33 167
oPr 154
Options 206
oS
Minor Fault 163
Output Phase Loss Detection
Protective function 307
Output Phase Loss Detection Time

Alarm Settings 311
ov
Fault 154
Minor Fault 163
ovEr 163
Overtorque detection
Parameter 300

P

Parameter
Access Level Selection 250
Automatic selection 253
Backup (drive to keypad) 104
Changing setting values 104
Checking modified parameters 103
Delete Backed-up Parameters 106
Restore (keypad to drive) 105
Setup mode 103
User-set 253
Verify (keypad and drive) 105
Parameter Setting Errors 142, 166
Parameter Setup 99
PASS 163
Password
Setting 252
Verification 252
PC
Connection procedure 63
Peripheral Devices 206
PF 155, 163
Protection Functions 307
PF dv/dt Level 311
PF Min Current 311
PGo
Fault 155
Phase Order Selection 255
Power loss 30
Programming mode 84
Protection Functions
DC bus undervoltage 295
Drive Overheating 306
GF 308
Ground Fault Detection 308
HCA 310–311
Input Phase Detection 307
Motor Overload 293
oH 306
oH2 306
Output Current Overload 310–311
Output Phase Loss Detection Time 311
PF 307
Software Current Limit Selection 309
Uv1 295
Protective function
LF 307
oL2 308
Output Phase Loss Detection 307
Overload 308

R

Rating (400 V)
Drive 200
rdEr 171
Remove
Front cover 31
Keypad 33
Rescue Operation 98–99
RESET key 79
Residual Current Monitor/Residual Current Device
RCM/RCD 66
Wiring 66
Restore
Parameters (keypad to drive) 105
rF 155
rH 155
RJ-45 connector 79
Rotational Auto-Tuning
Precautions 107
rr 156
RUN indicator 79
RUN key 79

S

SC 156
SCF 156
SE 164
Serial communication terminals
MEMOBUS/Modbus Communications 55
Setup Procedure for Elevator Applications 94
Accel Ramp and Jerk Settings 96
Battery 100
Brake Sequence 97
Decel Ramp and Jerk Settings 96
Down Commands 94
Inspection Operation 97
Multi-Function Terminal Setup 96
Parameter Setup 99
Rescue Operation 98–99
Speed Reference Selection 94
Speed Selection 94
Terminal Setup 96
Uninterruptable Power Supply 99
Up Commands 94
UPS 99
Wiring Examples 99
Side-by-side
Derating 309–310
Slip compensation
Parameter 262
Softcharge Relay Maintenance Set 320
Software Current Limit Selection
Protection Functions 309
Speed Agreement
Parameter 298
Speed Detection
Parameter 298
Speed reference
LOCAL/REMOTE Run selection 266
Switching between LOCAL/REMOTE 266
Stall Prevention function
Parameter 296
Stationary Auto-Tuning
Precautions 107
SToF 164
STOP key 79
STOP Key Function Selection 317
Stopping Method Selection 254
Surge protective device
Connection 70

T

Temperature
Environment 25
Terminal block 55
Configuration of main circuit terminal block 42
Control circuit terminal block functions 53

I/O terminals function selection switches 58
Terminal function selection 60
Terminal A2 ..... 58, 60
Terminal AM 58, 61
Termination resistor
Setting switch 61
Test run
Fine tuning 109
Procedure 88–89
Thermal overload relay
Connection 67
Tightening torque
Control circuit terminals 56
Main circuit terminals 45
TiM
Fault 156
Minor Fault 164
Timer function
Parameter 255
Torque Compensation
Parameter 263
Torque limit function
Parameter 304
Troubleshooting
Code Displayed 143
No Code Displayed 173
Troubleshooting without Fault Display 173
TrPC 164
Tuning 258

U

U2, U3 Initialization 320
UL3
Fault 156
Minor Fault 164
UL4
Fault 156
Minor Fault 164
Undertorque detection
Parameter 300
Uninterruptable Power Supply 99
Unit of measurement setting
Acceleration and deceleration times 260
Up/Down command 254
UPS 99
USB port 79
Connecting a PC 63
User Monitor Select afterPowerUp 315
User Parameter Default Value 317
User-Set Display Units Max Value 316
User-SetDisplayUnits Dec Display 316
Uv 164
Uv1 156
Detection level settings 295
Uv2 157
Uv3 157

V

vAEr 171
Verify
Parameters (keypad and drive) 105
vFyE 171
Vibration-resistant 25

W

WEEE 198
Wire gauge
Control circuit terminals 56
Main circuit terminals 45
Voltage drop 45
Wiring 66
AC reactor 69
Braking Resistor 64
Checklist 74
Control circuit terminal block 57
Control circuit terminals 52
DC reactor 69
Main circuit terminal block 48
Main circuit terminals 41
Motor 41
Noise filter 71
Thermal overload relay 67
Wiring distance
Drive and motor 45
Wiring Examples 99

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In the event that the end user of this product is to be the military and said product is to be employed in any weapons systems or the manufacture thereof, the export will fall under the relevant regulations as stipulated in the Foreign Exchange and Foreign Trade Regulations. Therefore, be sure to follow all procedures and submit all relevant documentation according to any and all rules, regulations and laws that may apply.
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